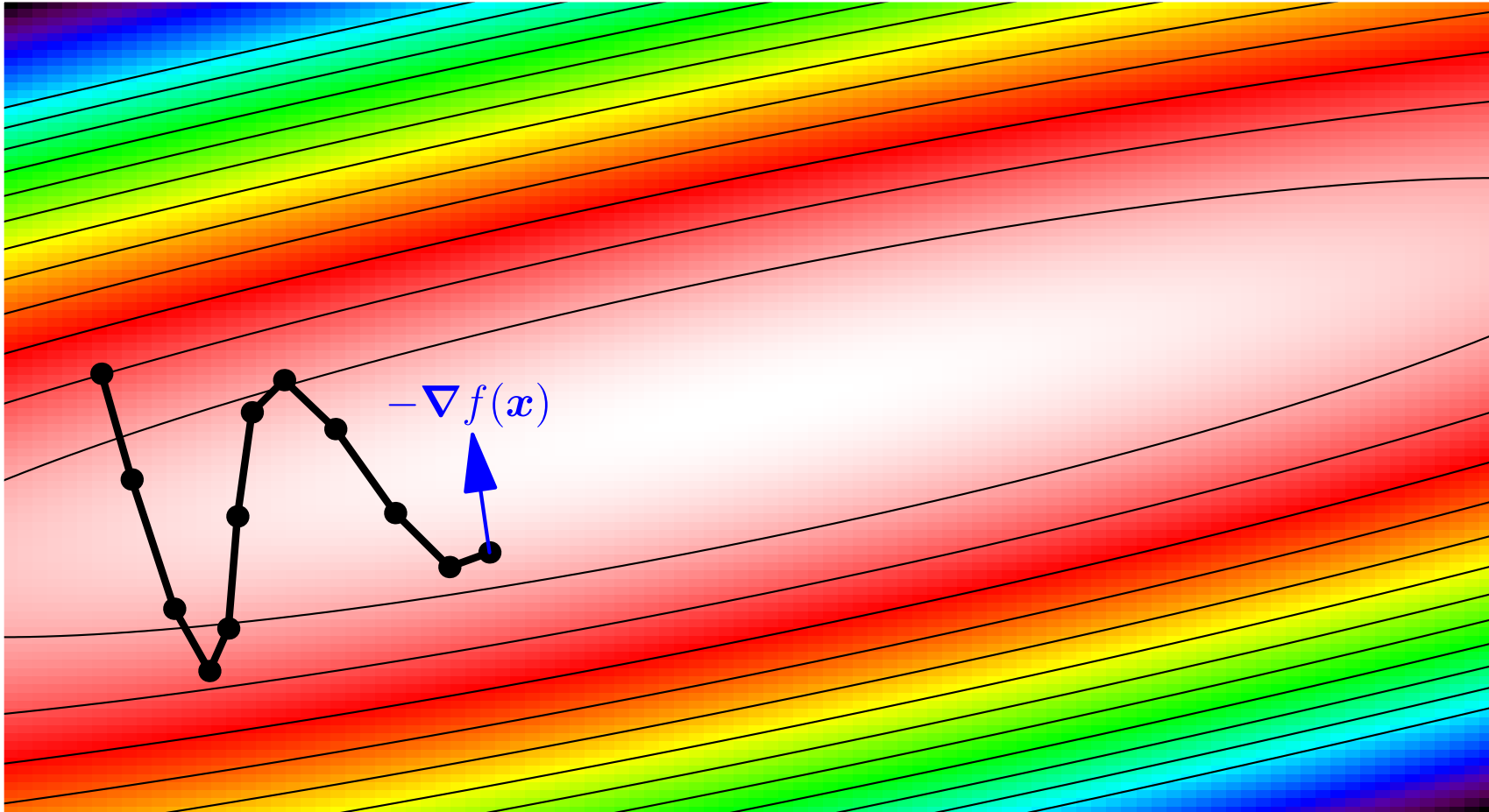


Mathematics of Machine Learning

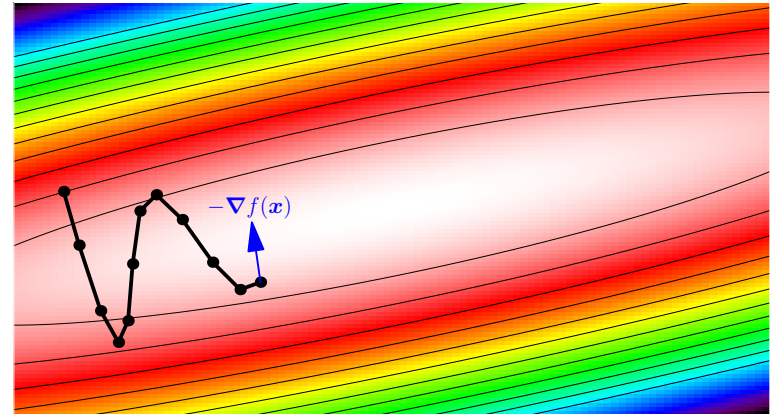
Stochastic Gradient Descent



SGD, momentum, step size, ADAM

Outline

1. **SGD**
2. Momentum
3. Loss Landscapes



The Next Step

- In most machine learning problems we design a network and a loss function
- The rest requires simple optimisation
- Although Newton and Quasi-Newton methods ensure rapid convergence for many tasks this apparent advantage is delusionary
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Mini-Batch Learning

- Rather than computing the gradient of the loss function, e.g.

$$\nabla L_{\mathcal{D}}(\boldsymbol{w}) = \nabla \sum_{(\boldsymbol{x}, y) \in \mathcal{D}} L(f(\boldsymbol{x}|\boldsymbol{w}), y)^2$$

- We often compute an approximation

$$\nabla L_{\mathcal{B}}(\boldsymbol{w}) = \nabla \sum_{(\boldsymbol{x}, y) \in \mathcal{B}} L(f(\boldsymbol{x}|\boldsymbol{w}), y)^2$$

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Automatic Differentiation

- For gradient descent we need to compute the gradient
- For most of my life this was just a pain
- You guys have it easy, modern deep learning frameworks do this for you automatically
- This is a *game changer*! We are no longer afraid of large models
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Convergence

- Asymptotic convergence of gradient descent (and SGD) is slower than quasi-Newton methods
- But there are three reasons why we don't really care
 - ★ For complex models we spend a lot of time before we are close to a quadratic minima where asymptotic convergence kicks in
 - ★ These days we often use ReLUs (rectified linear units) which are non-analytic. The convergence we discussed depends on a Taylor expansion that assumes analyticity
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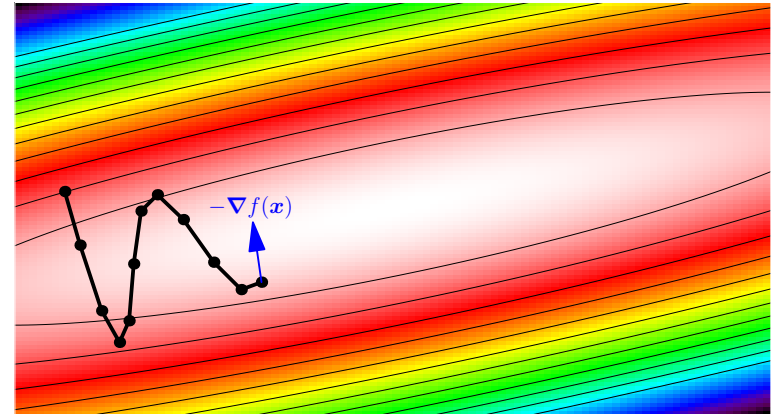
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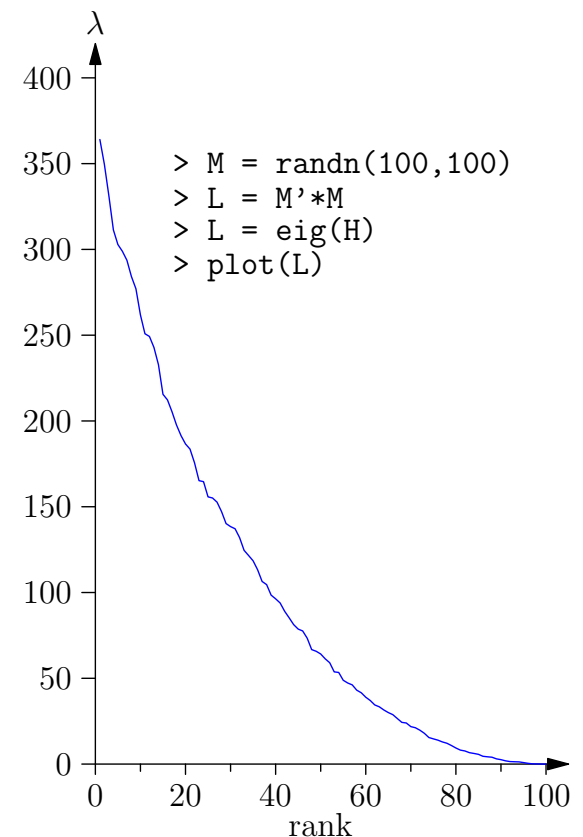


Step Size

- The optimal step size depends on the gradient and the curvature (second derivative)
- For a quadratic minima the minimum is given by

$$x^* = x - \frac{f'(x)}{f''(x)}, \quad \mathbf{x}^* = \mathbf{x} - \mathbf{H}^{-1} \nabla f(\mathbf{x})$$

- In high dimensions the Hessian $\mathbf{H}$ will have a spectrum of eigenvalues
- This means there are different scales

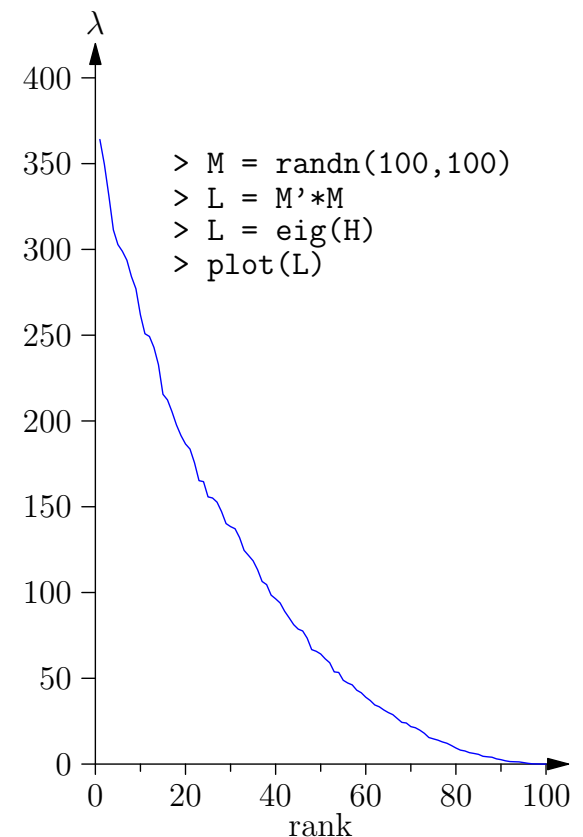


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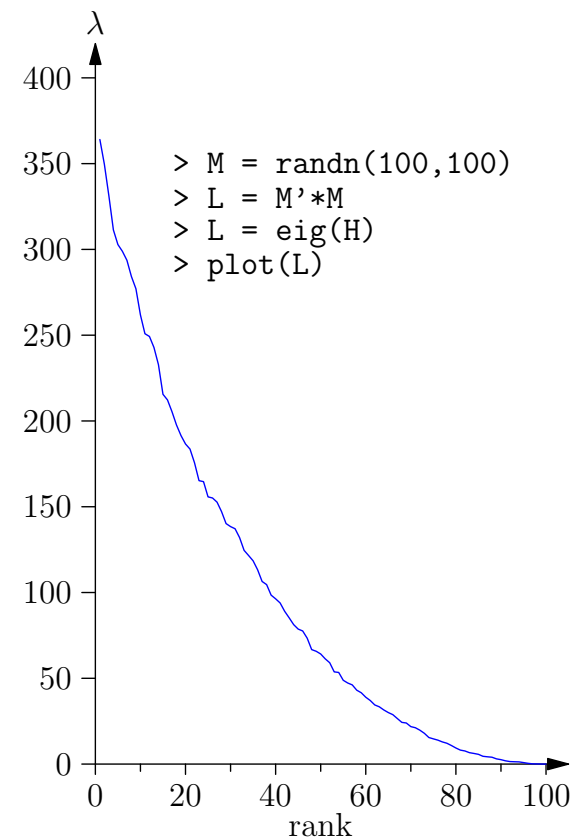


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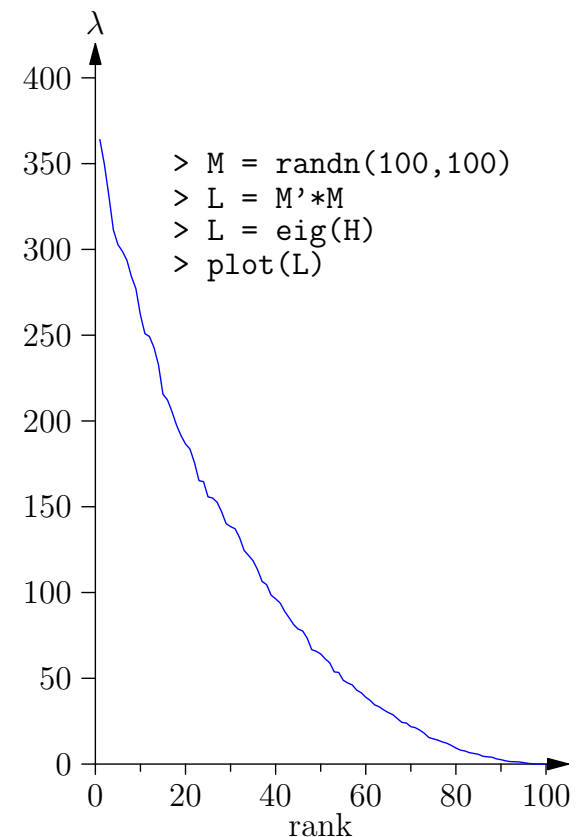
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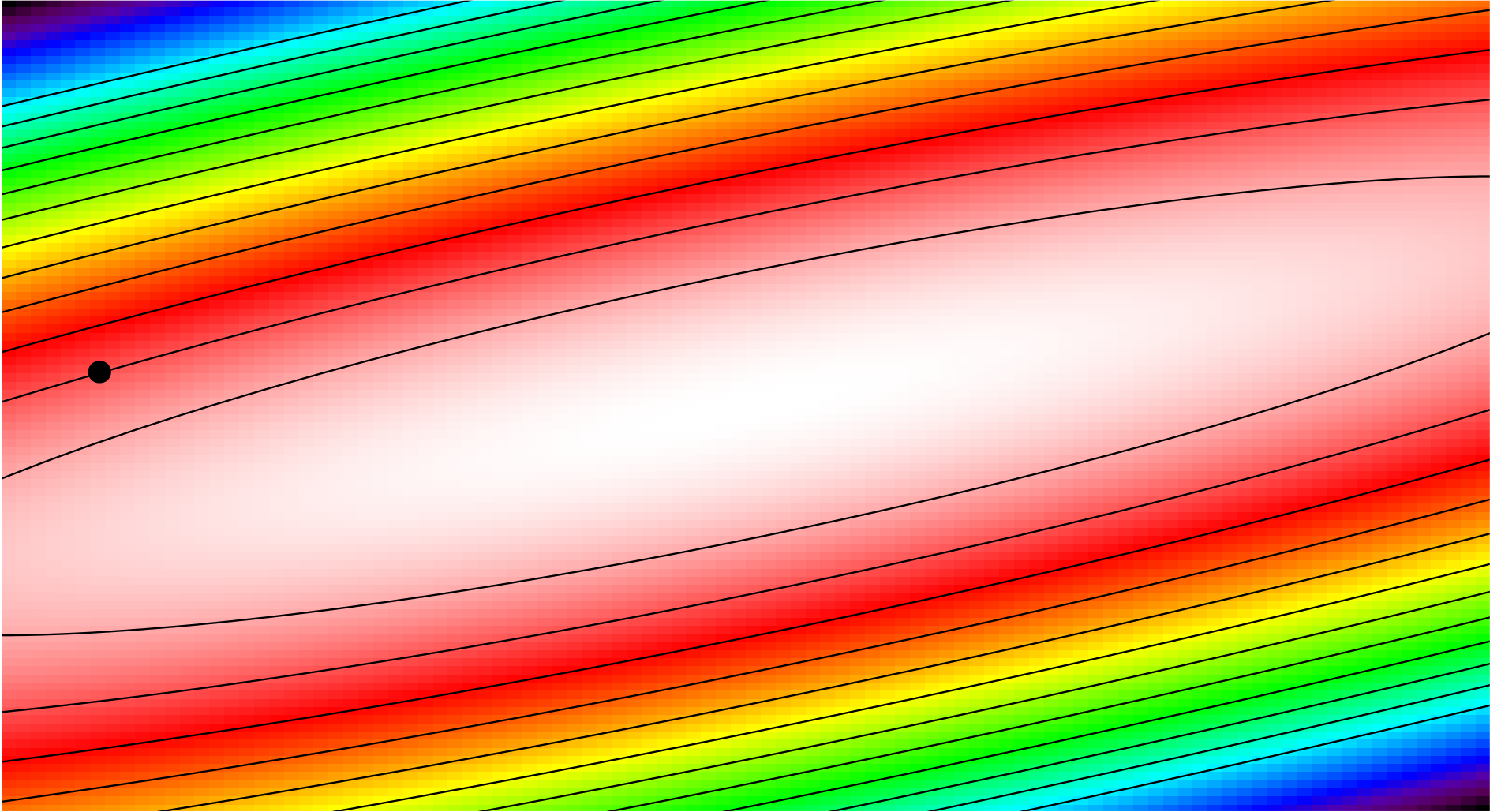
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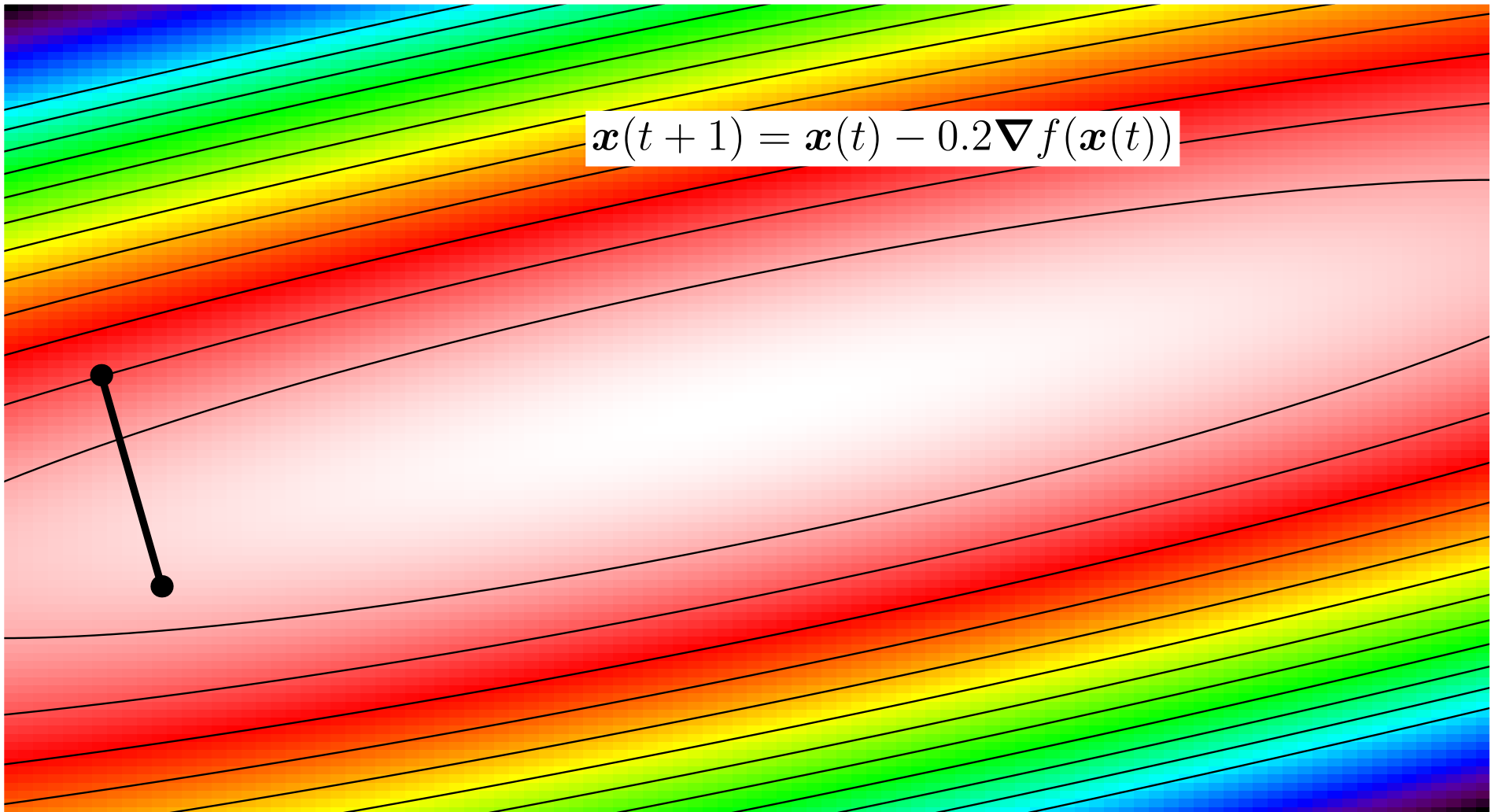
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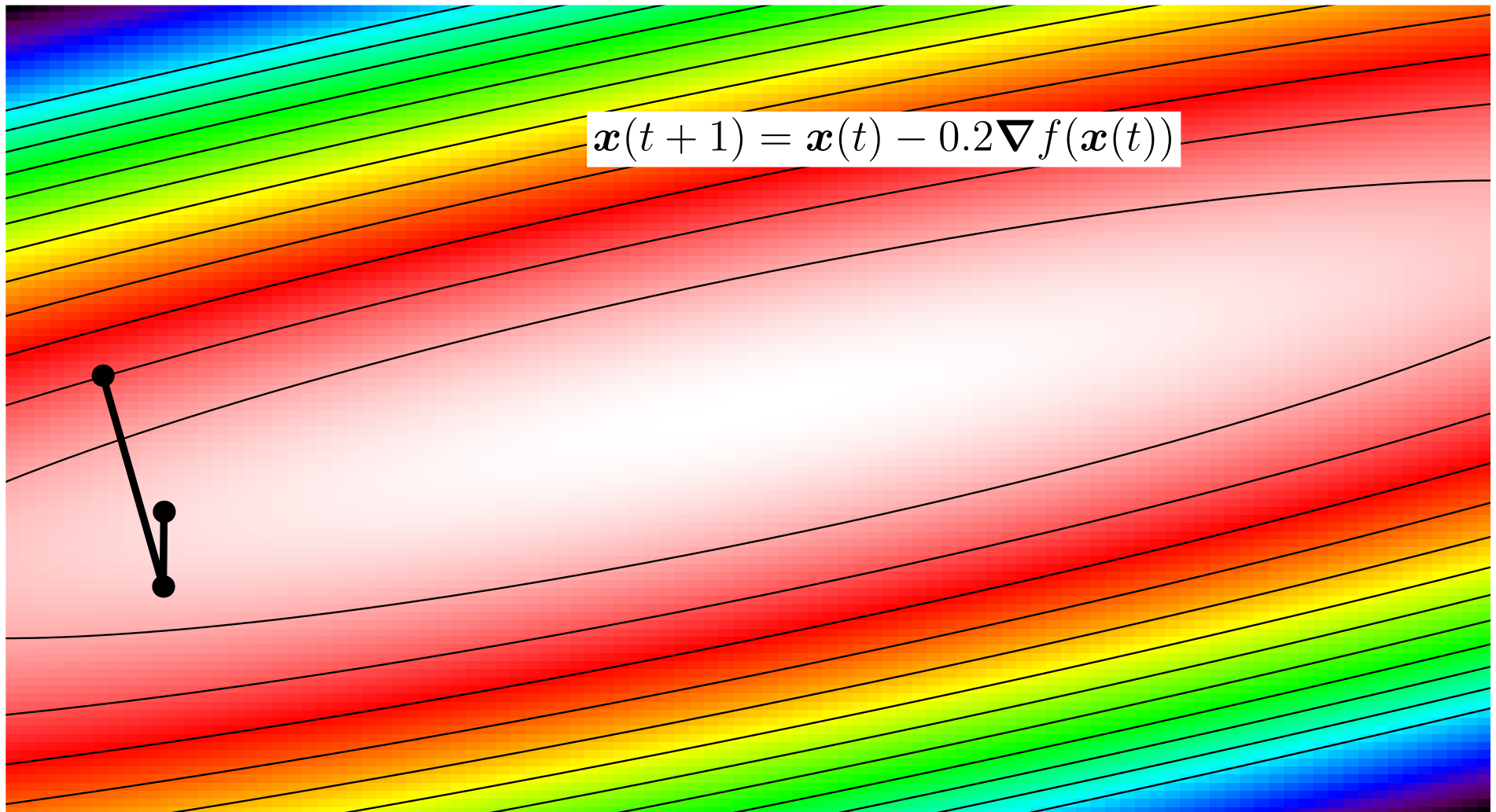
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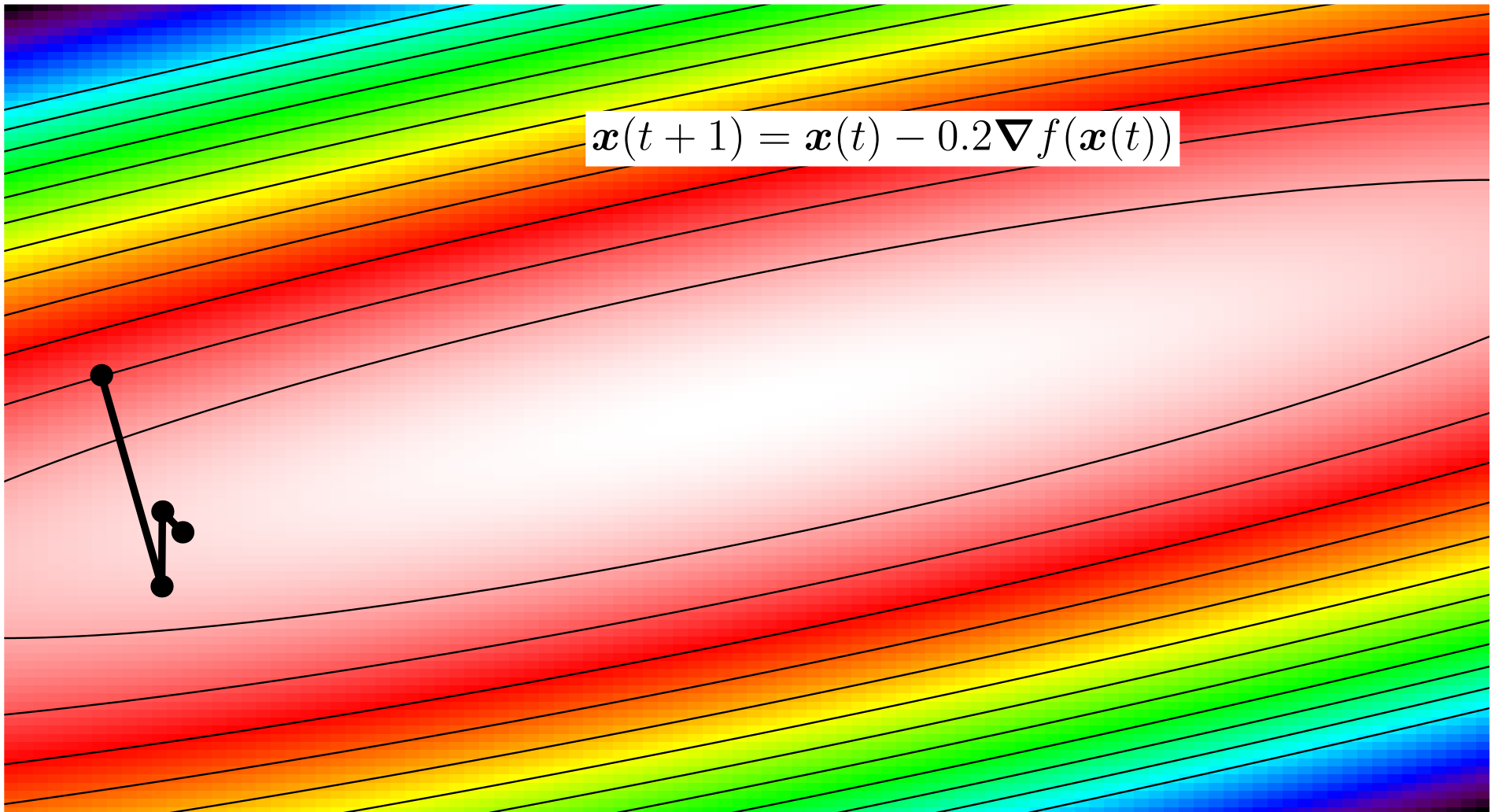
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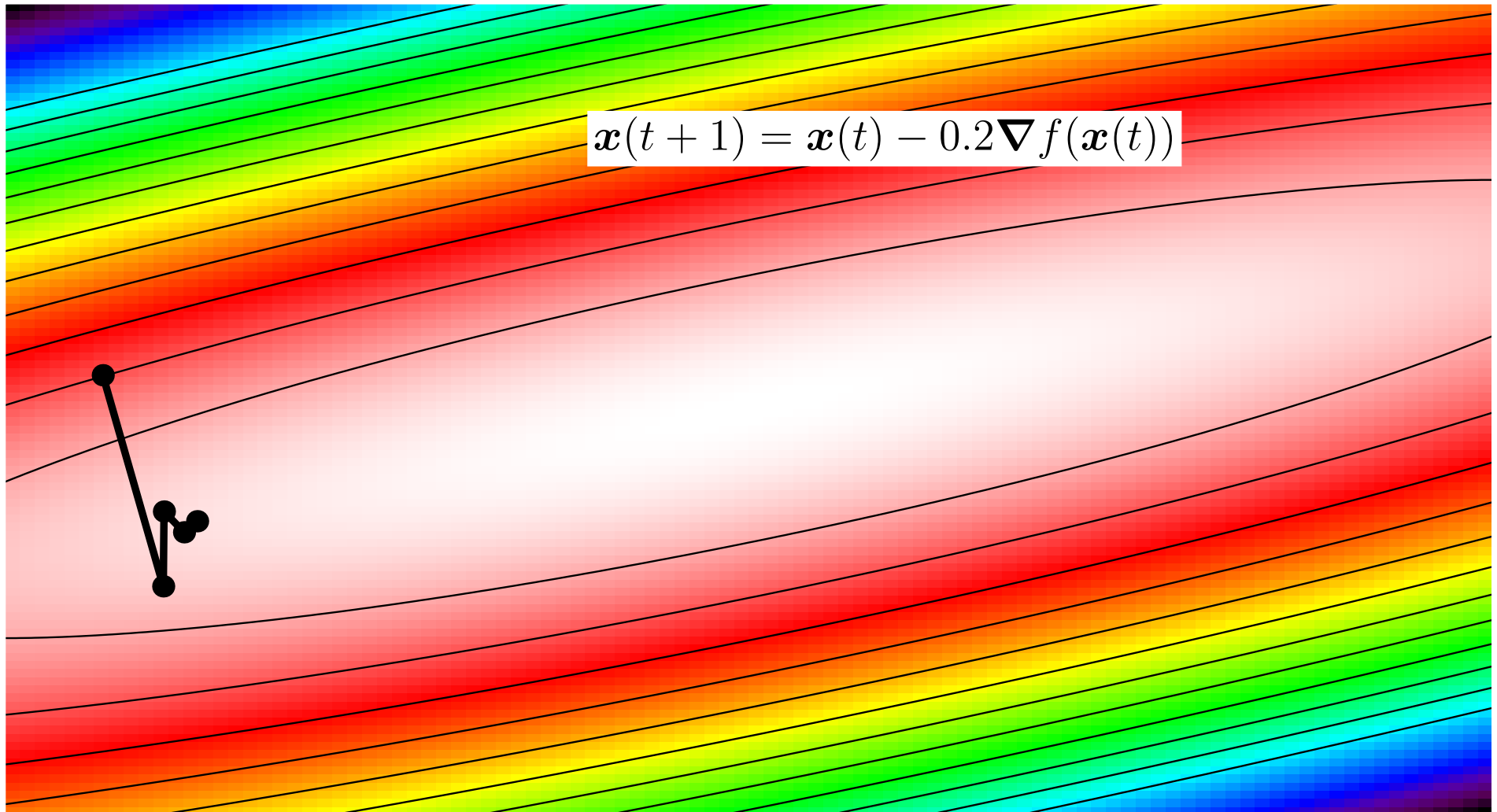
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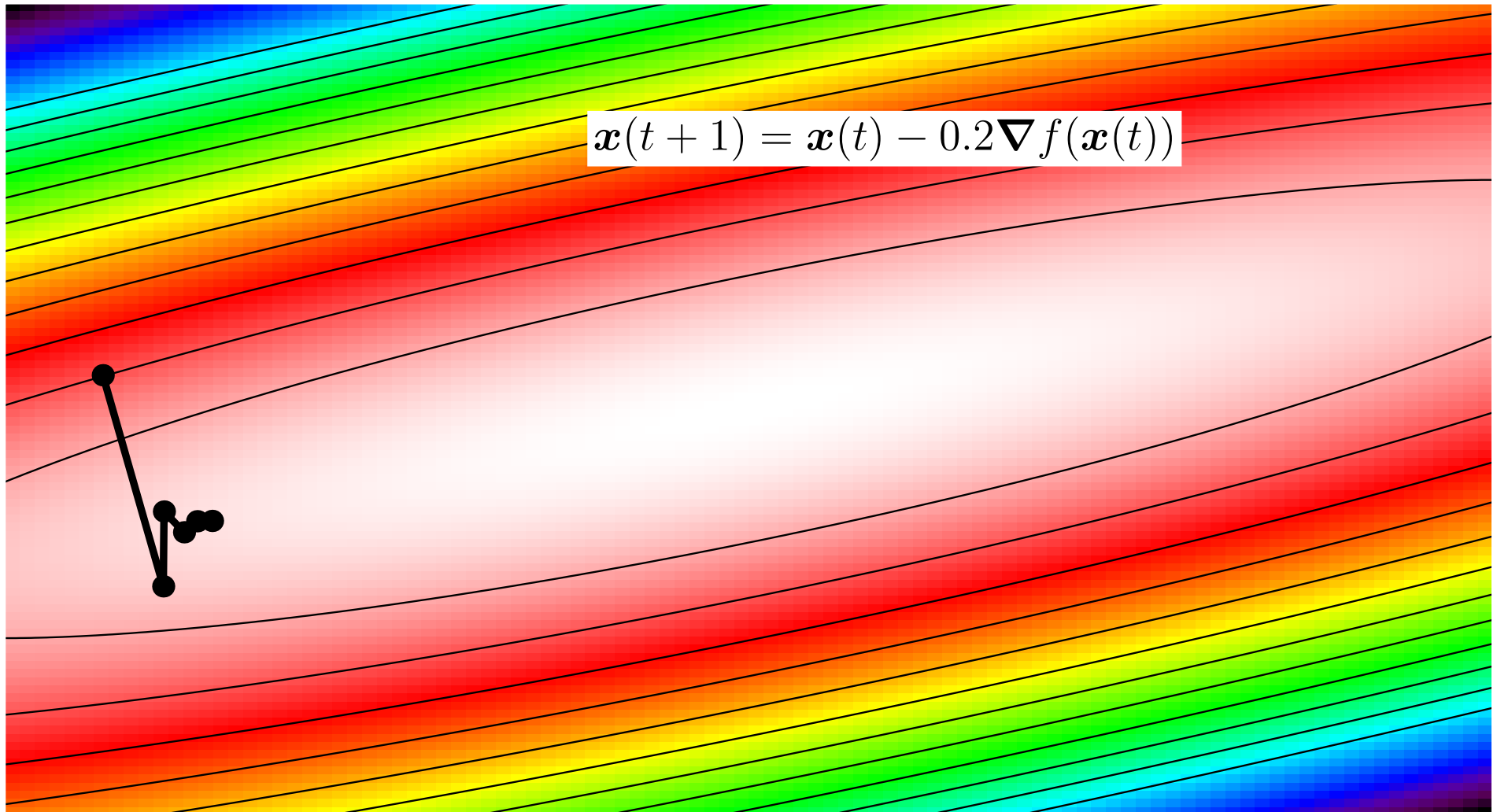
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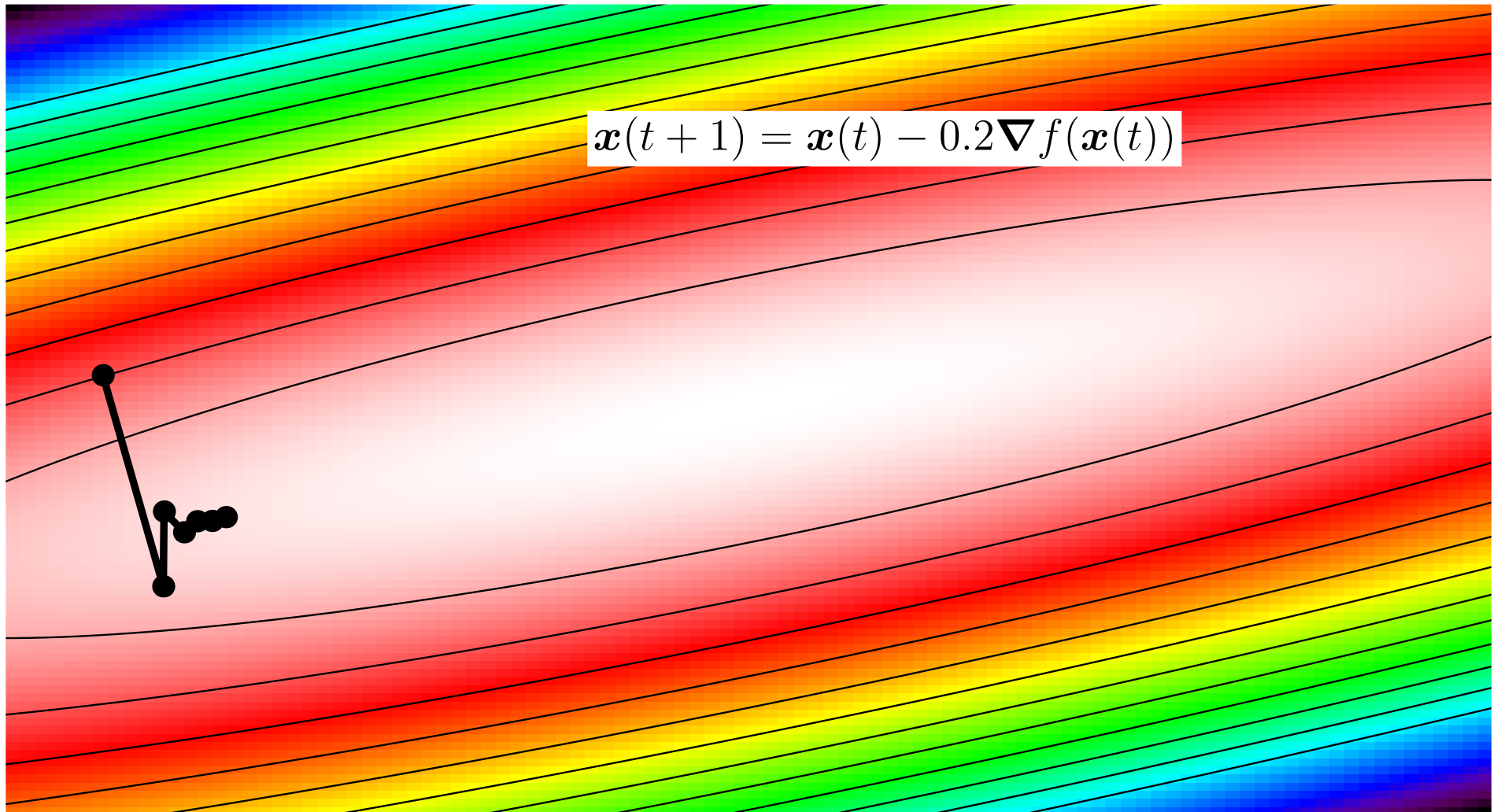
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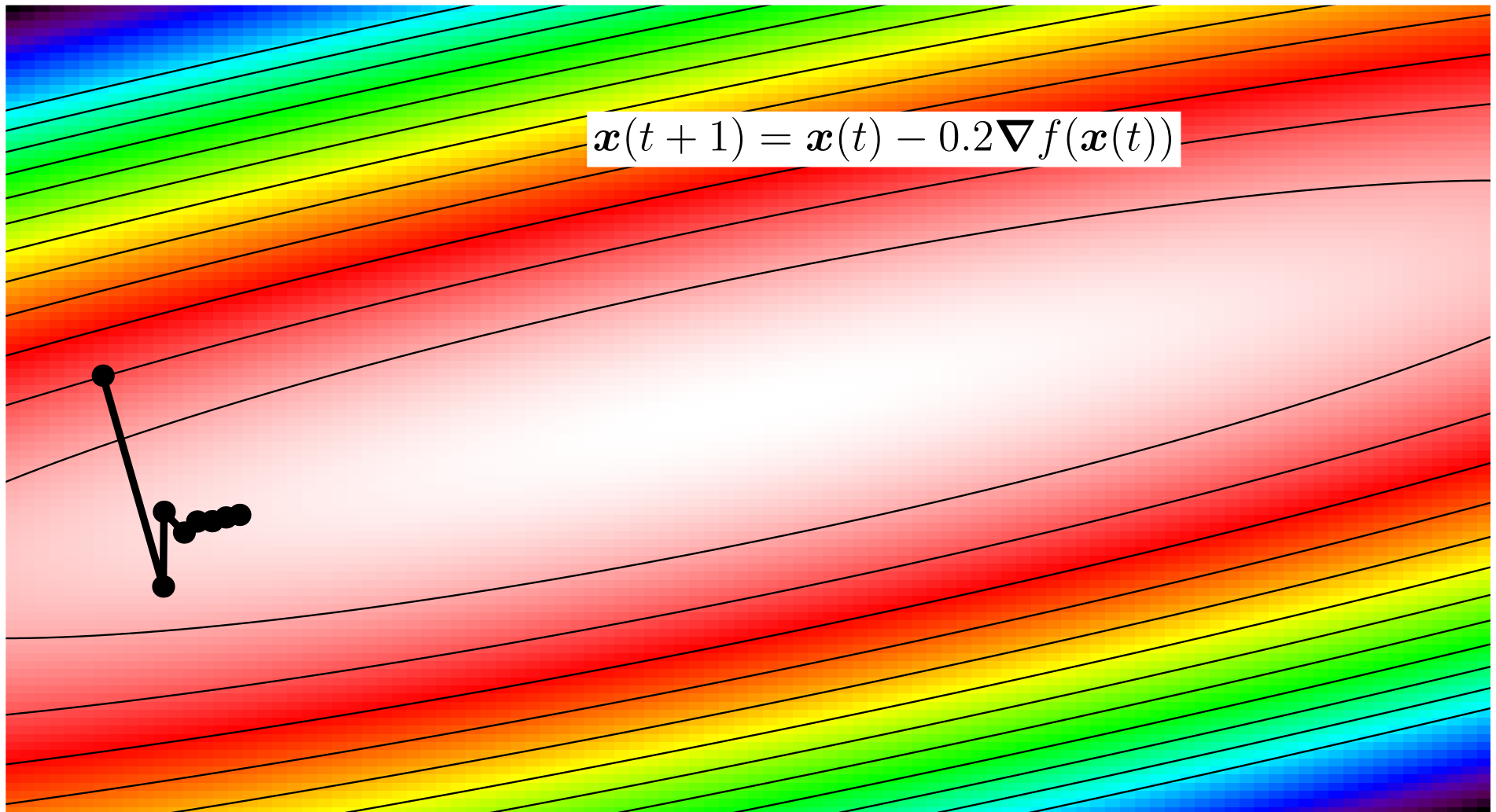
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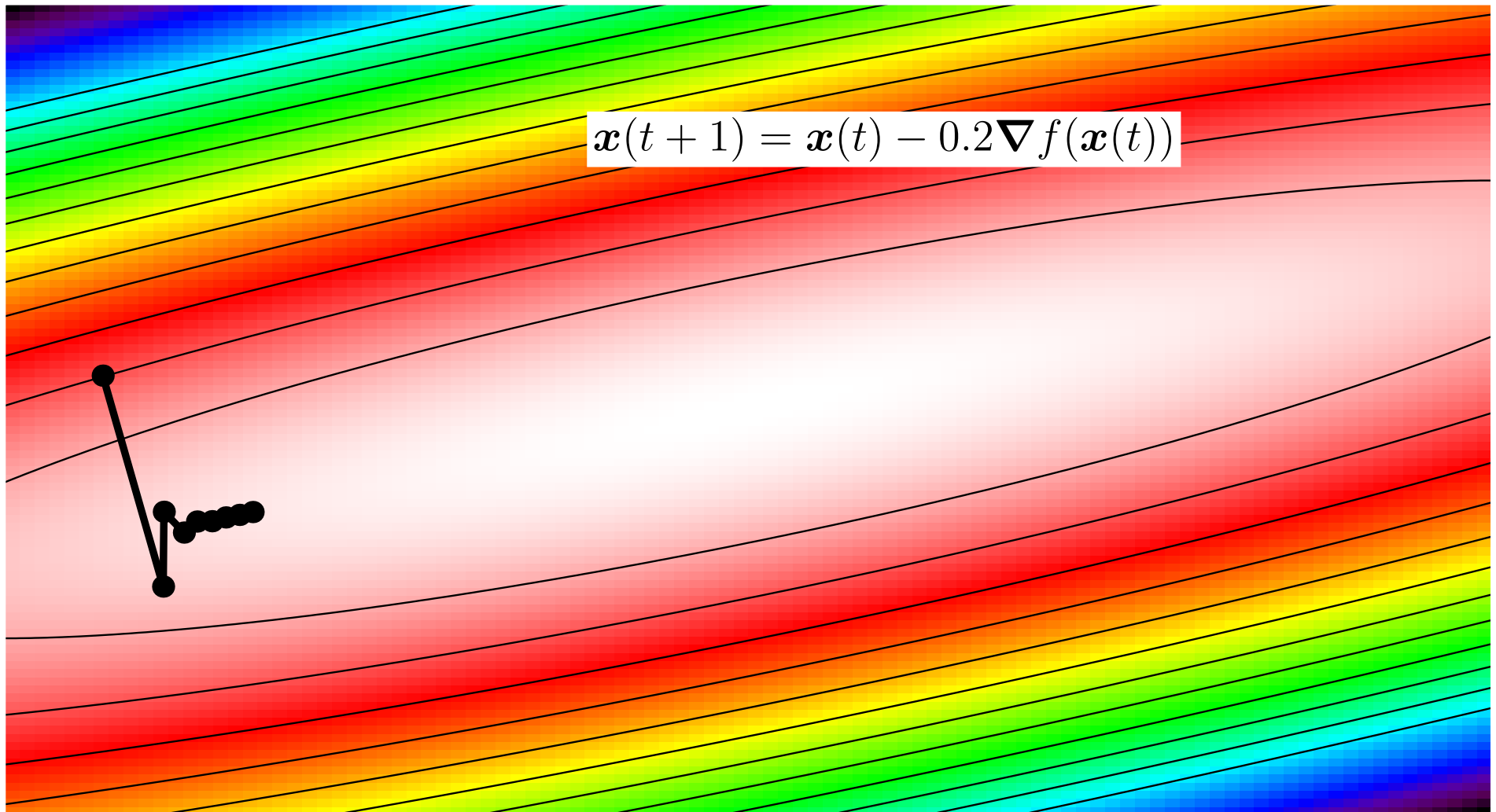
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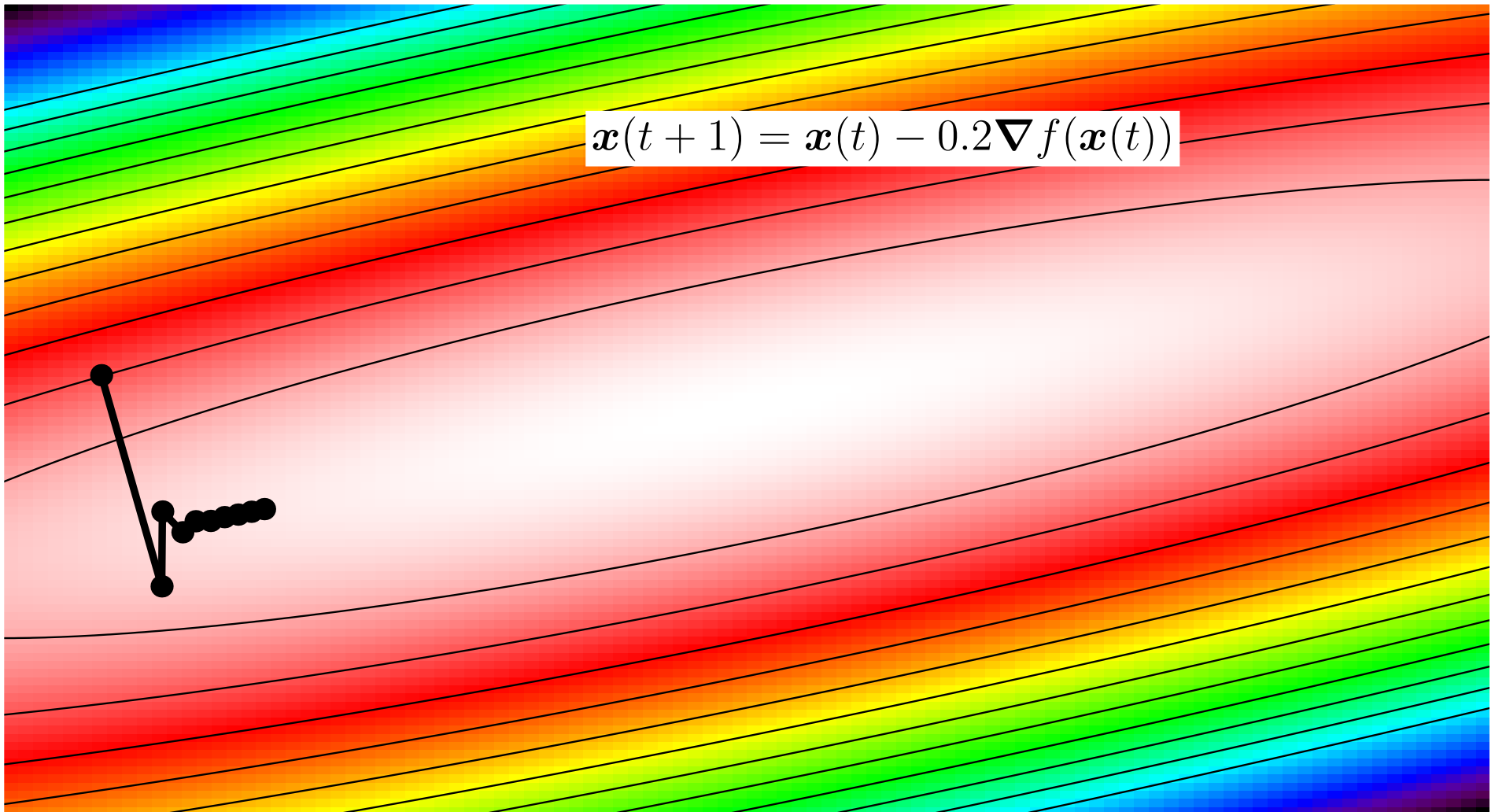
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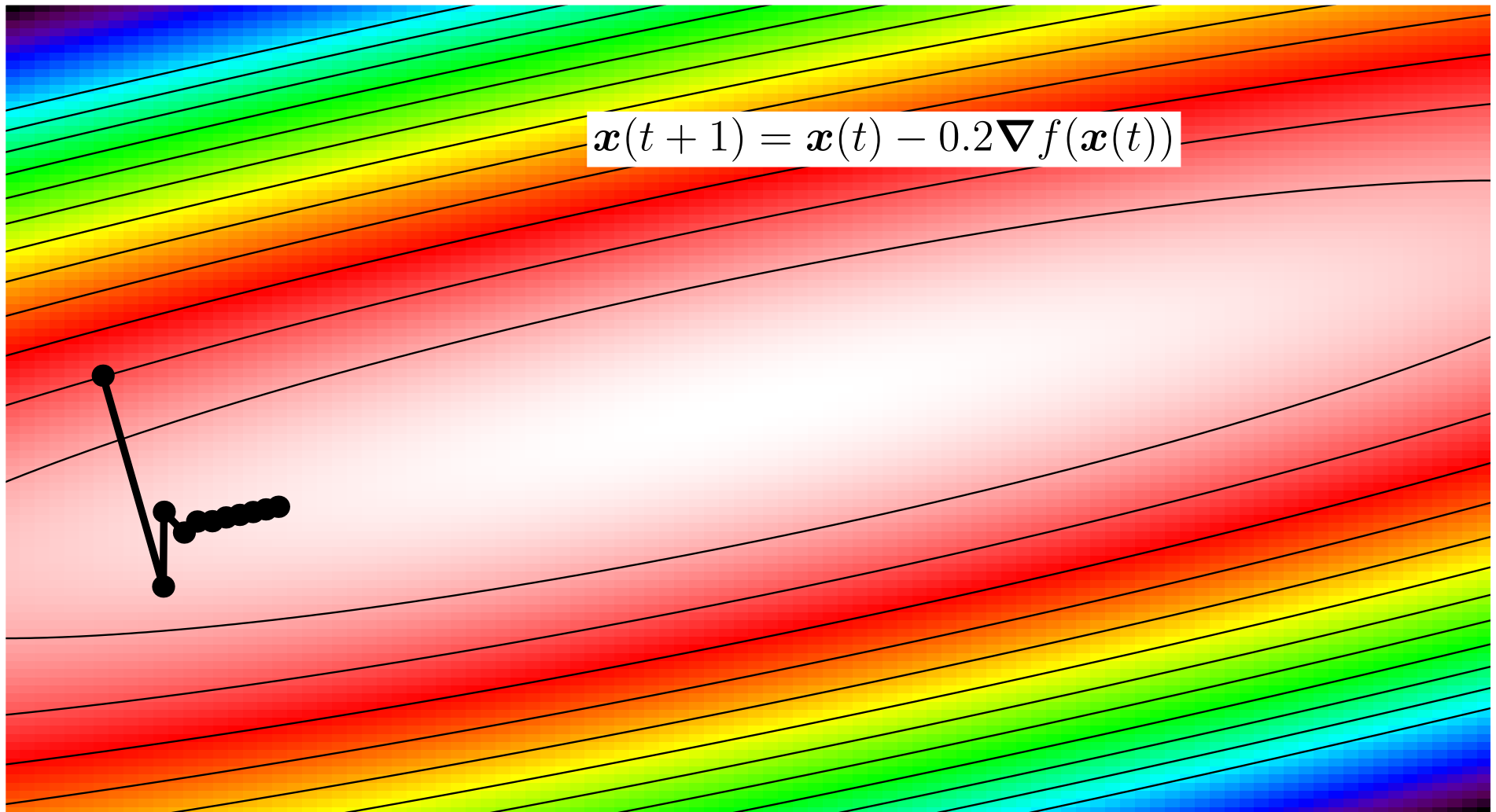
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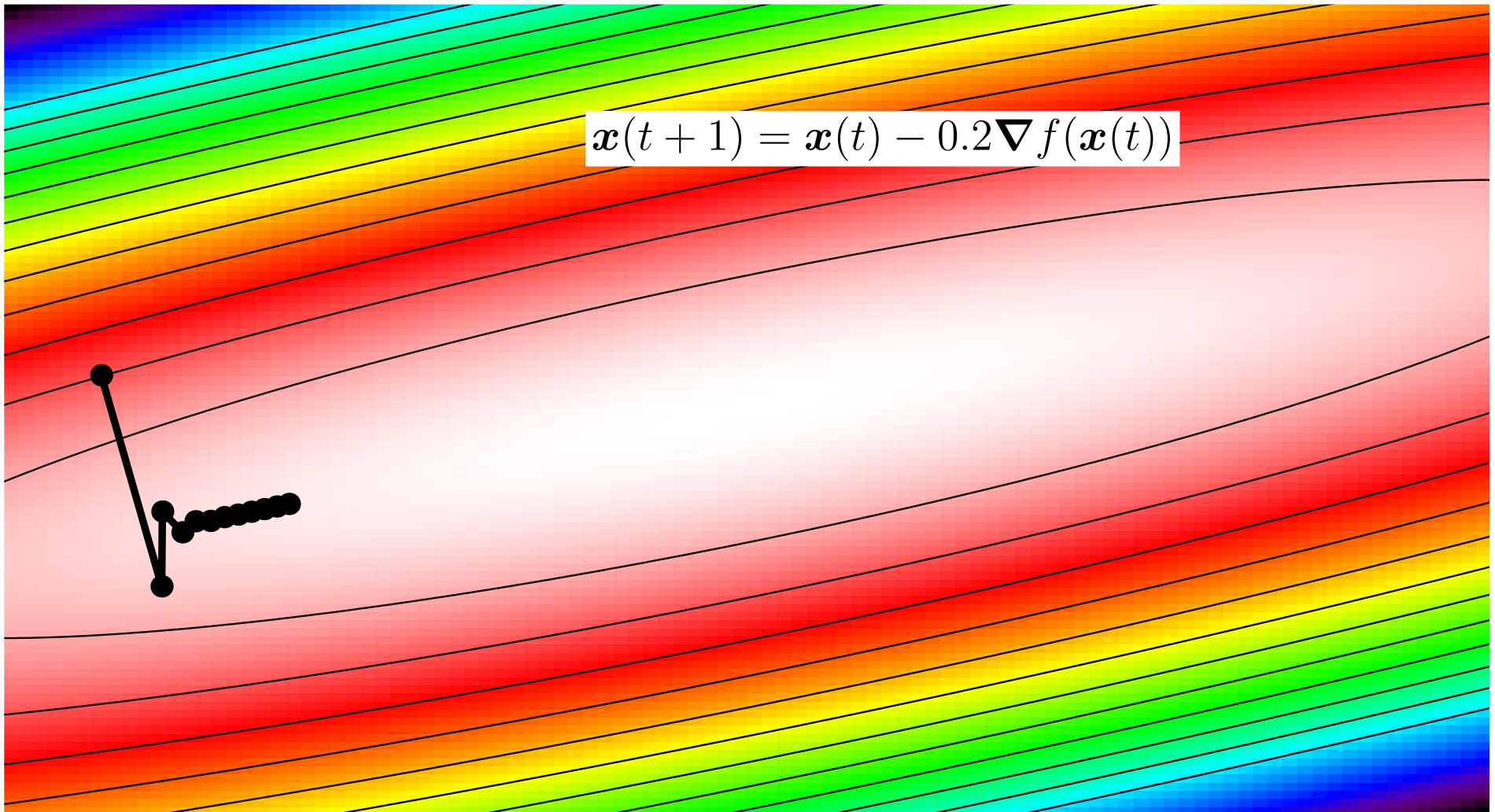
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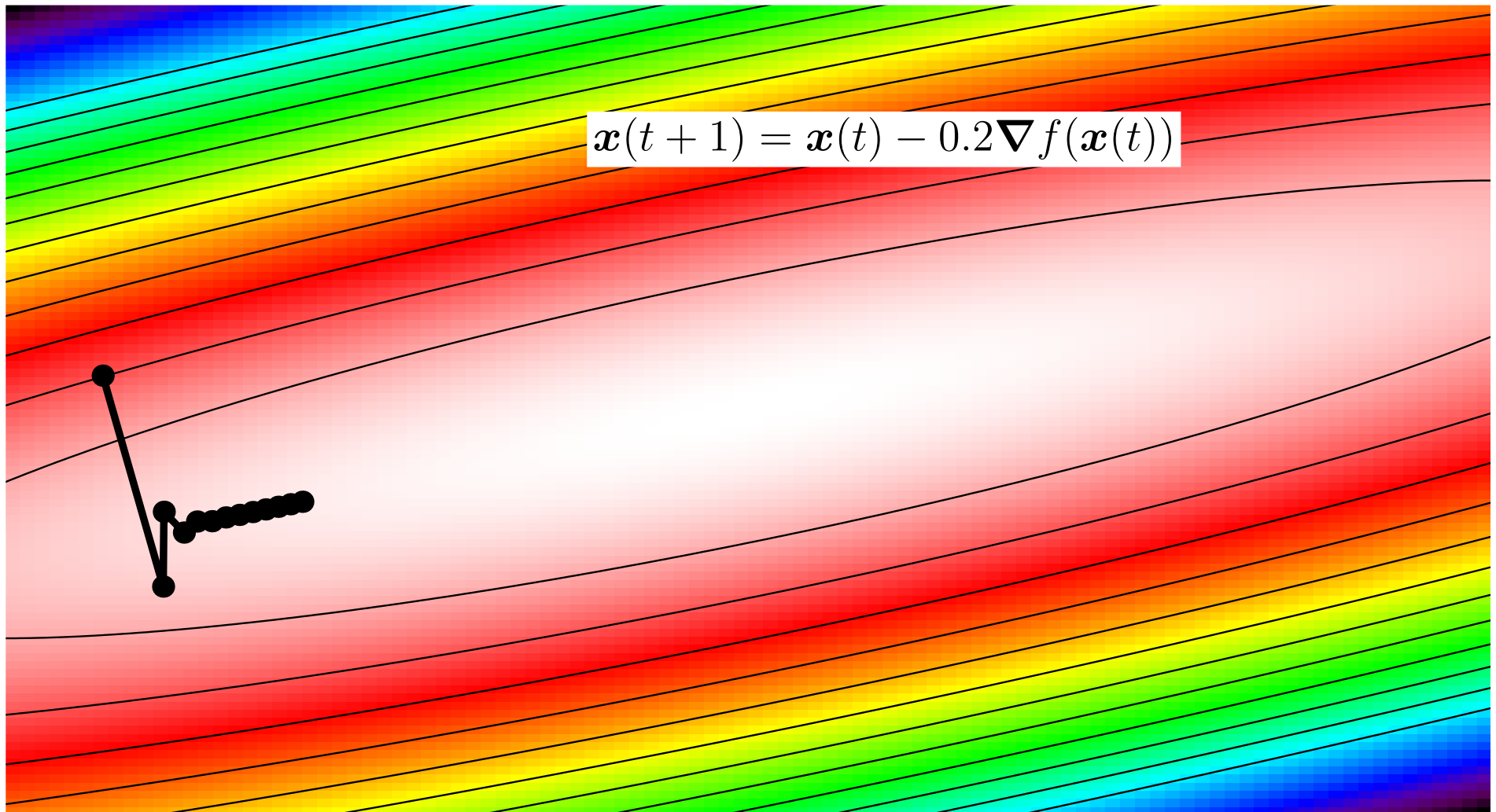
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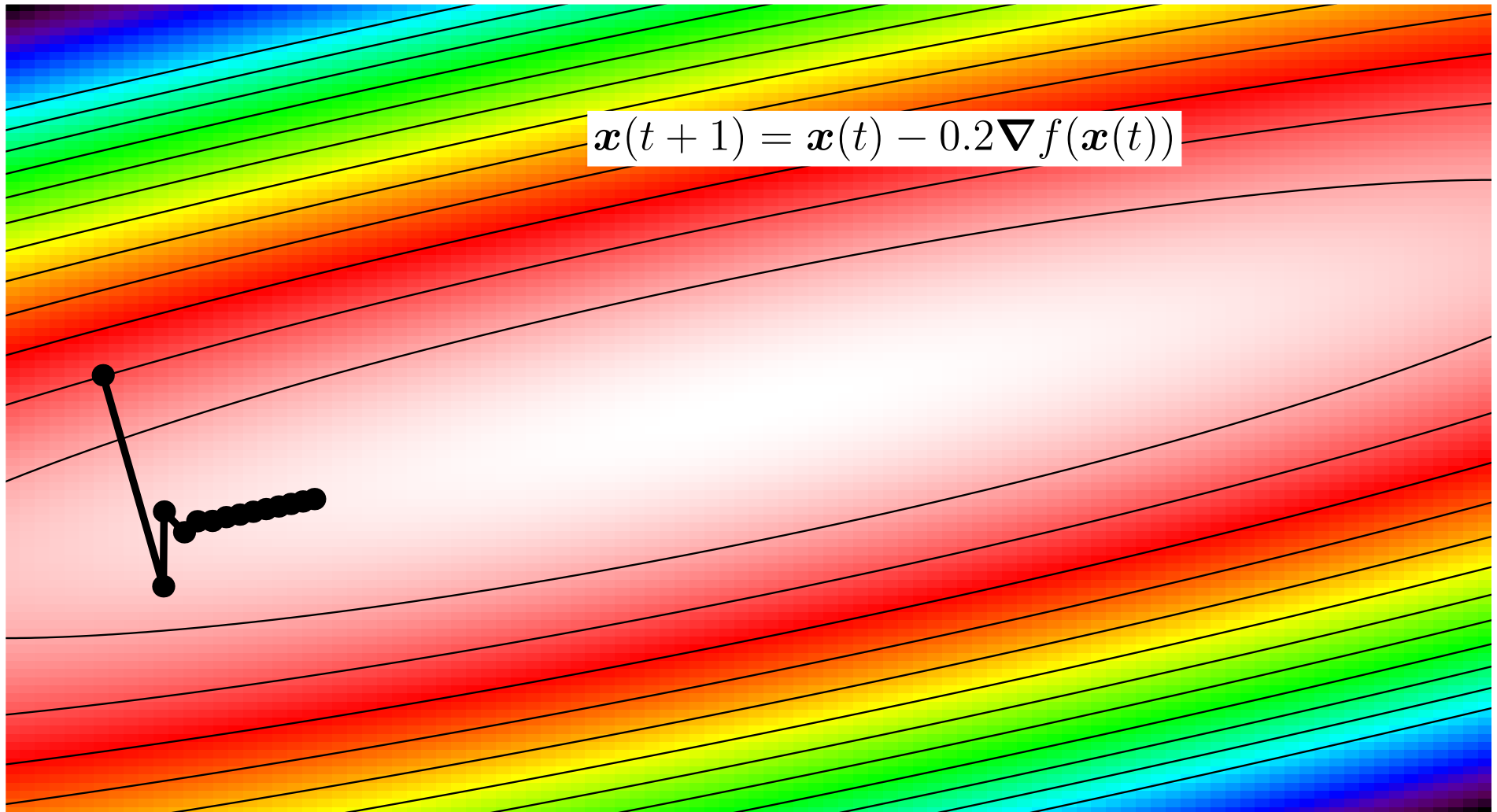
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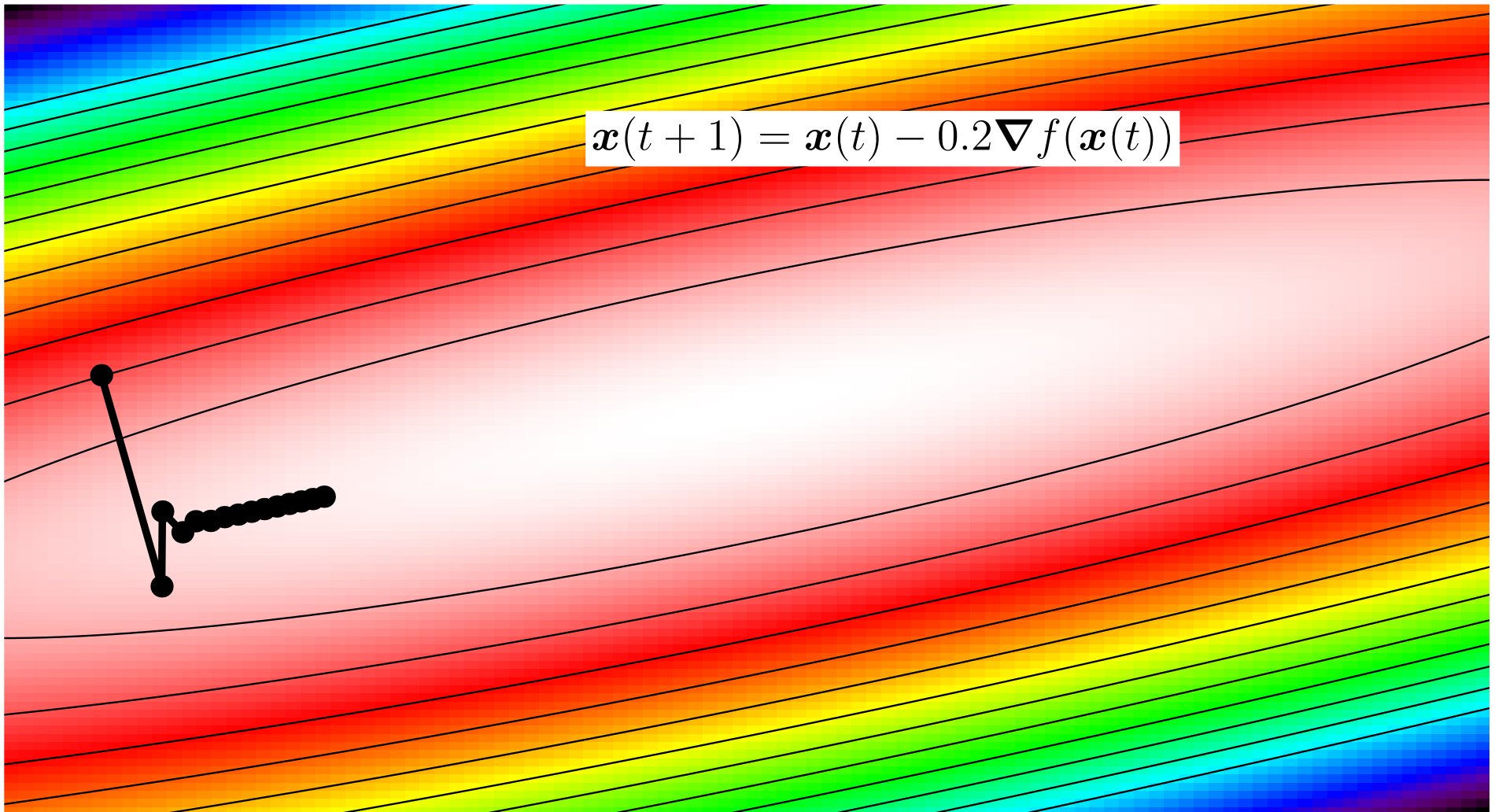
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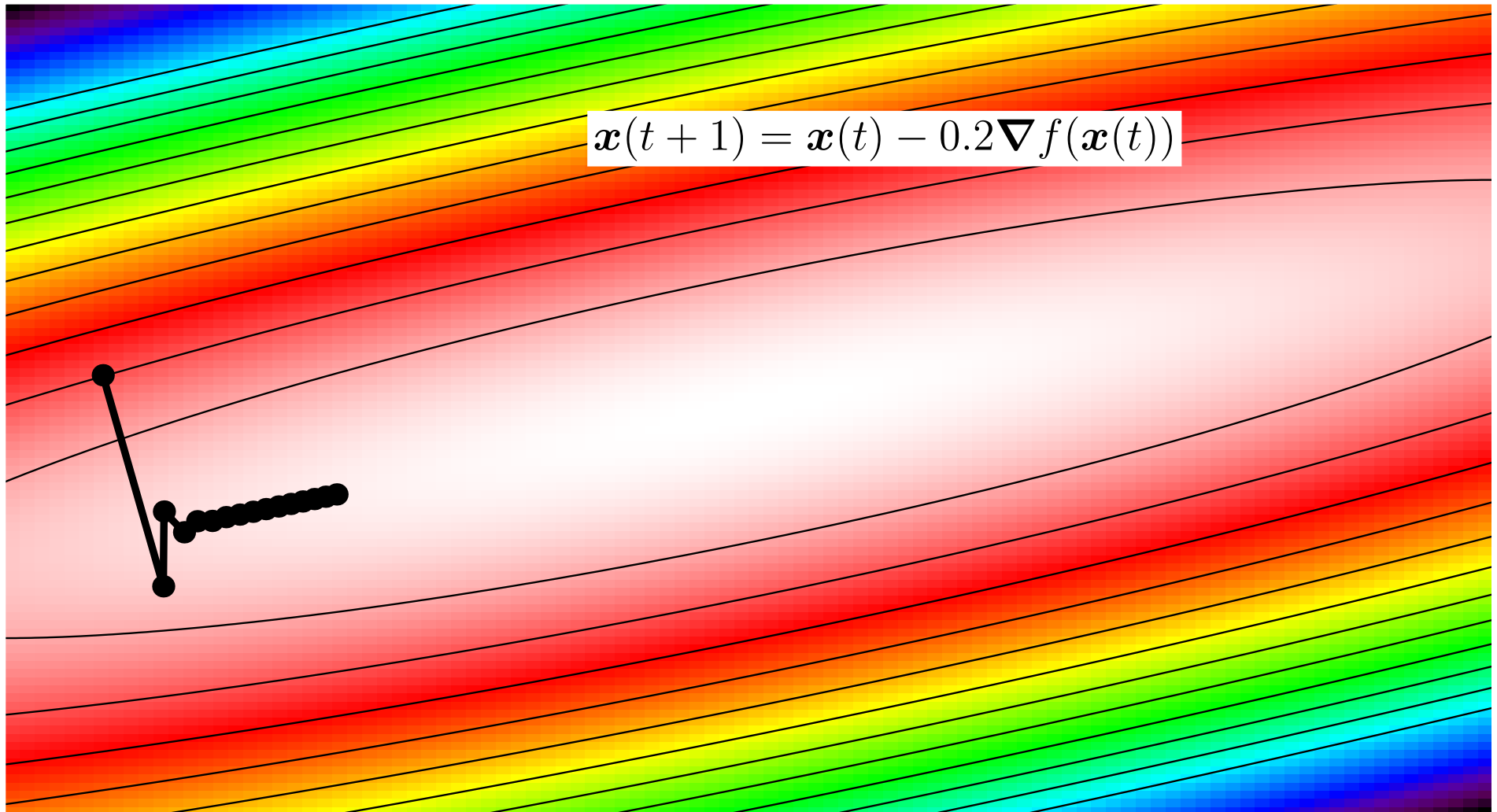
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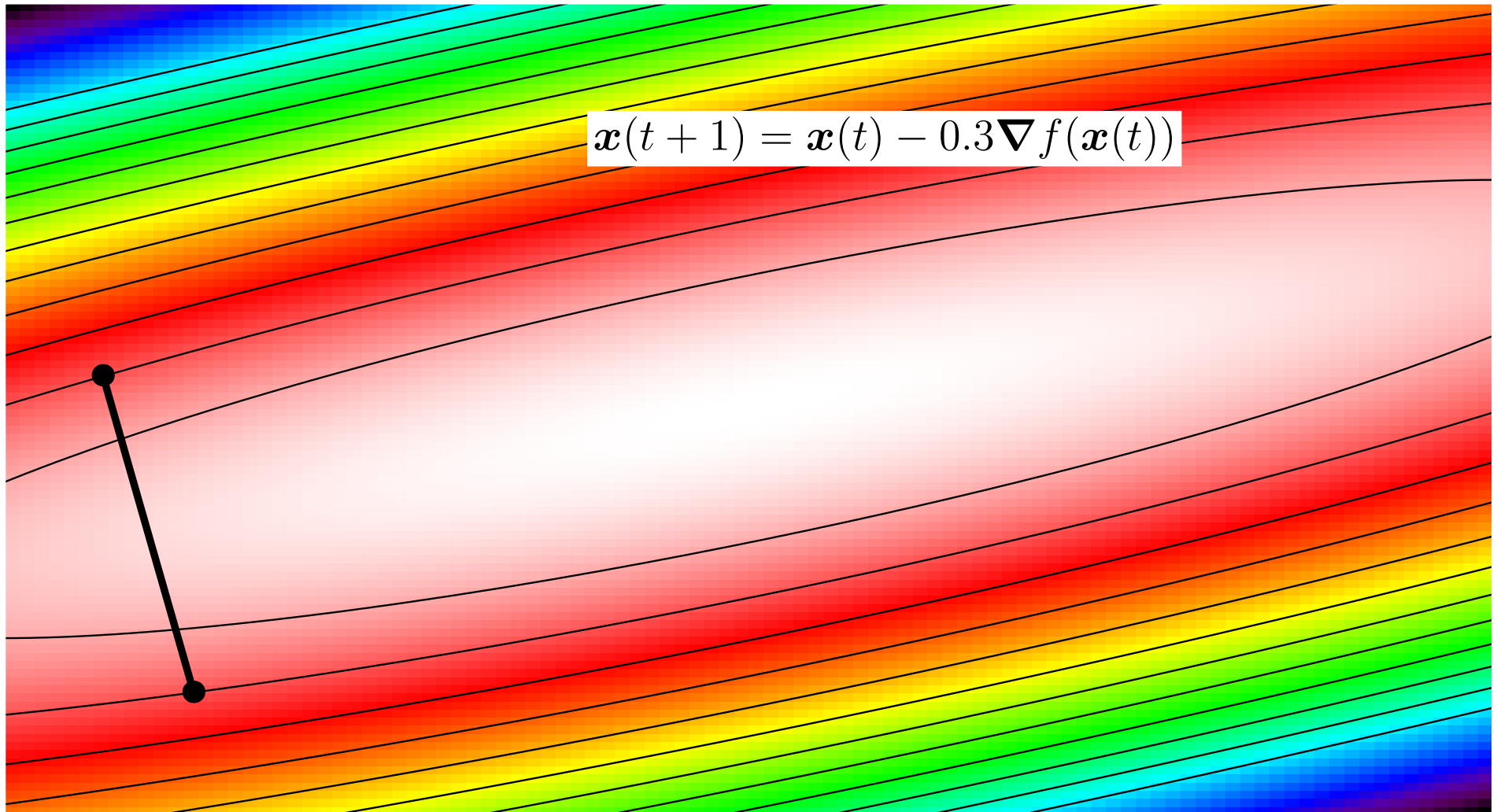
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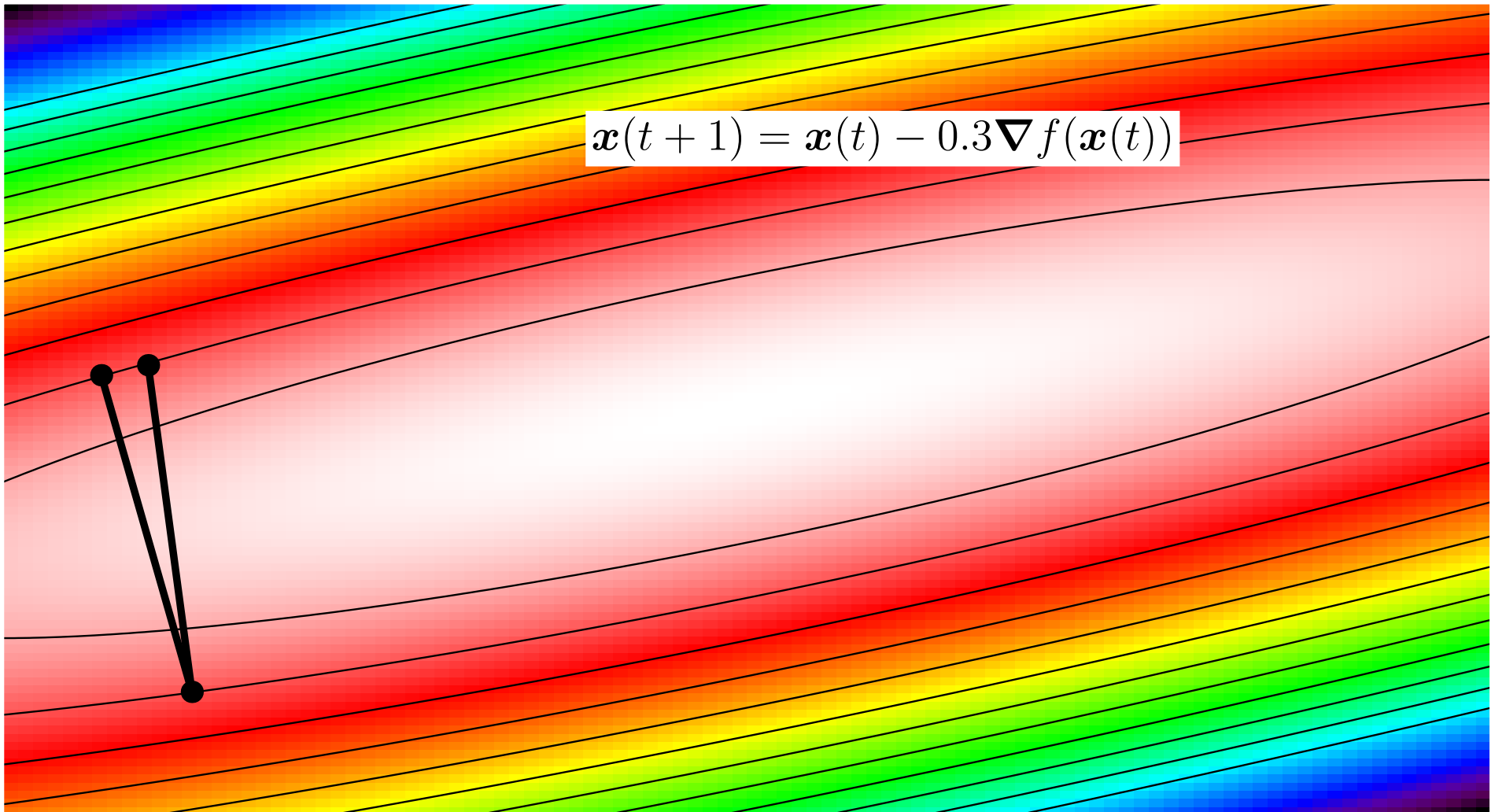
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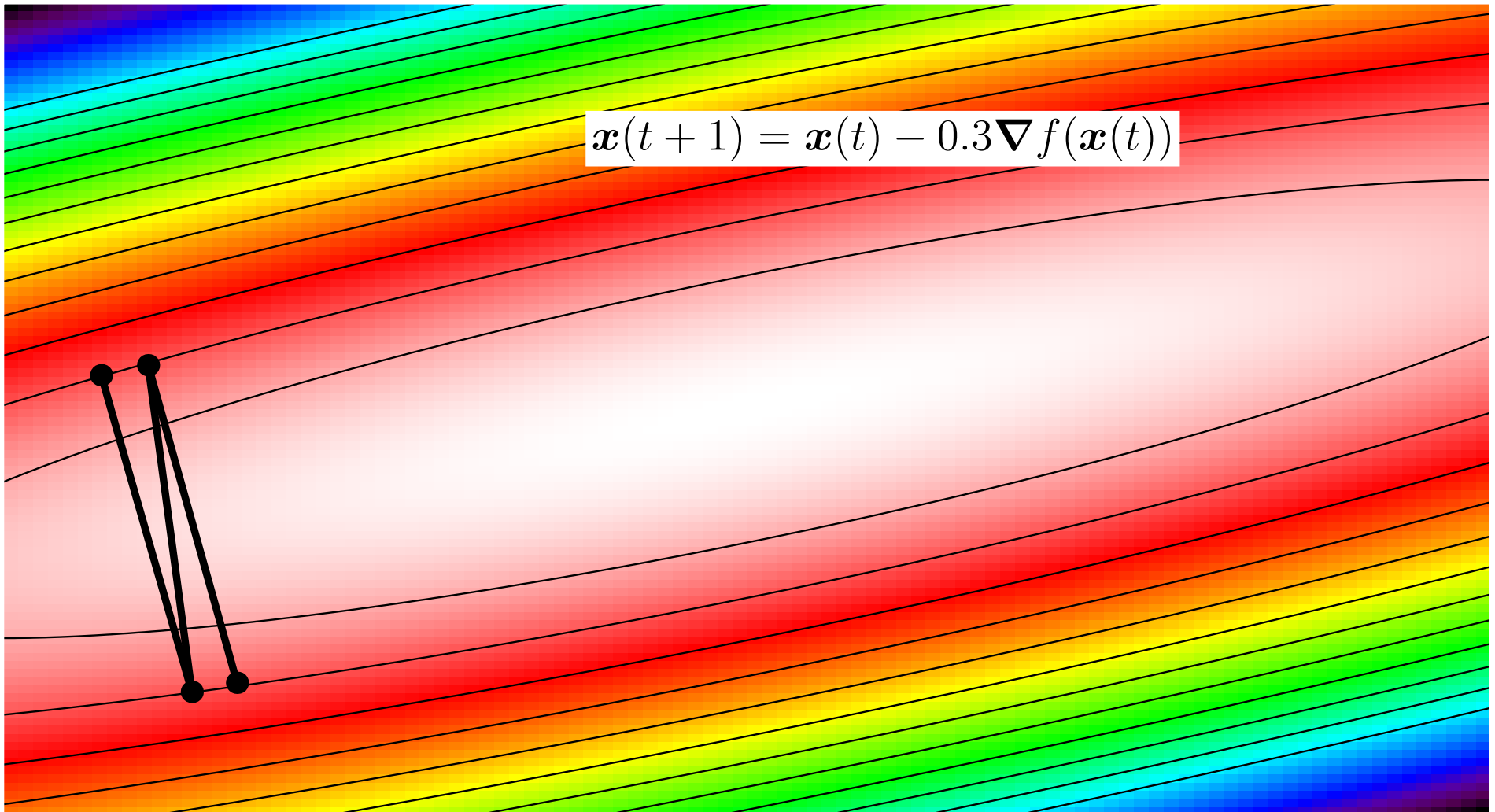
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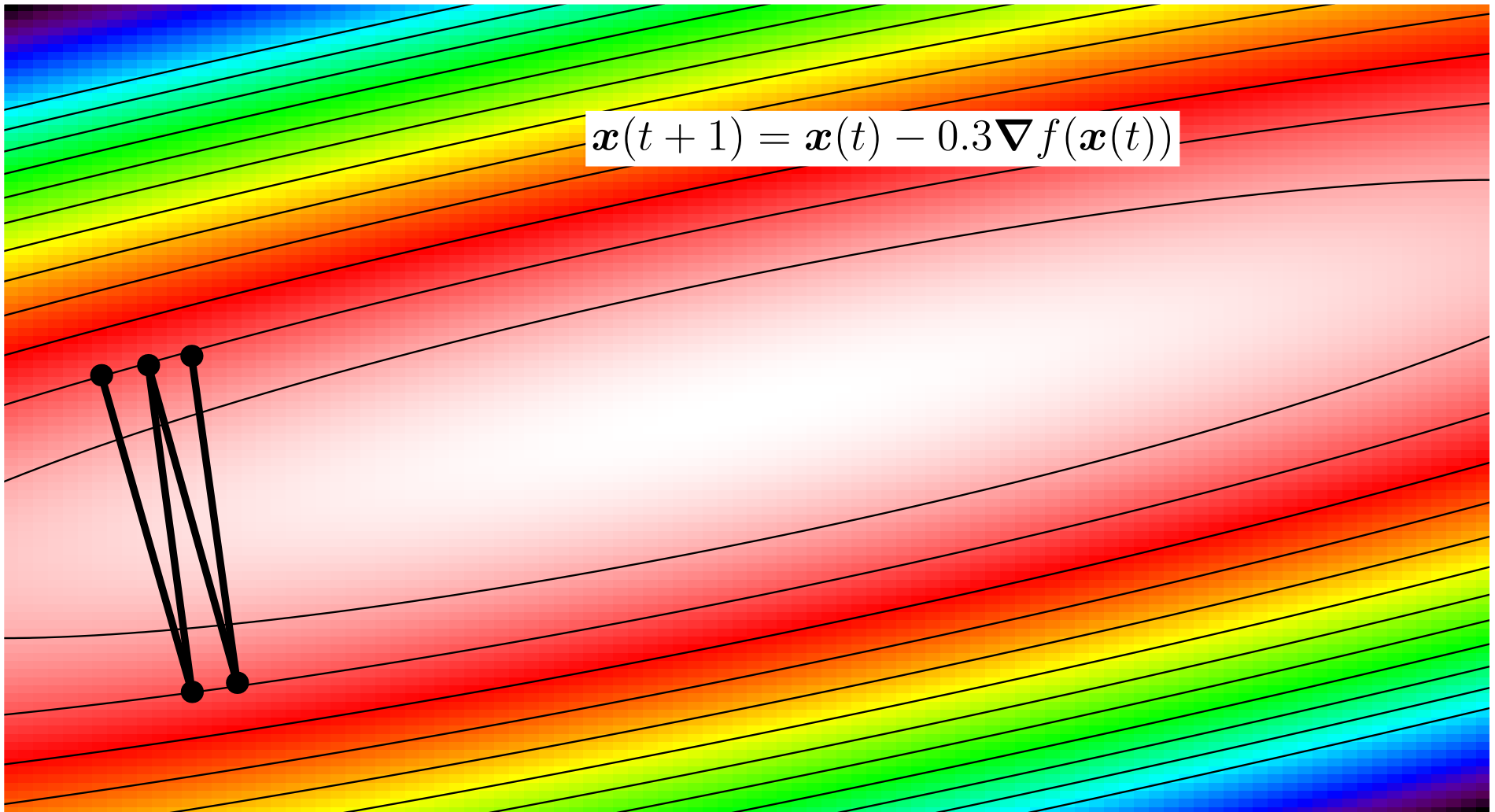
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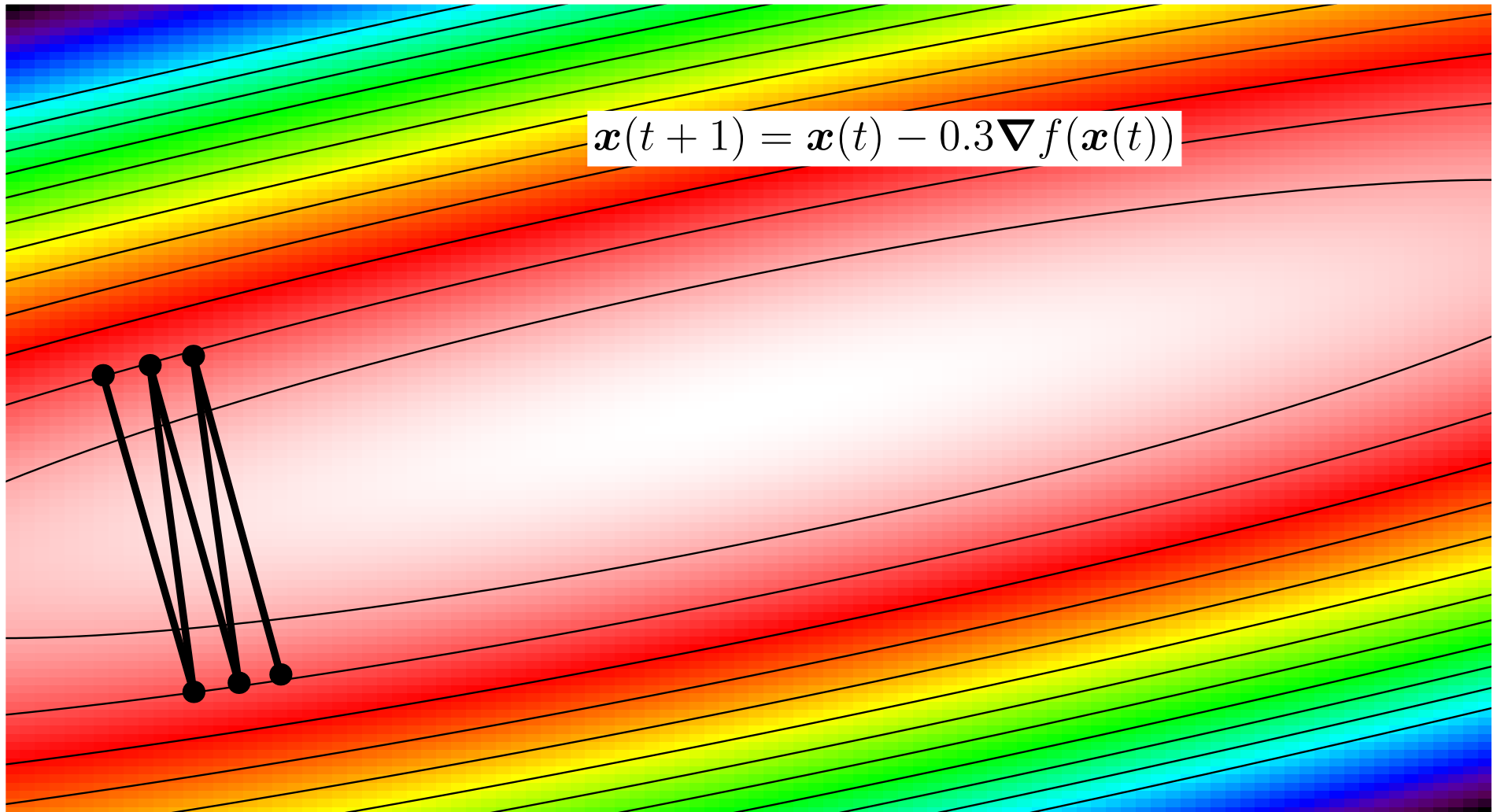
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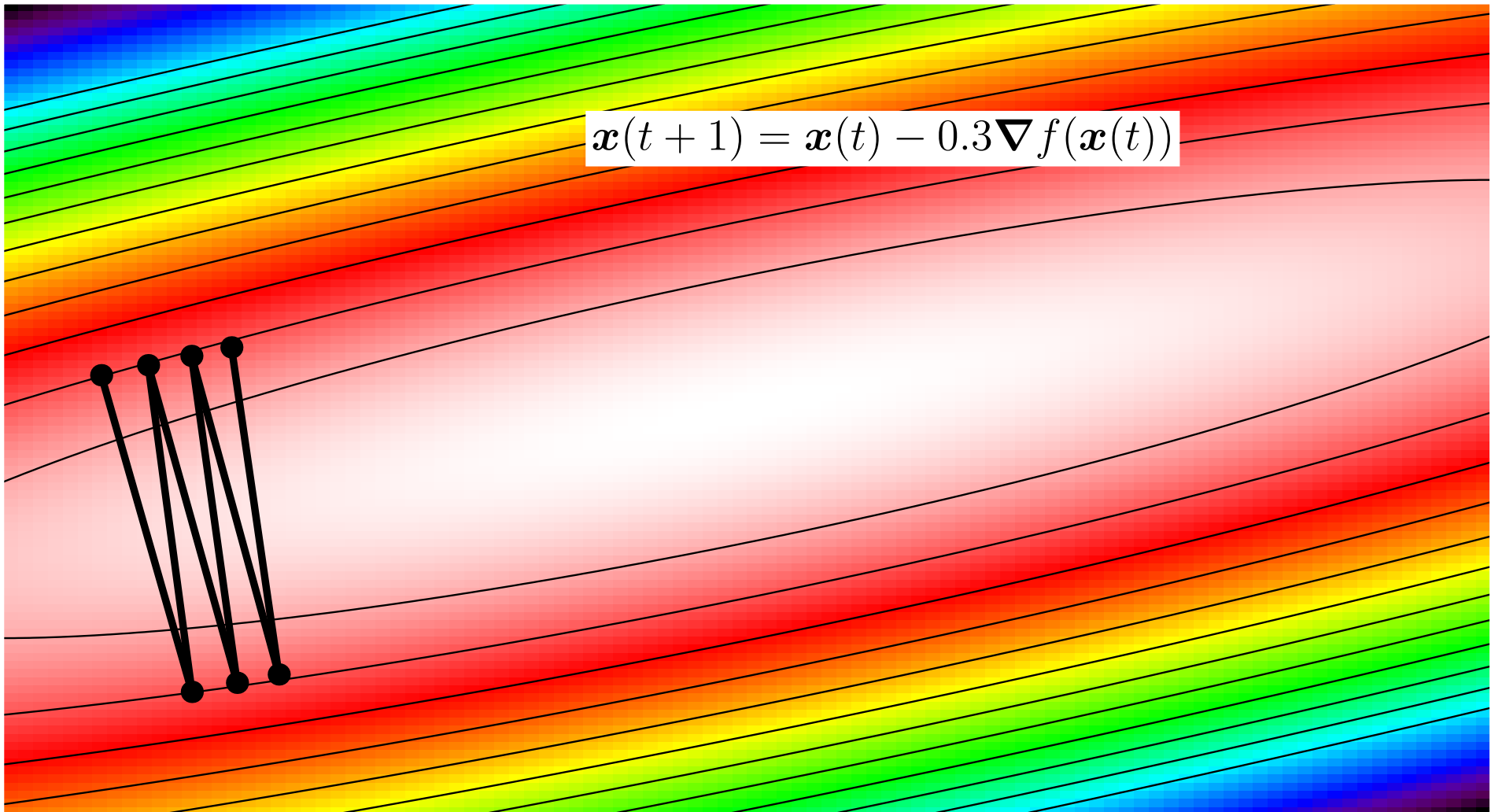
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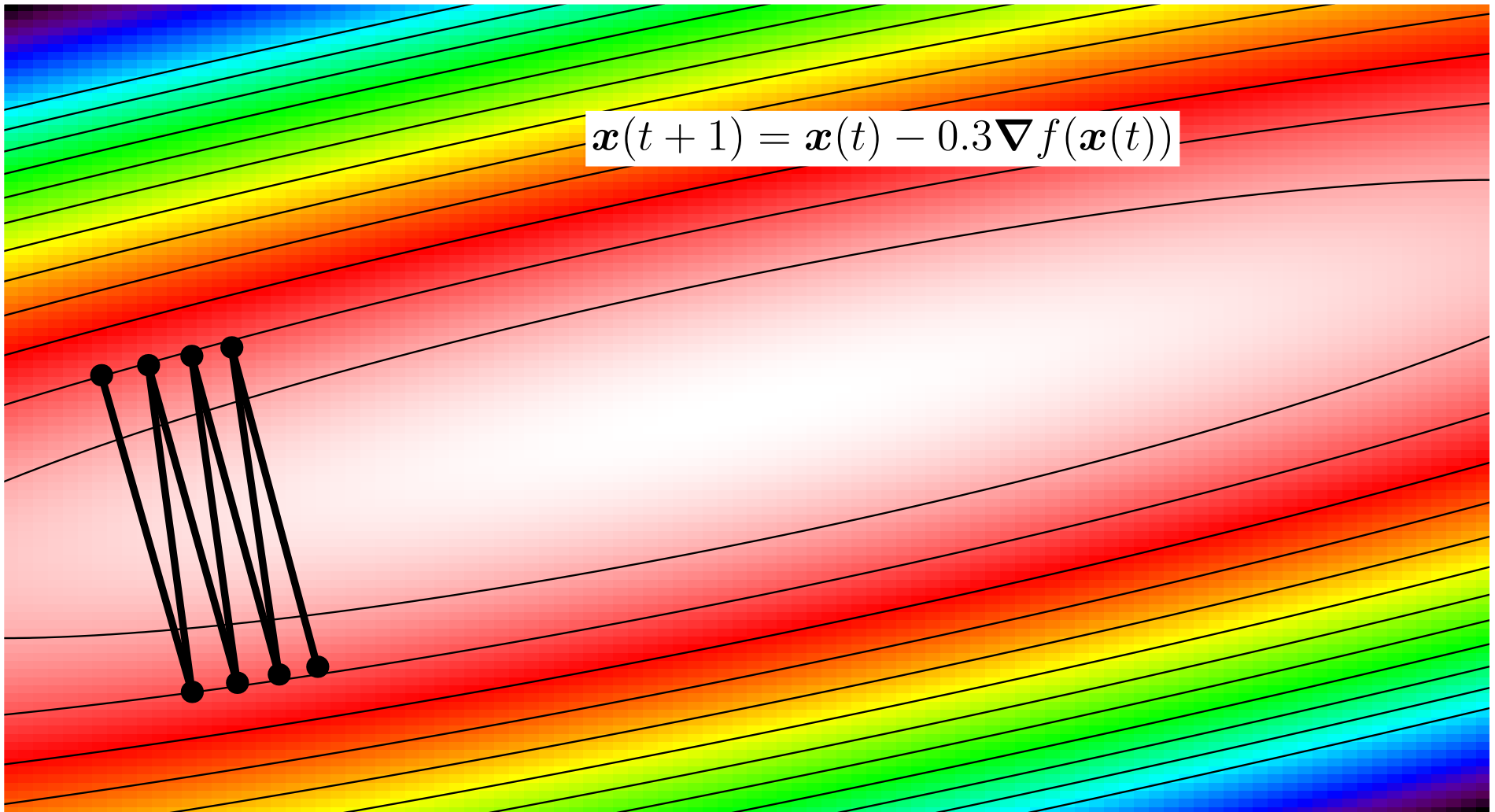
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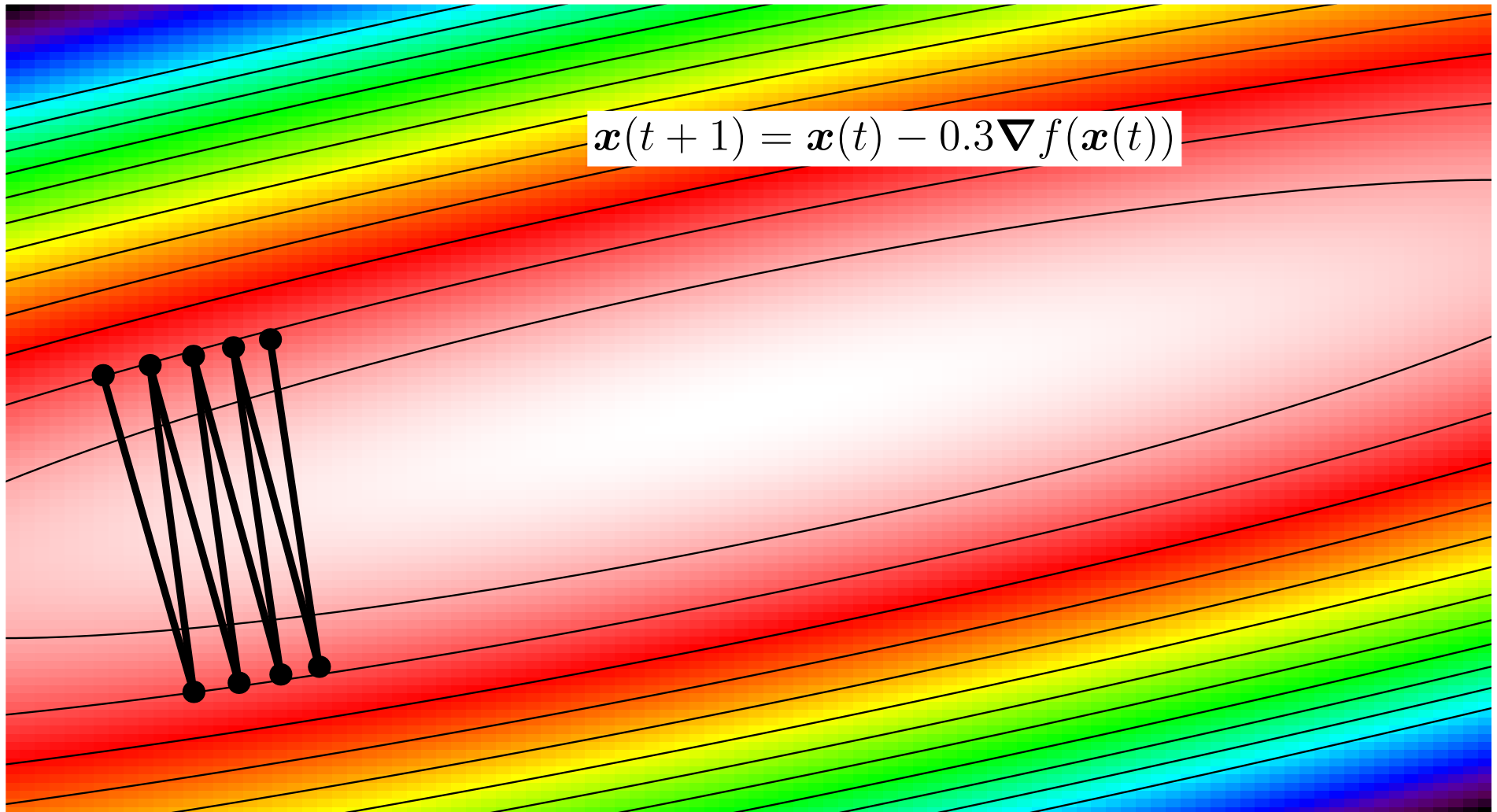
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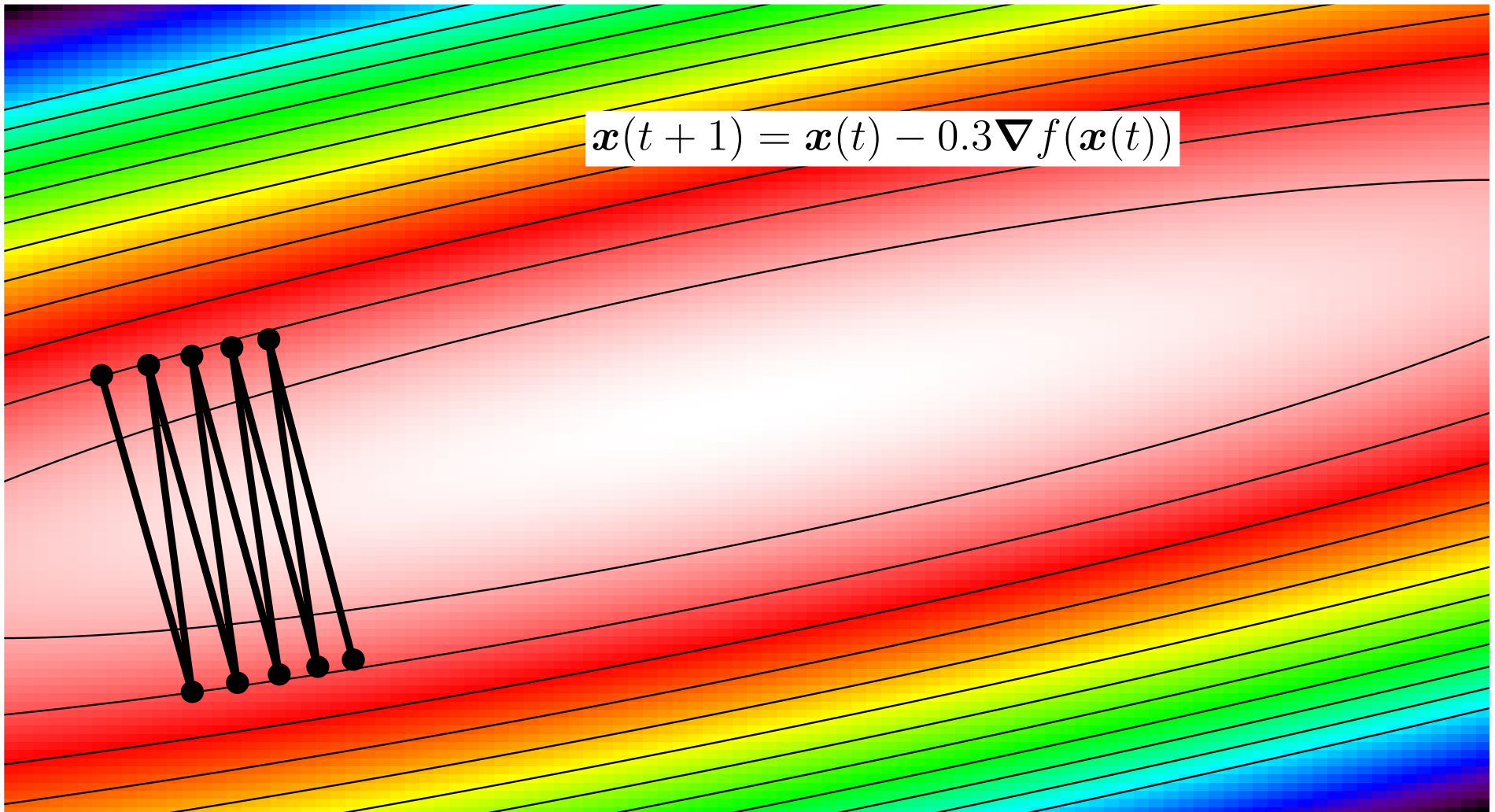
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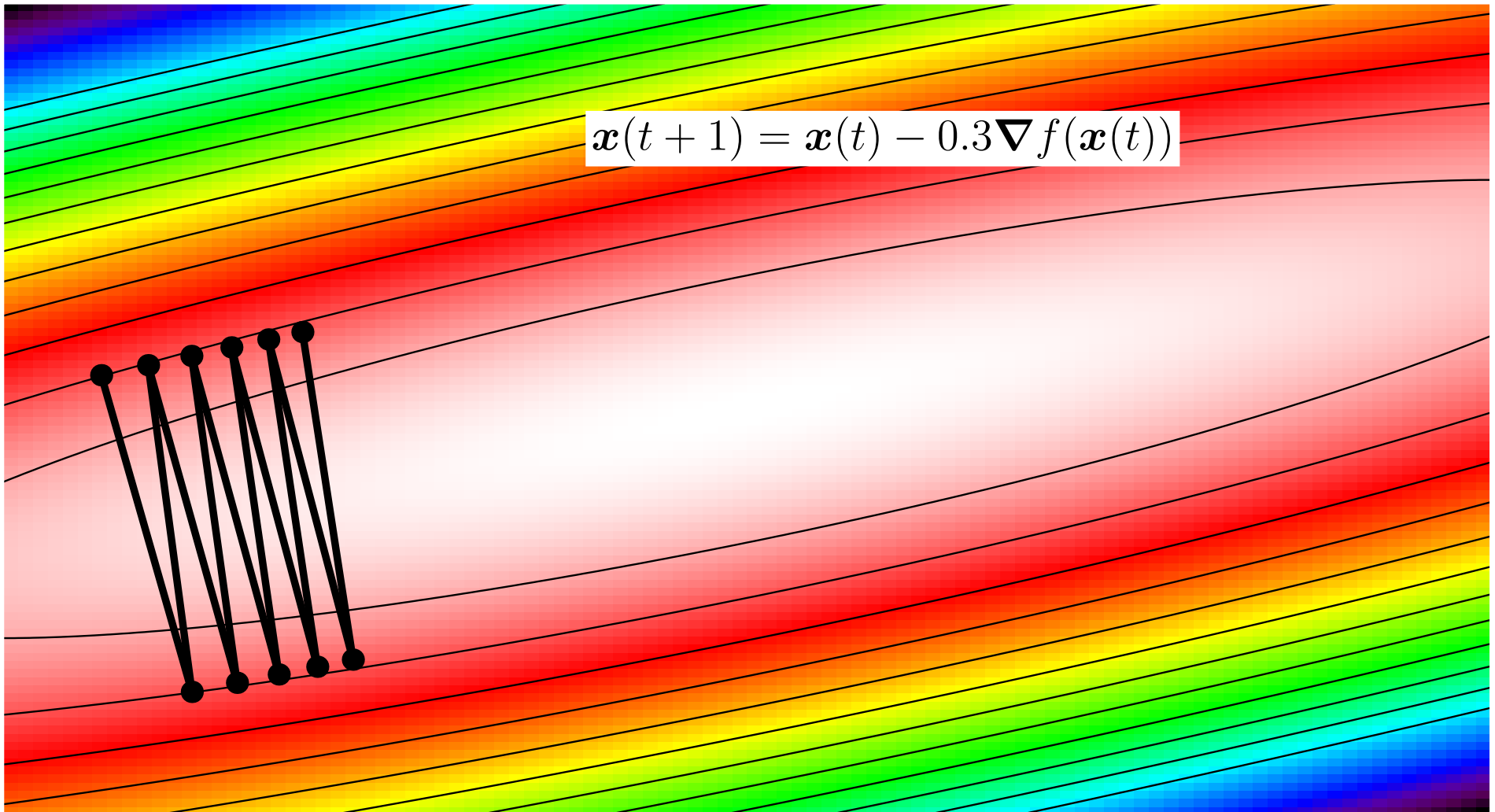
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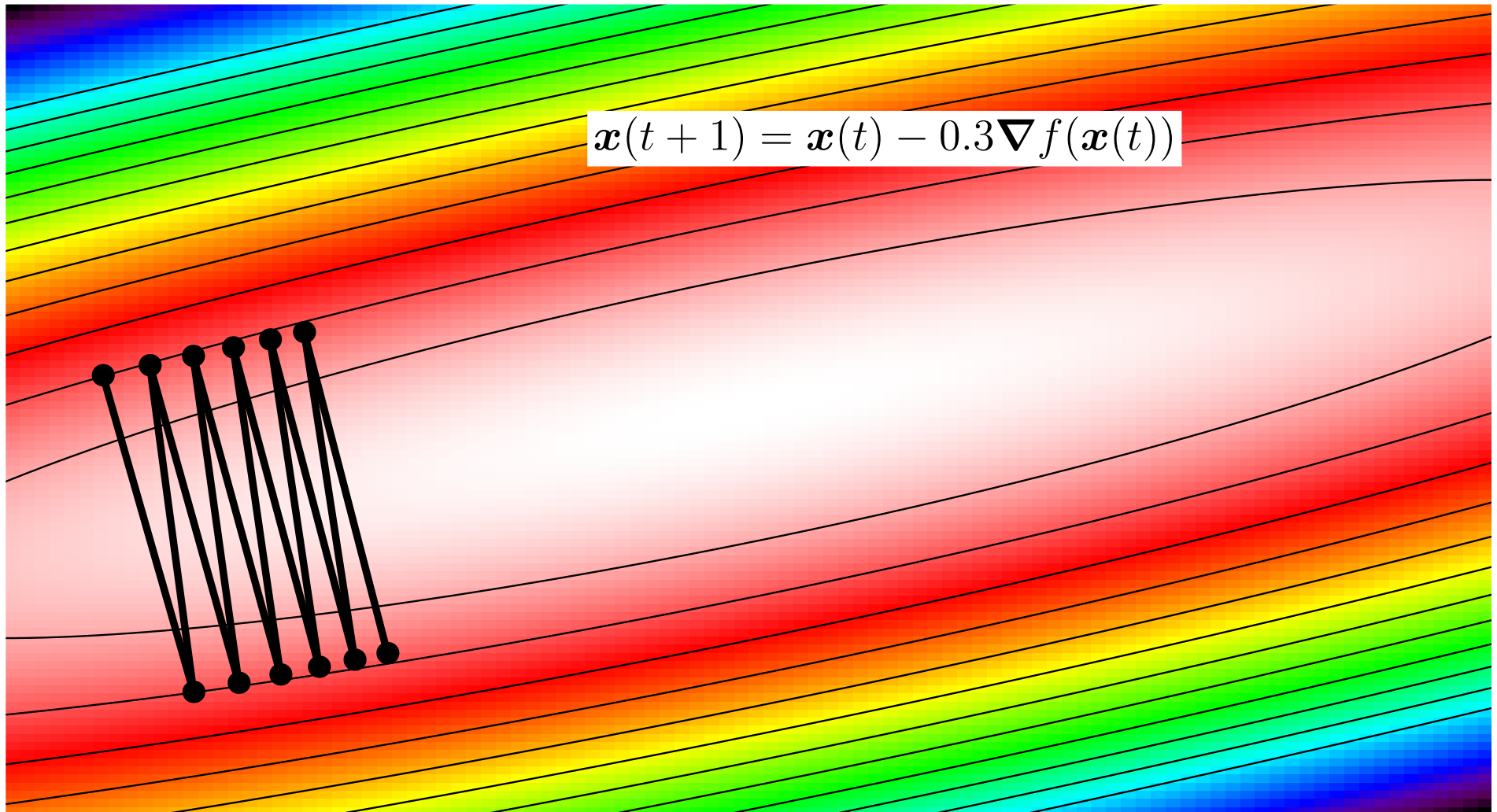
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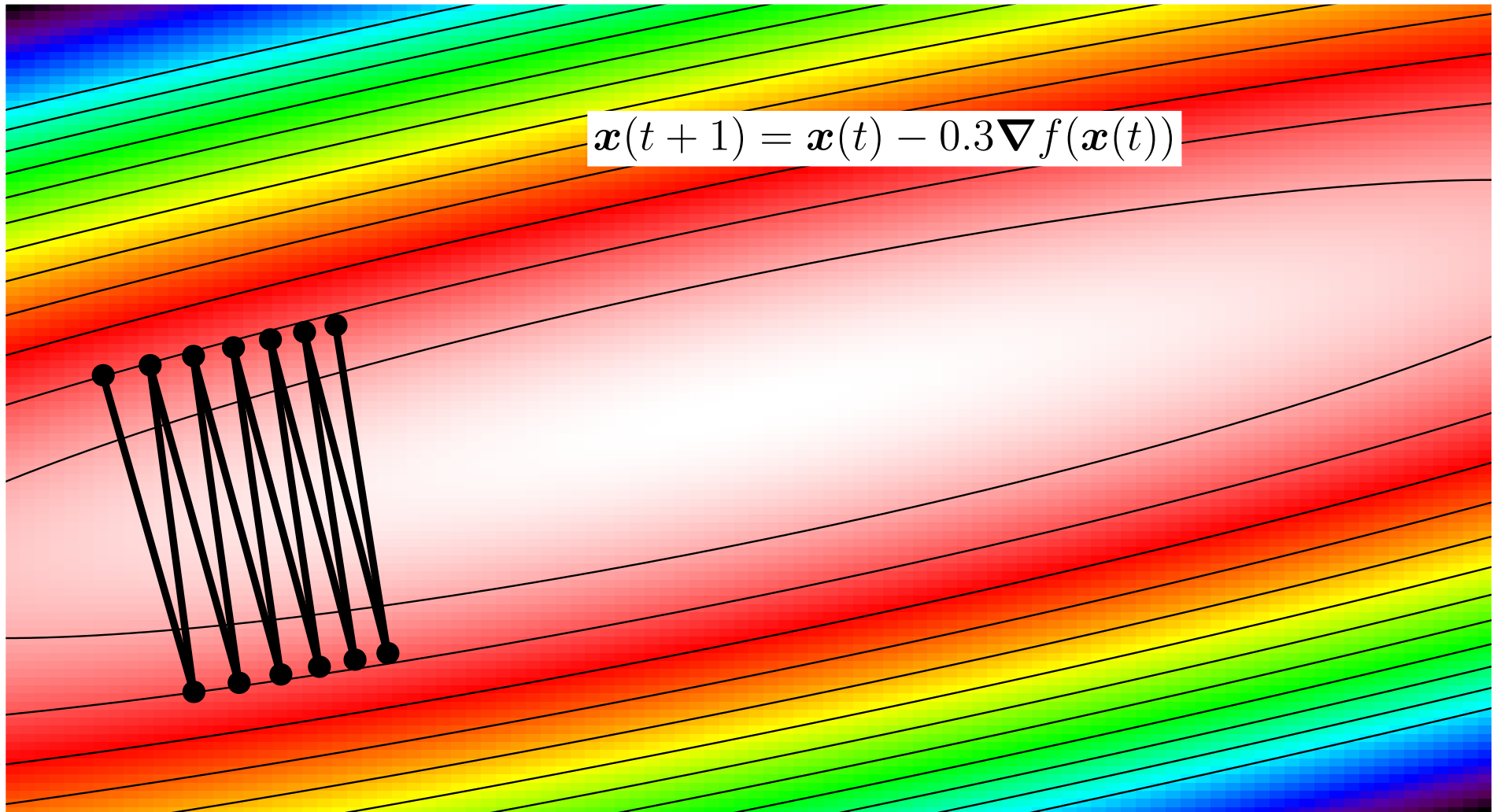
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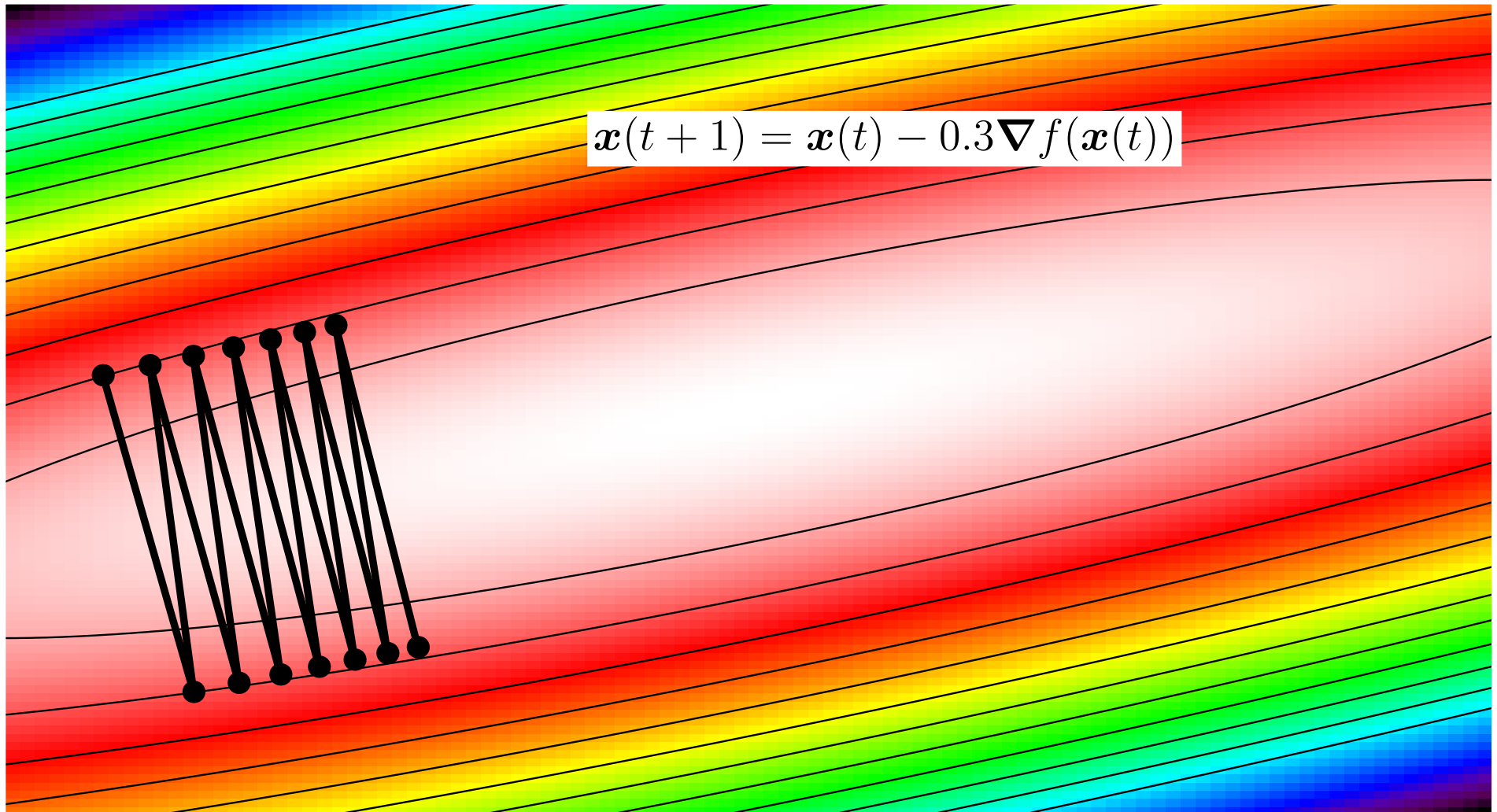
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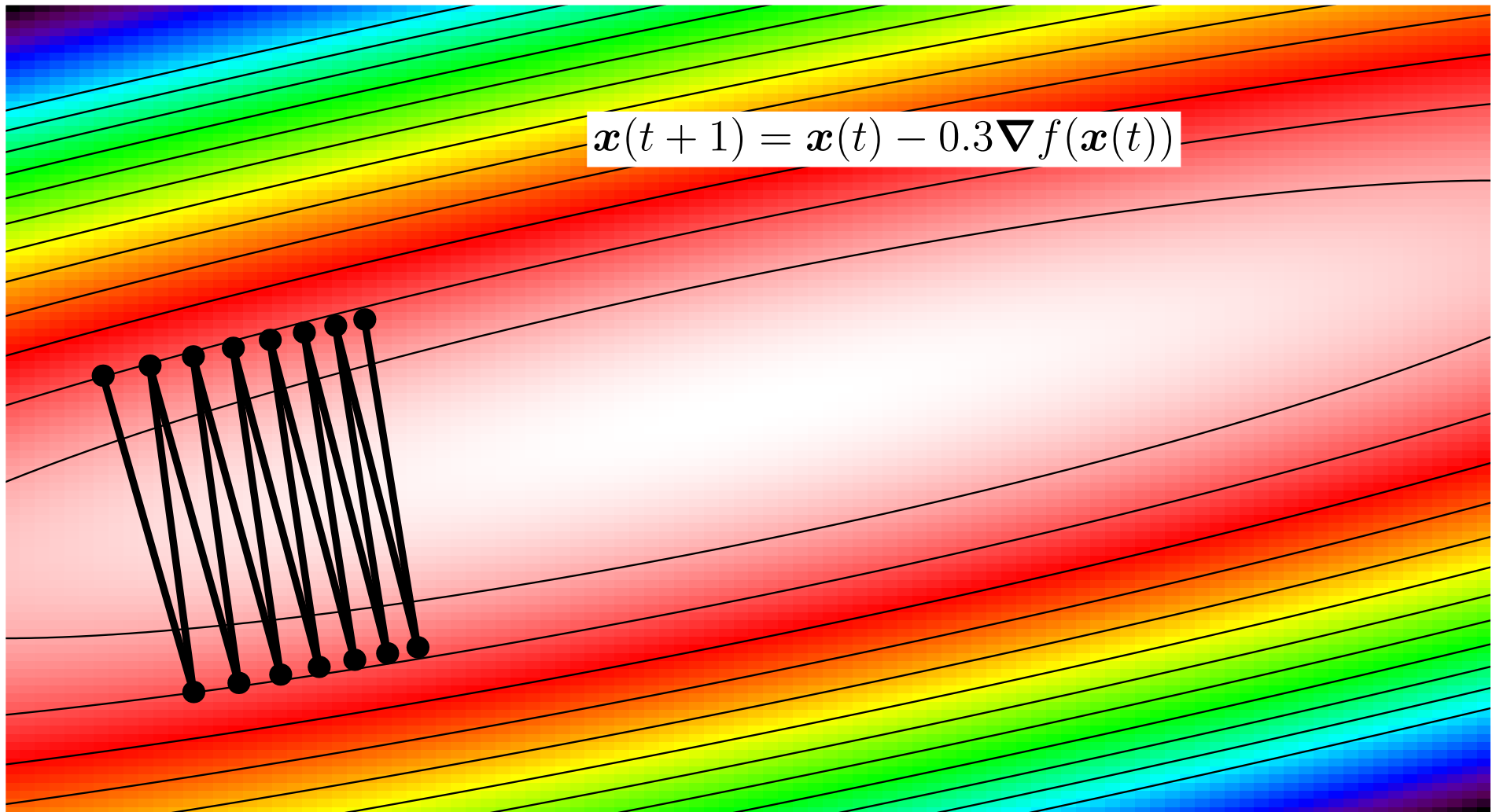
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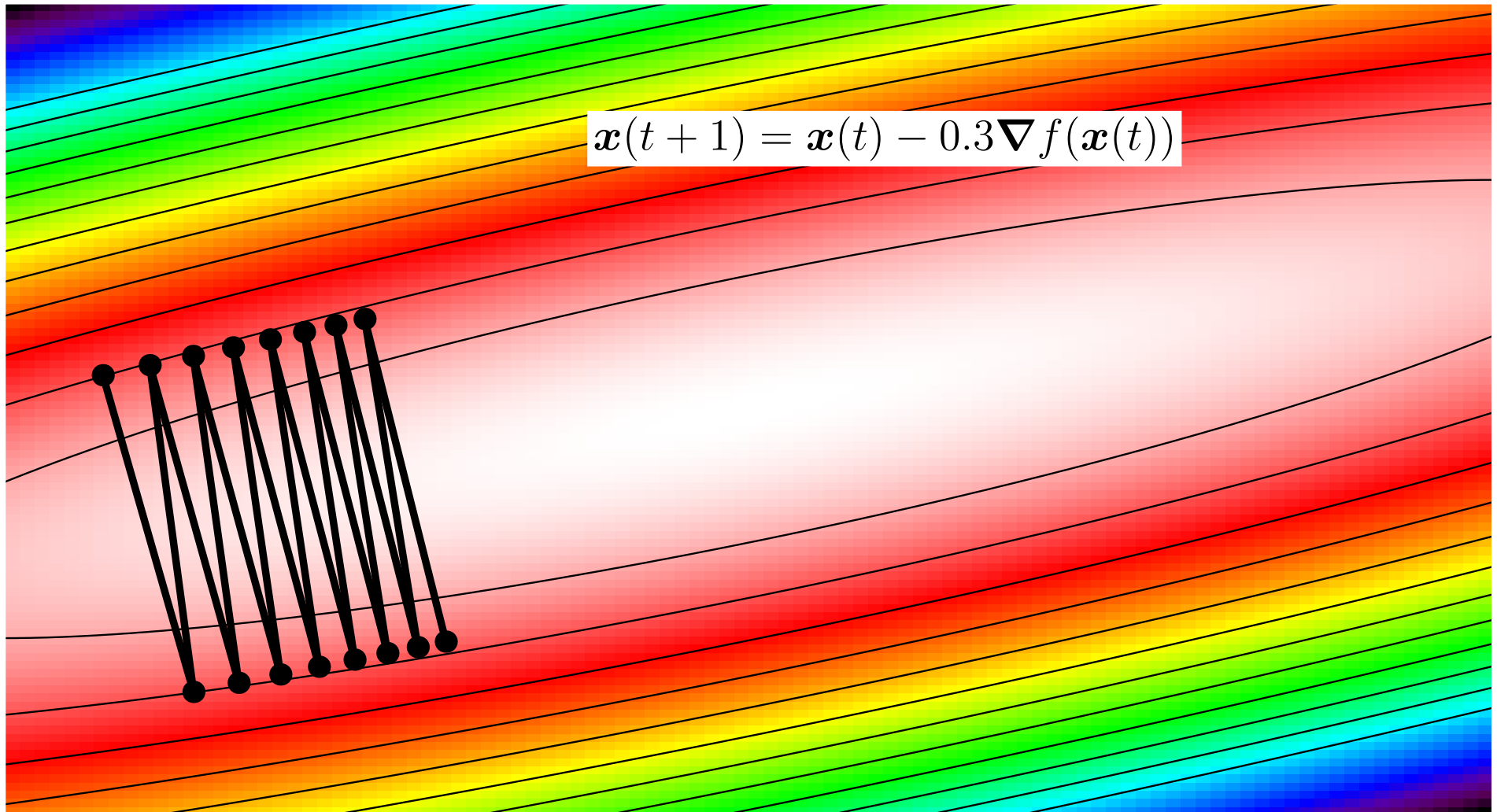
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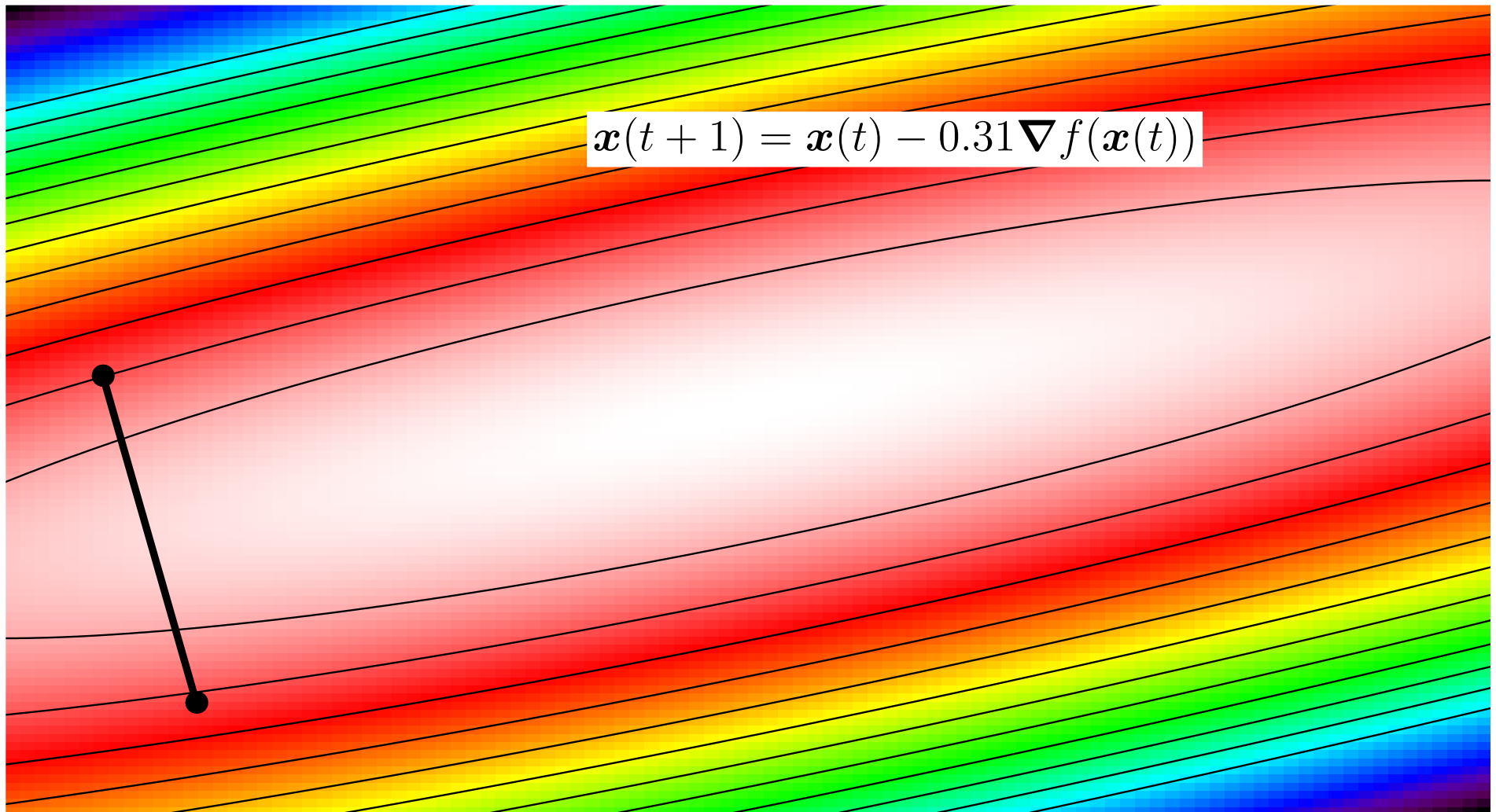
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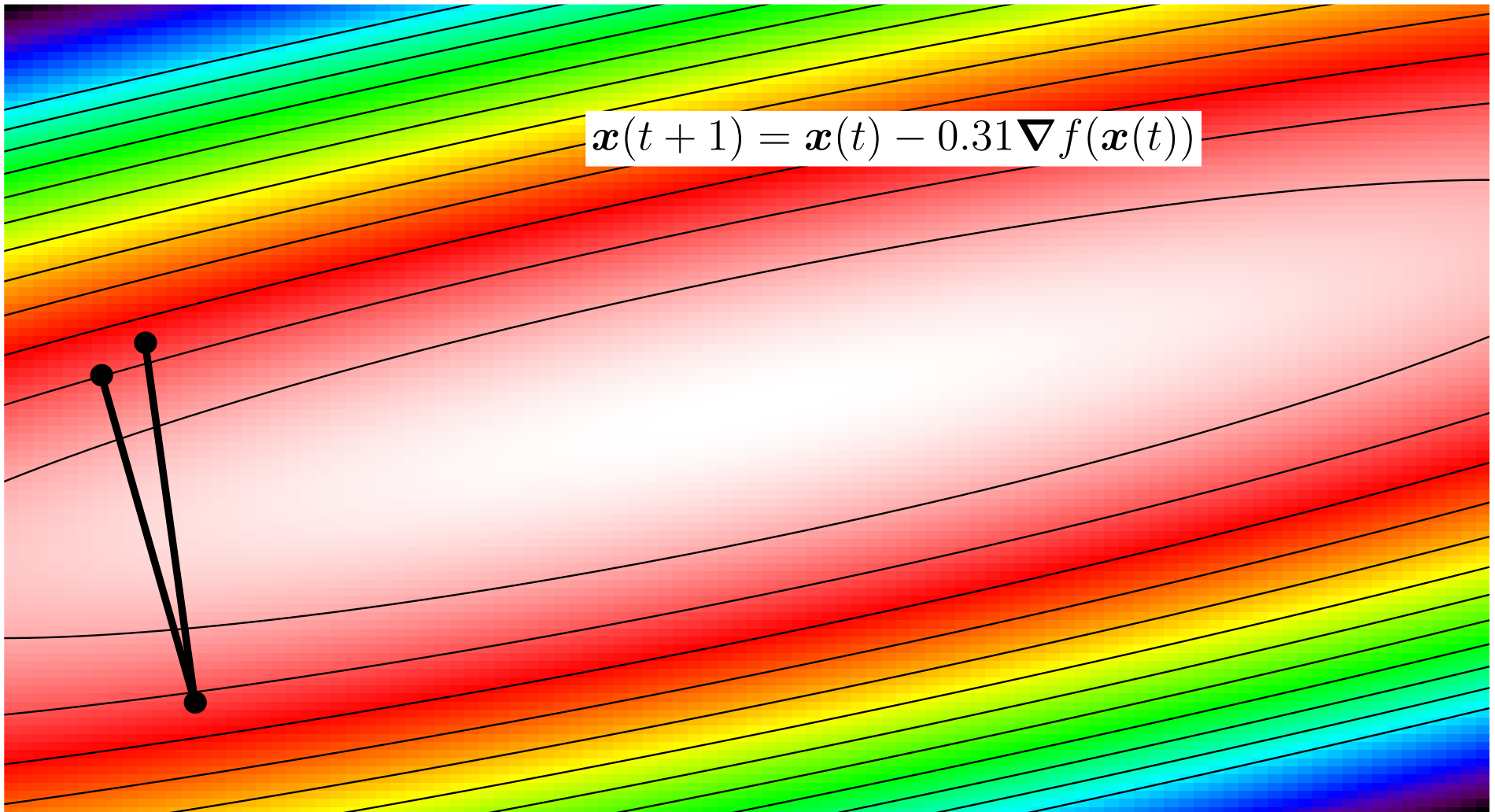
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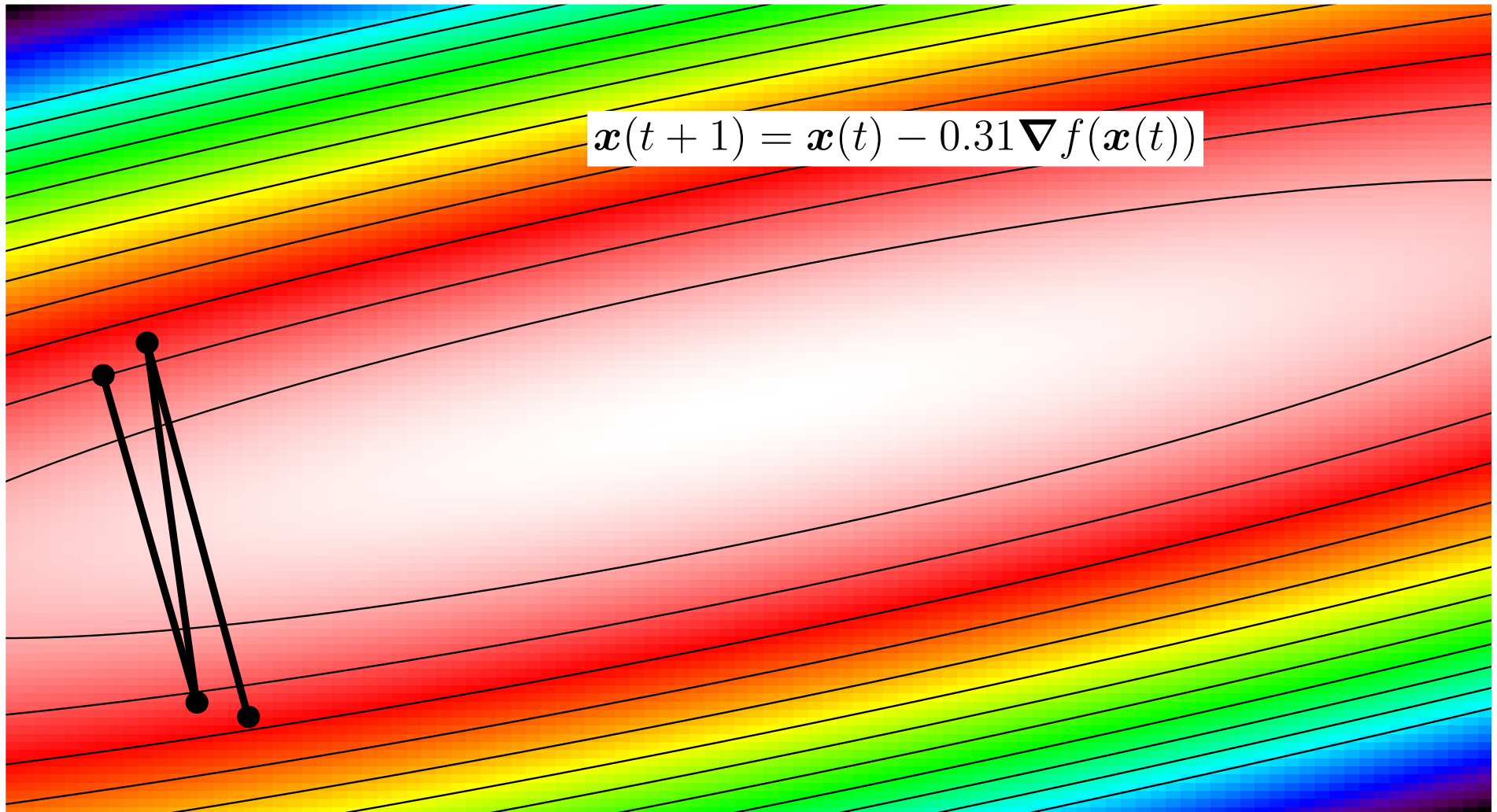
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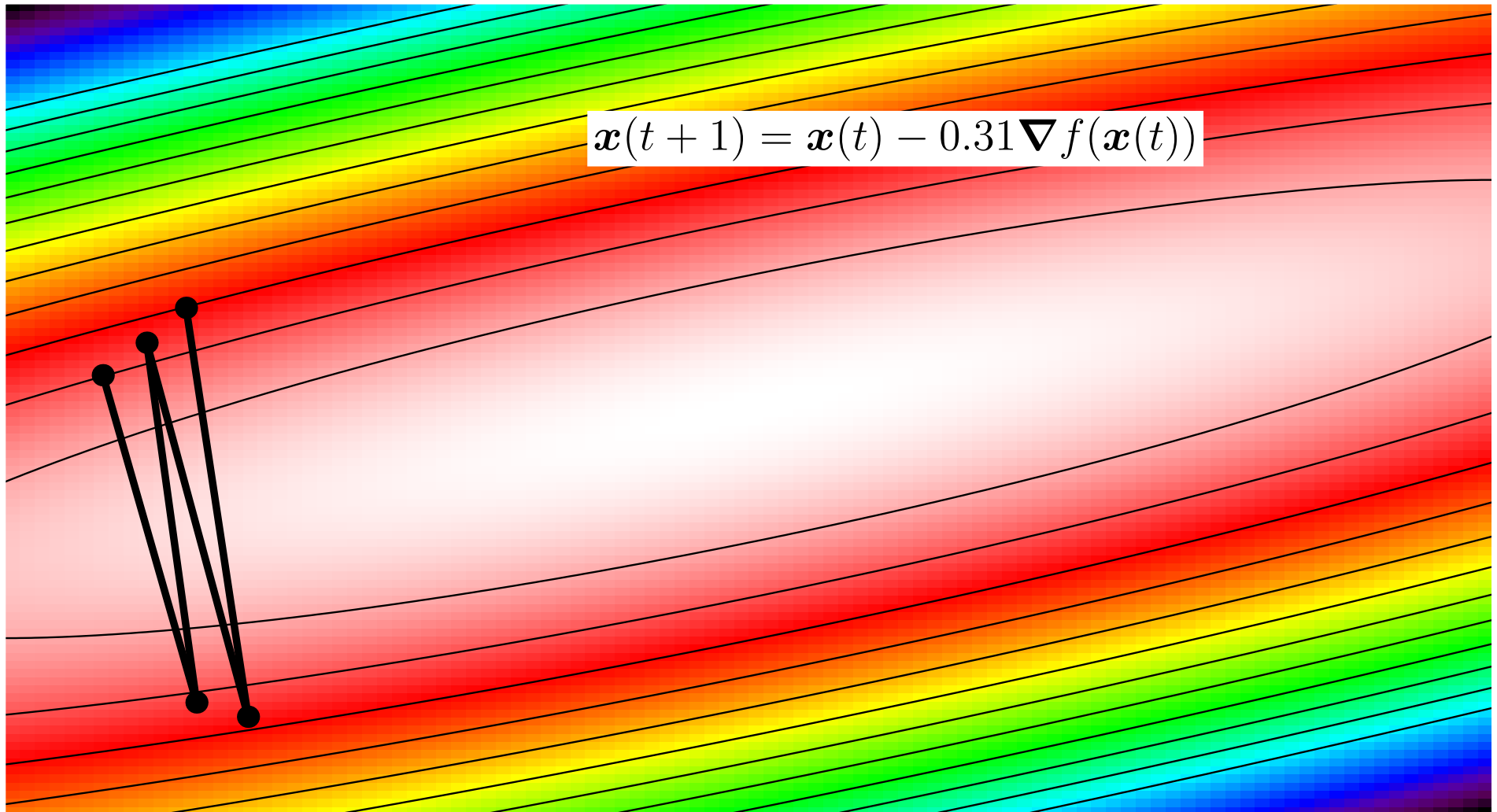
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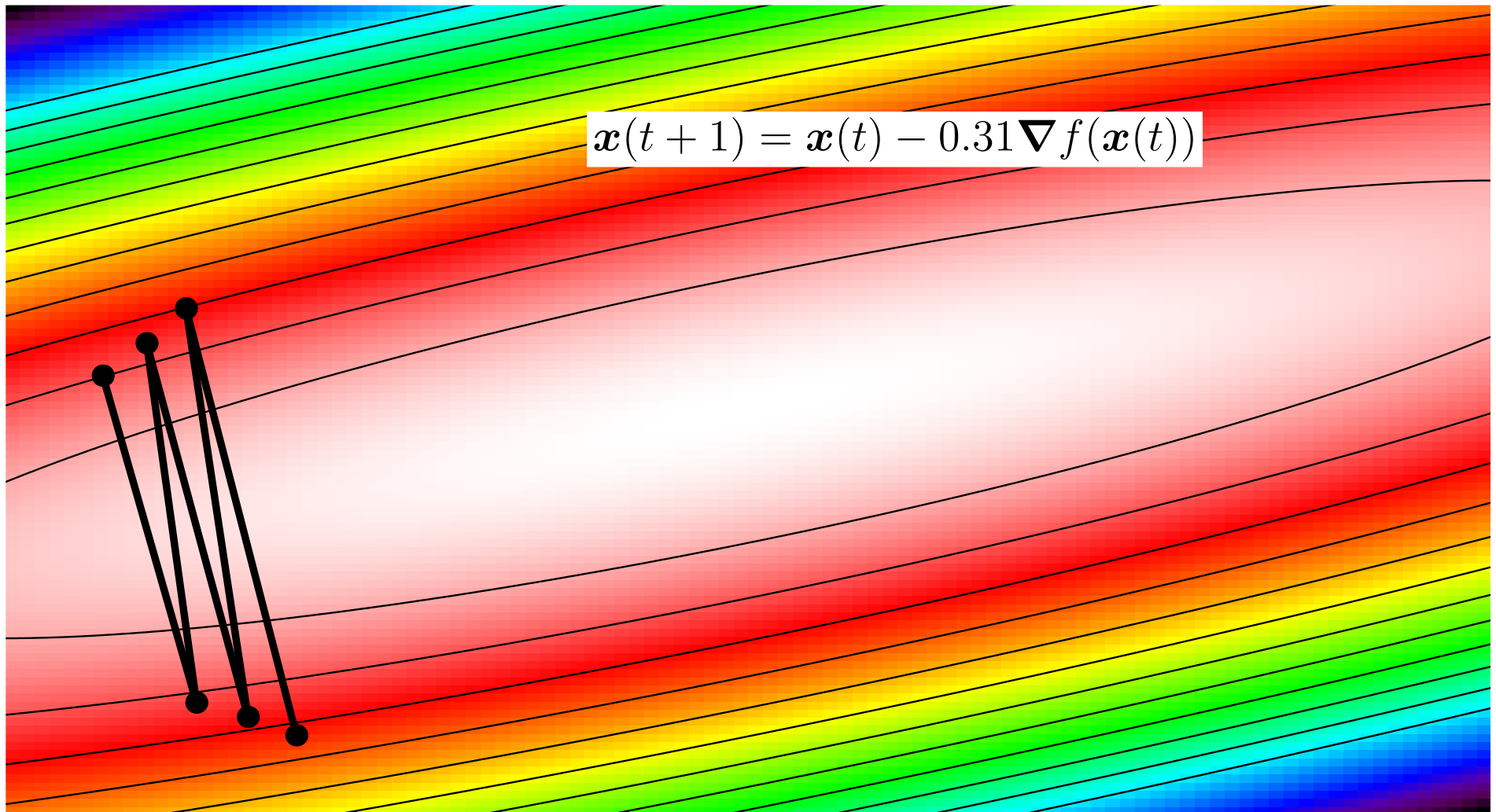
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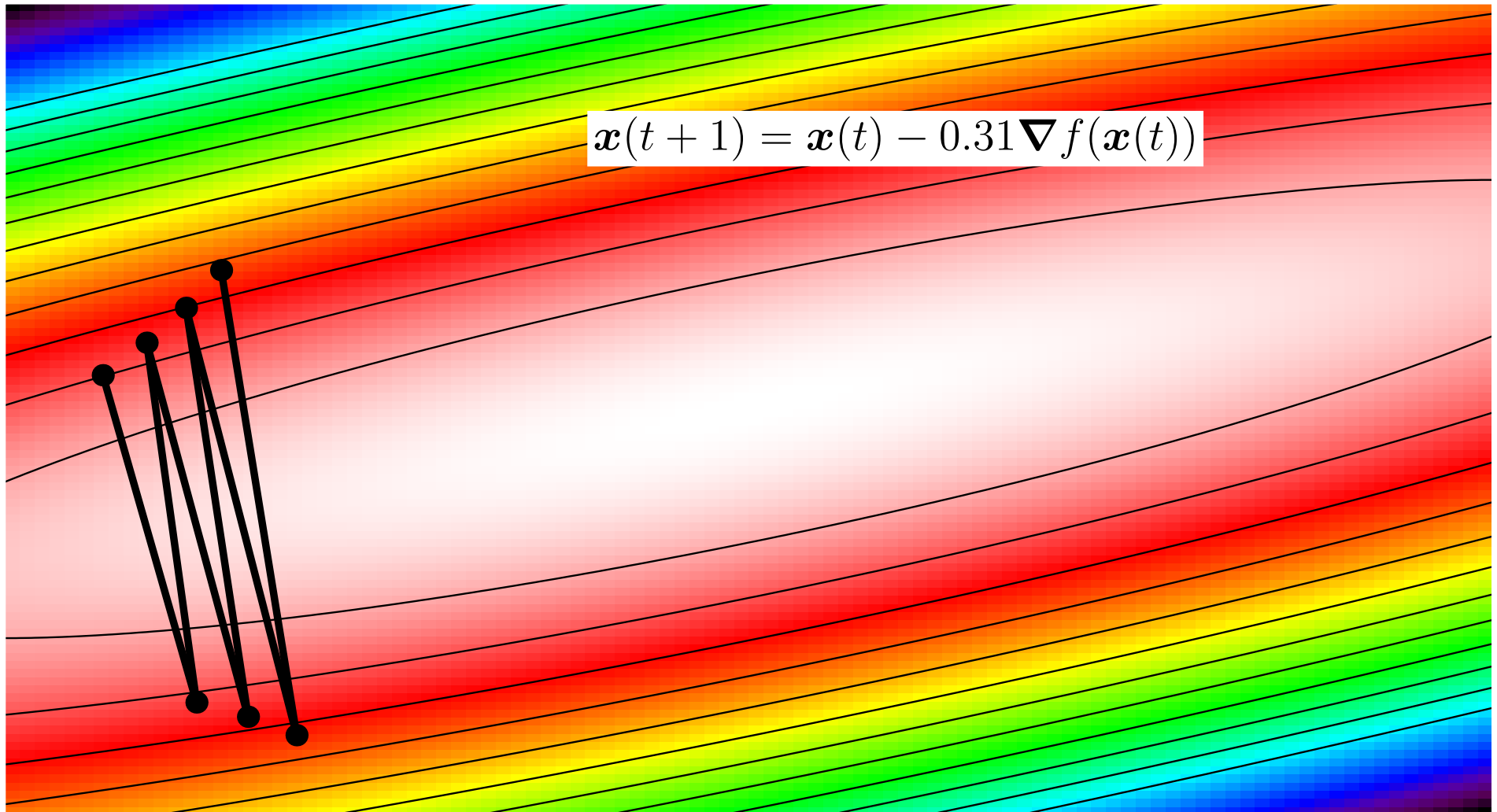
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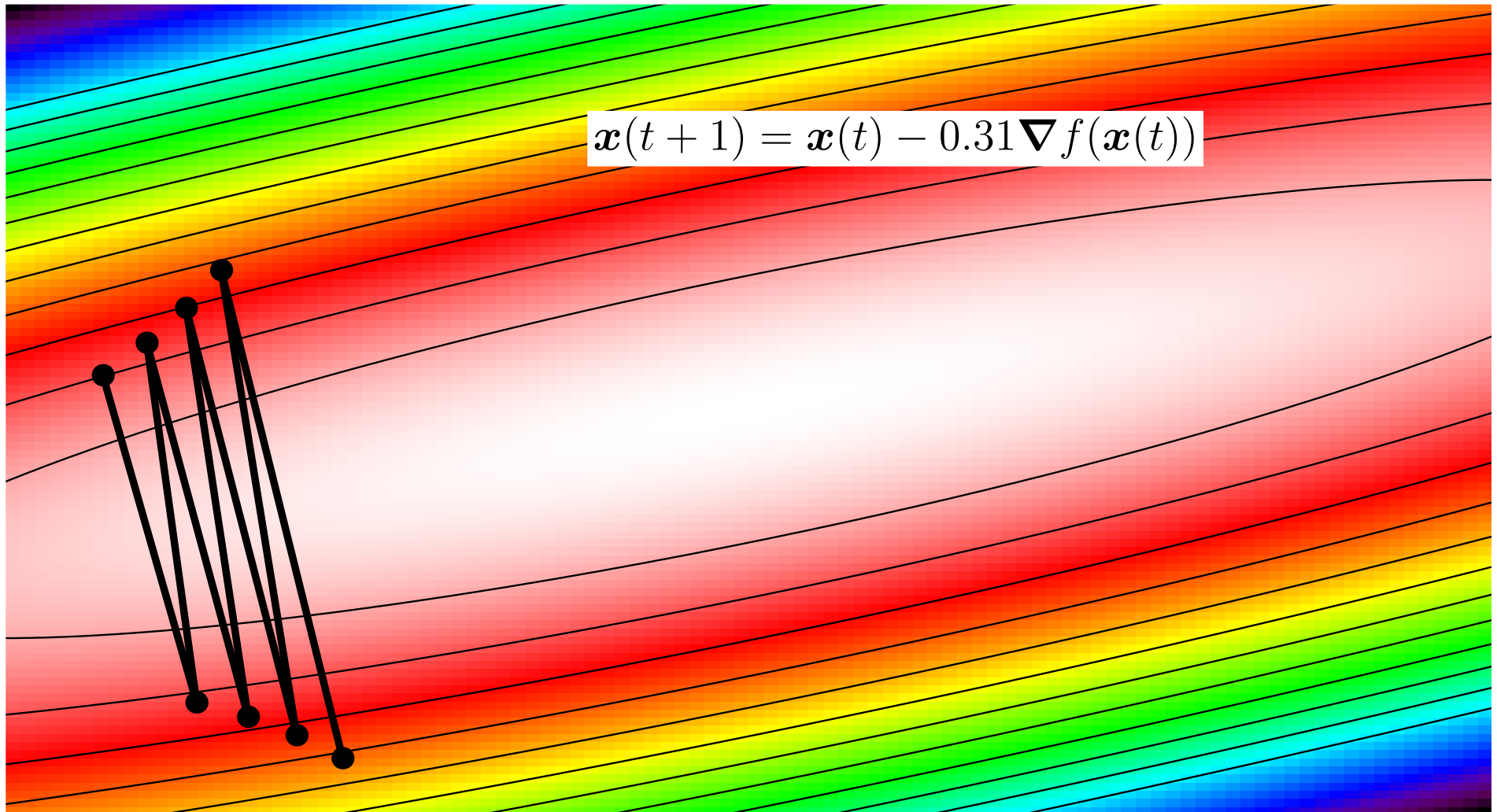
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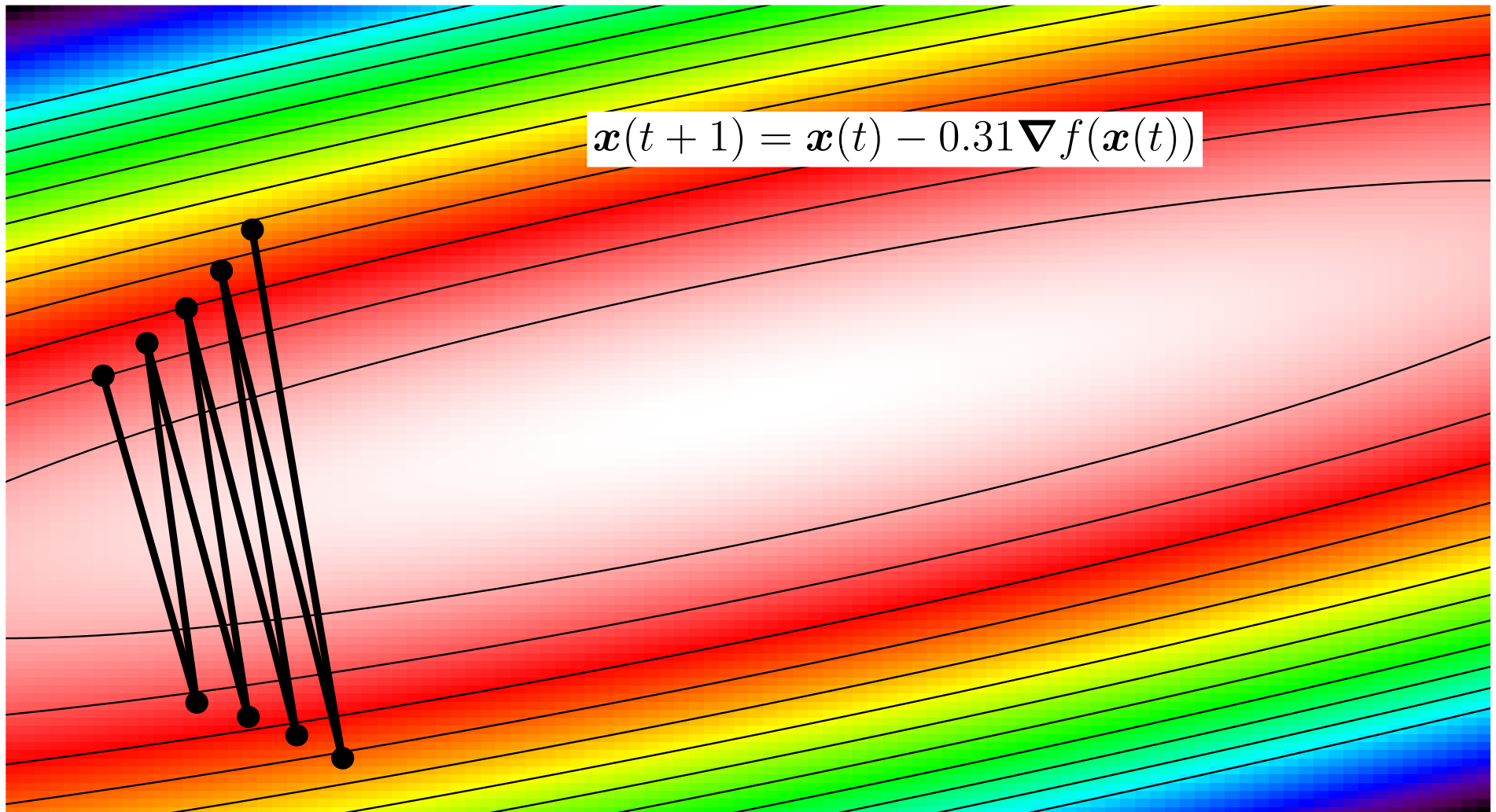
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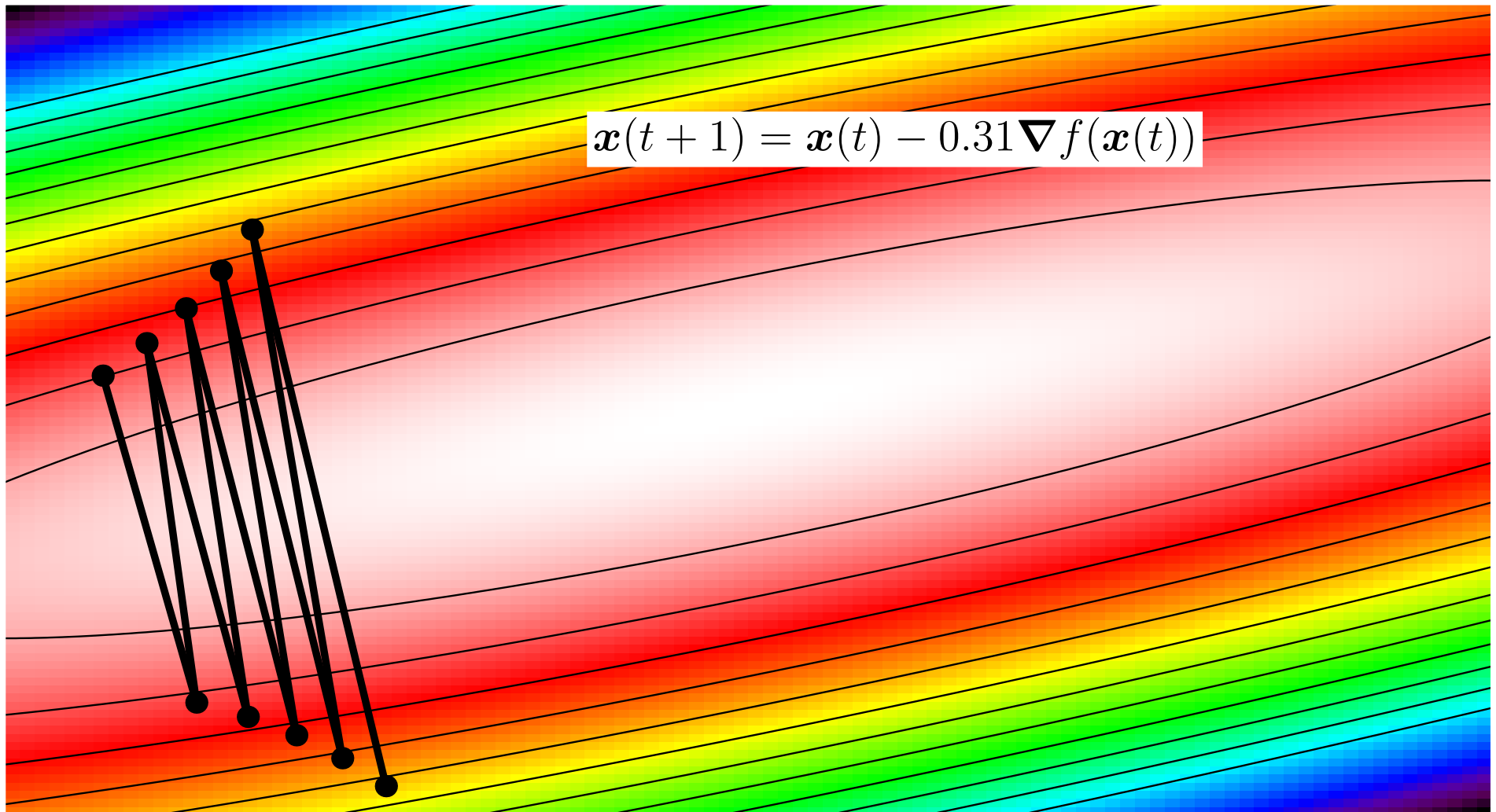
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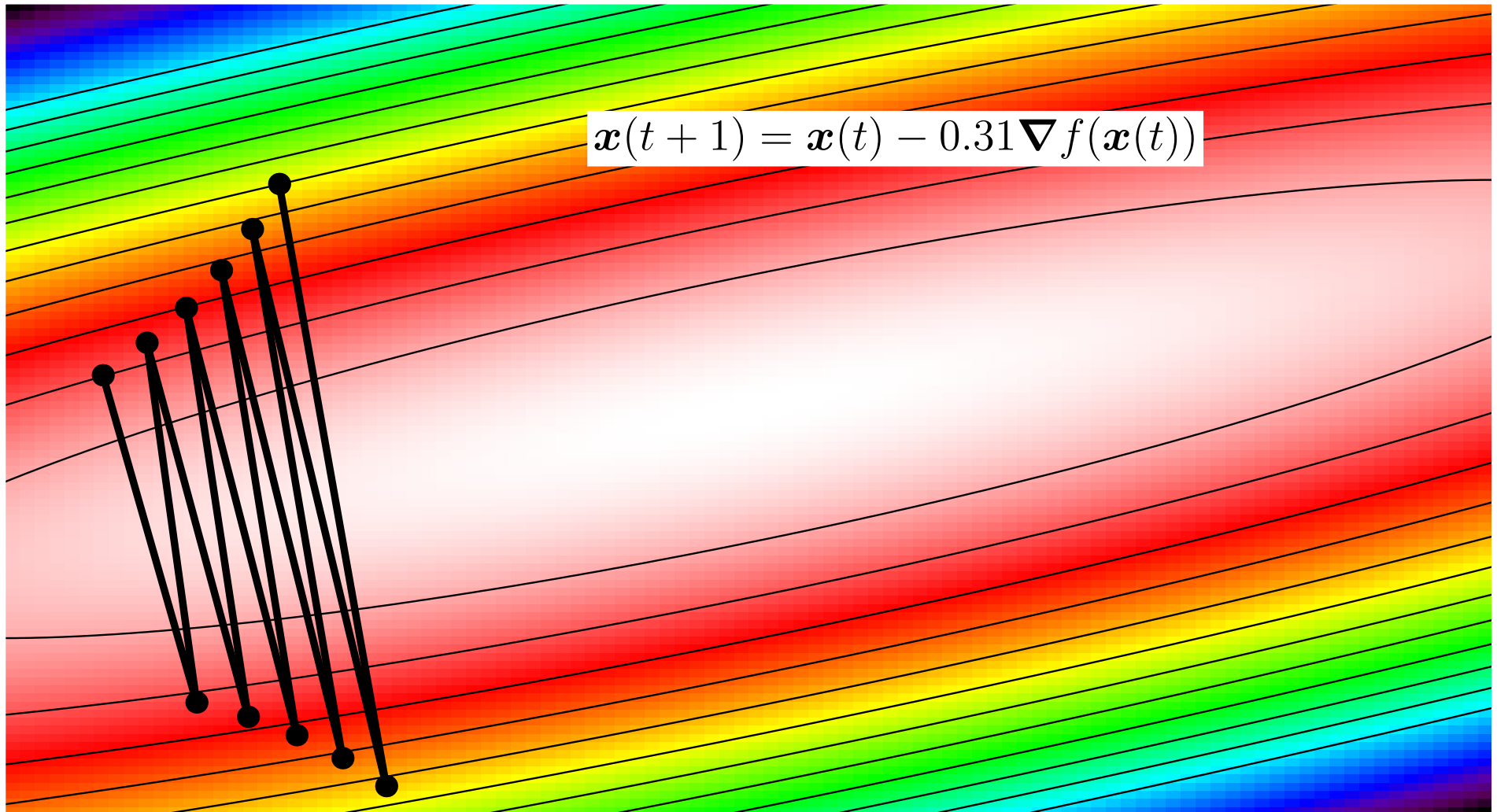
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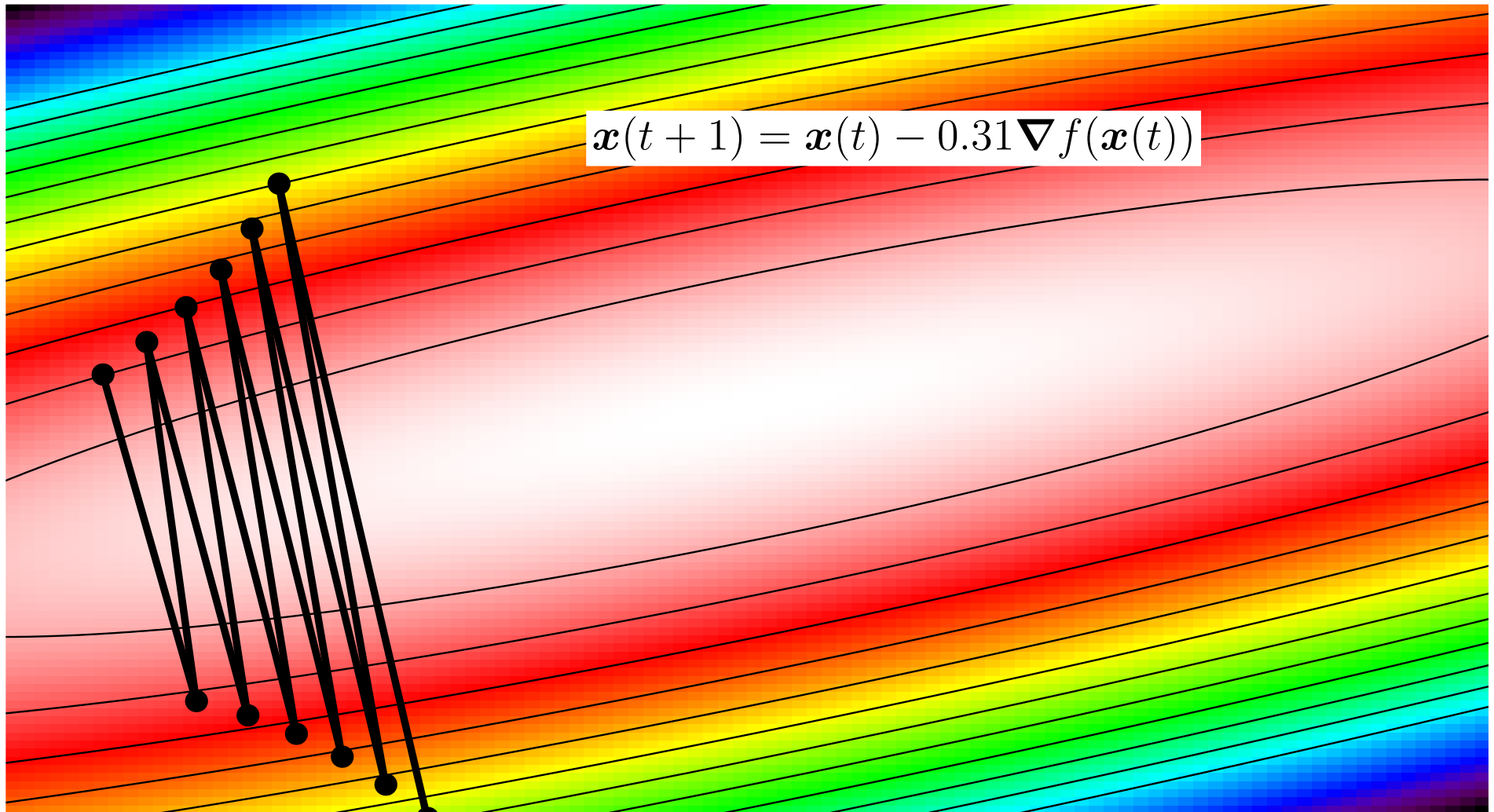
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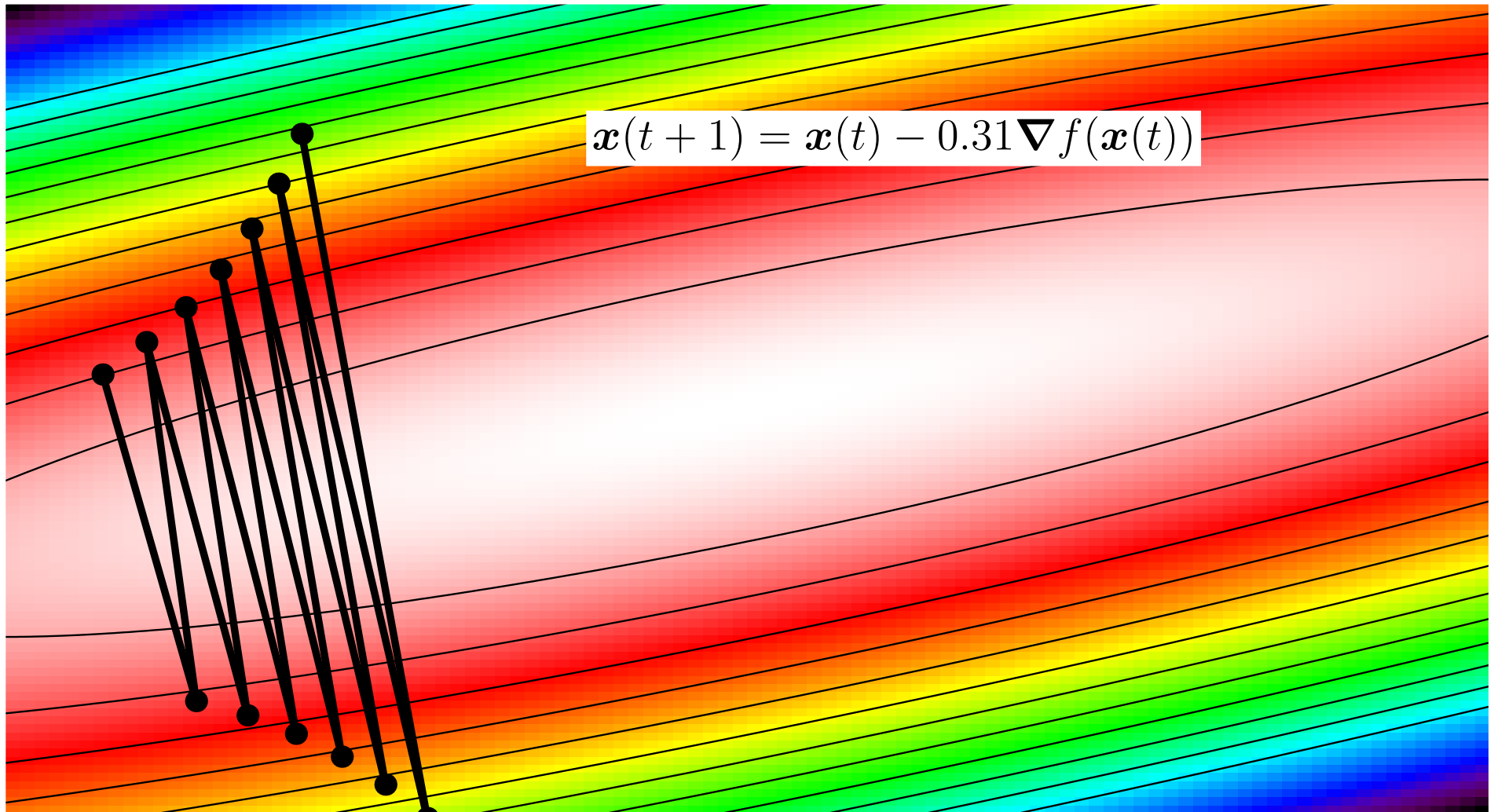
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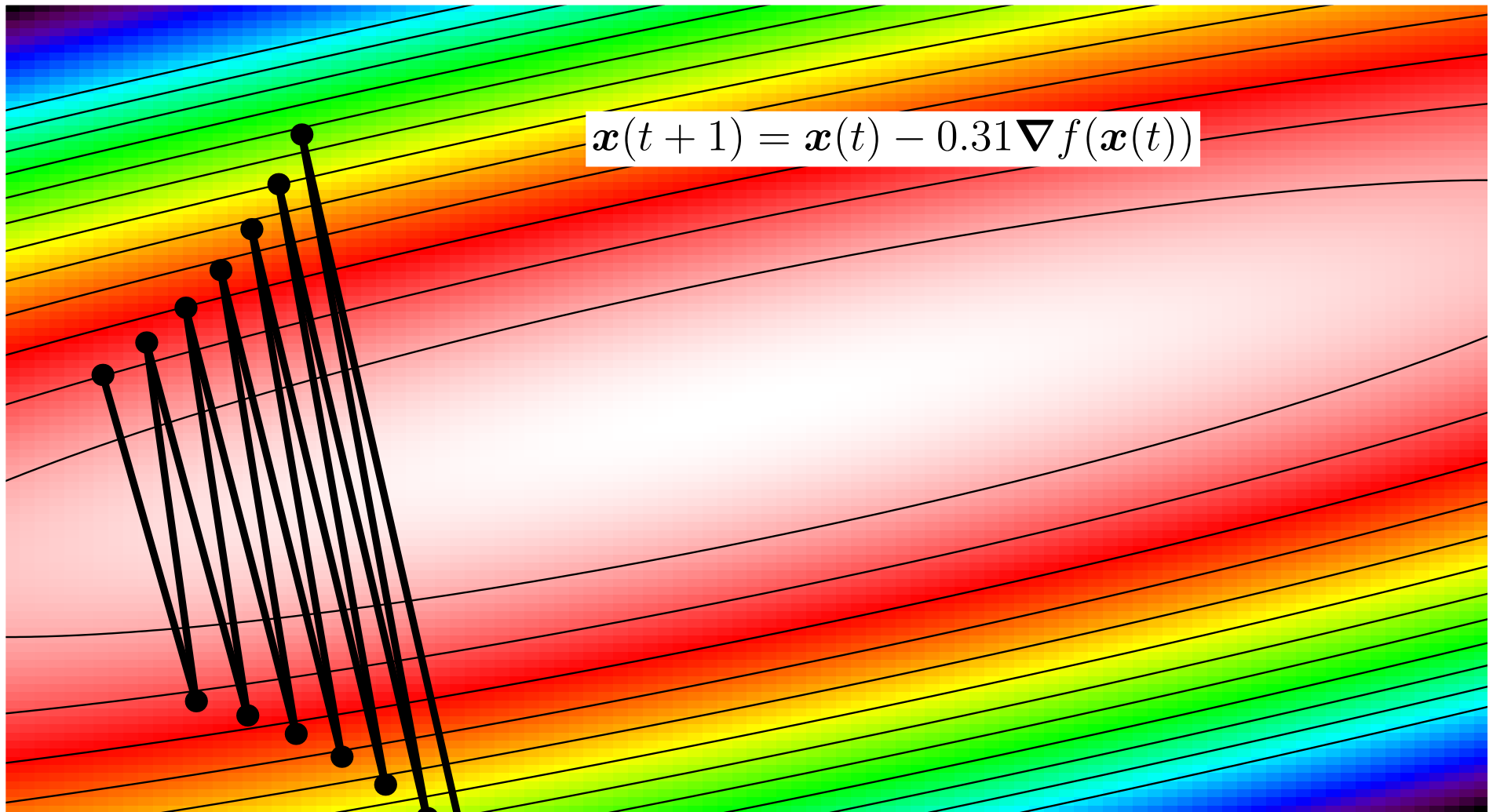
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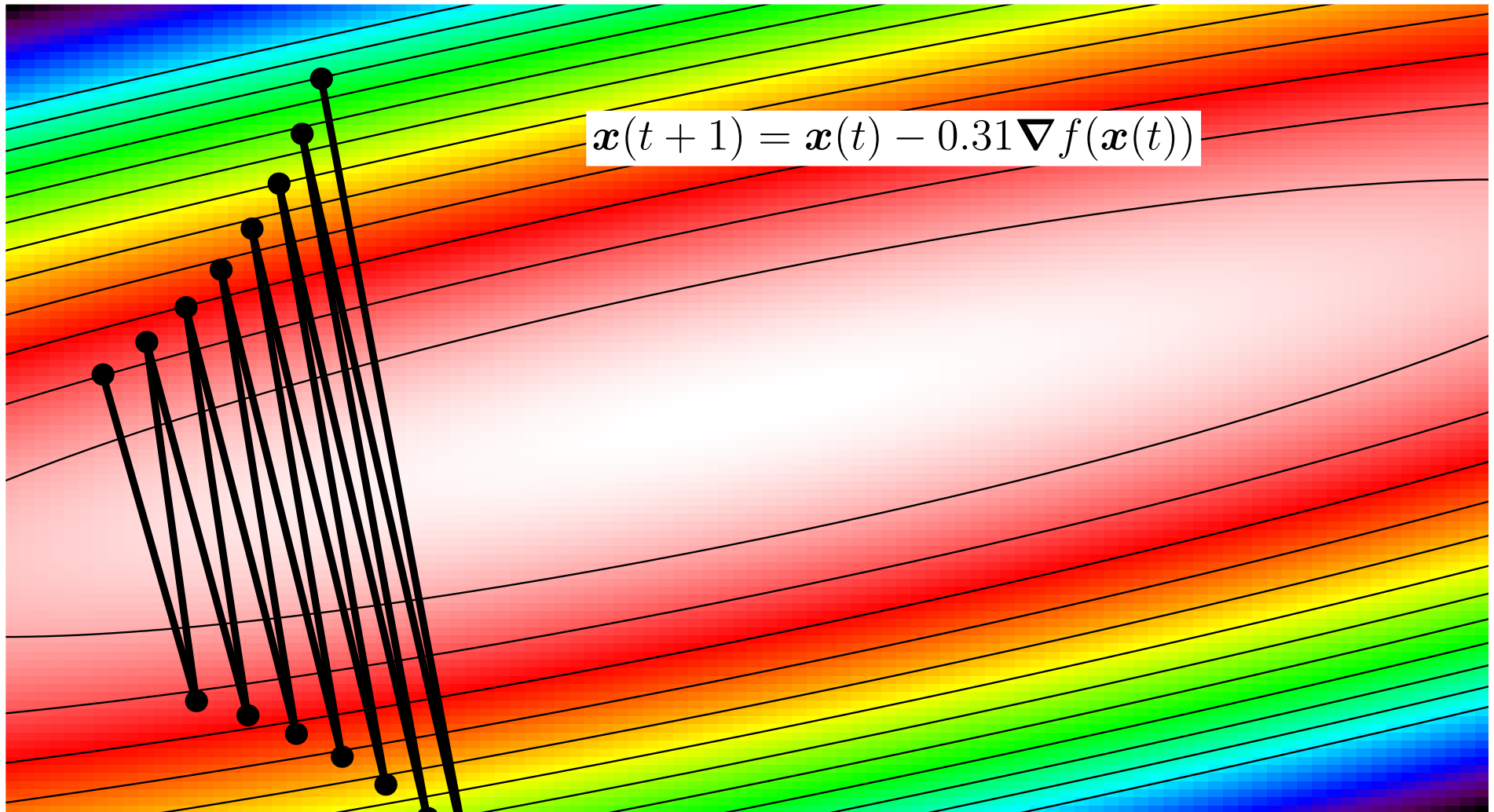
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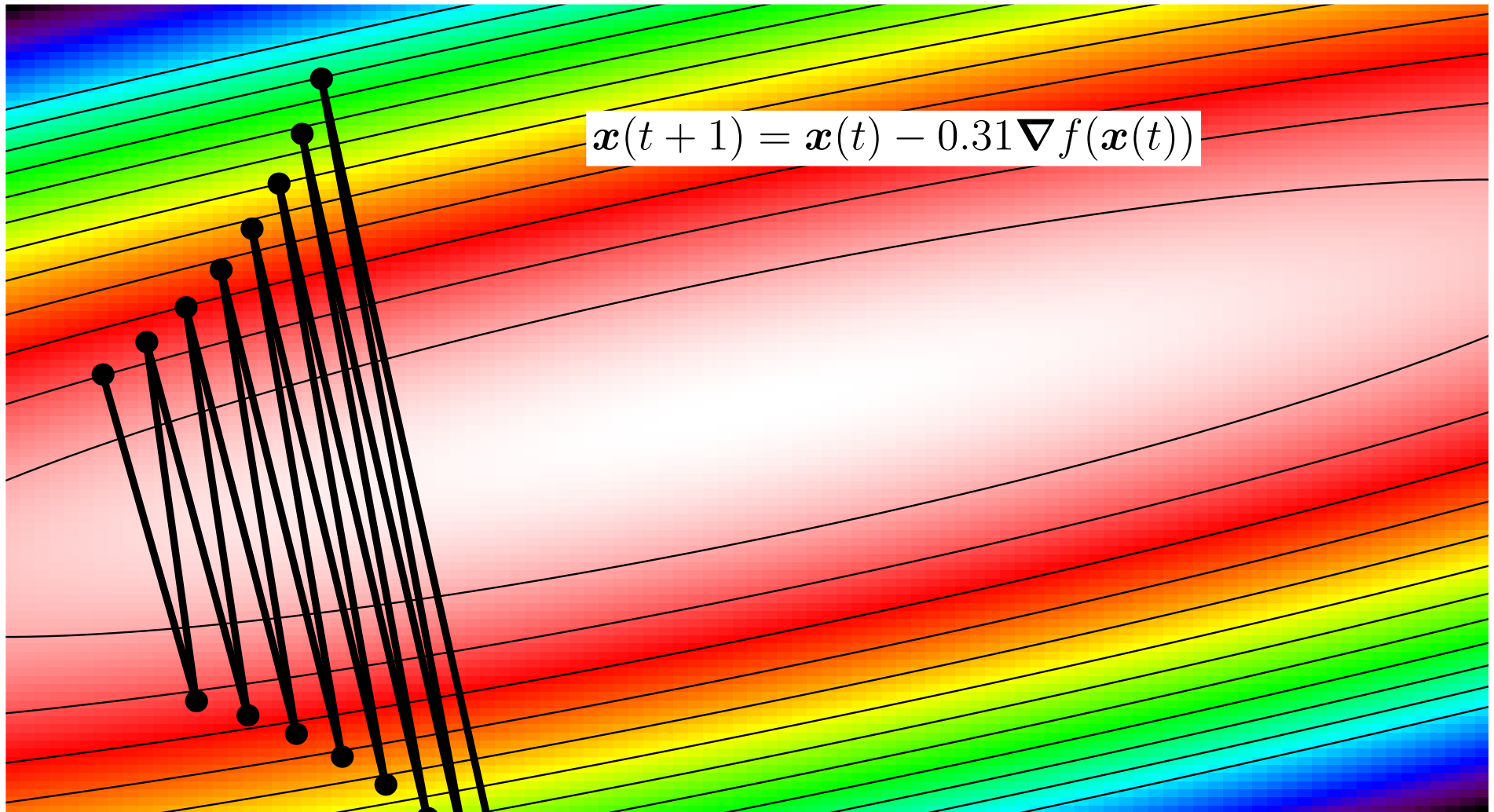
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- This seems impossibly slow, but weirdly straightforward SGD is often used in deep learning

Avoiding Divergence

- We saw in the last lecture that gradient descent can actually diverge from a local optimum if you have a too big step size
- If we use too large a step size we quickly get NaN errors
- If we only use SGD the step size is determined by the largest eigenvalue of the Hessian
- This seems impossibly slow, but weirdly straightforward SGD is often used in deep learning (although nearly always with momentum)

Momentum

- In high dimensions we zig-zag:
 - ★ stepping consistently in directions with low curvature
 - ★ jumping backwards and forwards past the minimum in directions with high curvature
- By introducing “momentum” we can increase our steps in low curvature directions and decrease it in high curvature directions

$$\mathbf{v}(t+1) = \gamma \mathbf{v}(t) - \eta \nabla f(\mathbf{x}(t))$$

$$\mathbf{x}(t+1) = \mathbf{x}(t) + \mathbf{v}(t+1)$$

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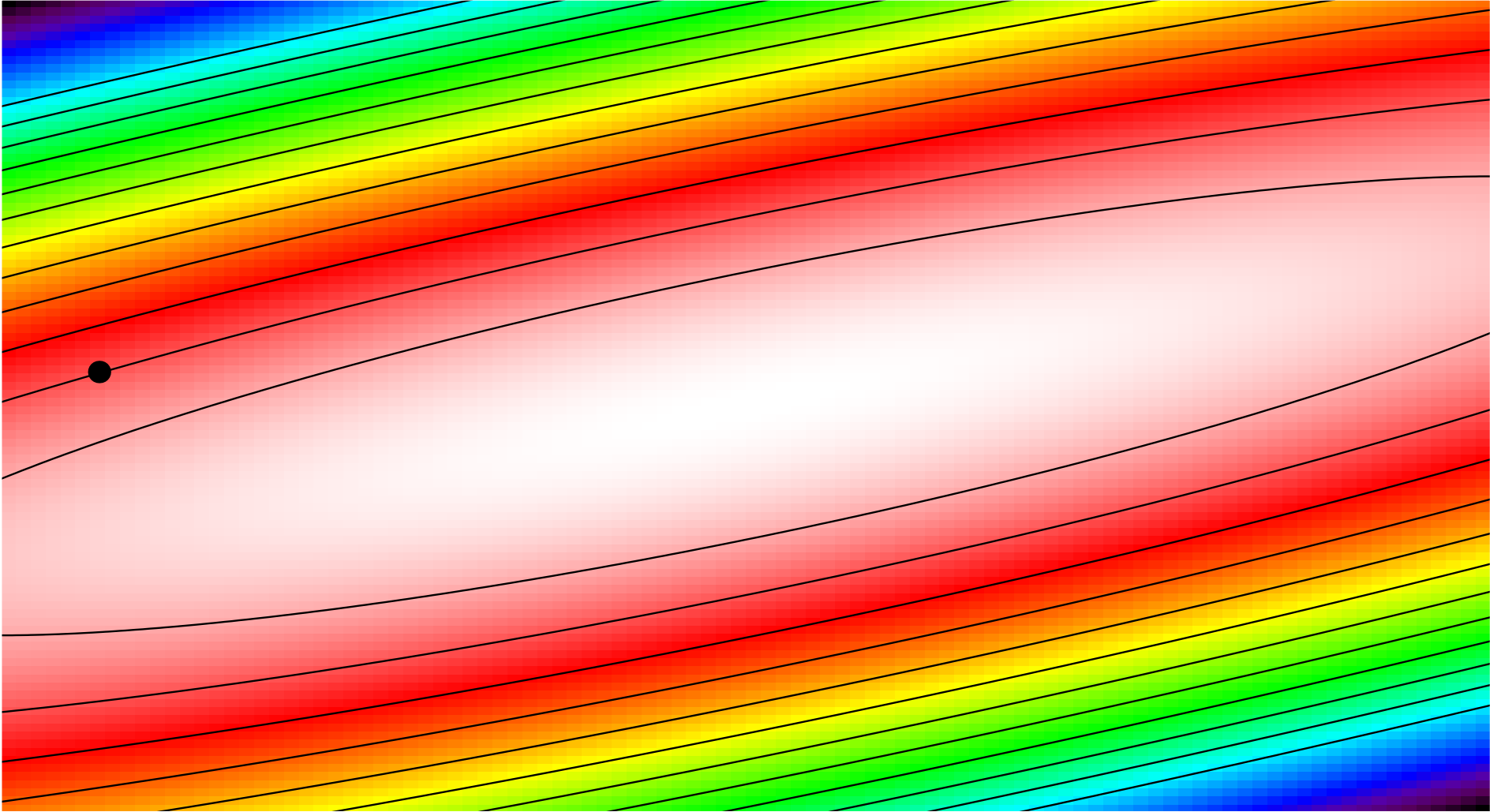
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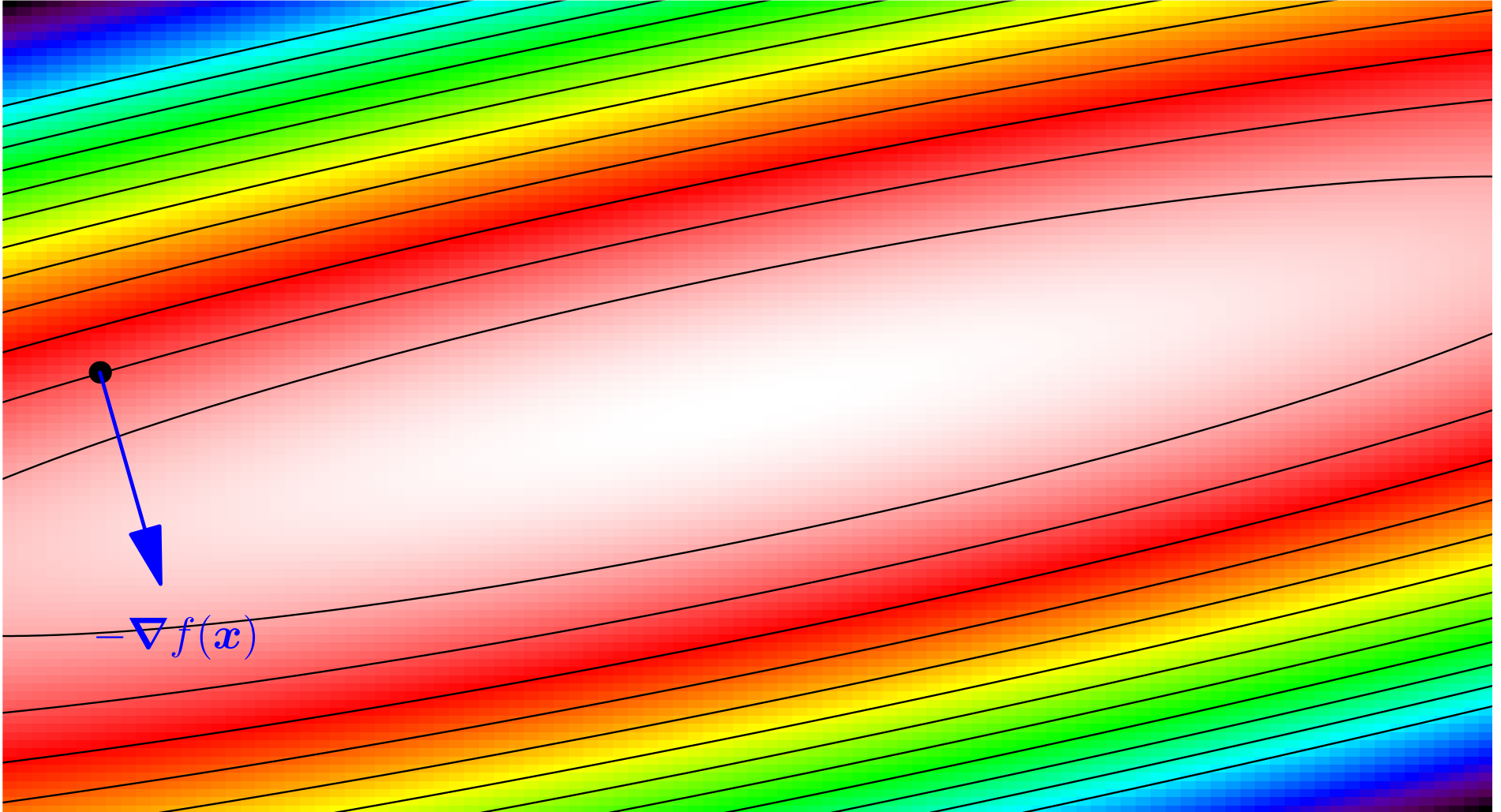
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$0 < \gamma < 1$ is a damping term

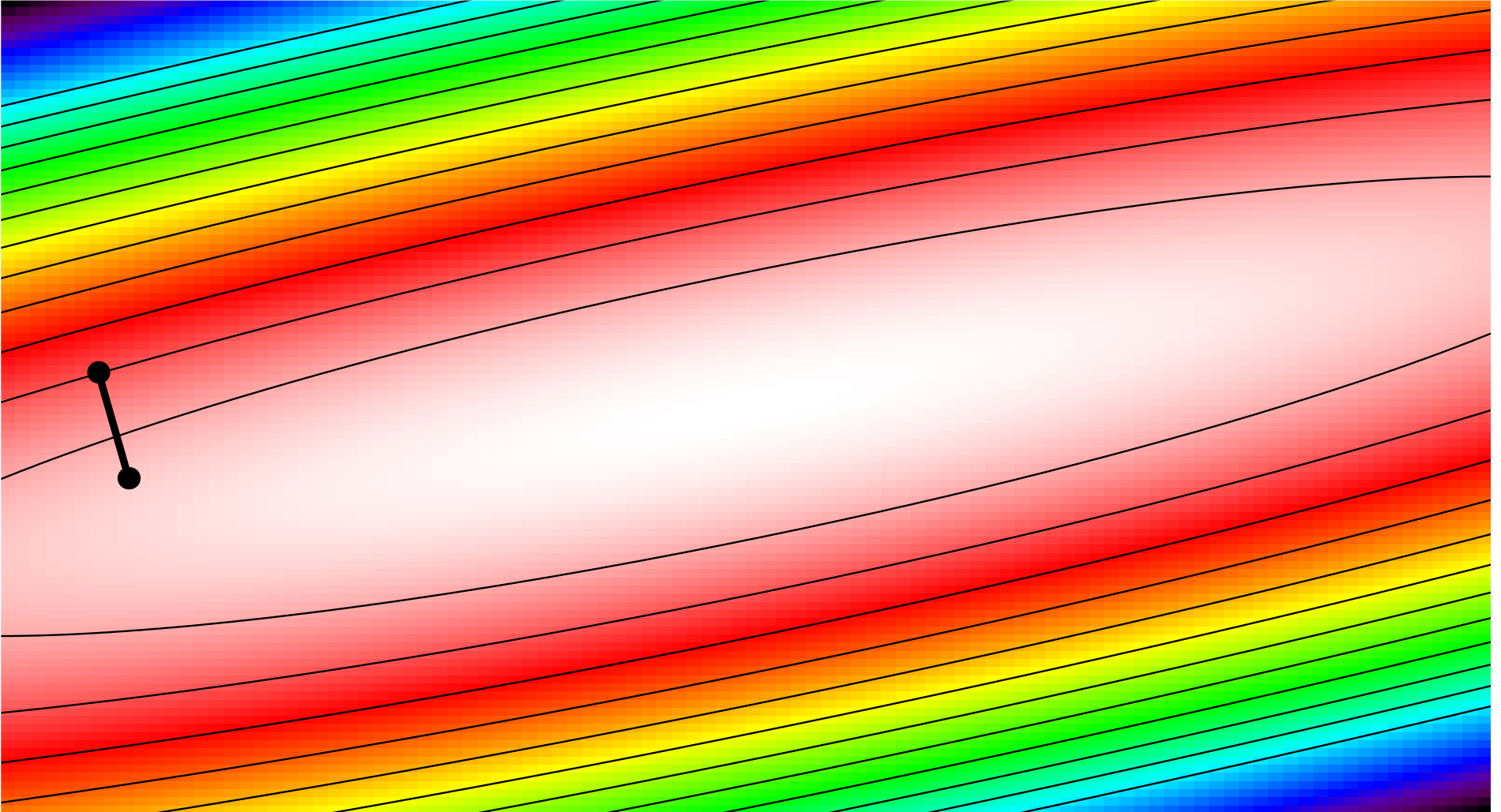
Gradient Descent with Momentum



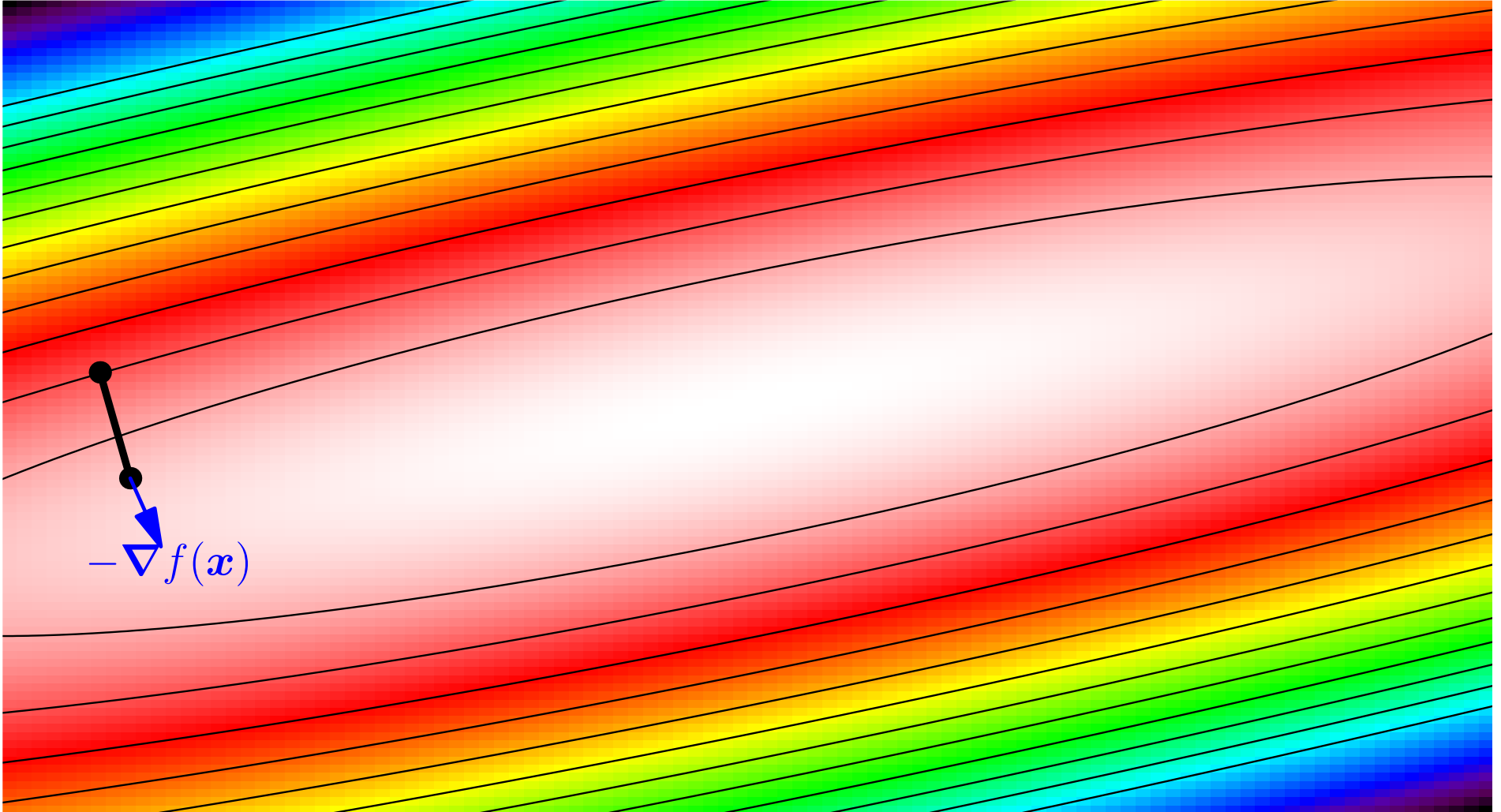
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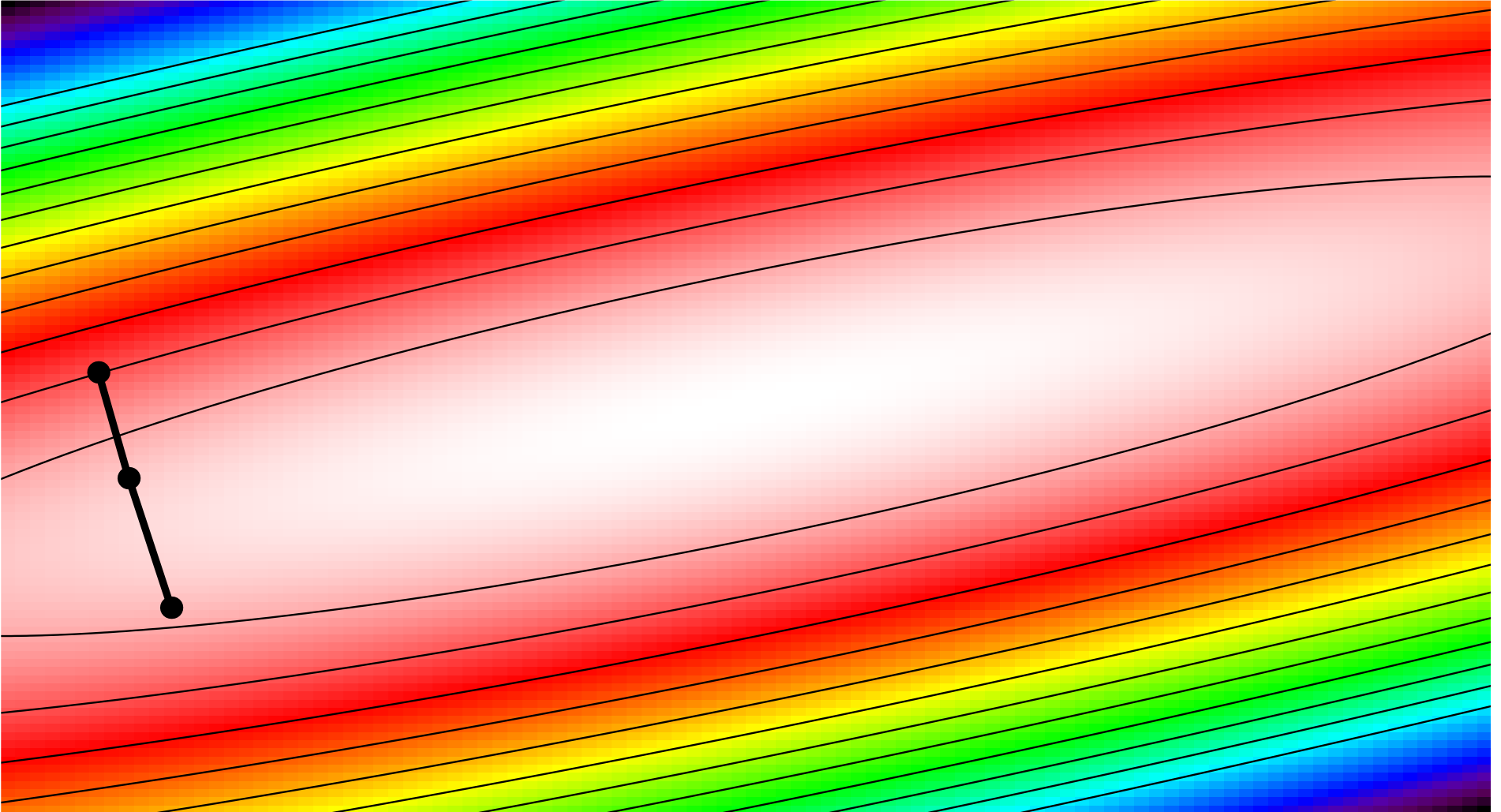
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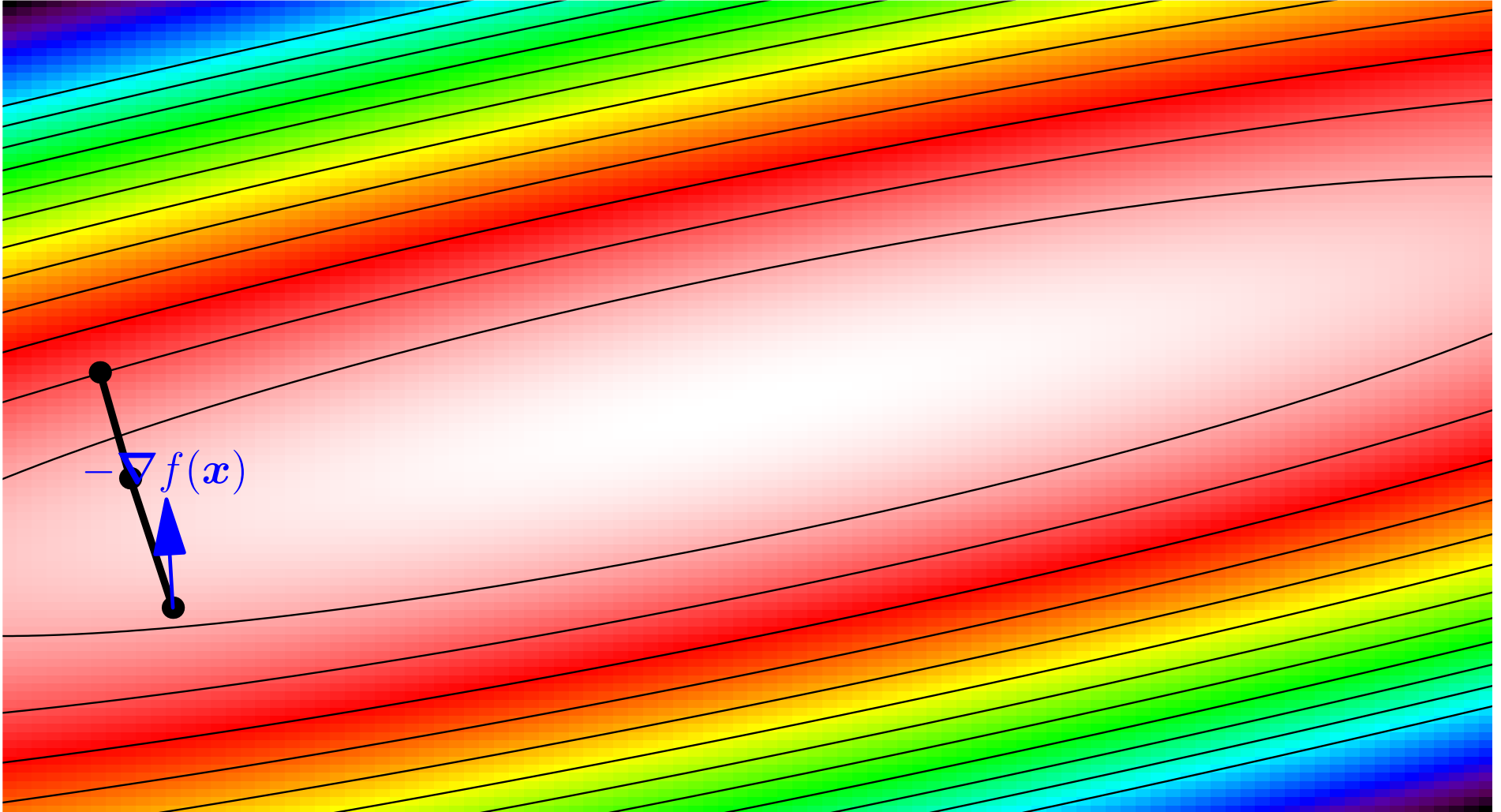
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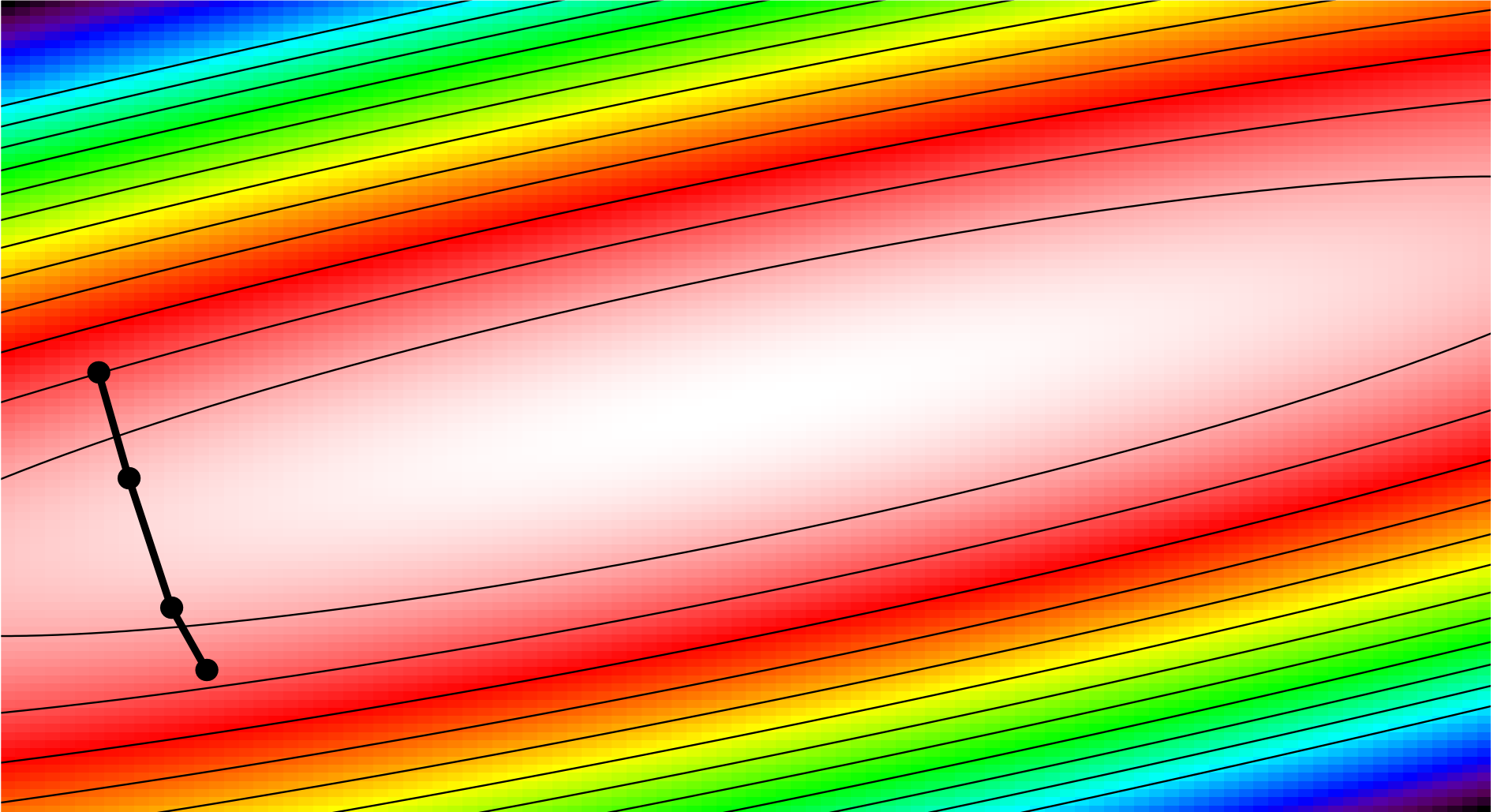
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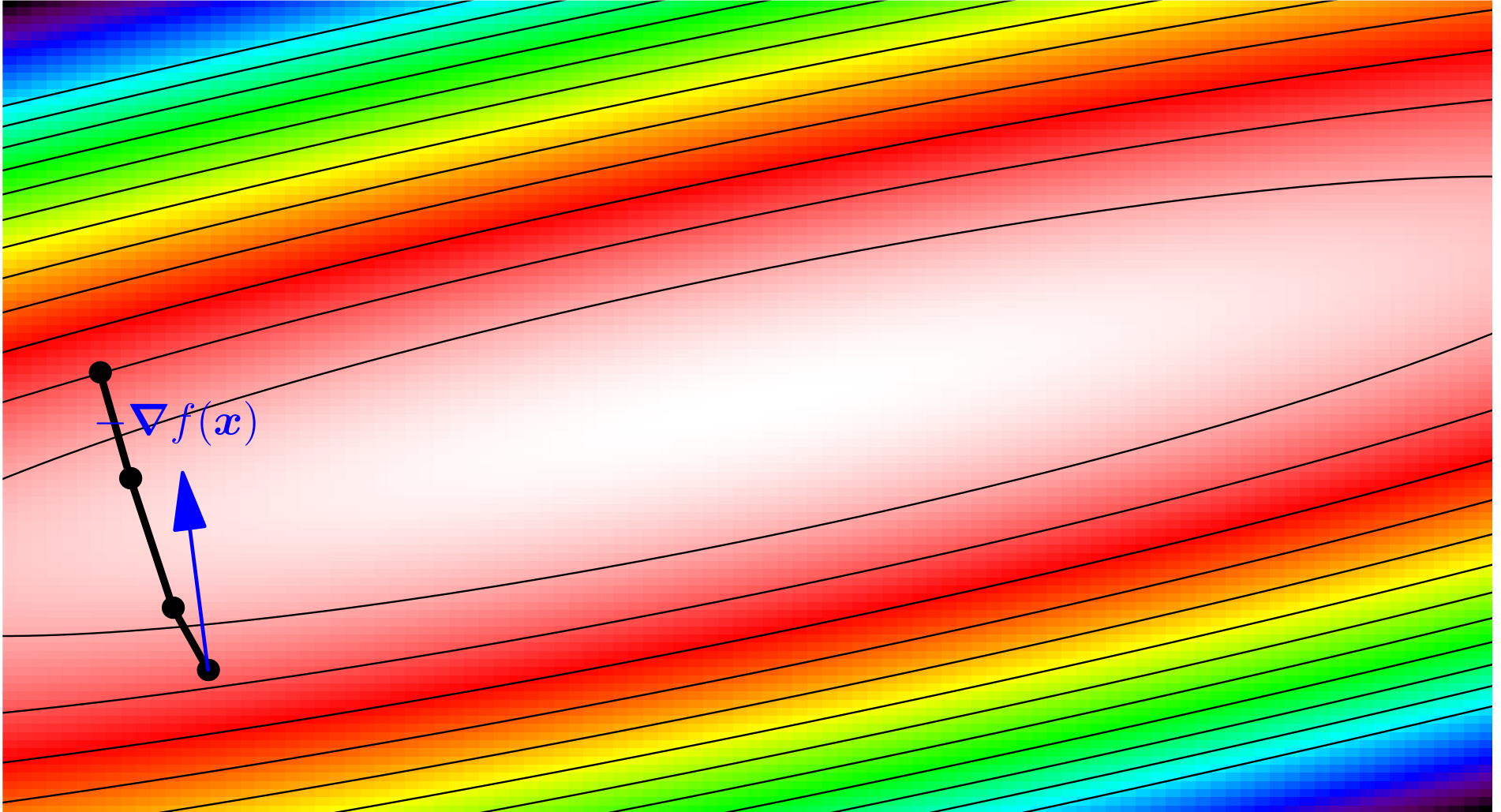
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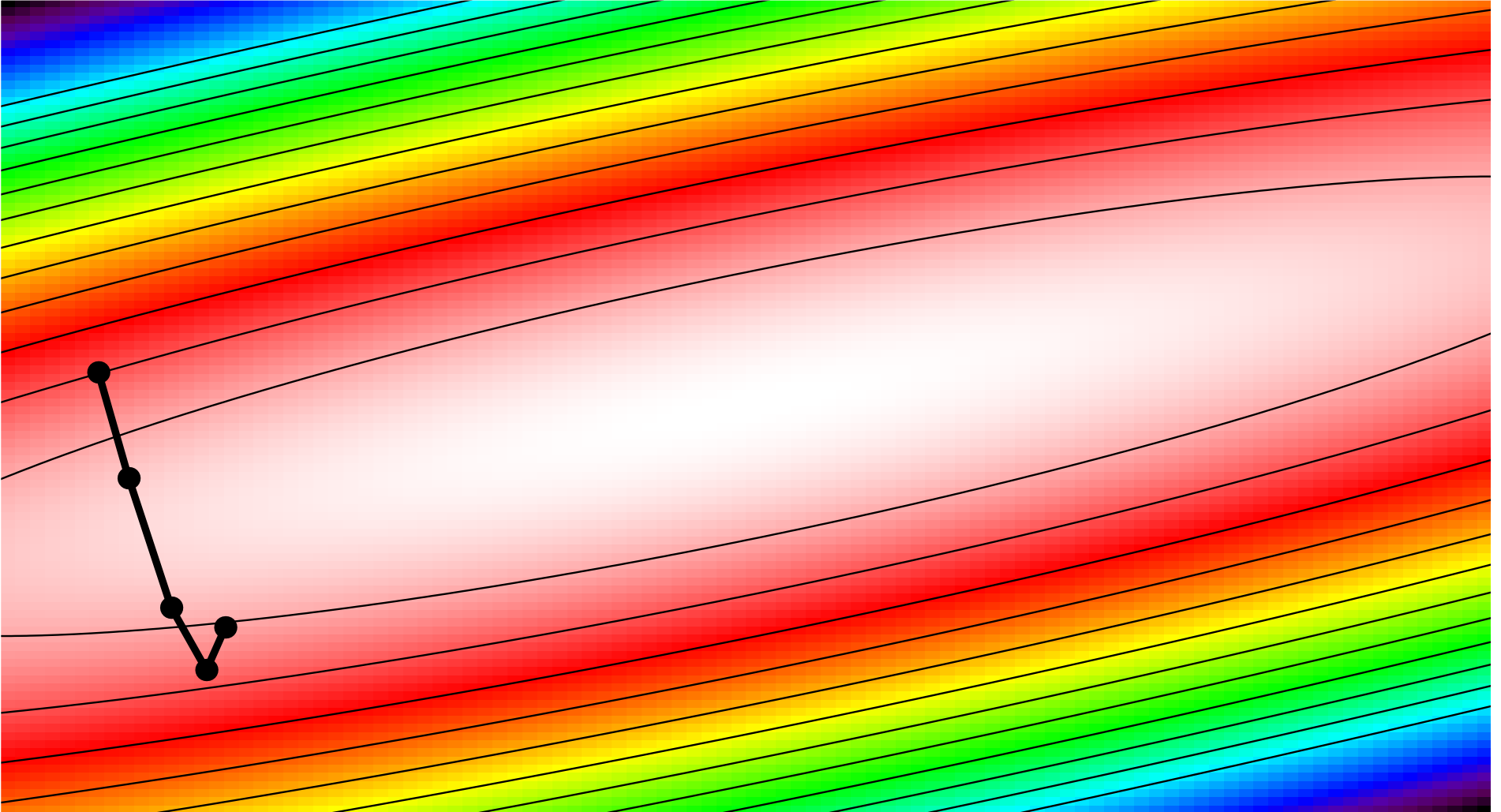
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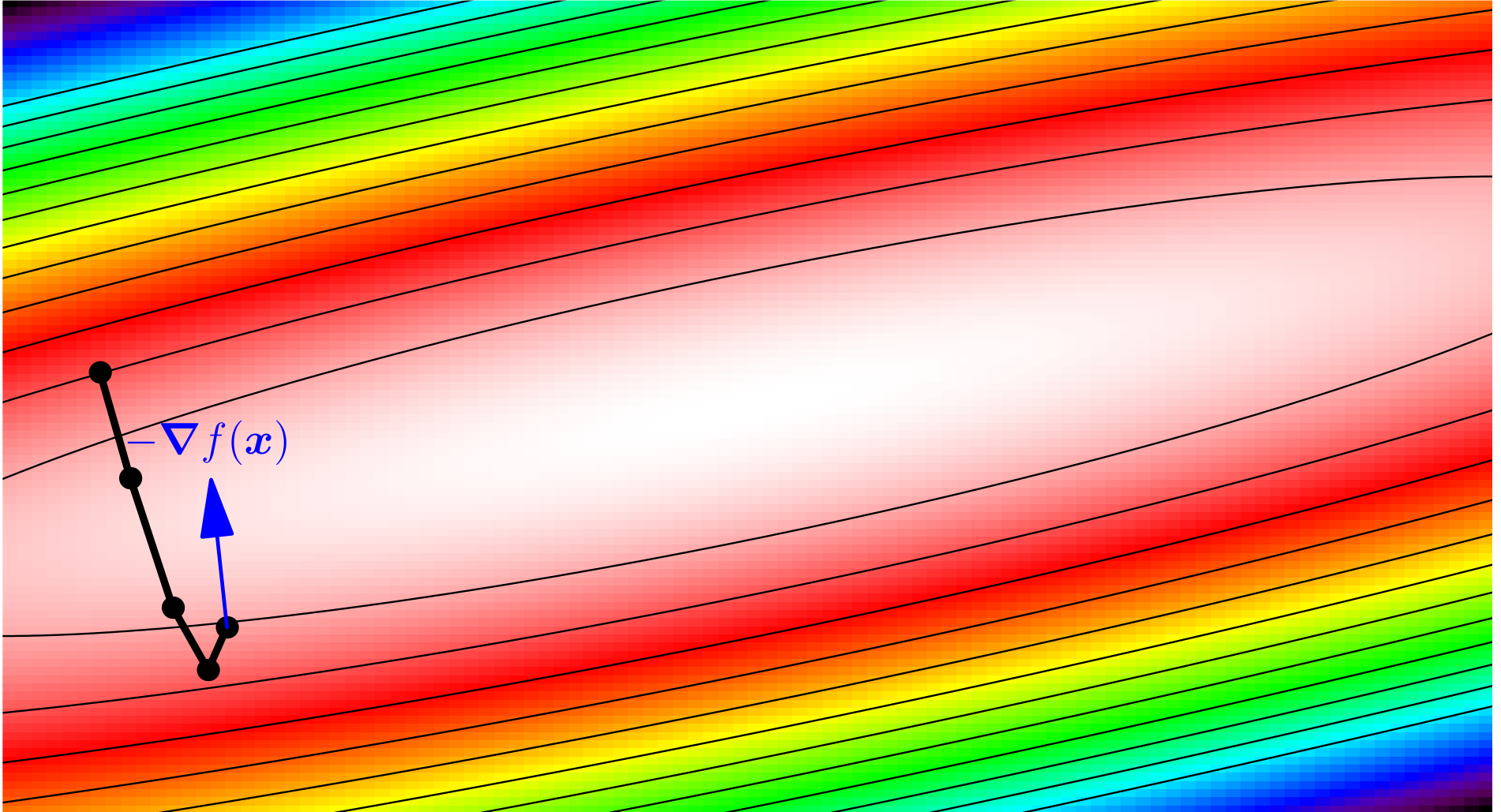
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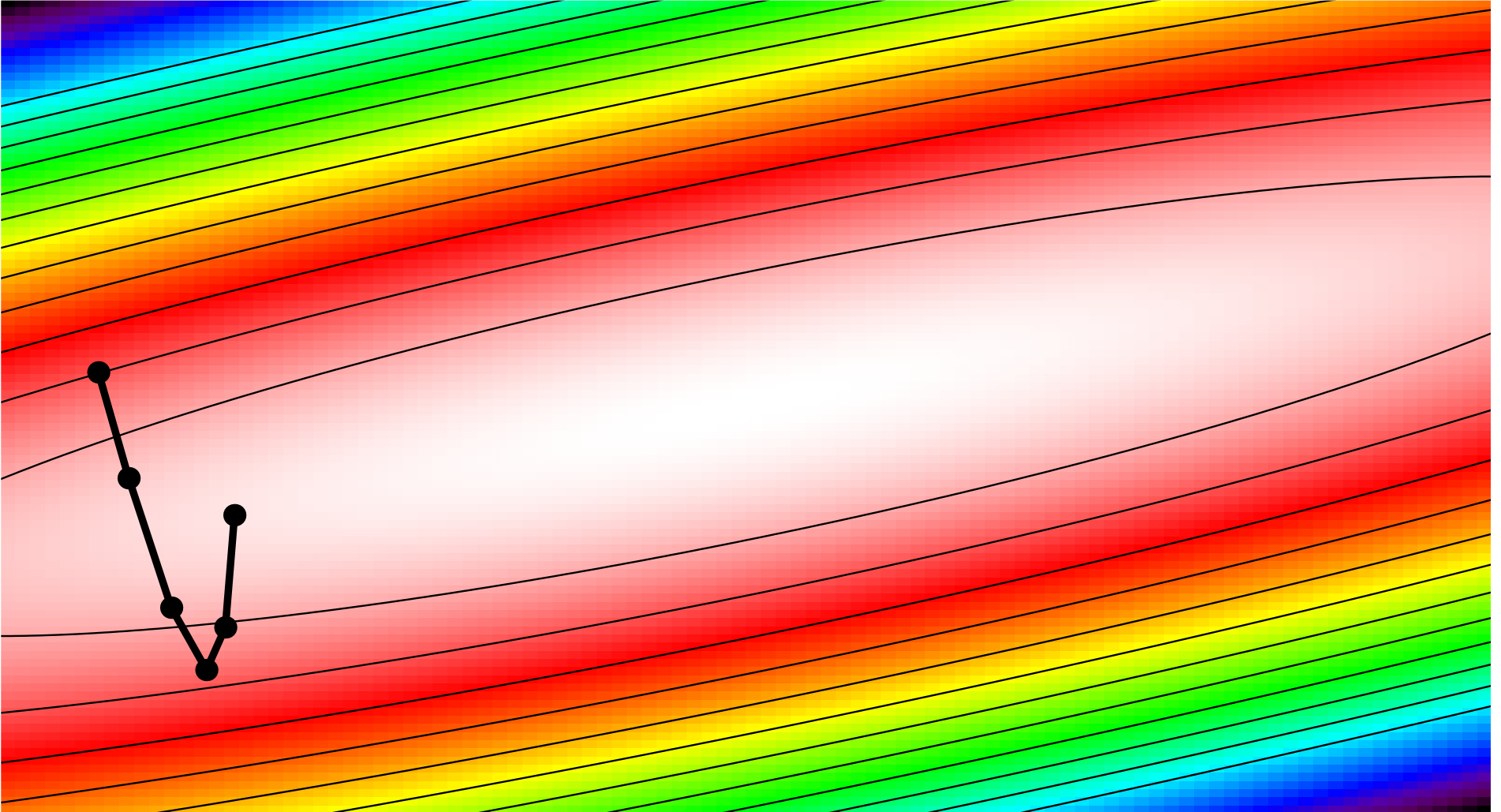
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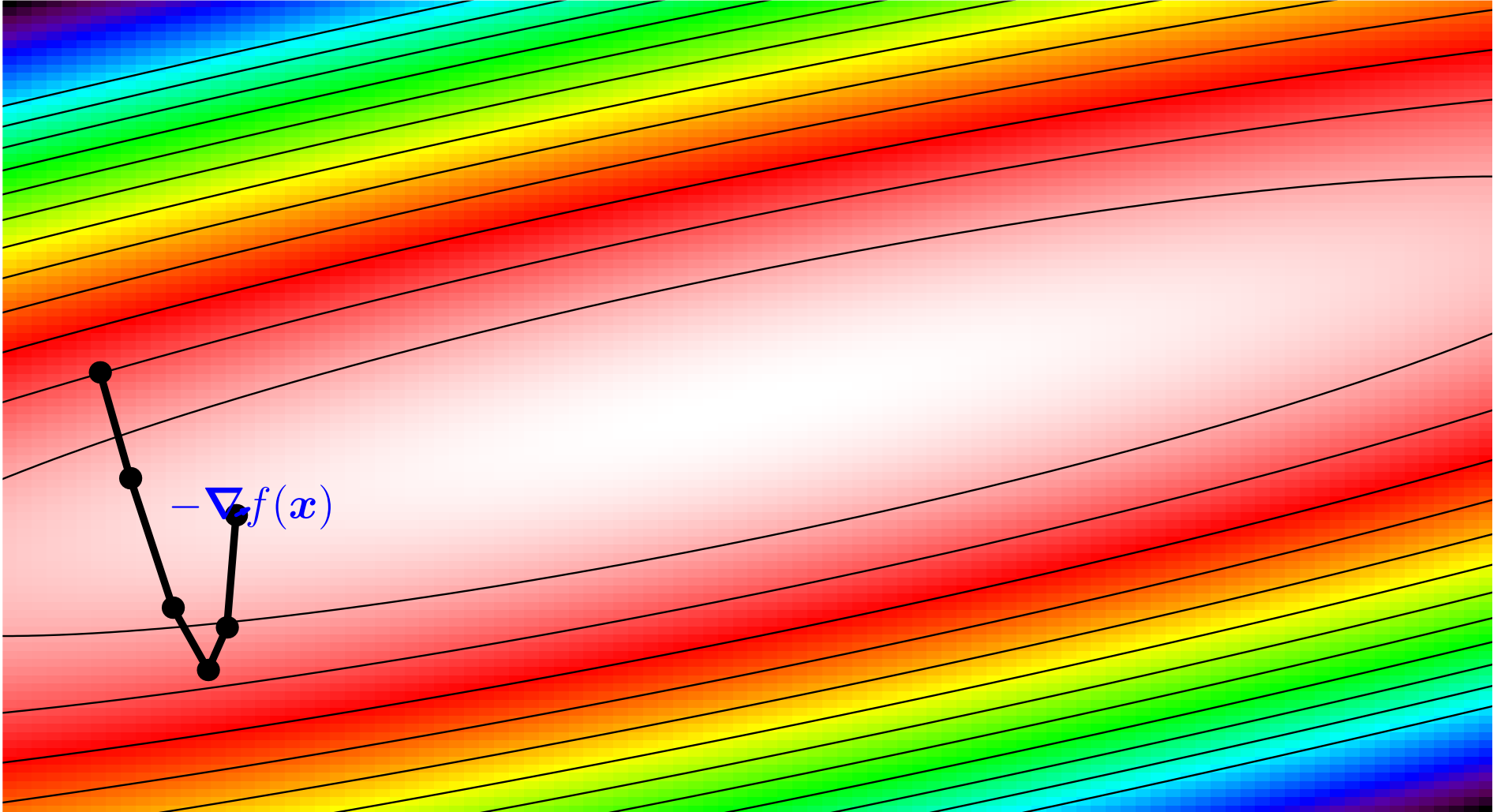
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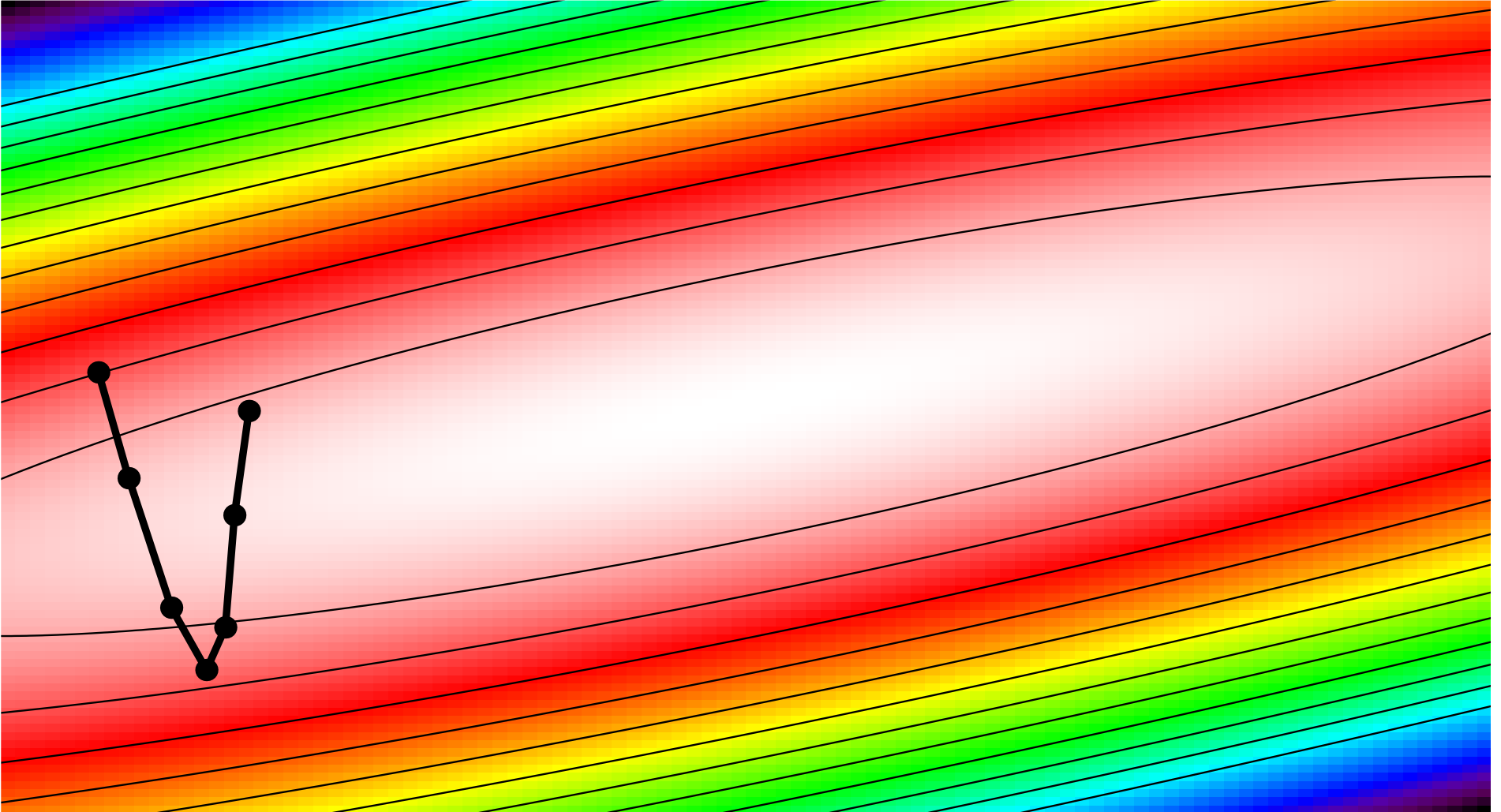
Gradient Descent with Momentum



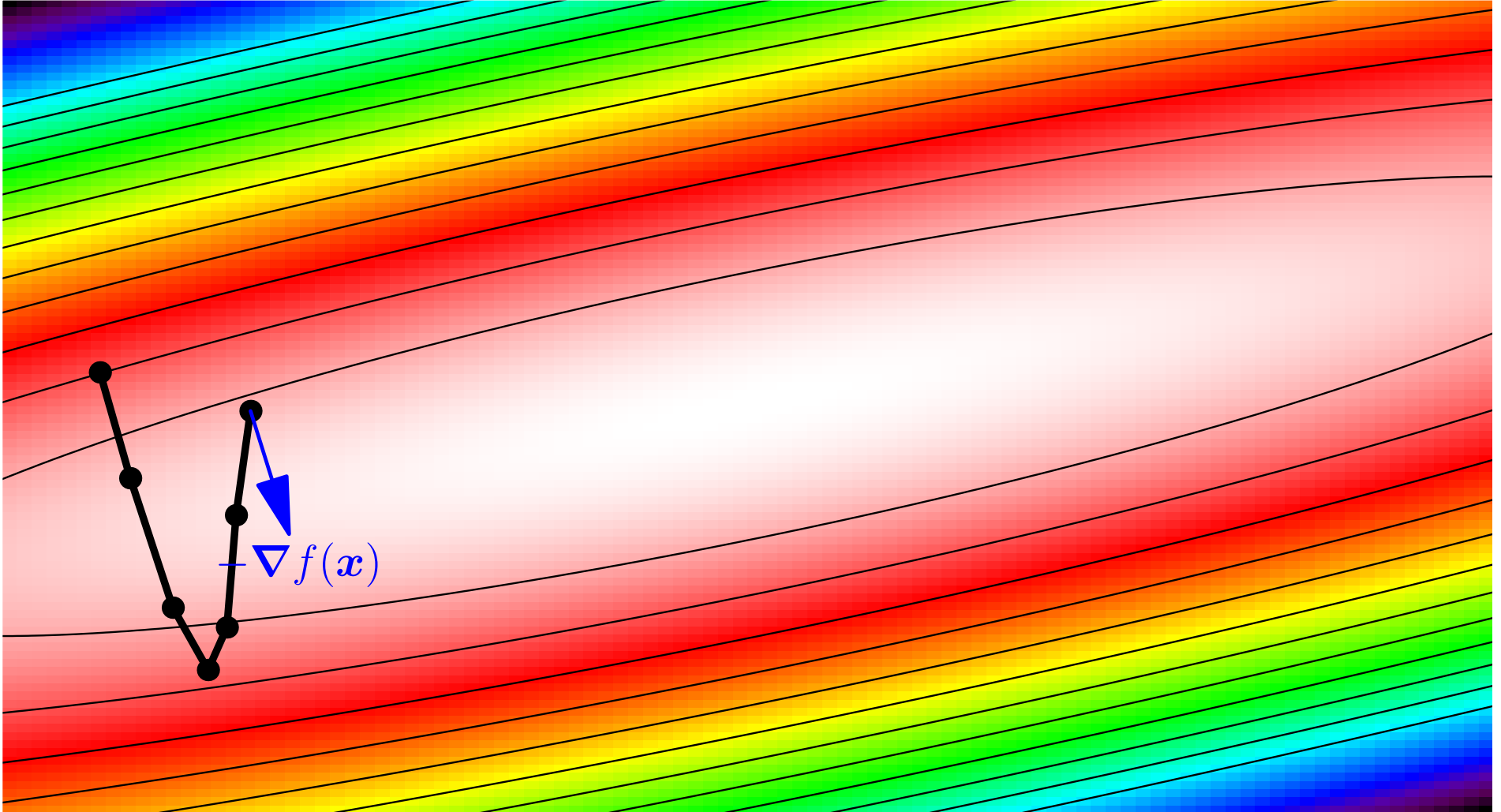
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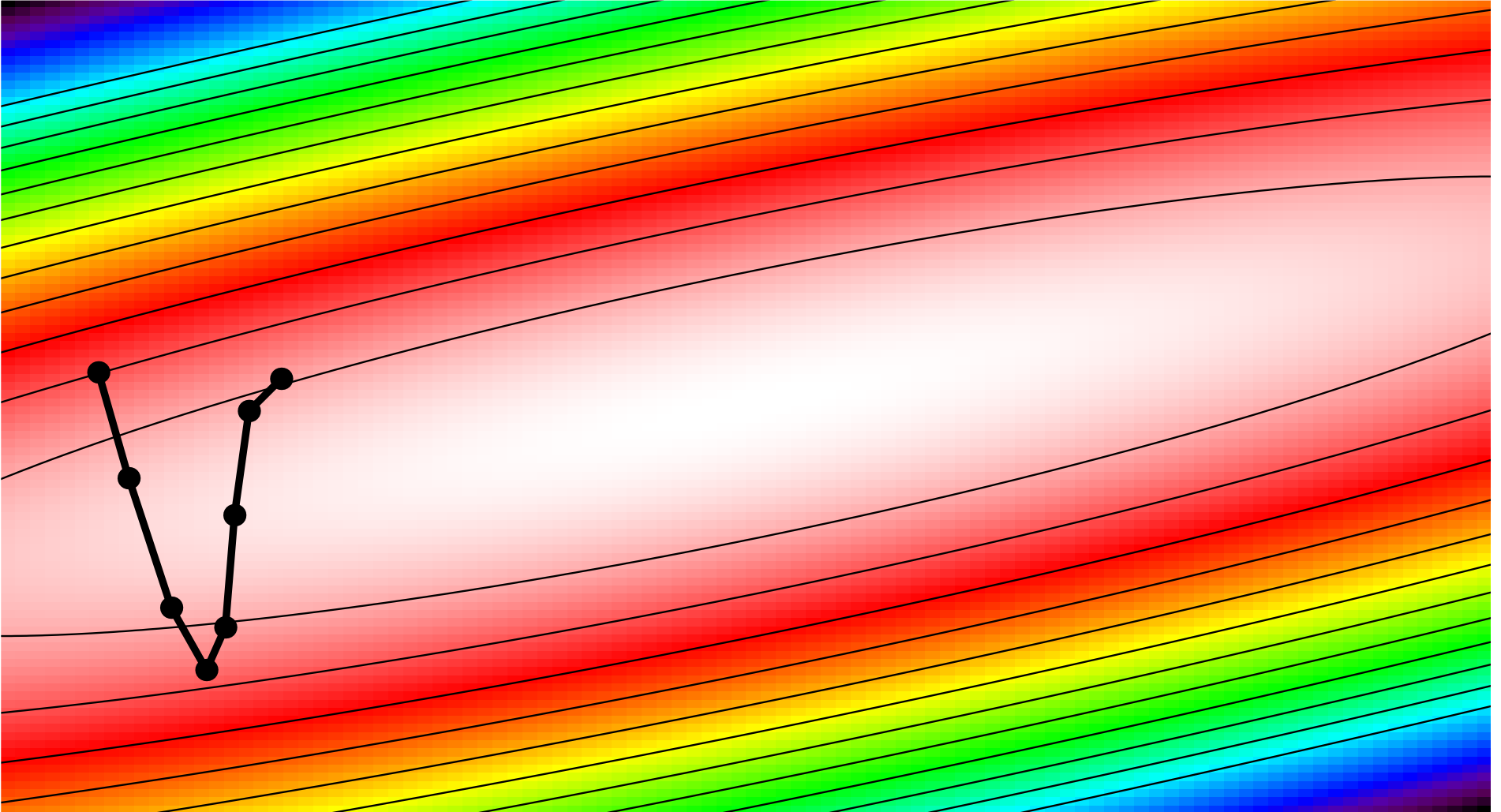
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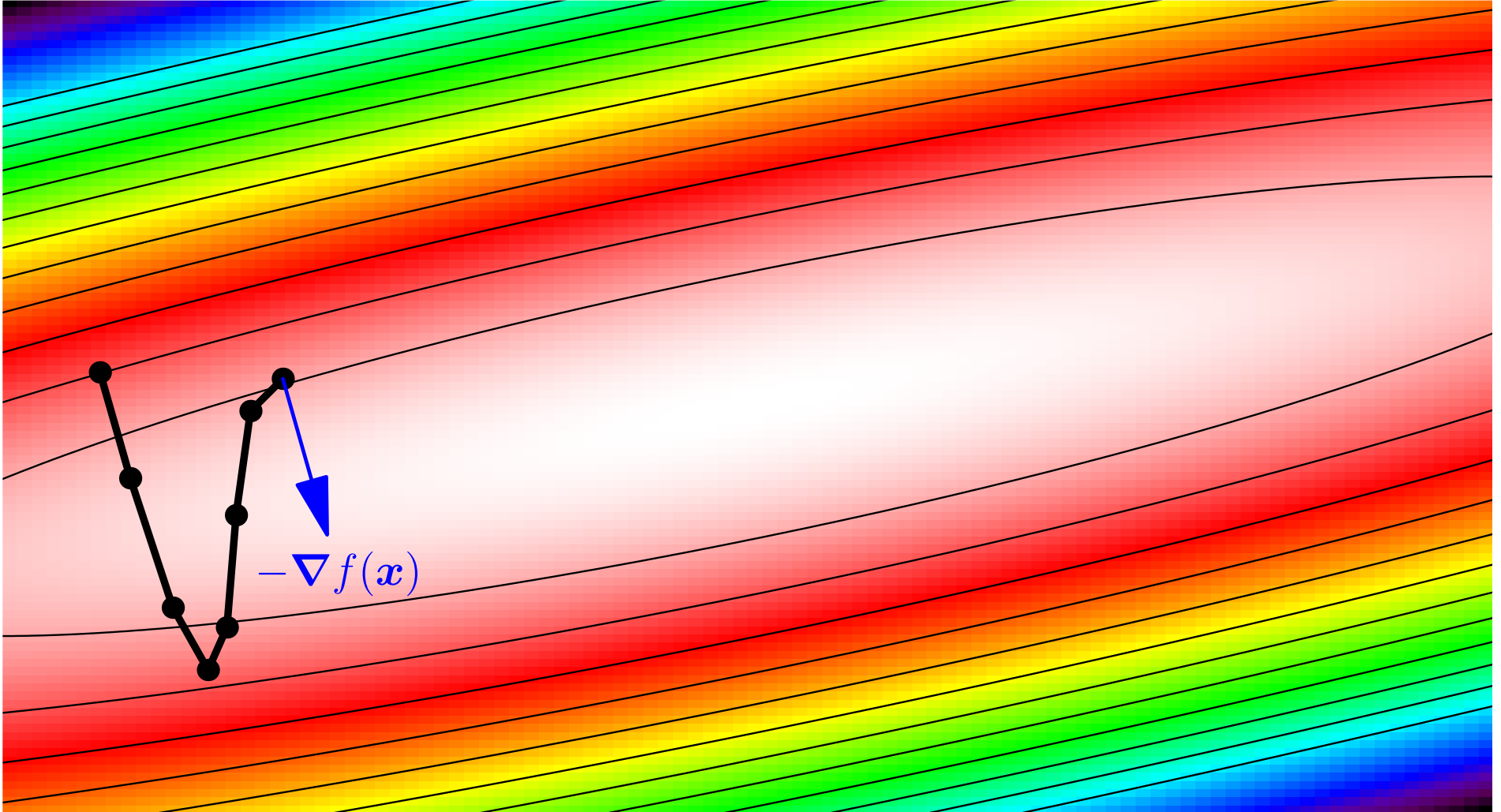
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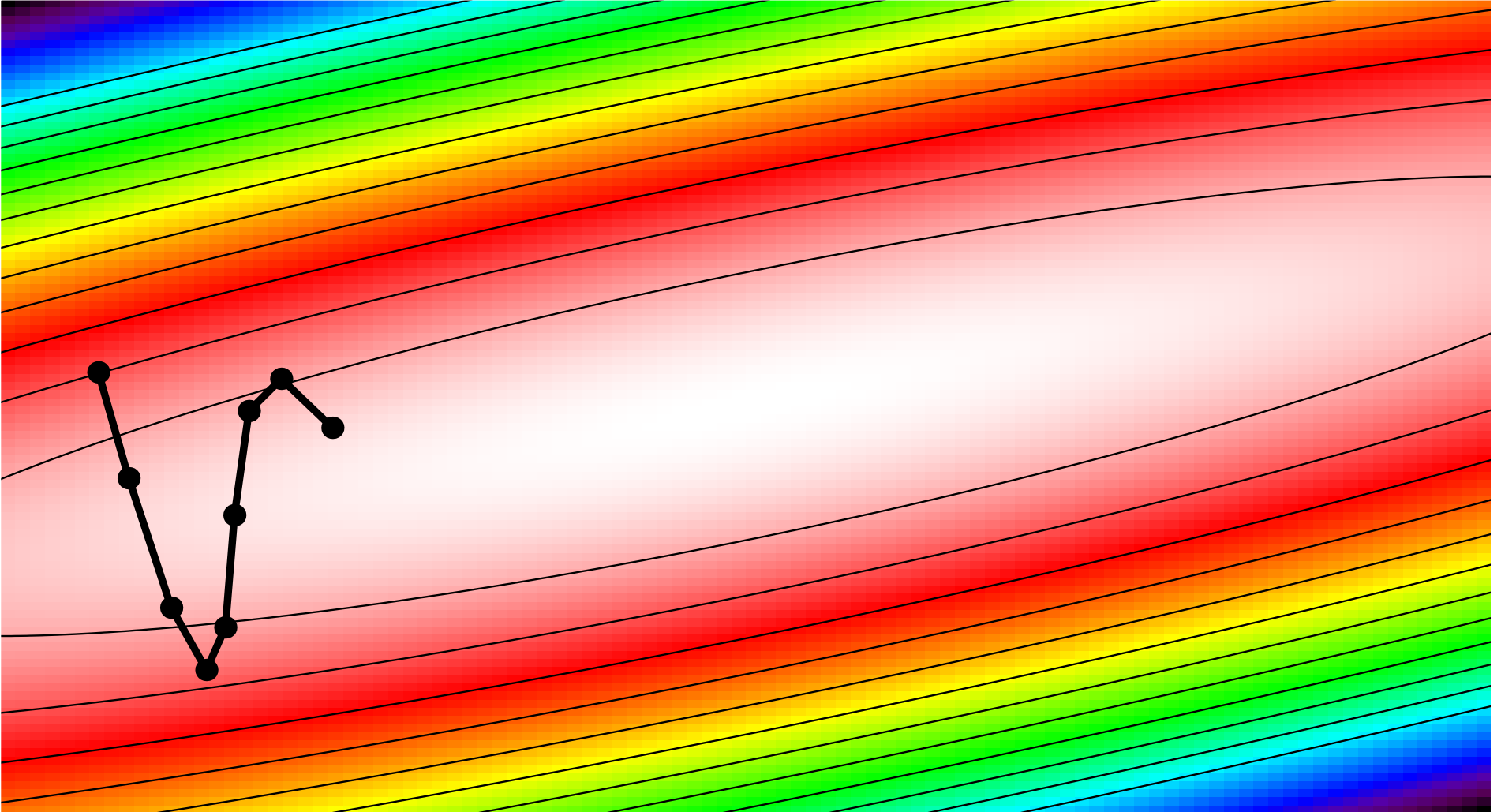
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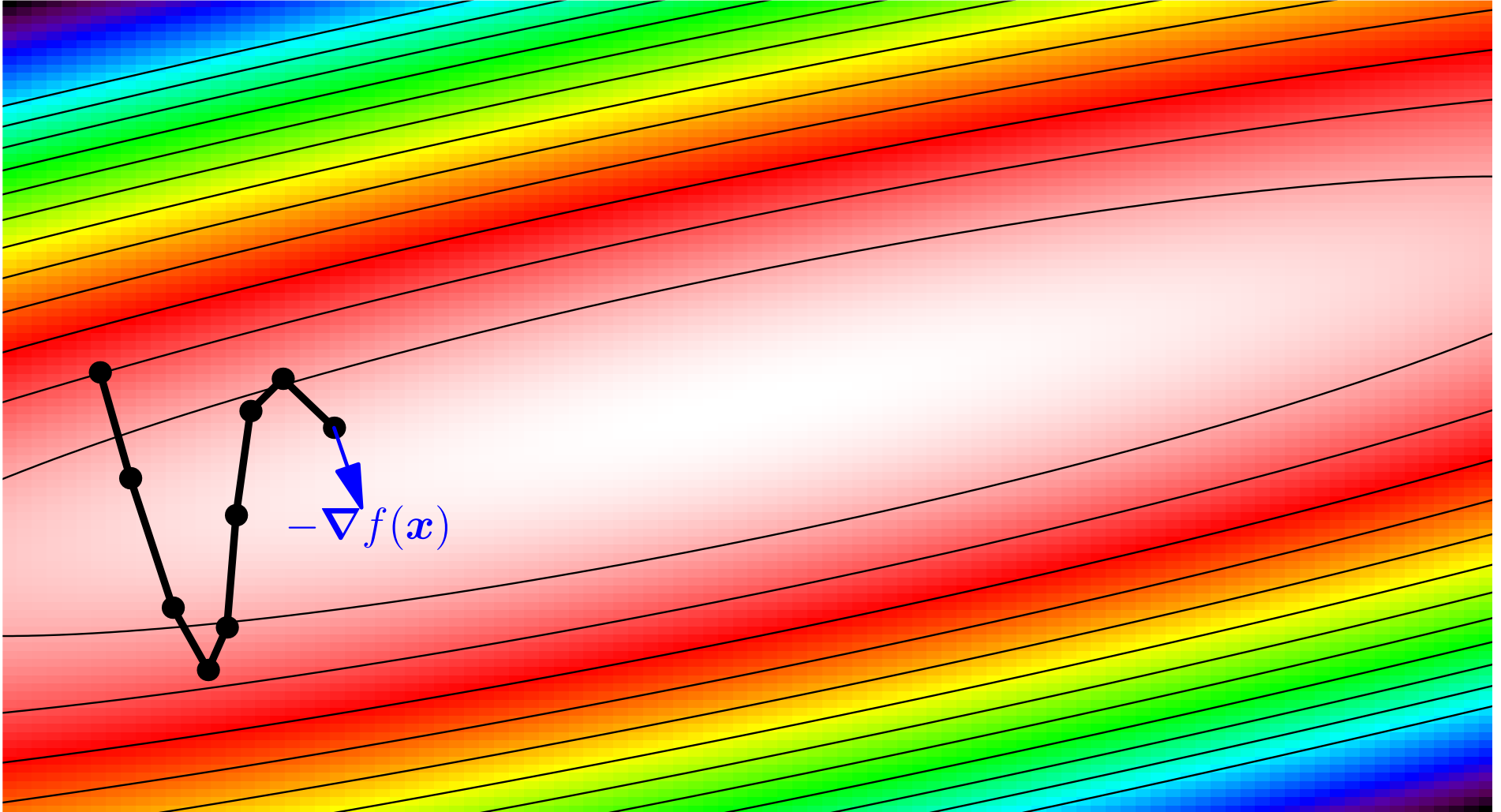
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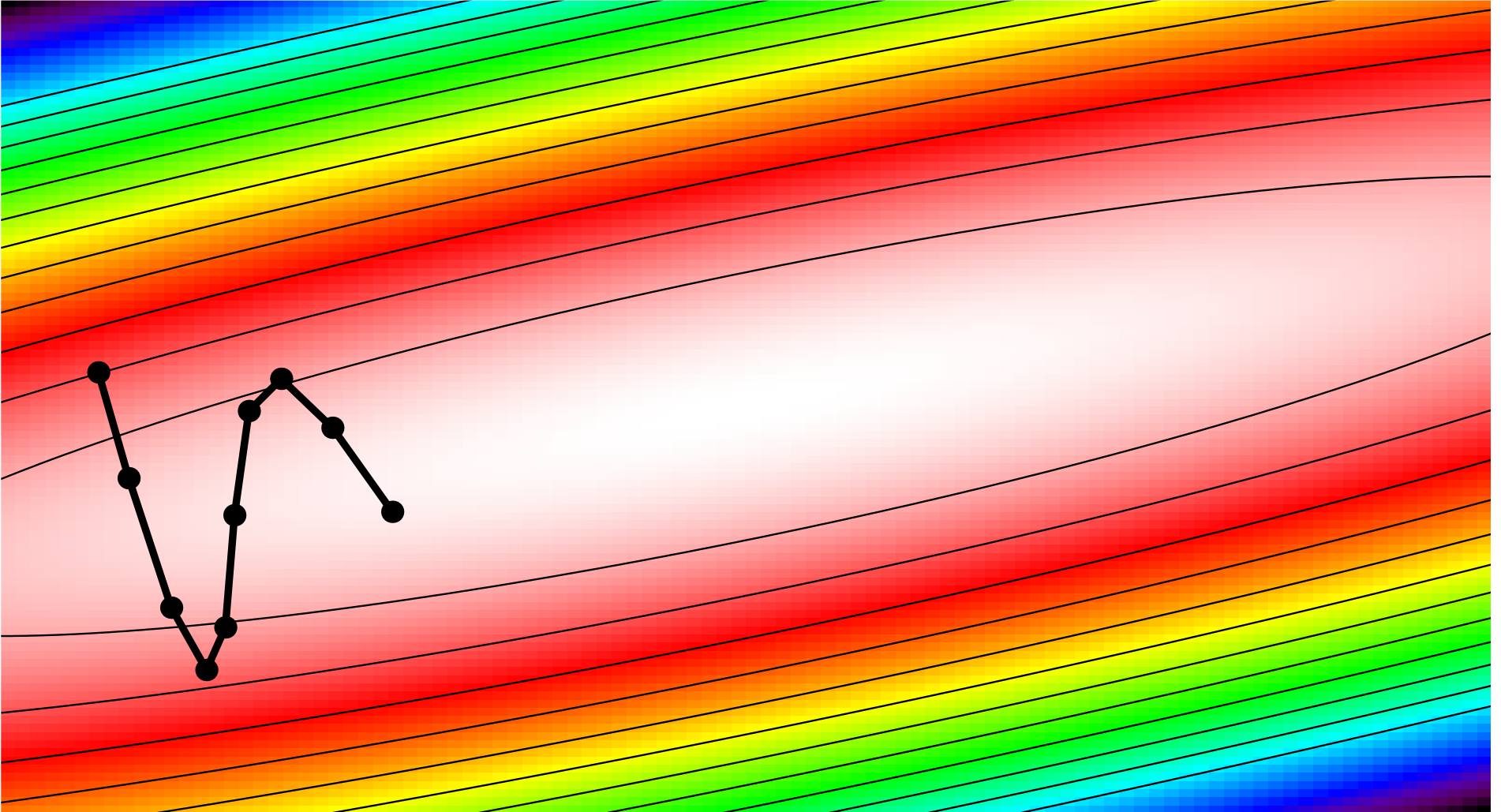
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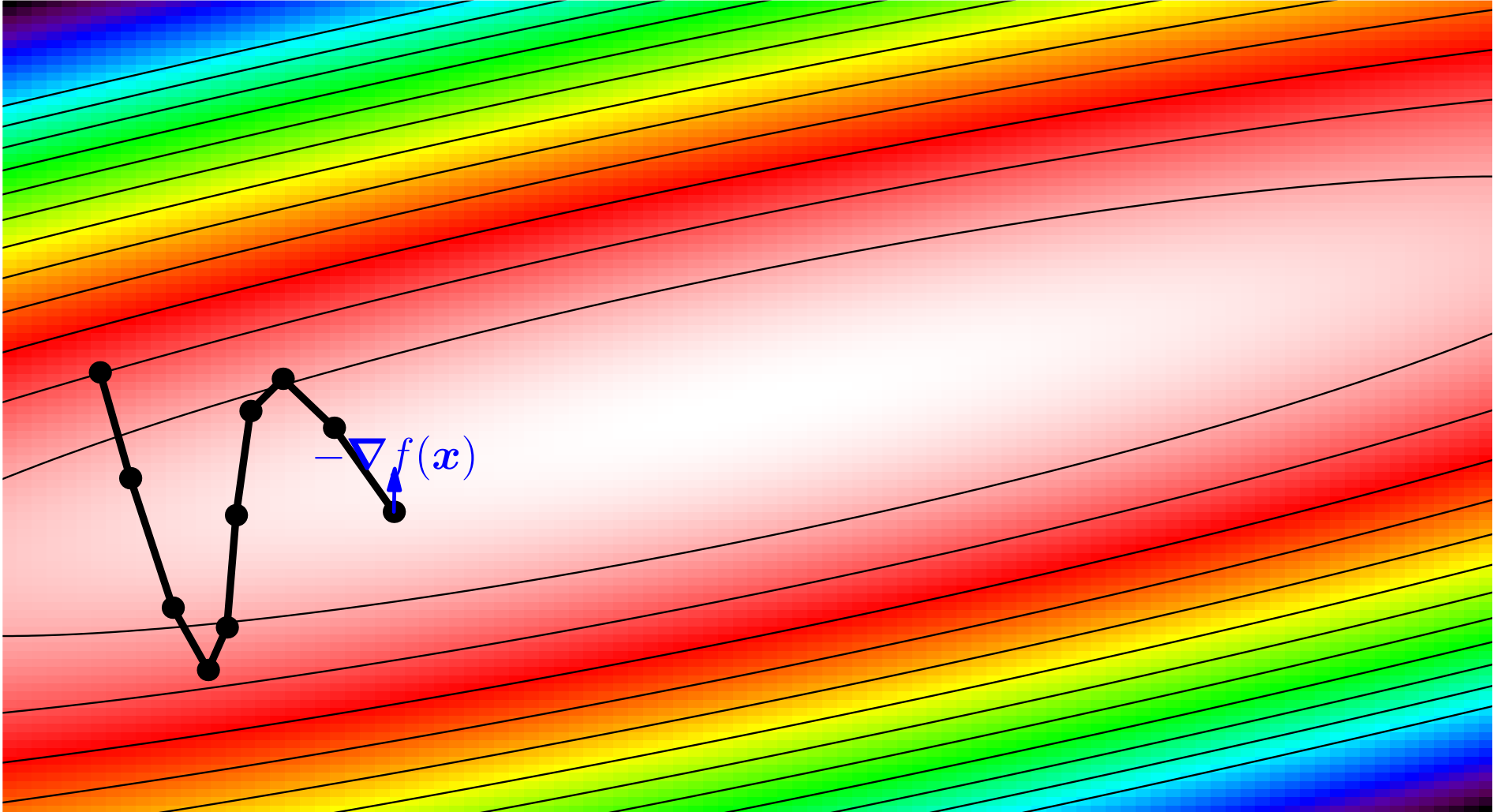
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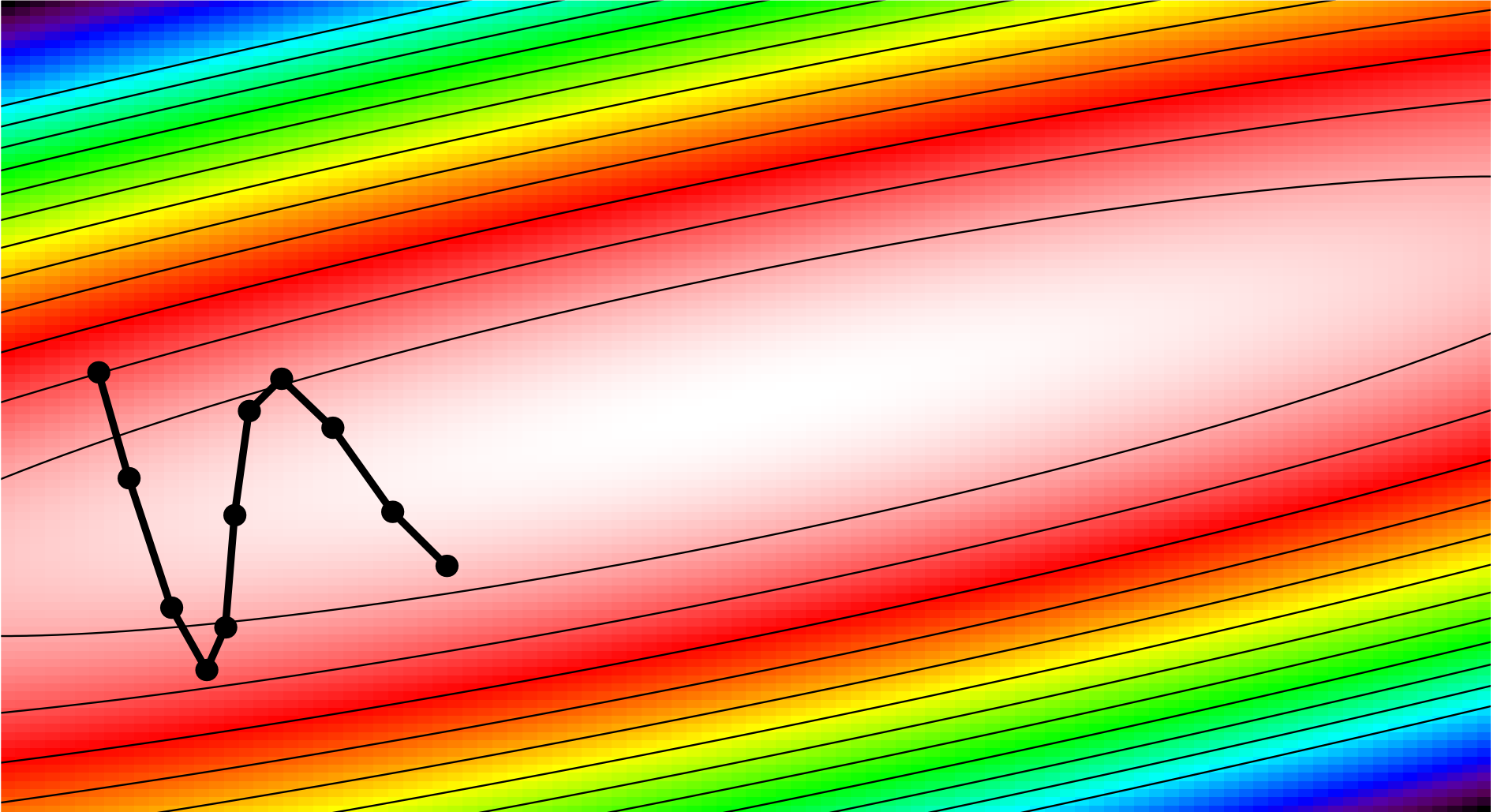
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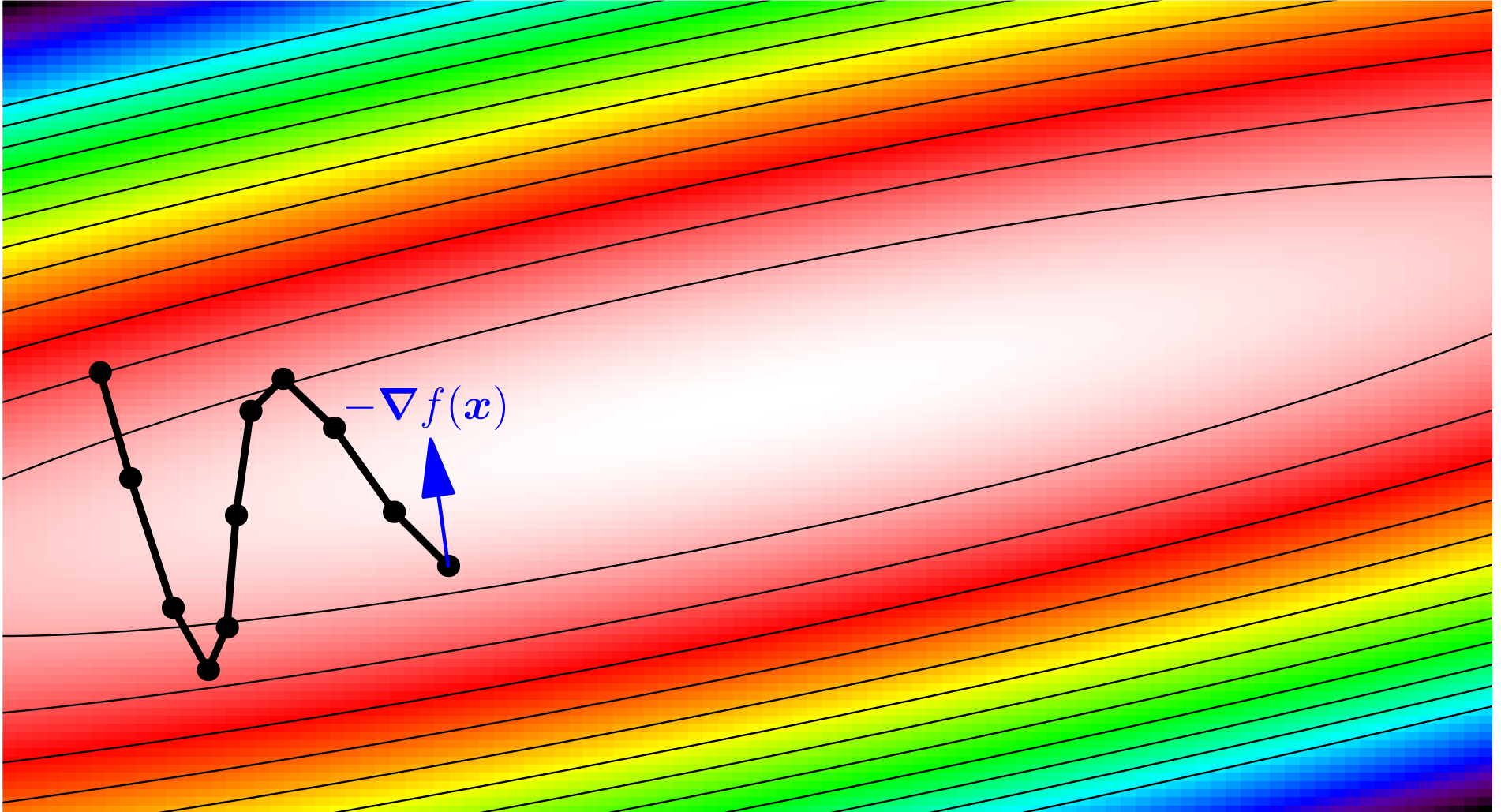
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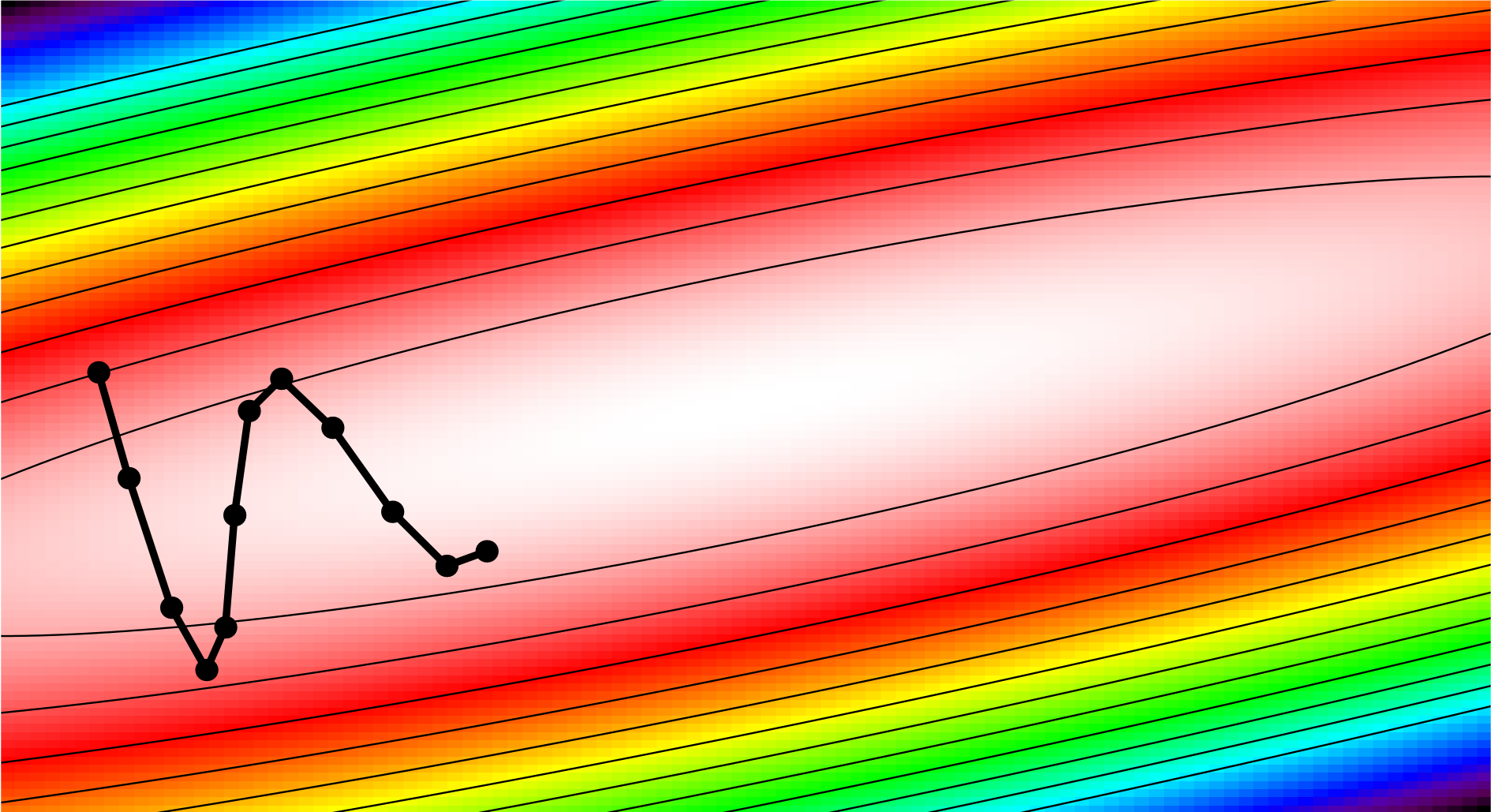
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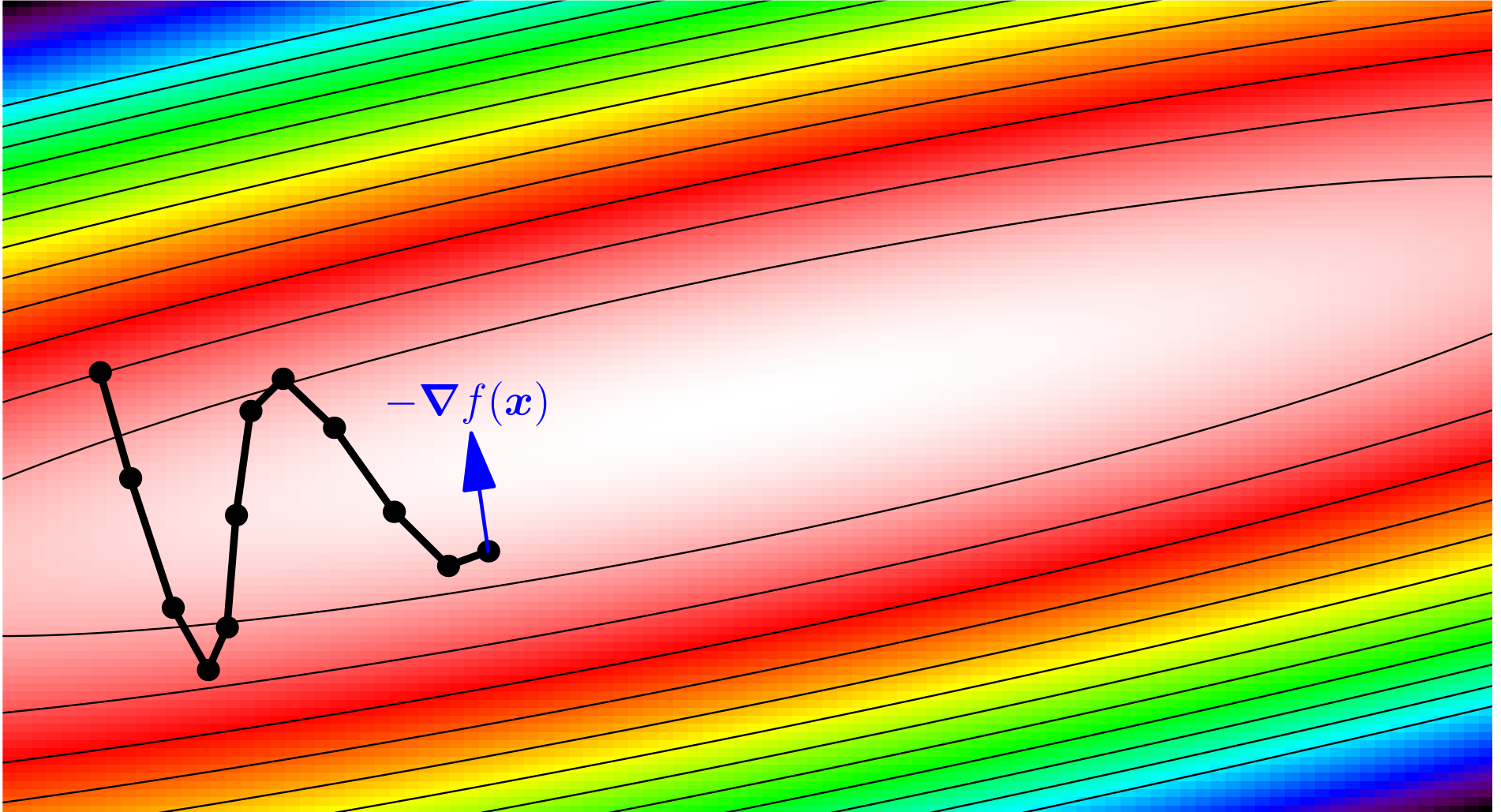
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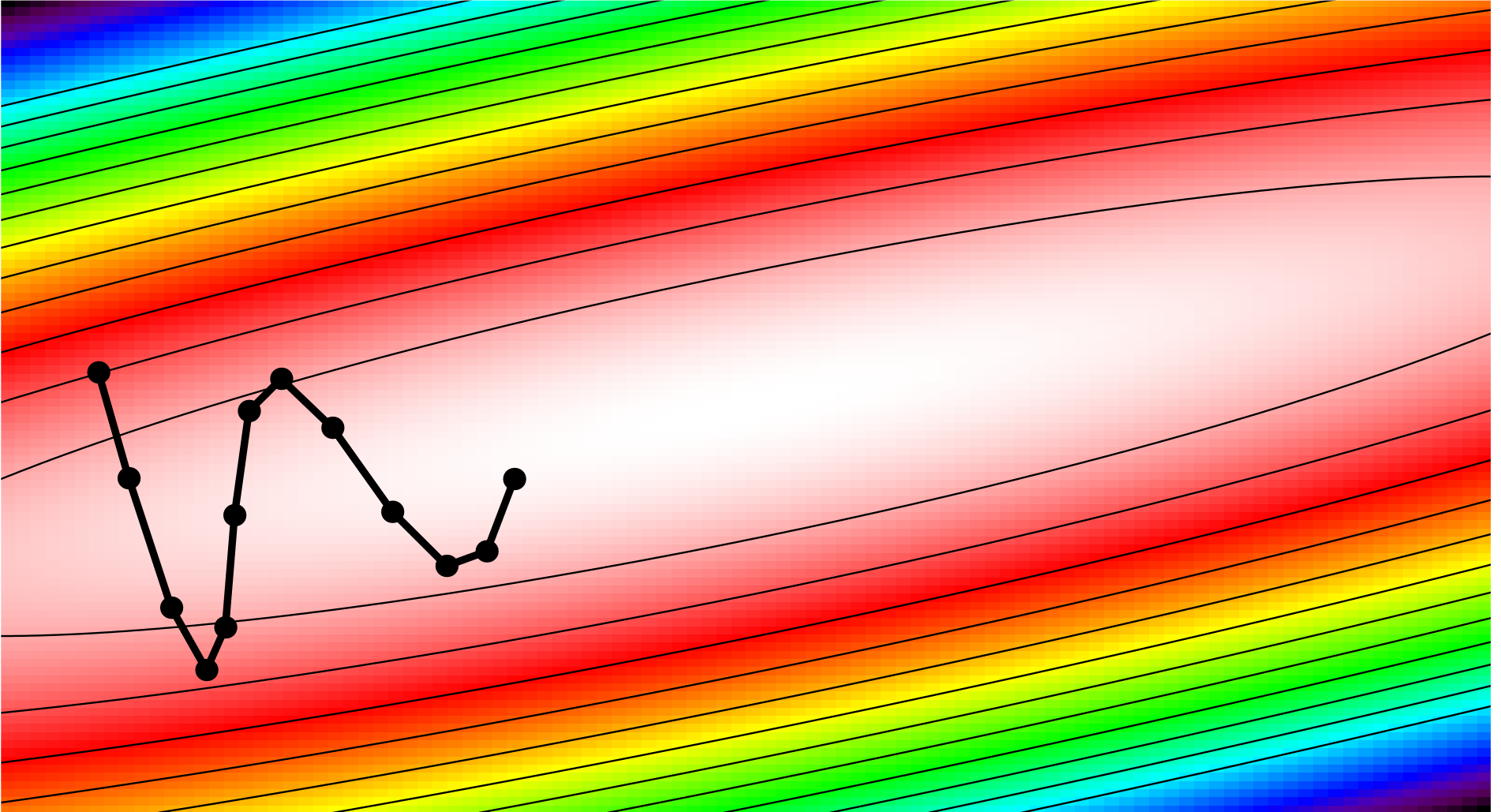
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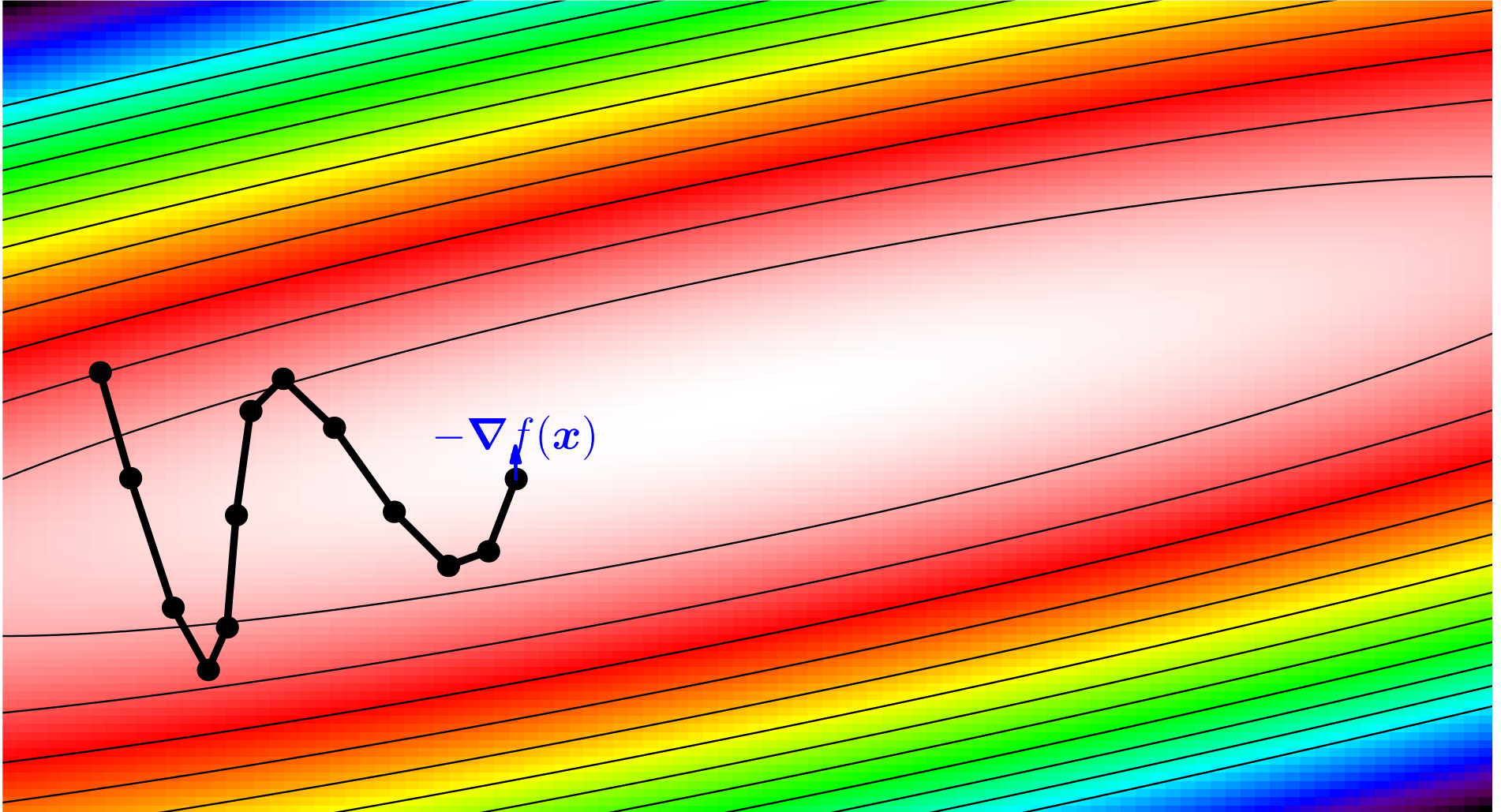
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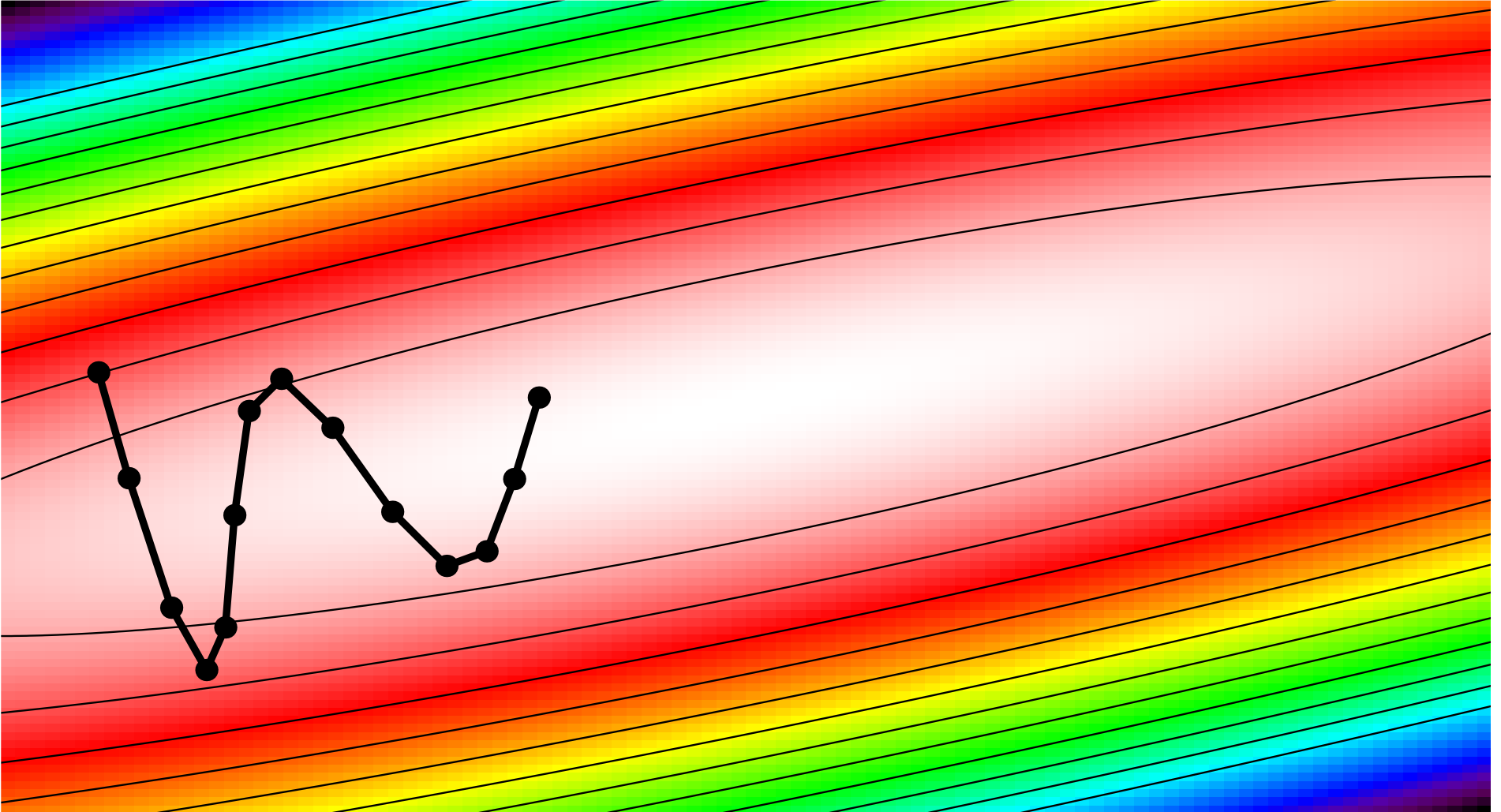
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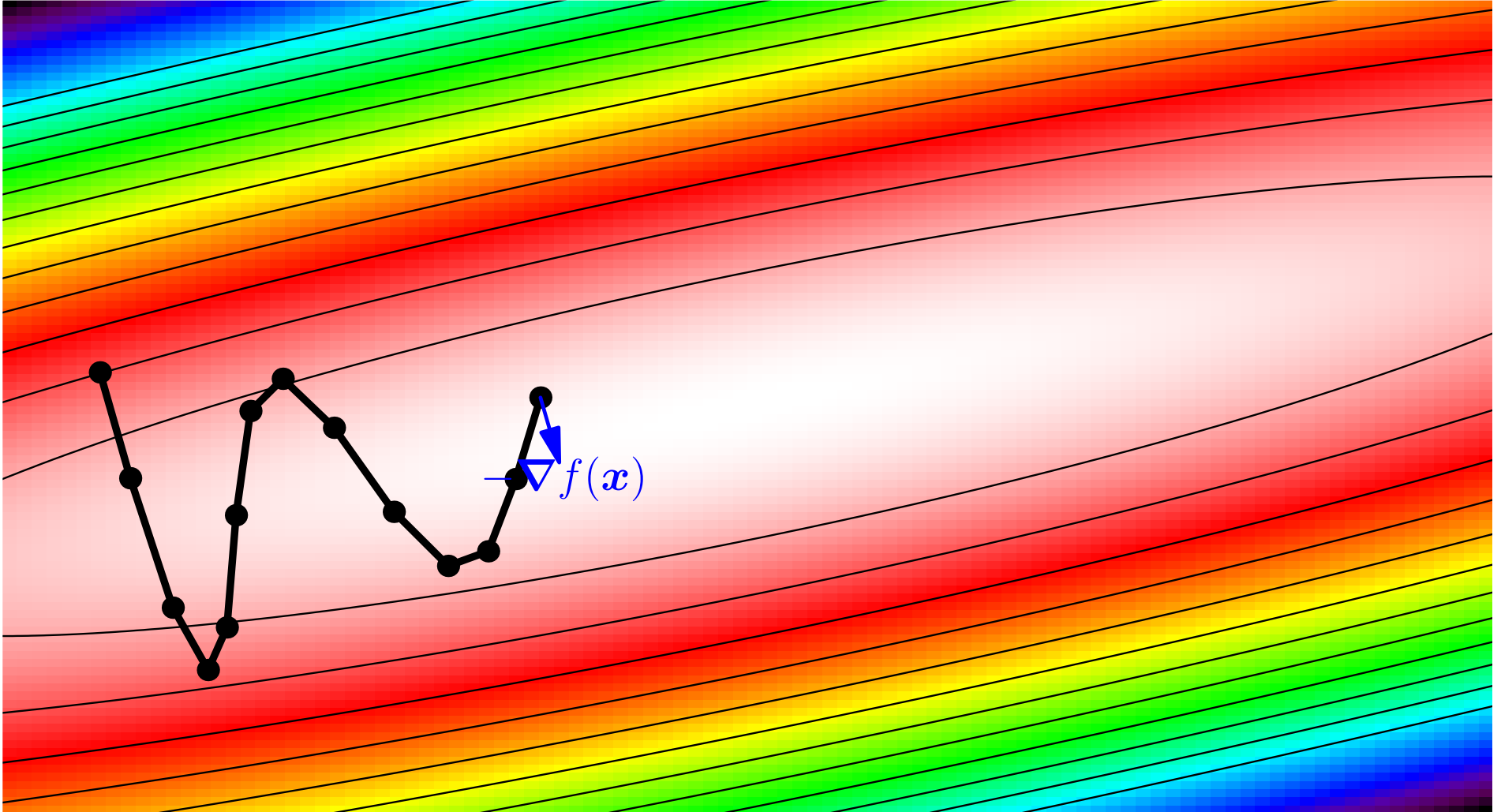
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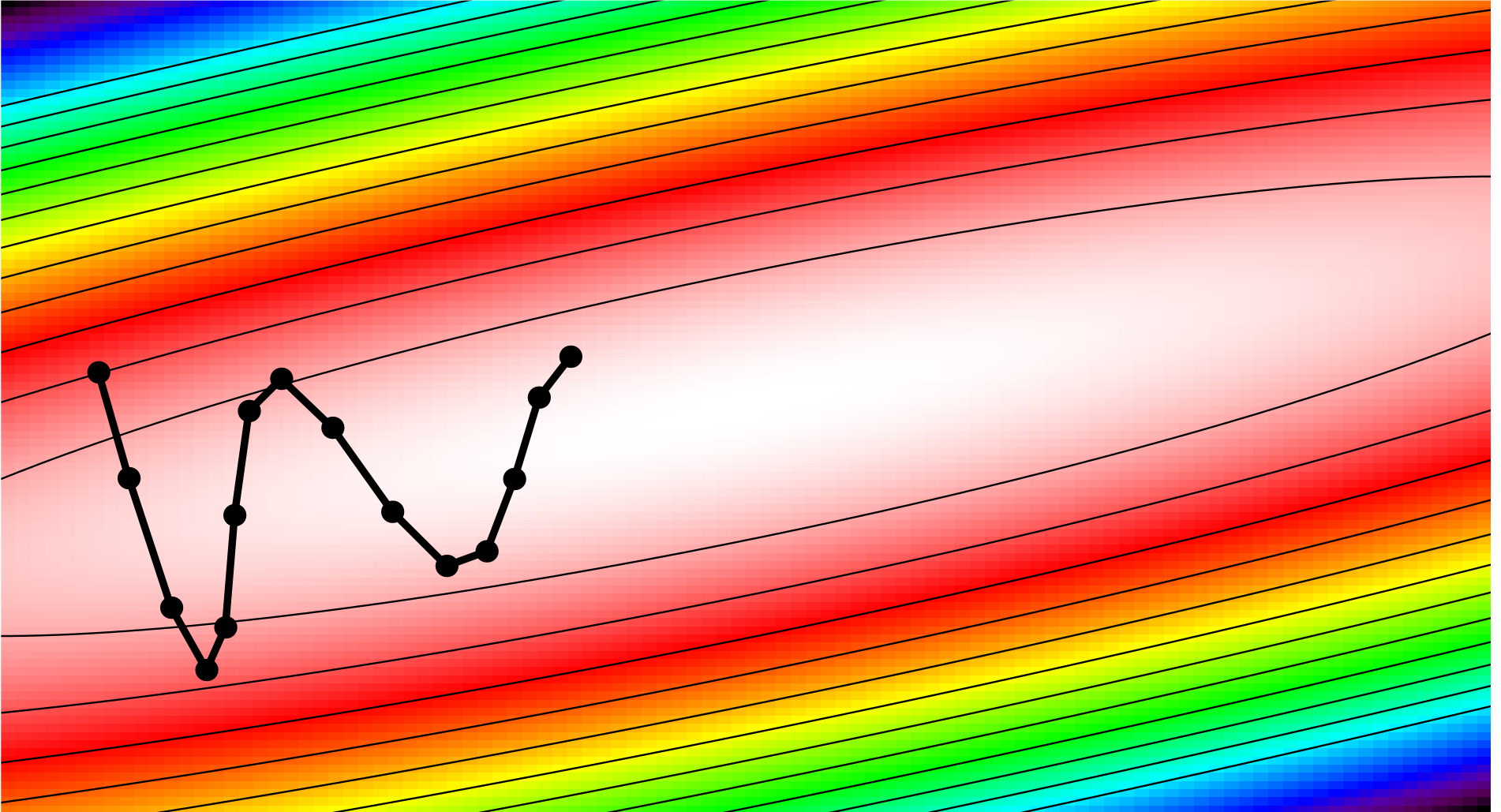
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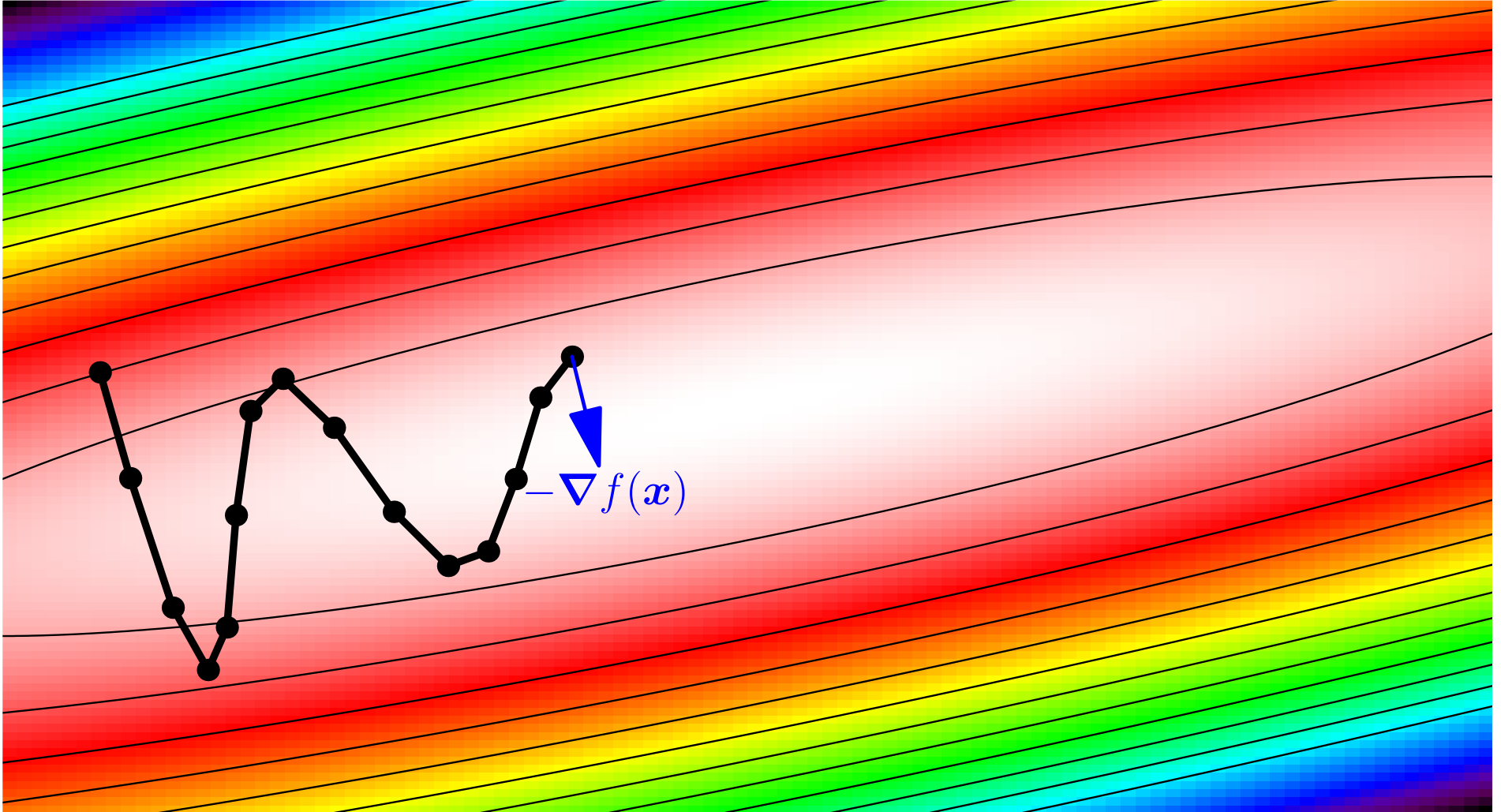
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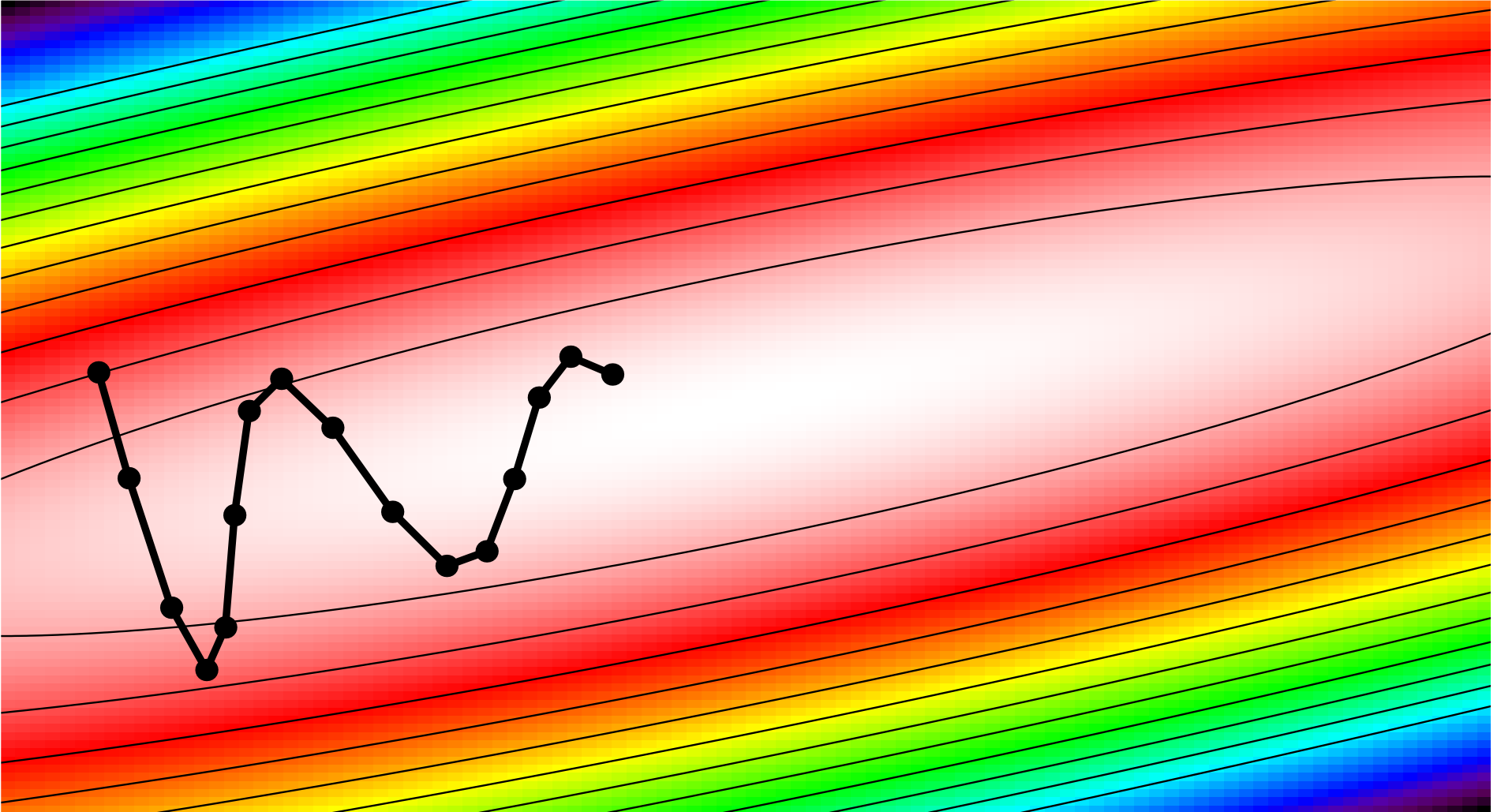
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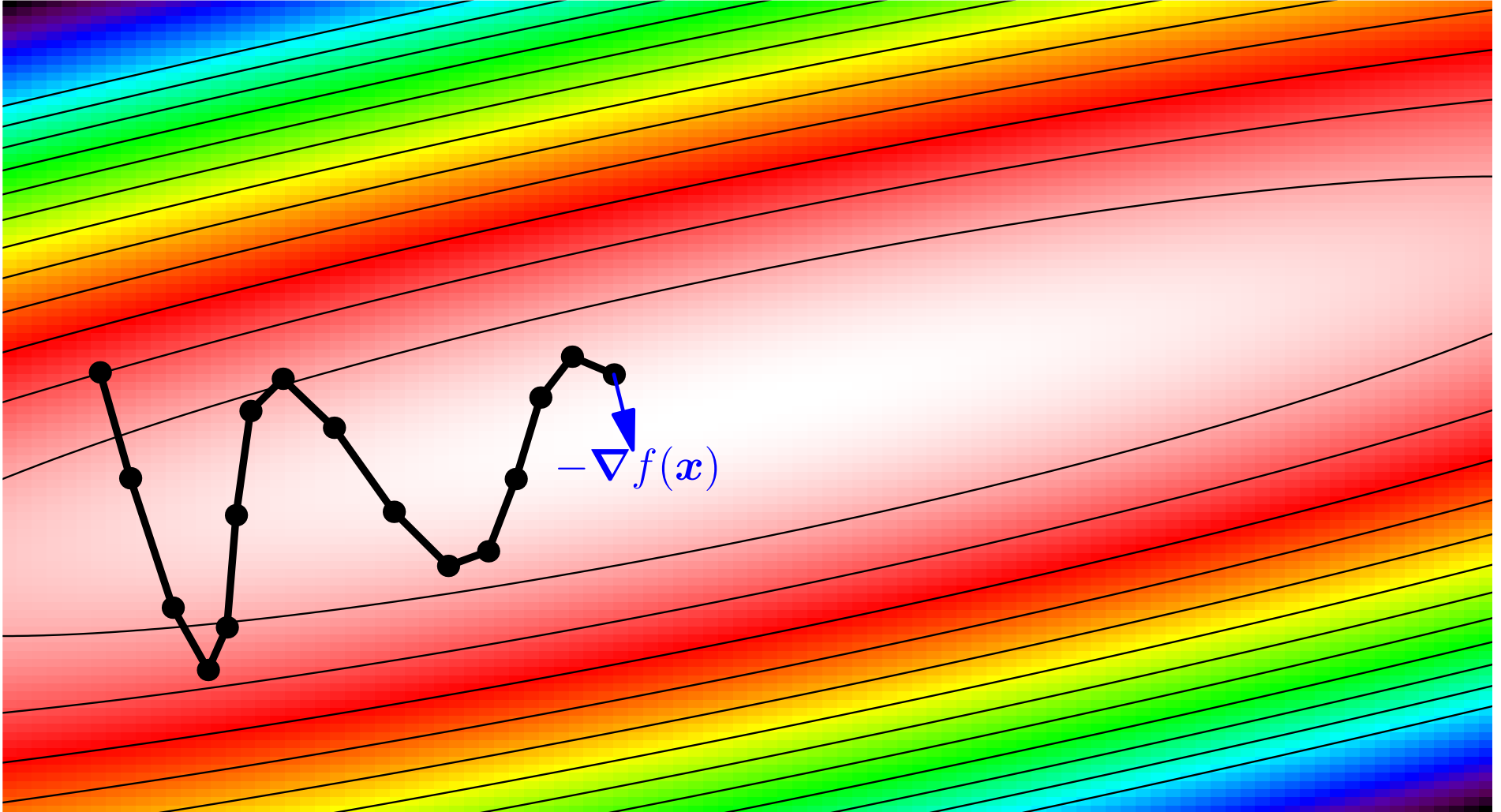
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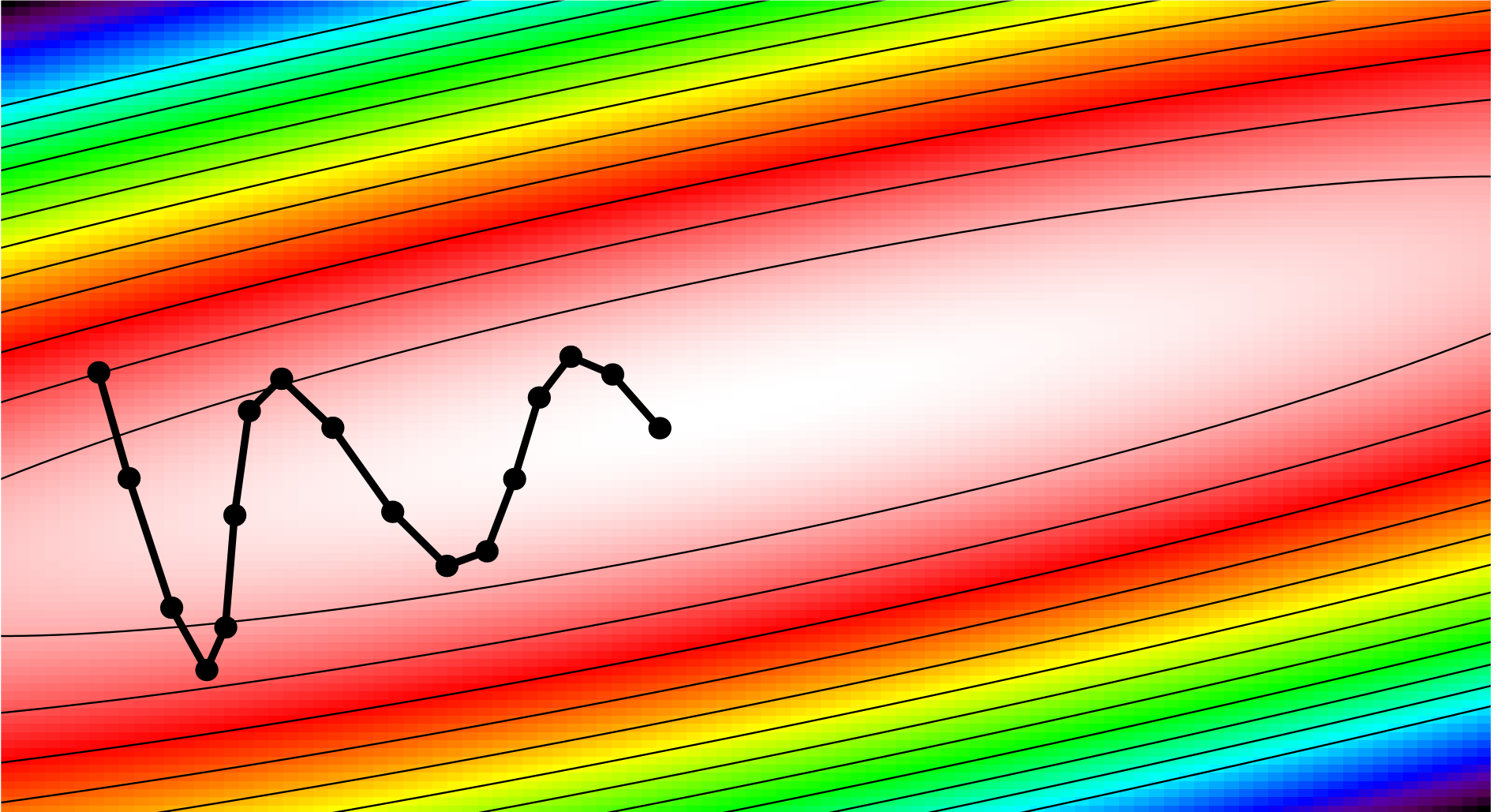
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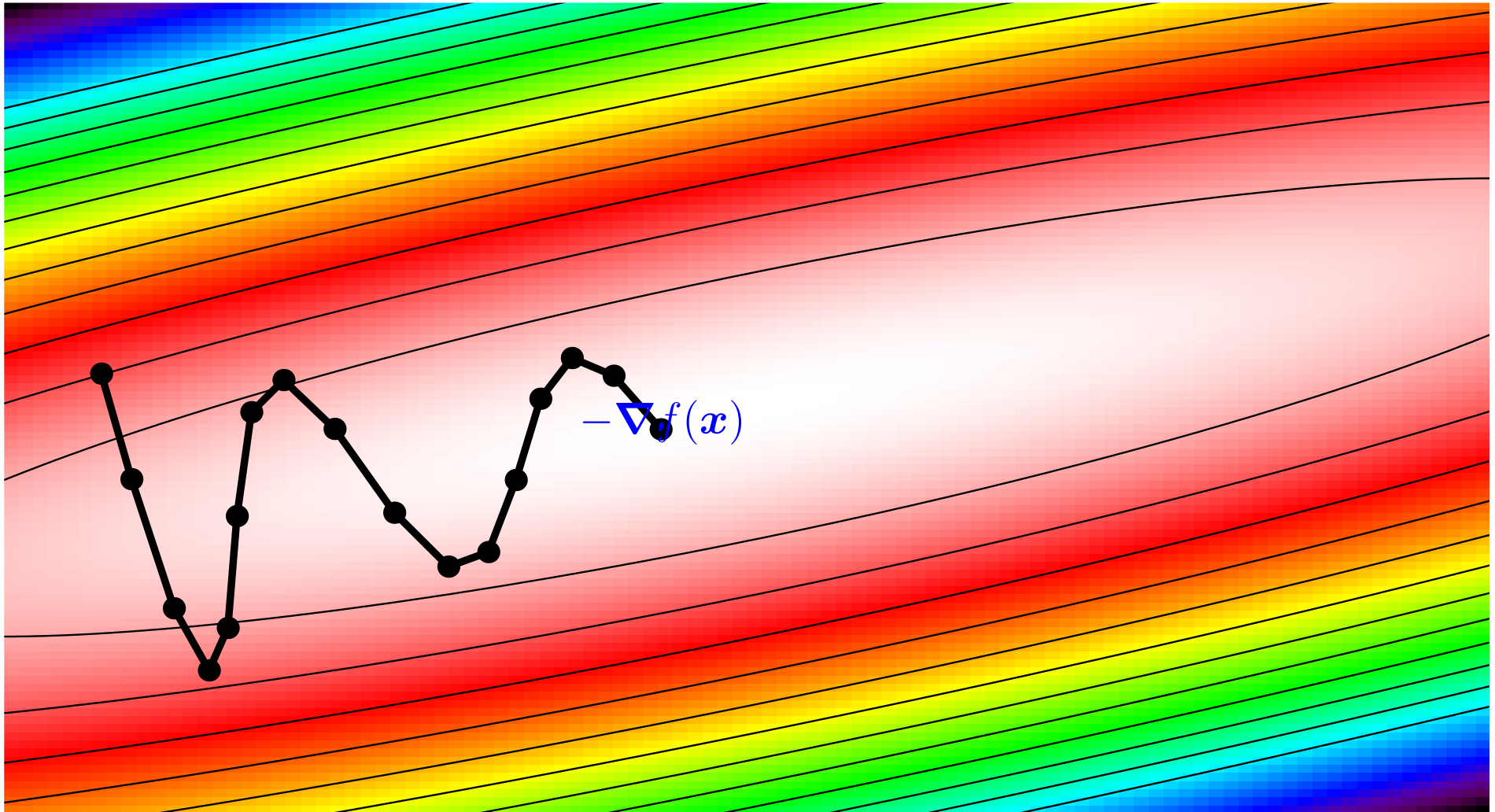
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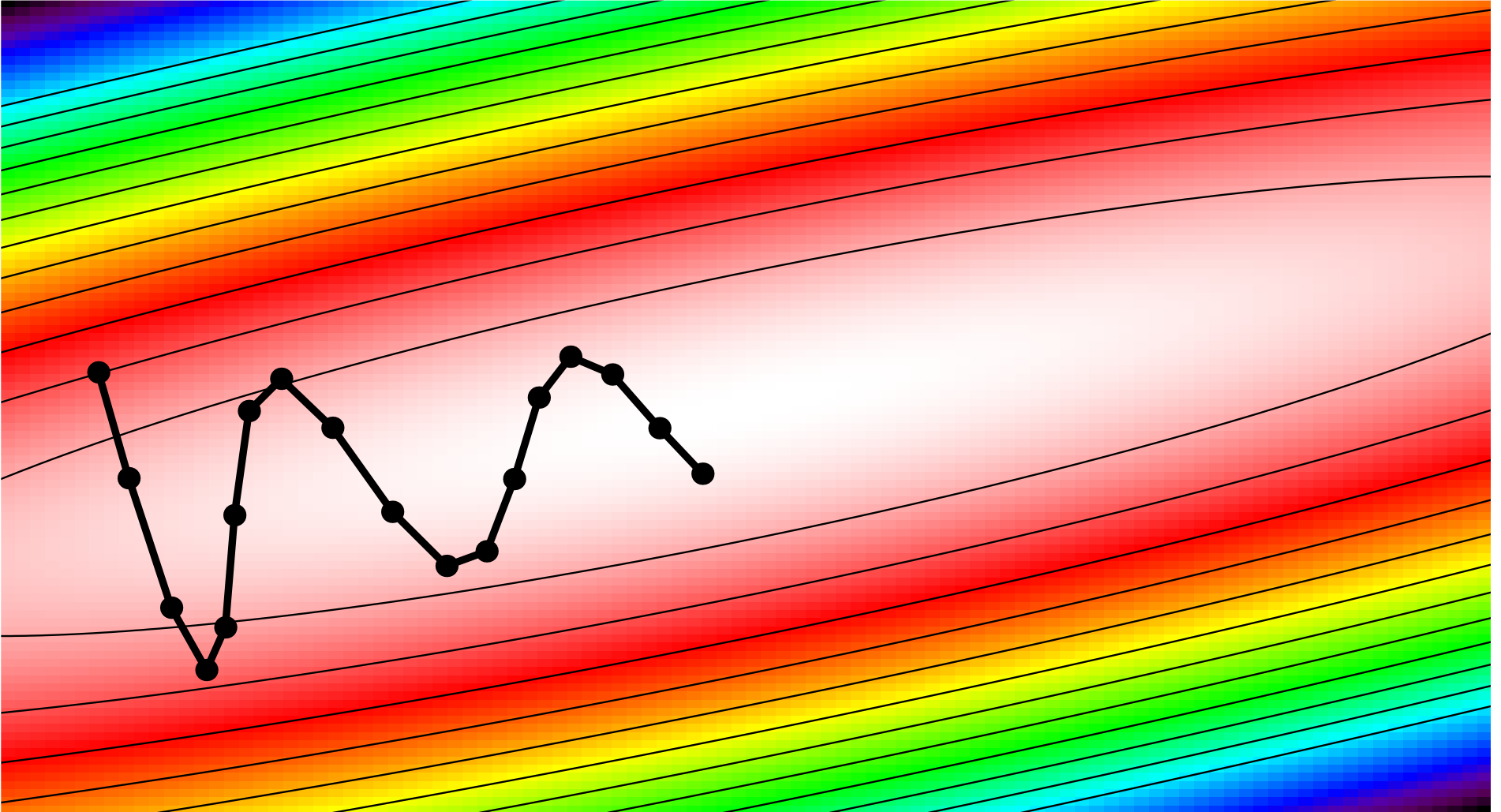
Gradient Descent with Momentum



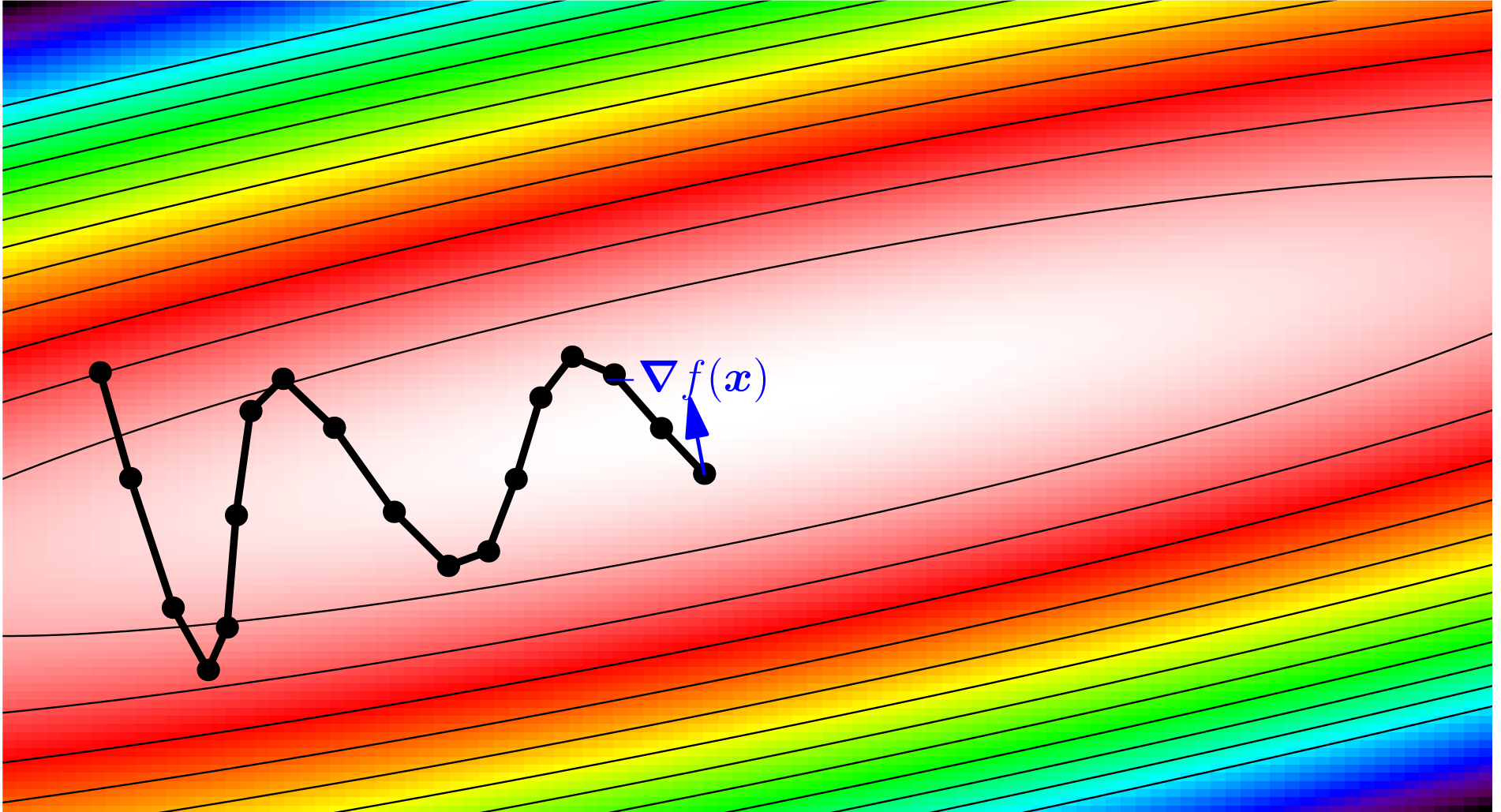
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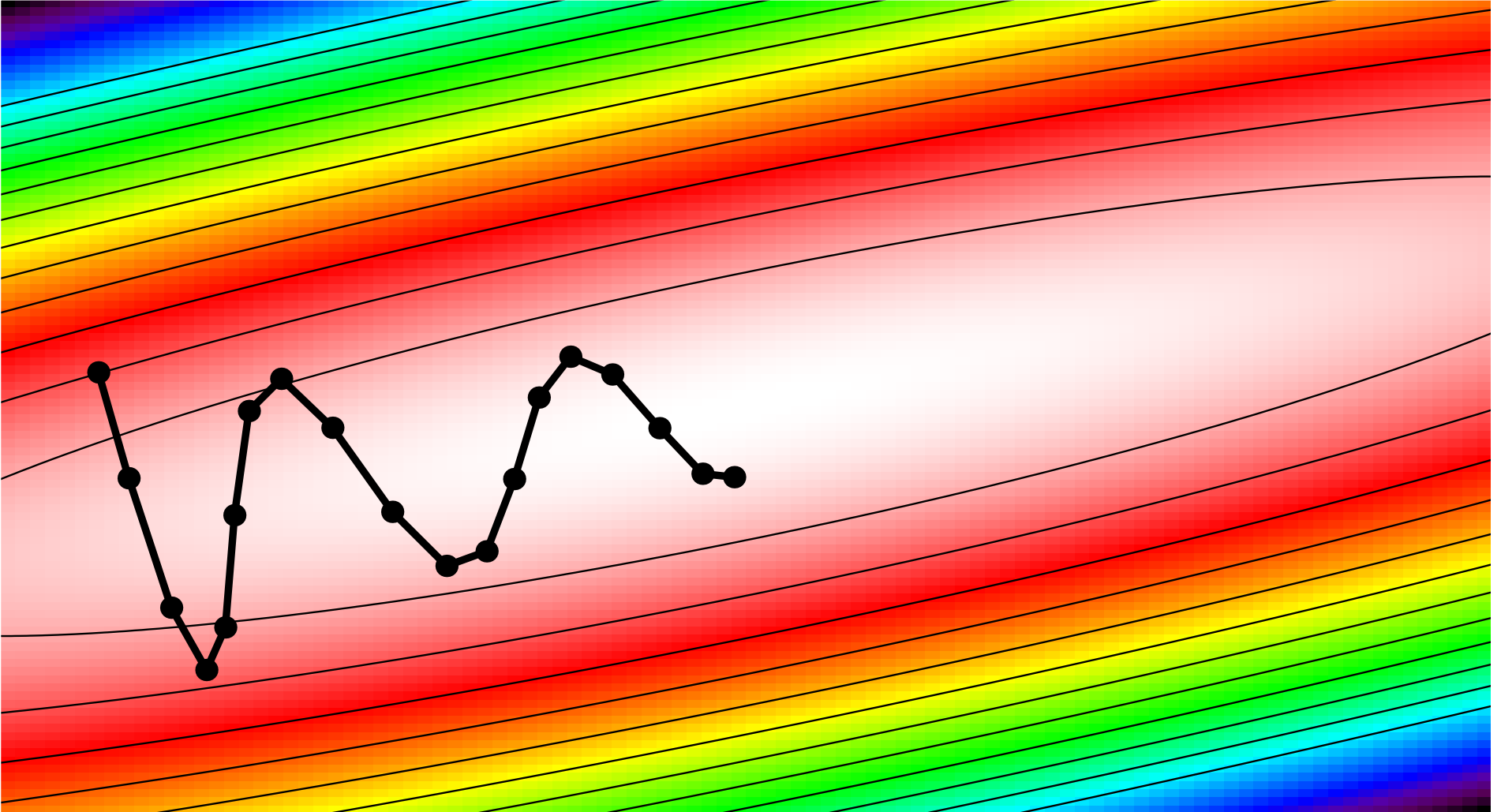
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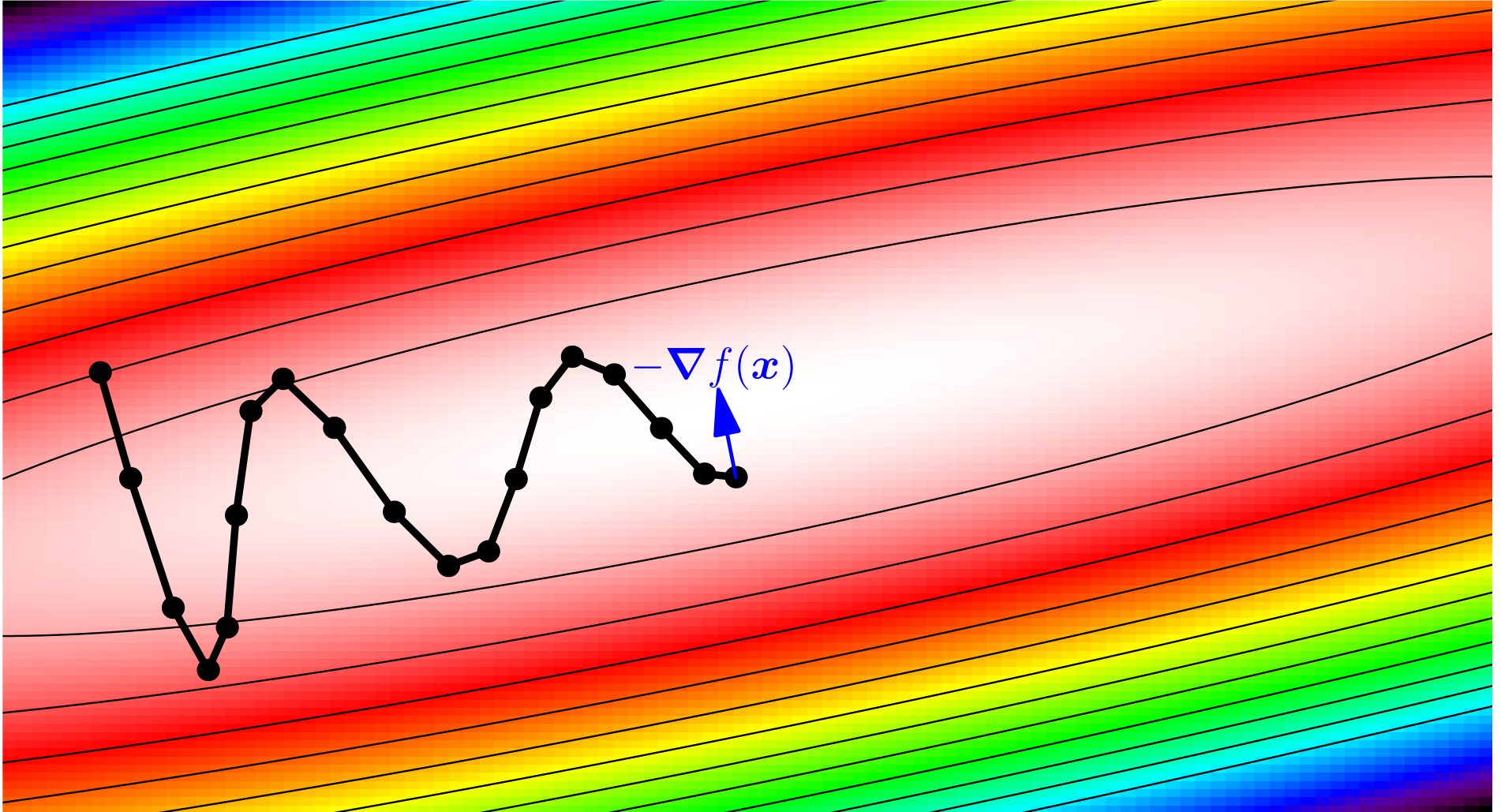
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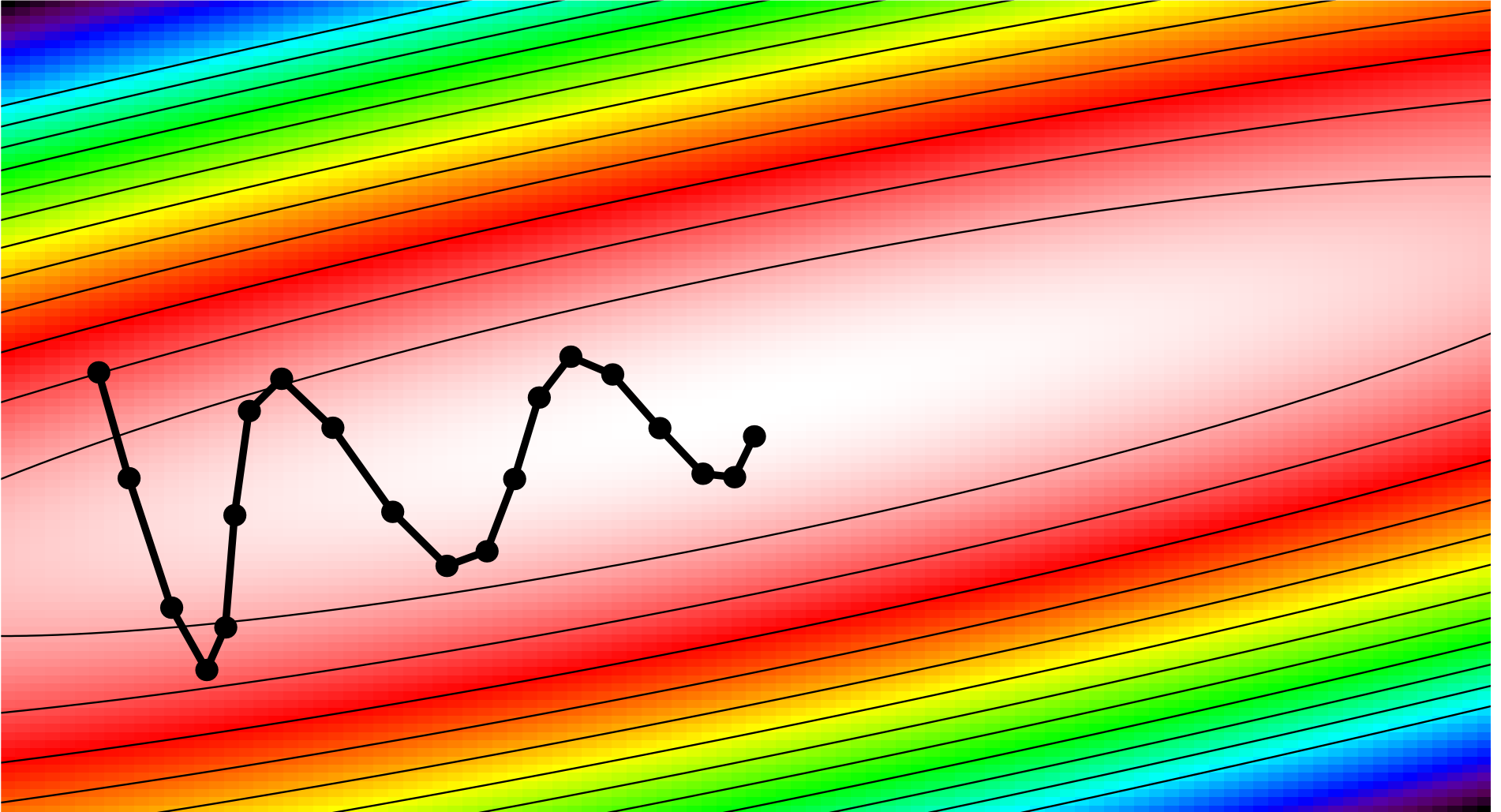
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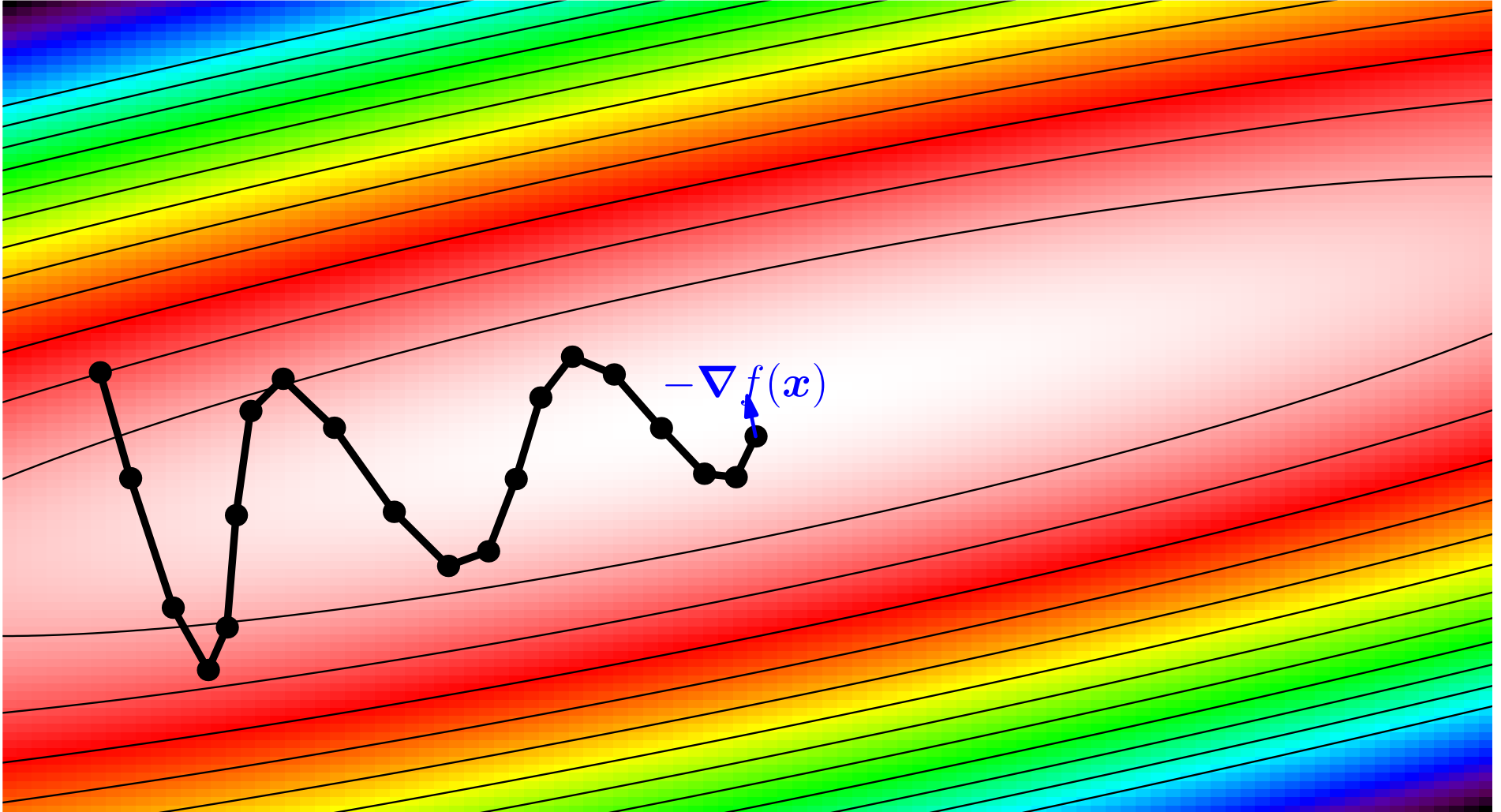
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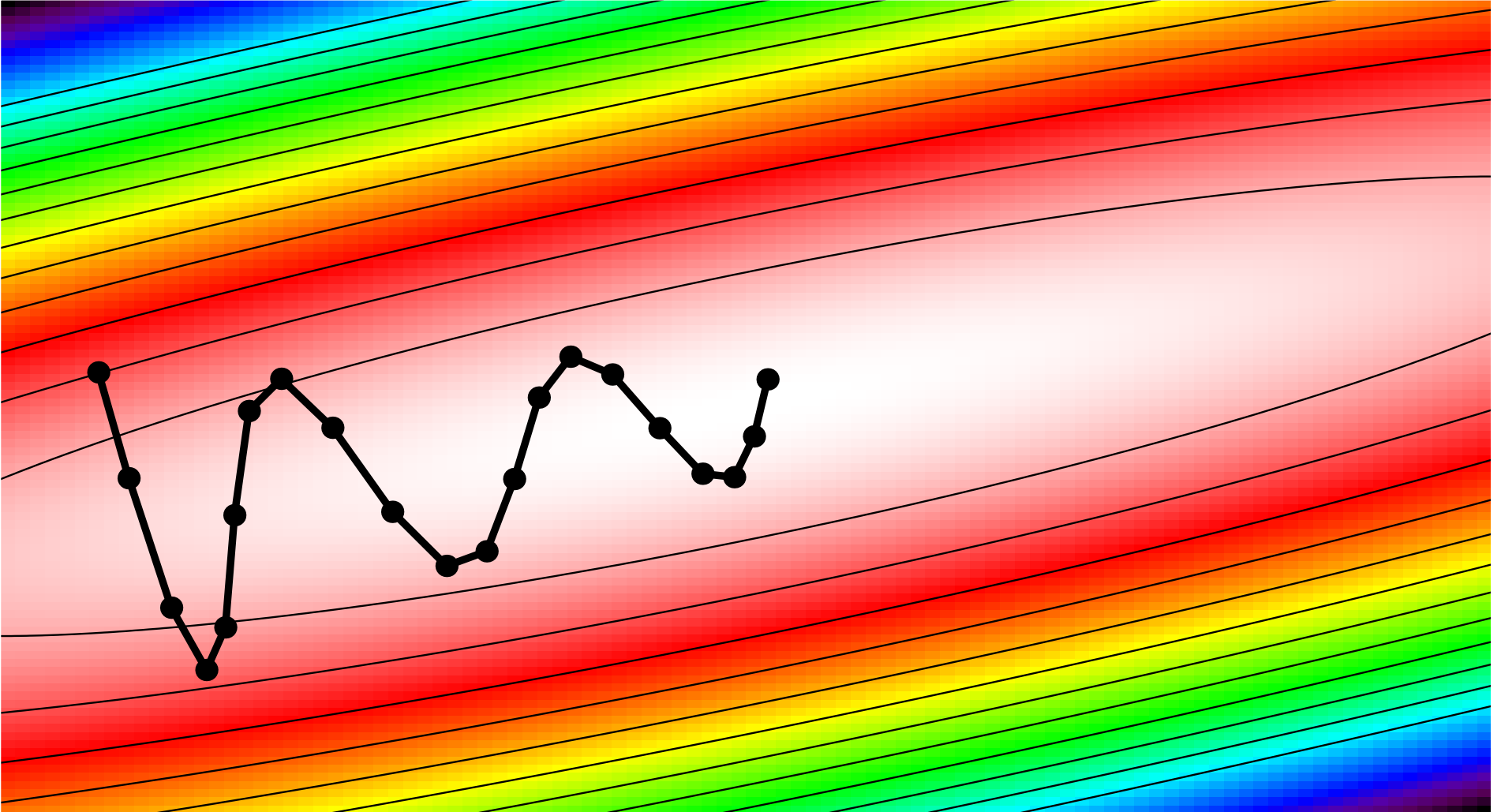
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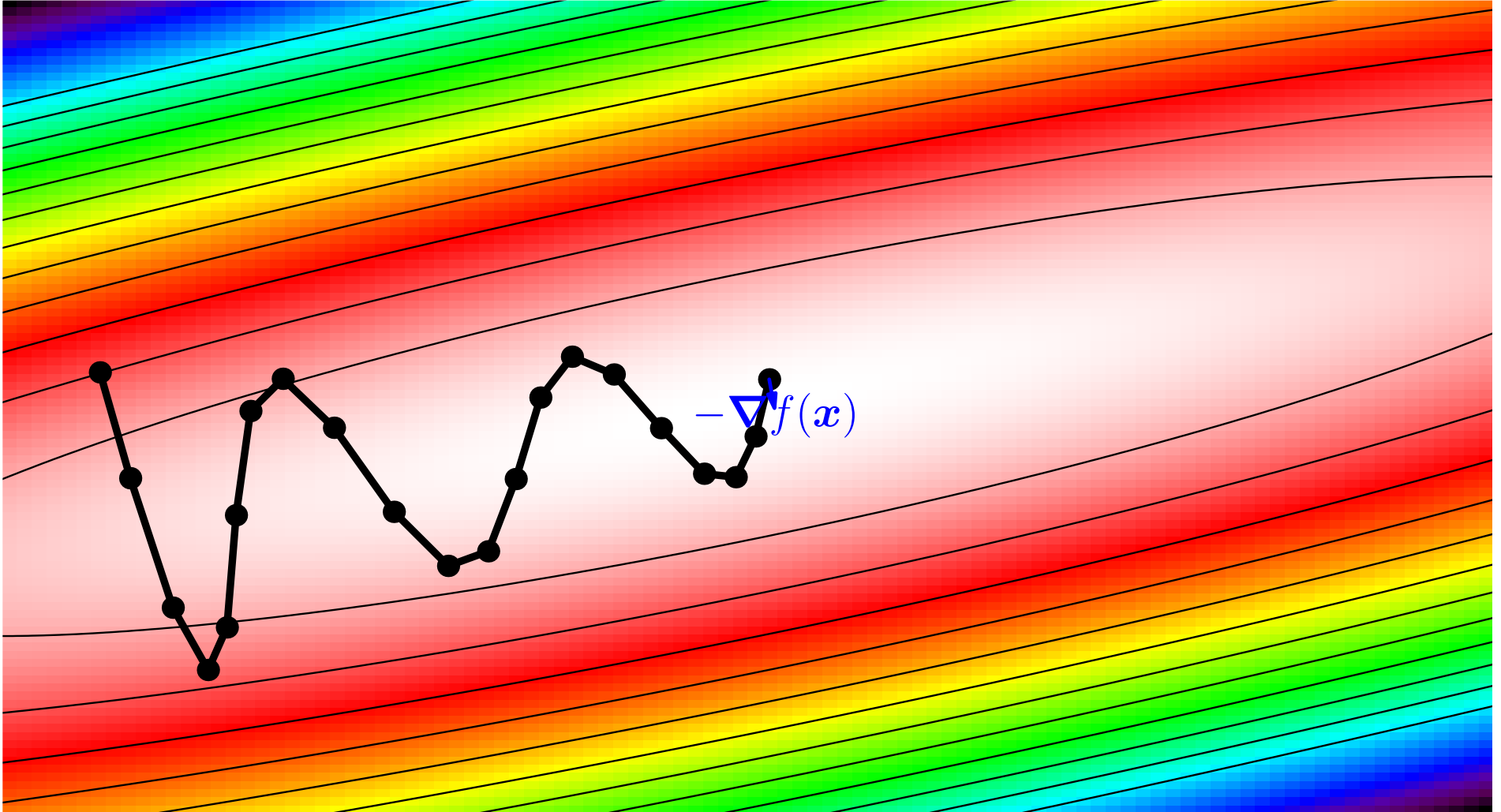
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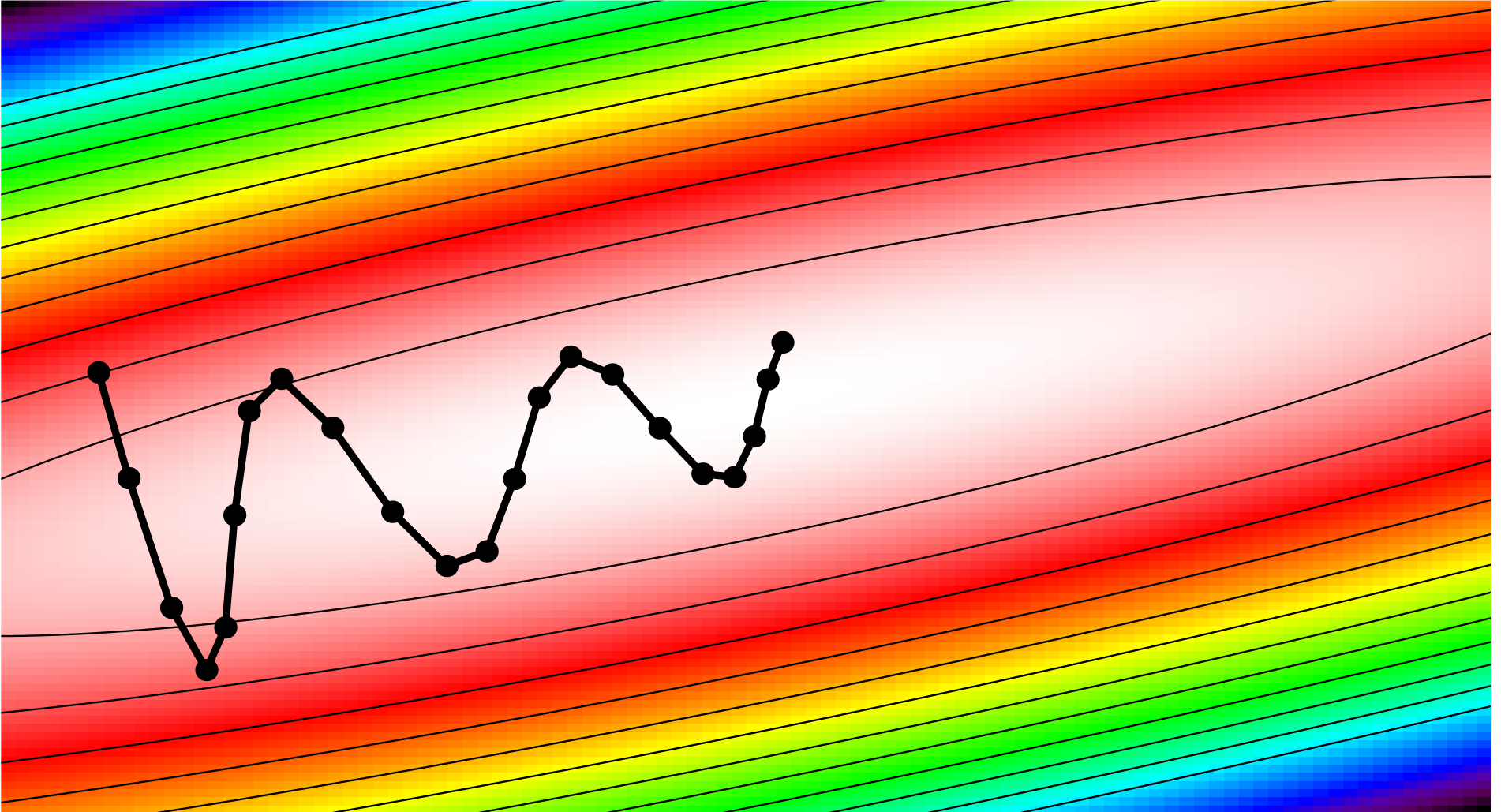
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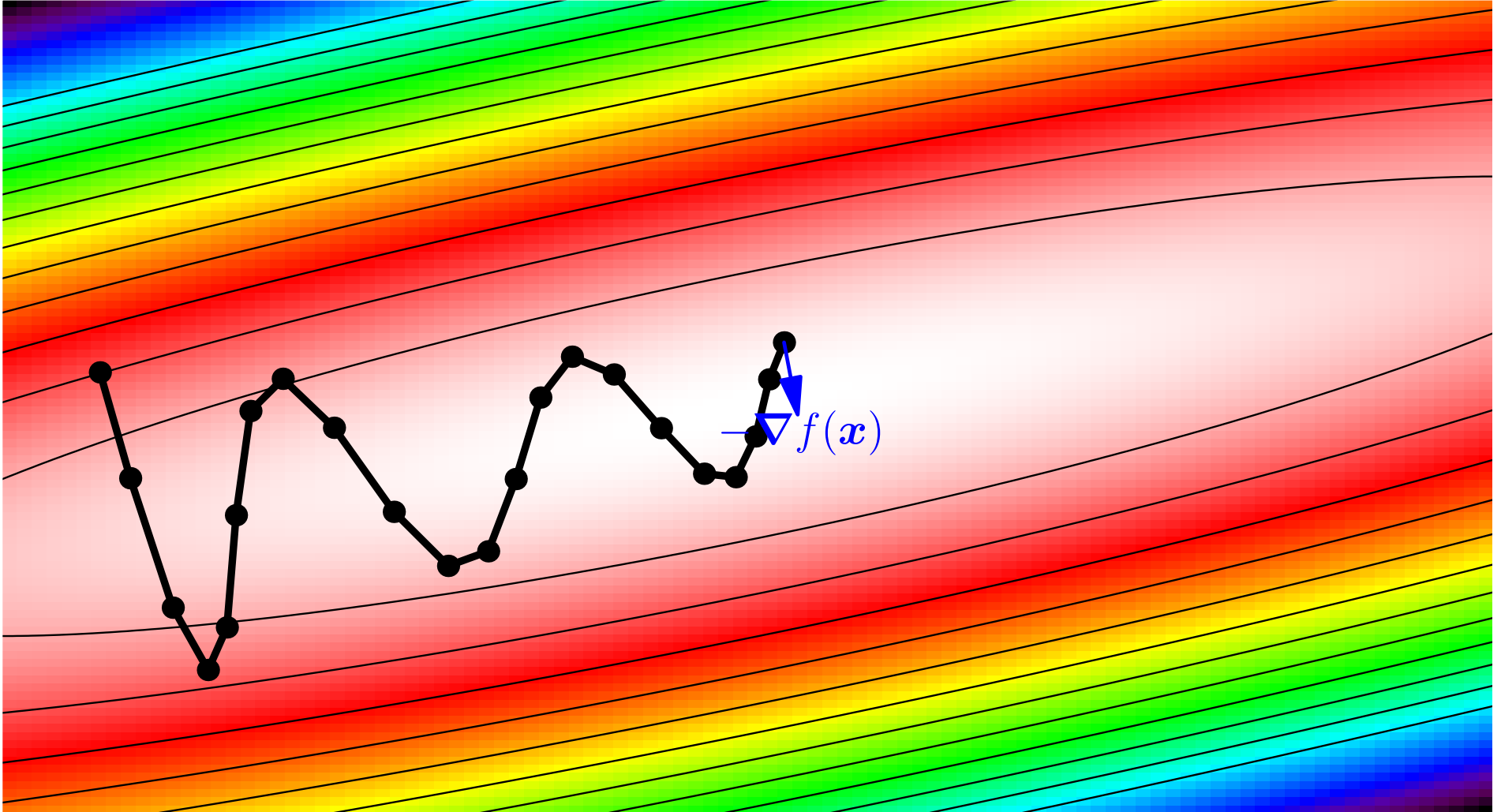
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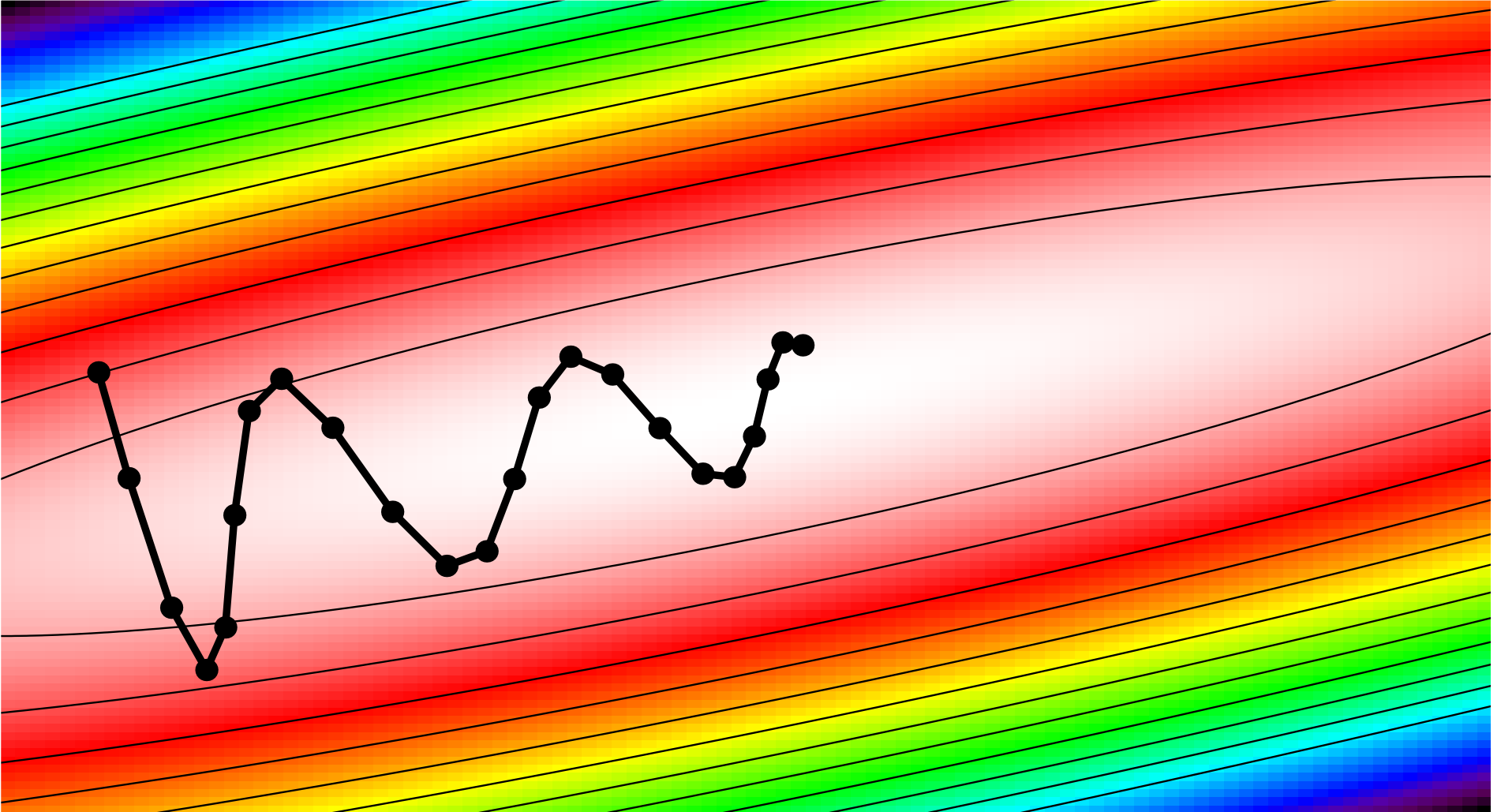
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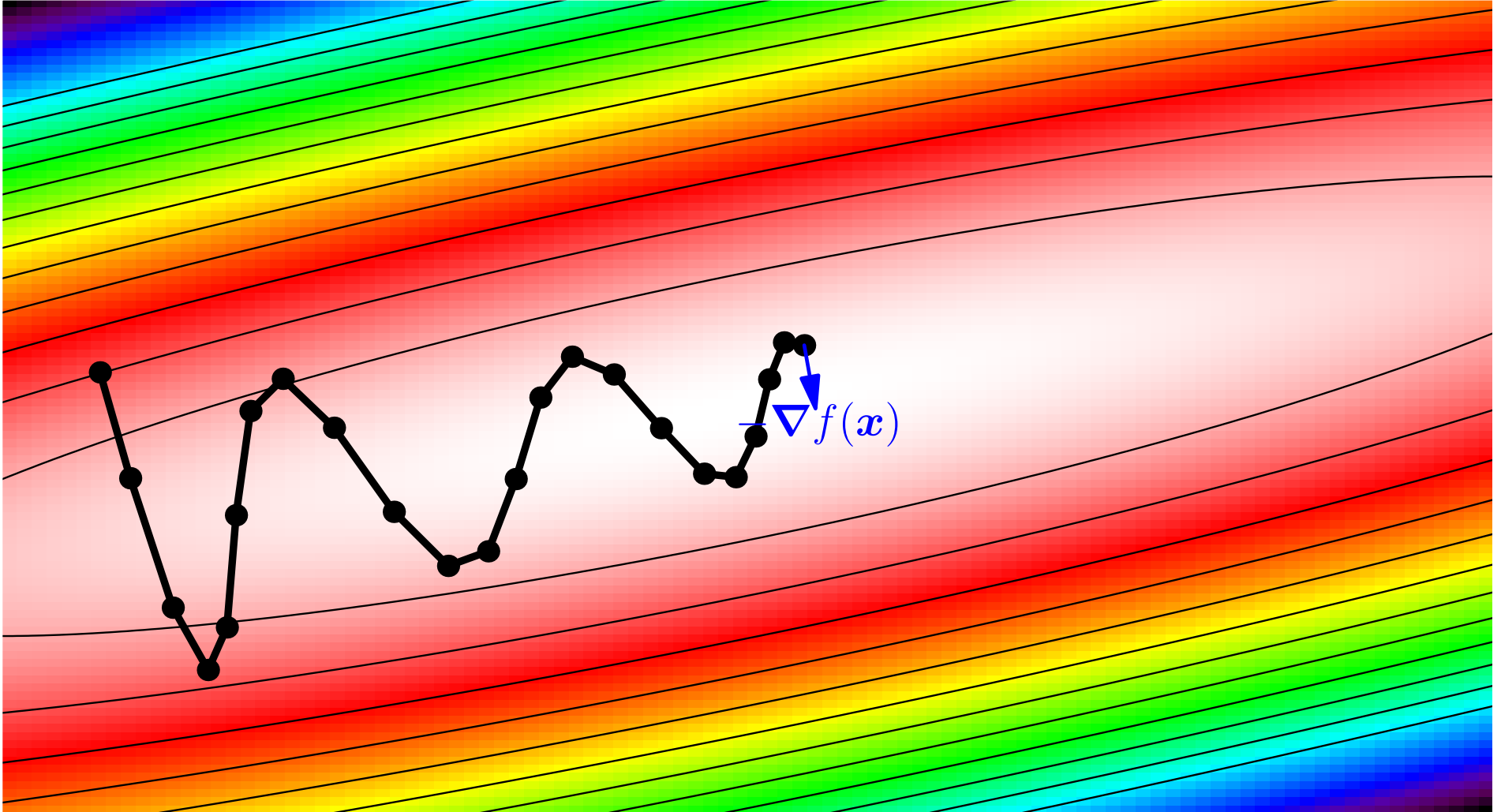
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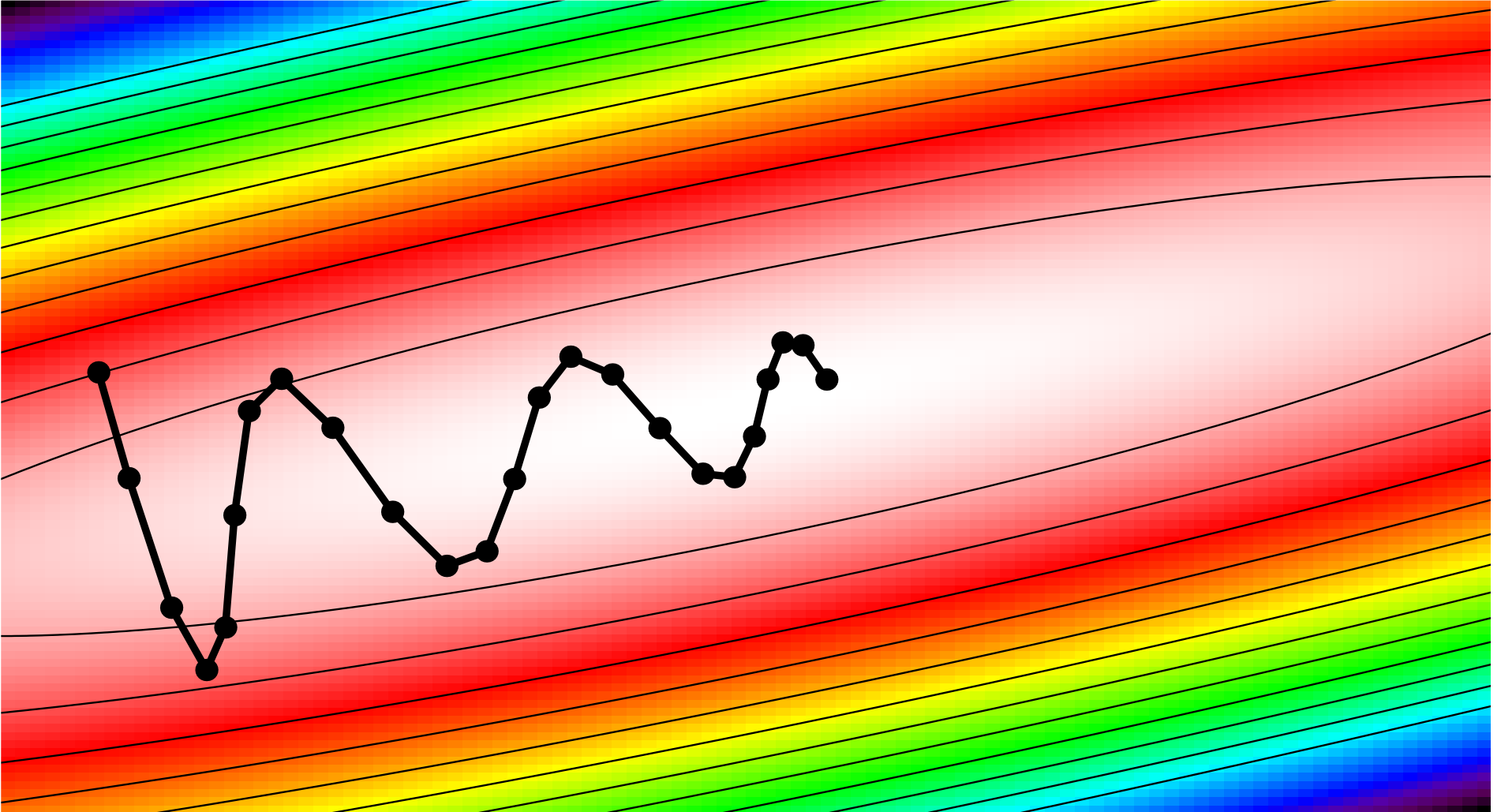
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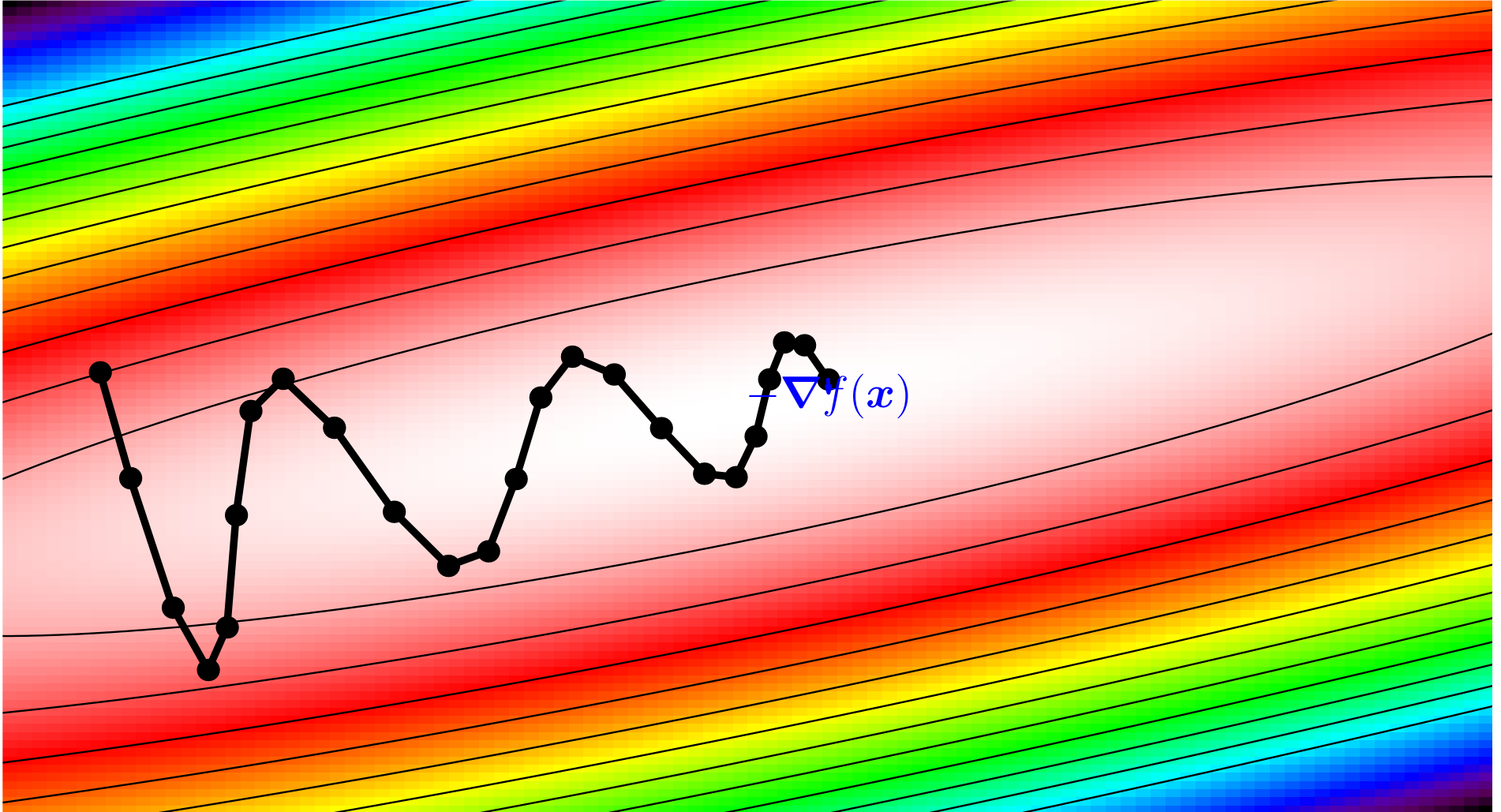
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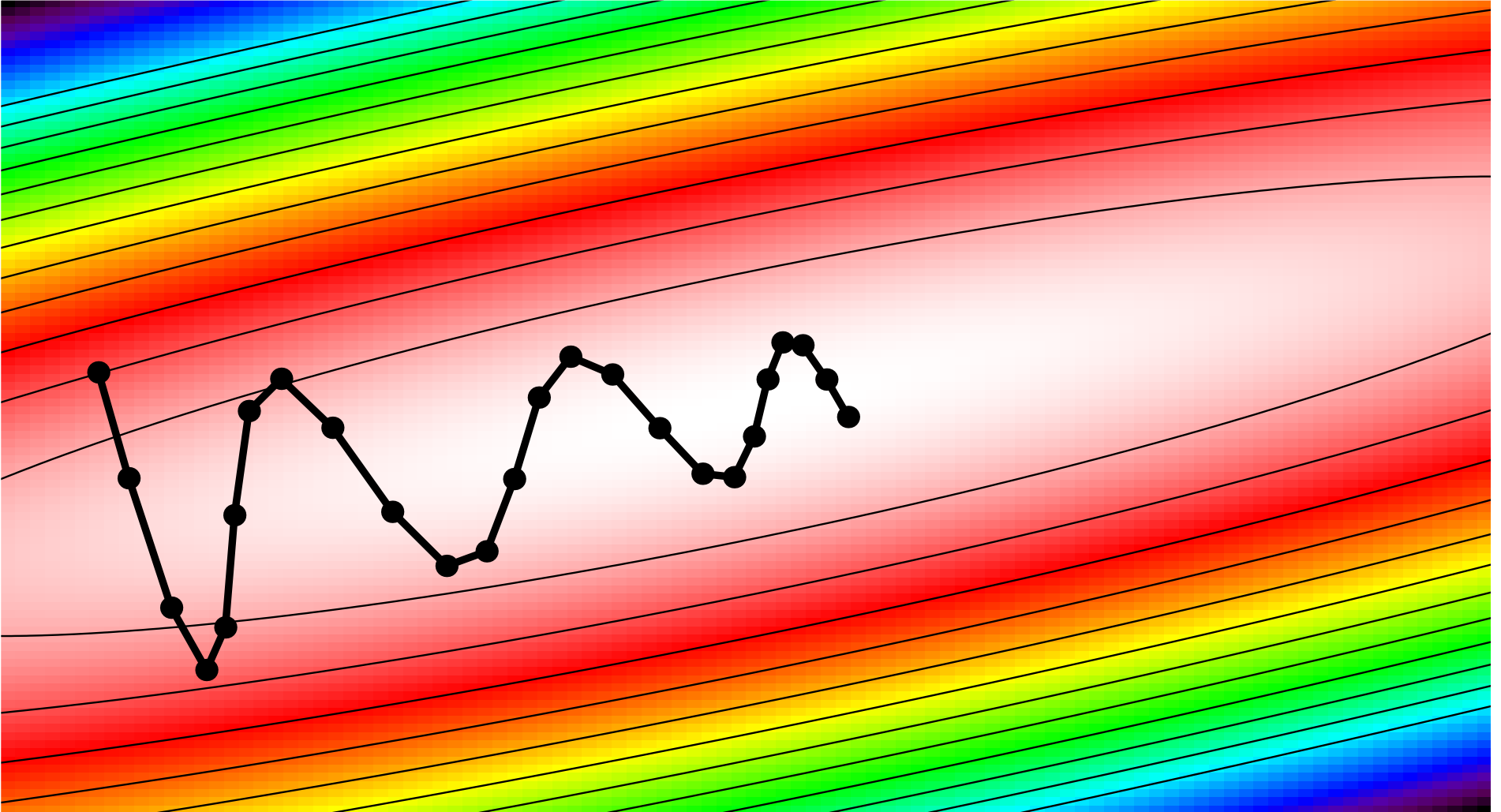
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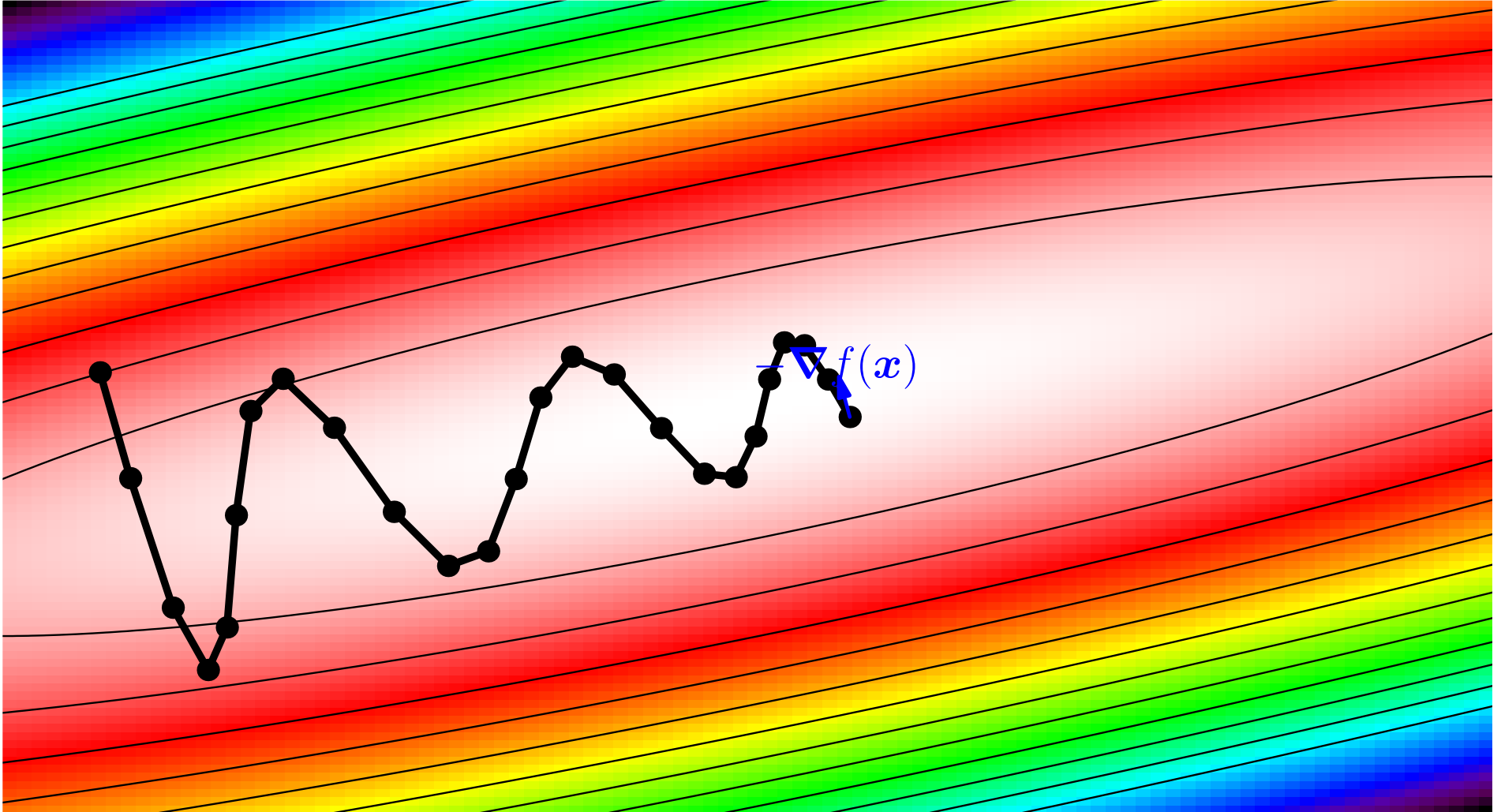
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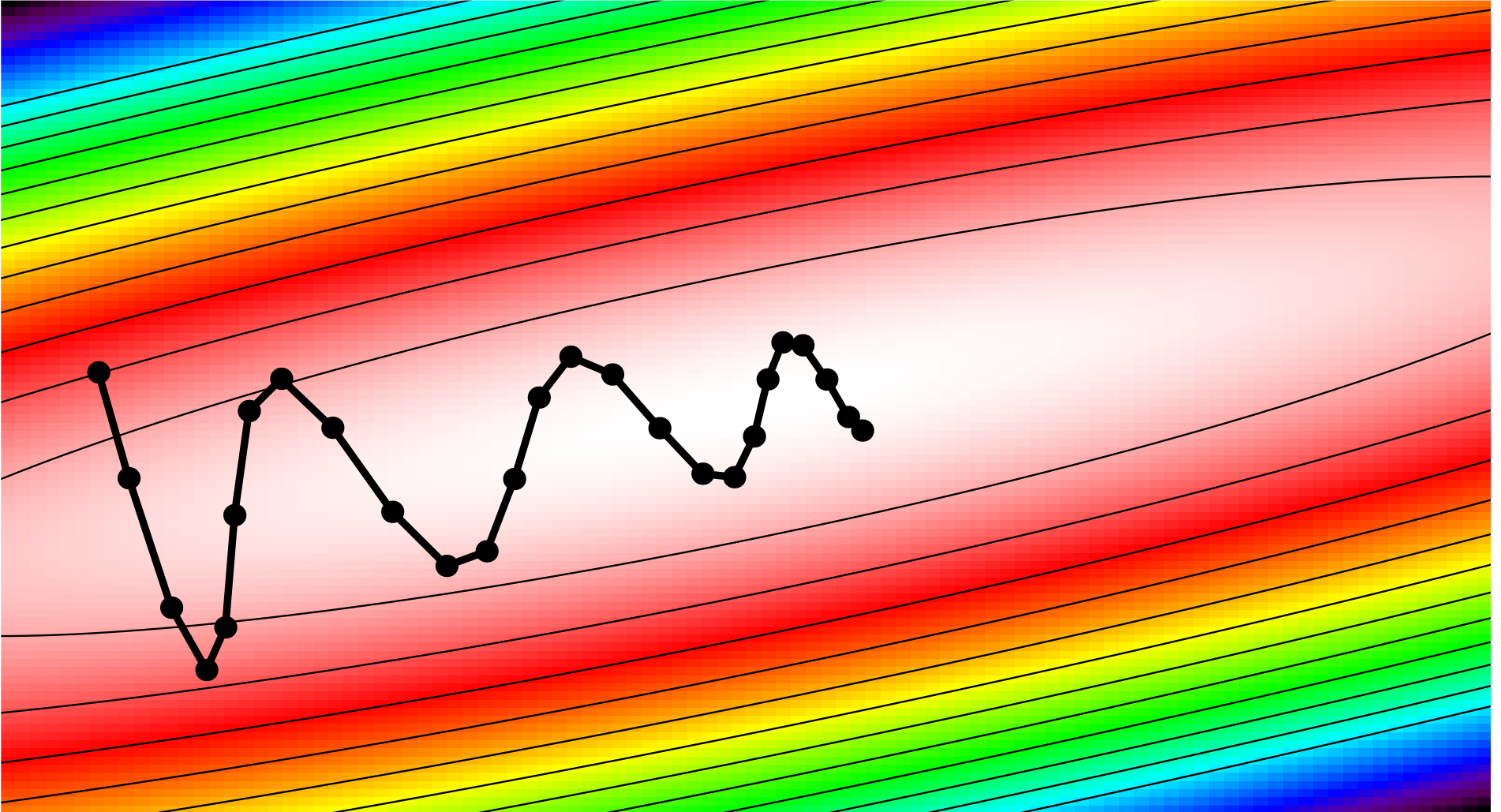
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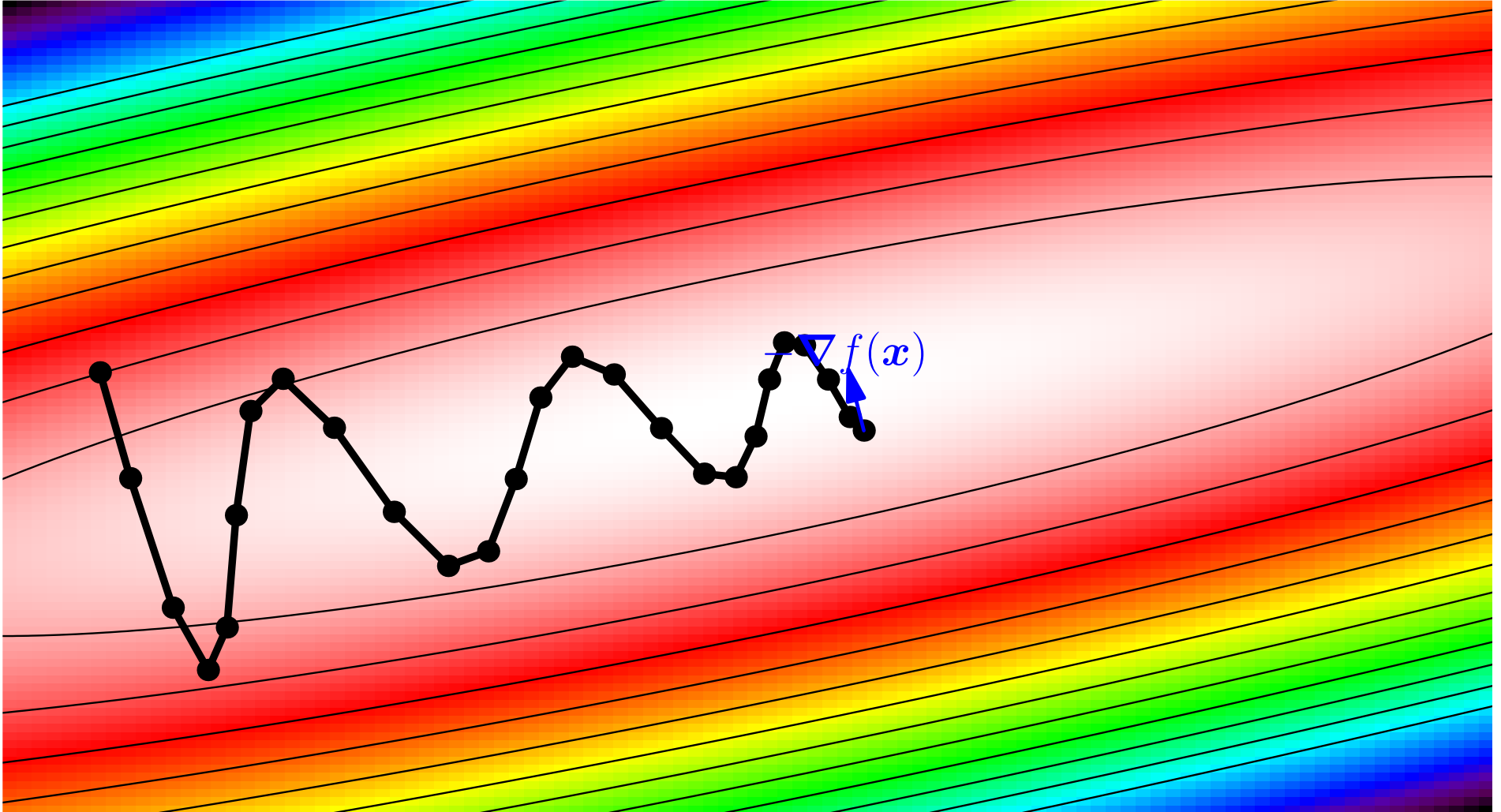
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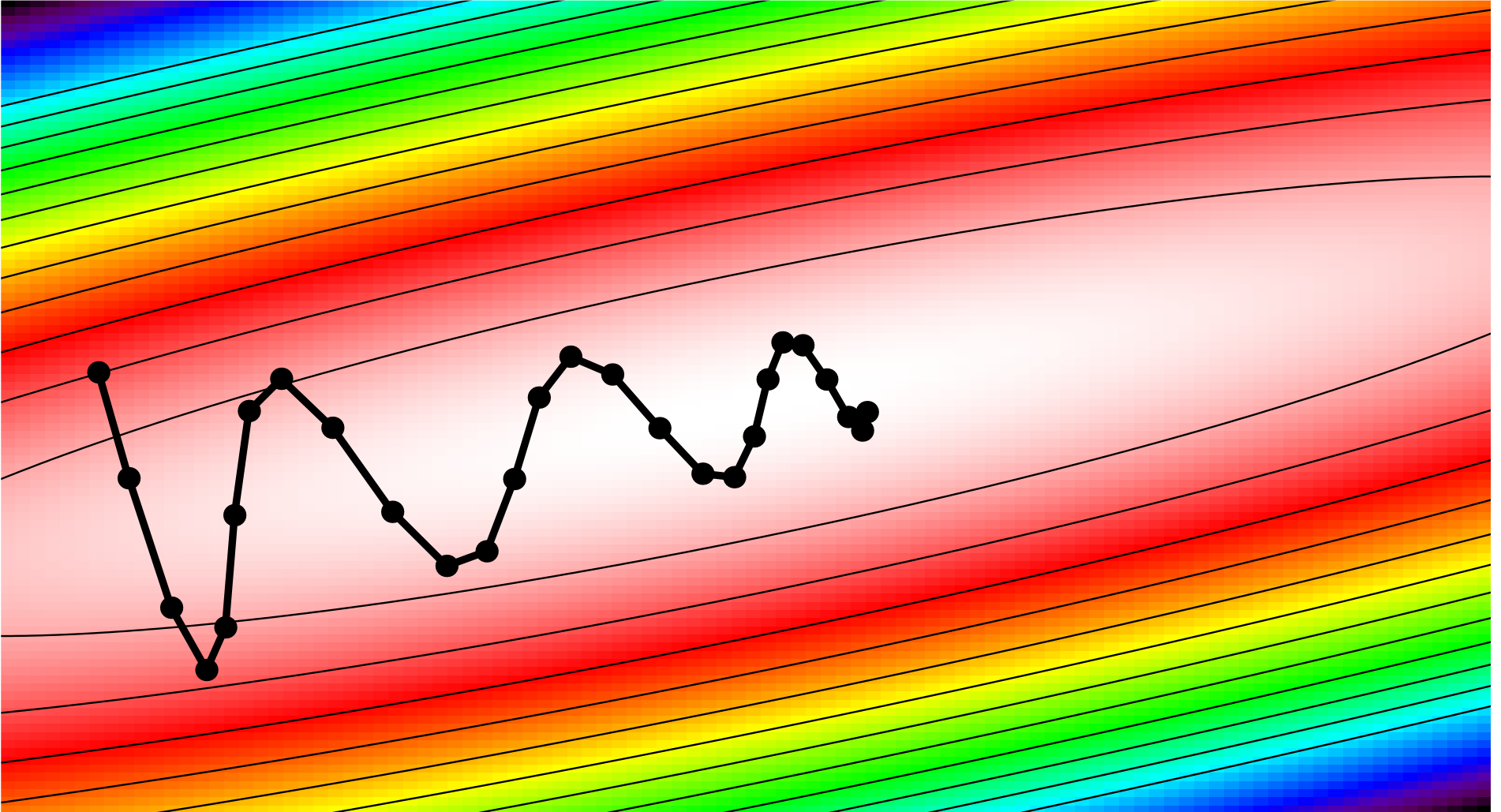
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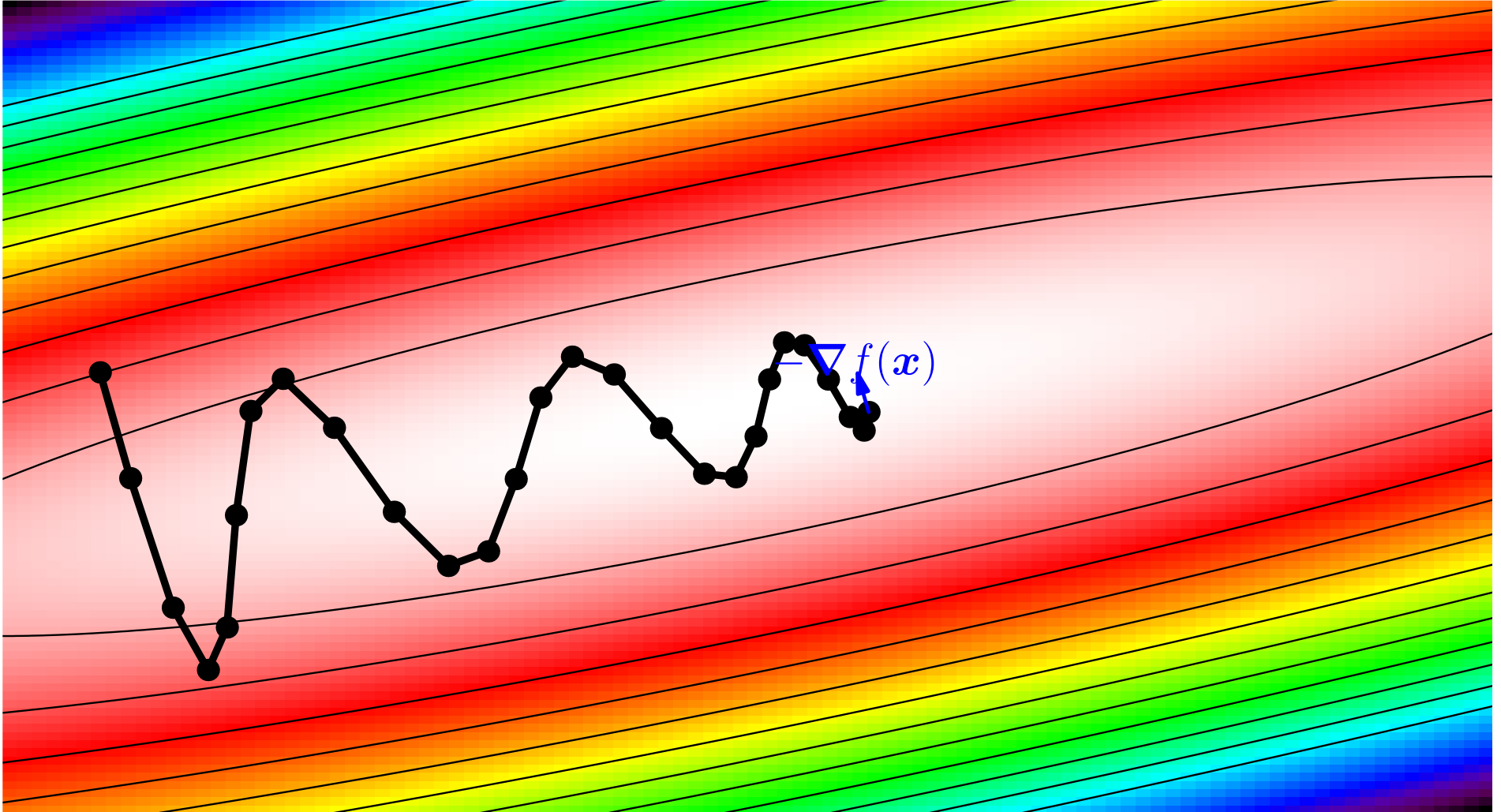
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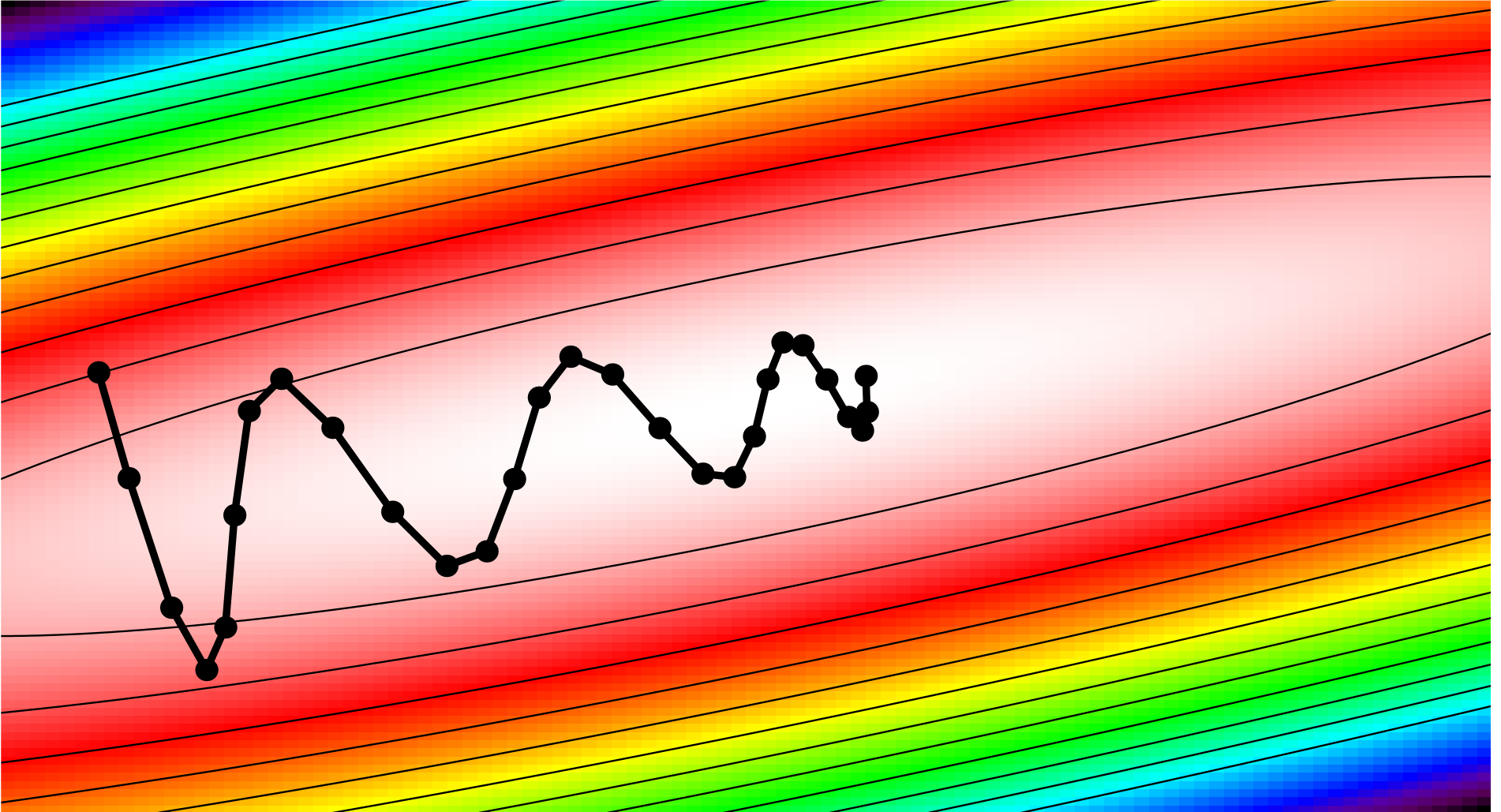
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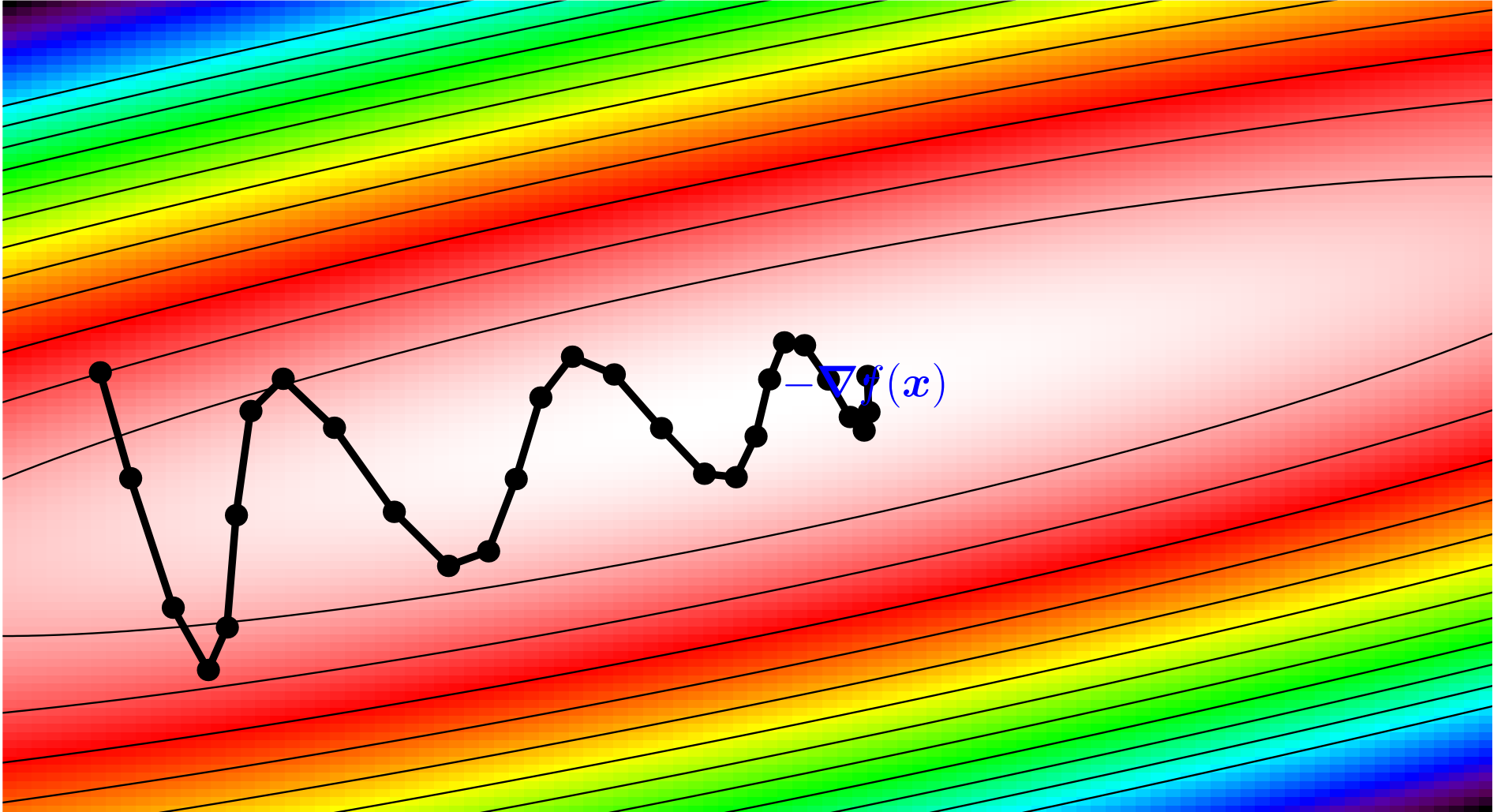
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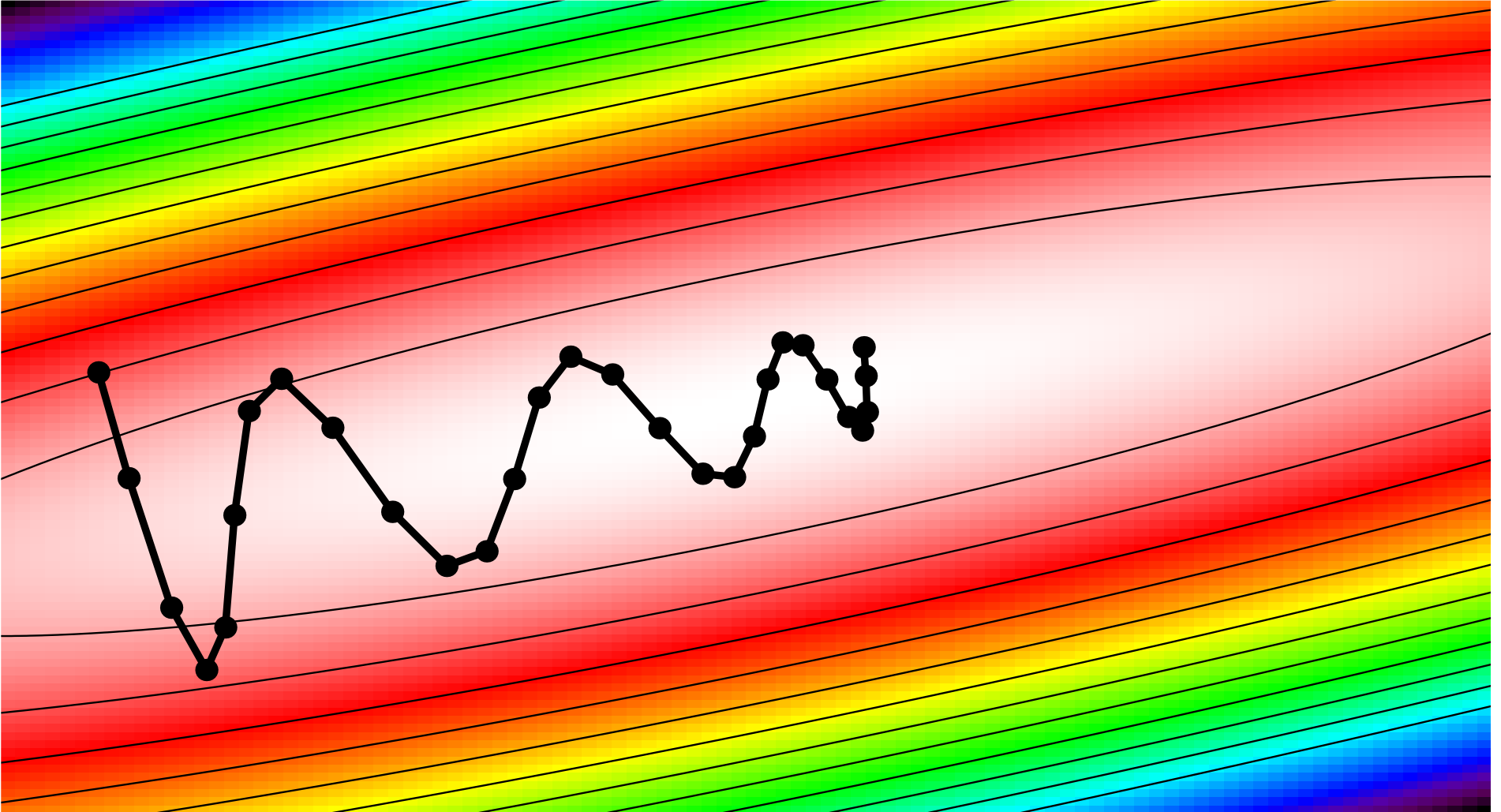
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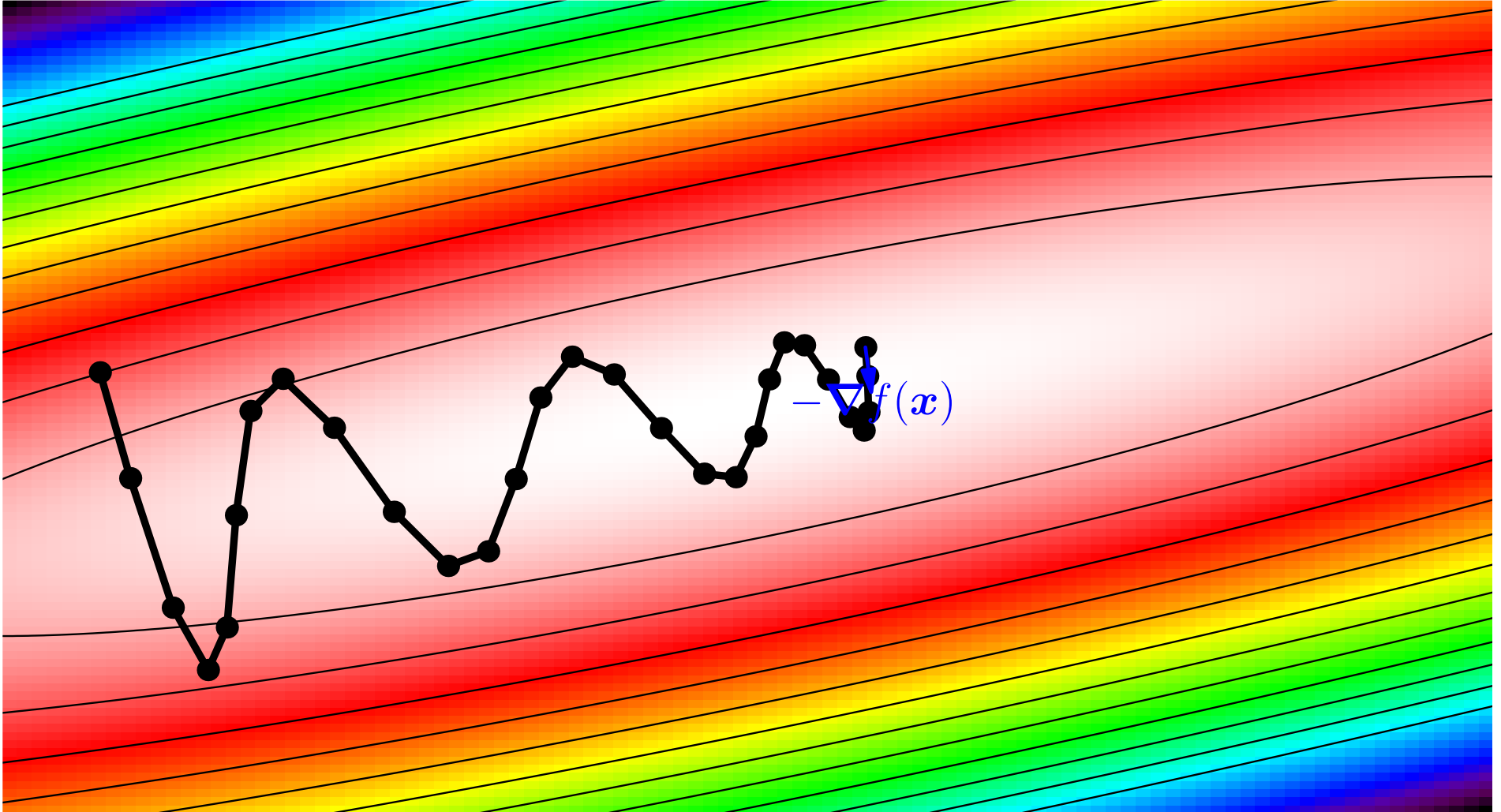
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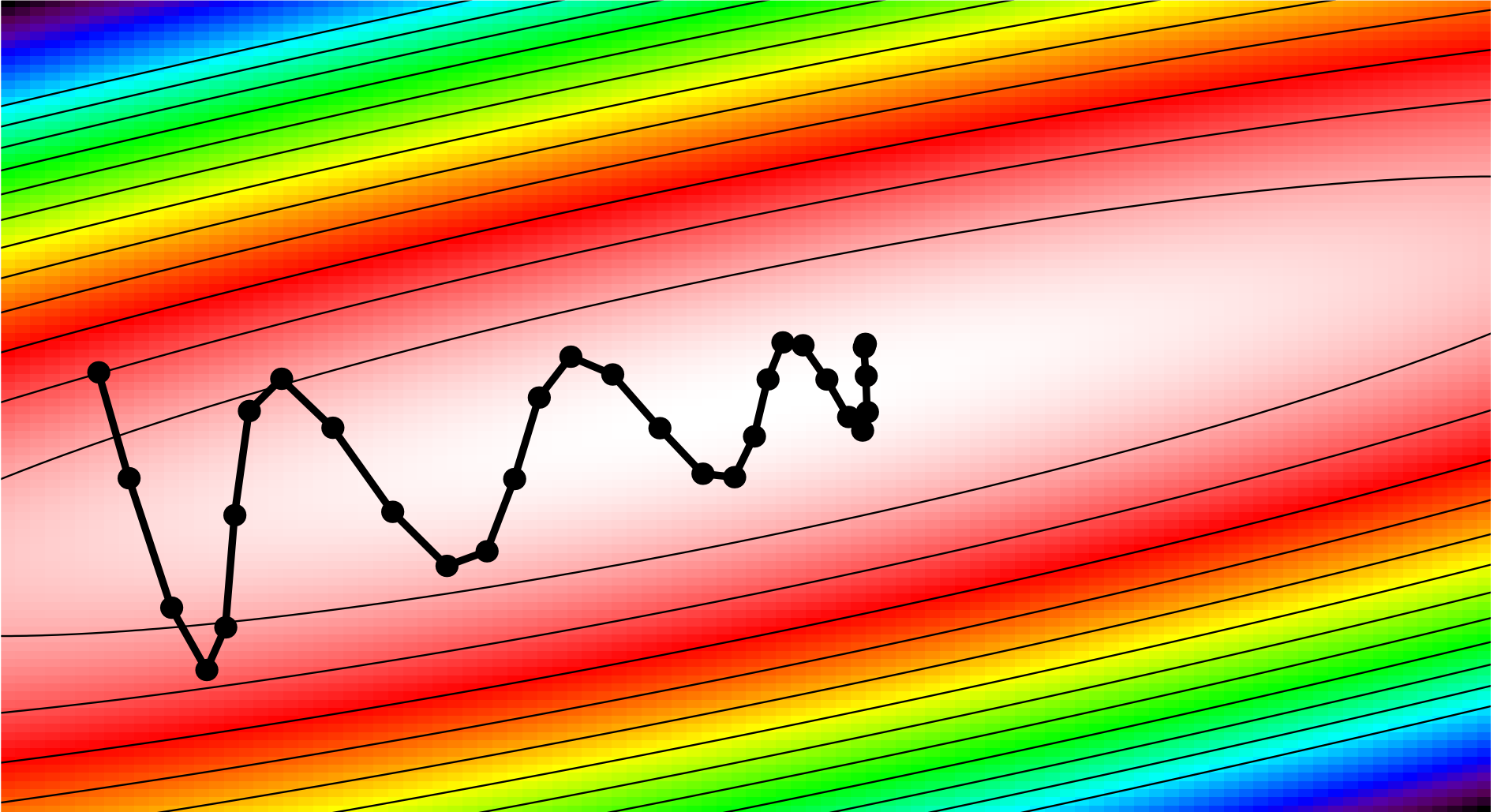
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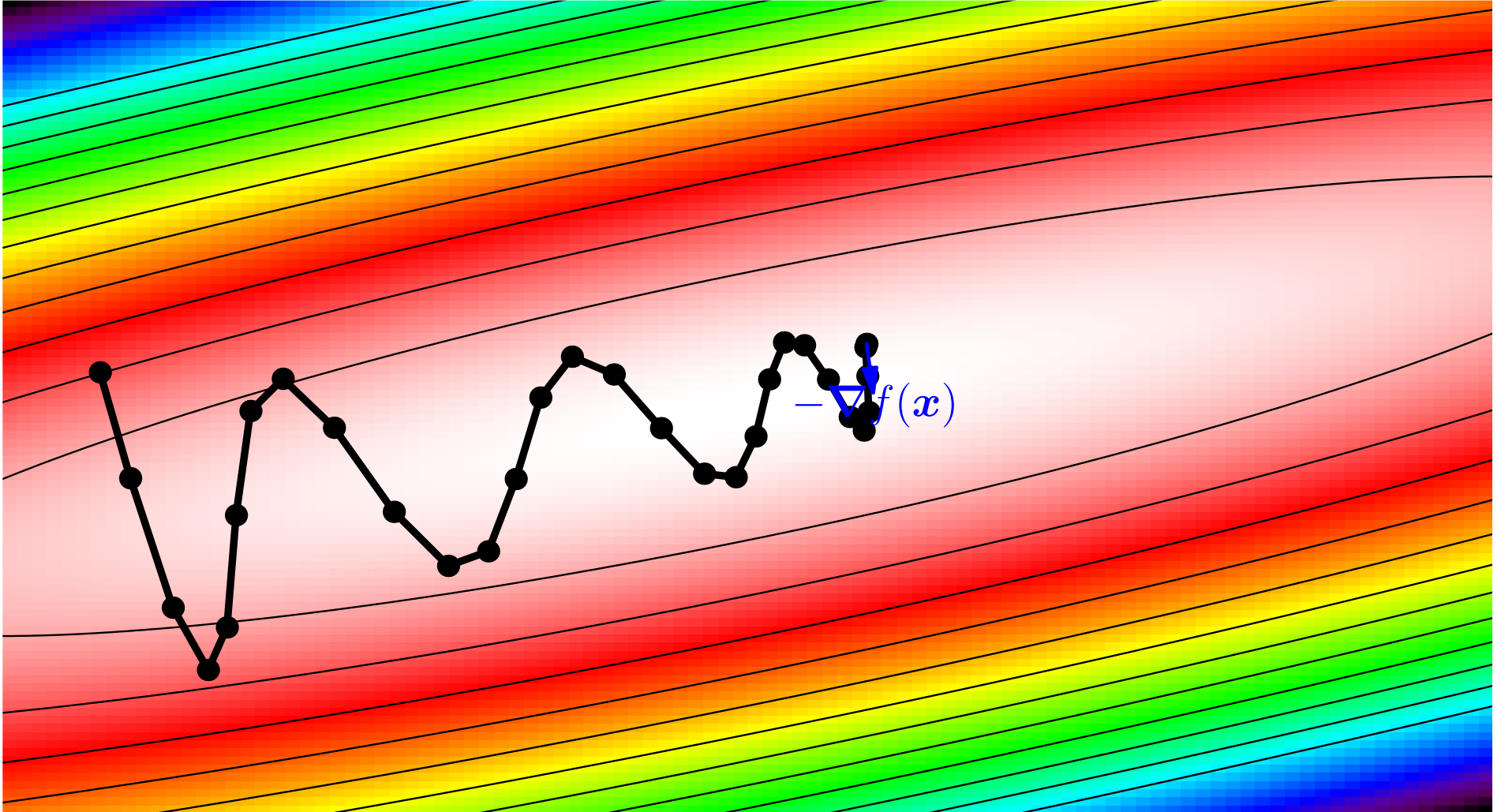
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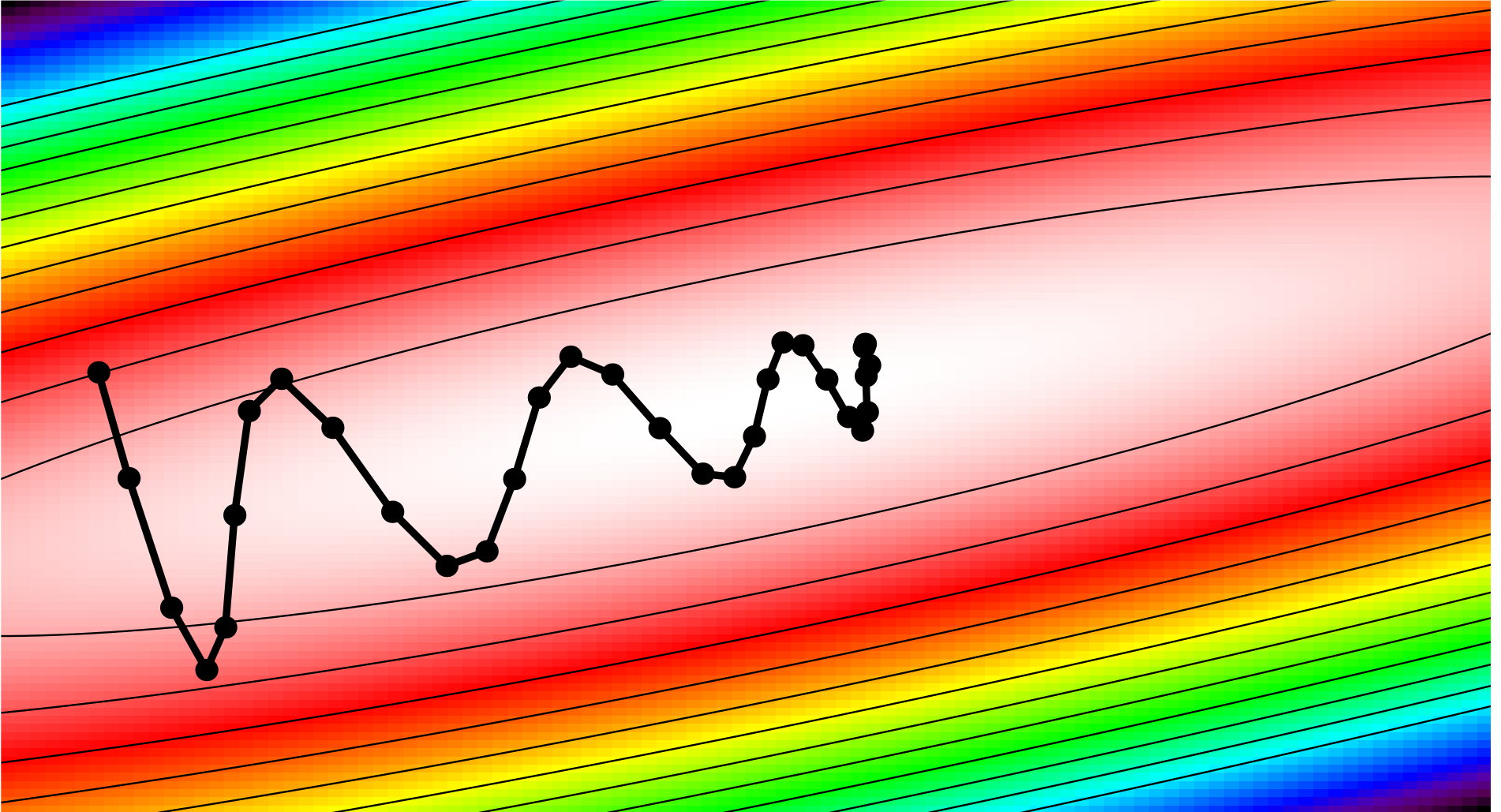
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Adaptive Methods

- A major difficulty of high dimensional optimisation is the existence of different scales
- That is, there are some directions where we have to move a lot (low curvature), while other directions we have to make small steps
- In adaptive methods we use a dynamic algorithm to rescale the step size of each parameter (weight)
- We can think of this as a “regularisation” that makes our basins of attraction more spherical

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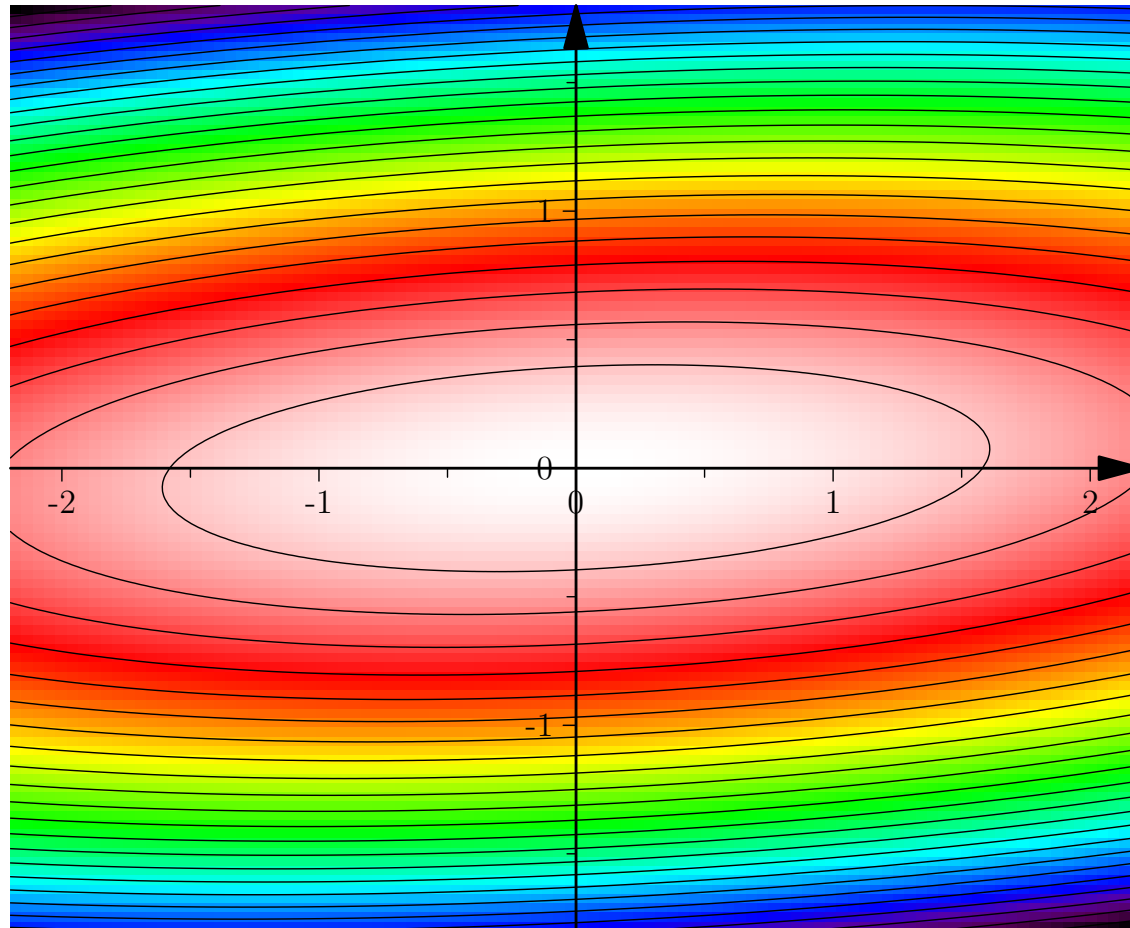
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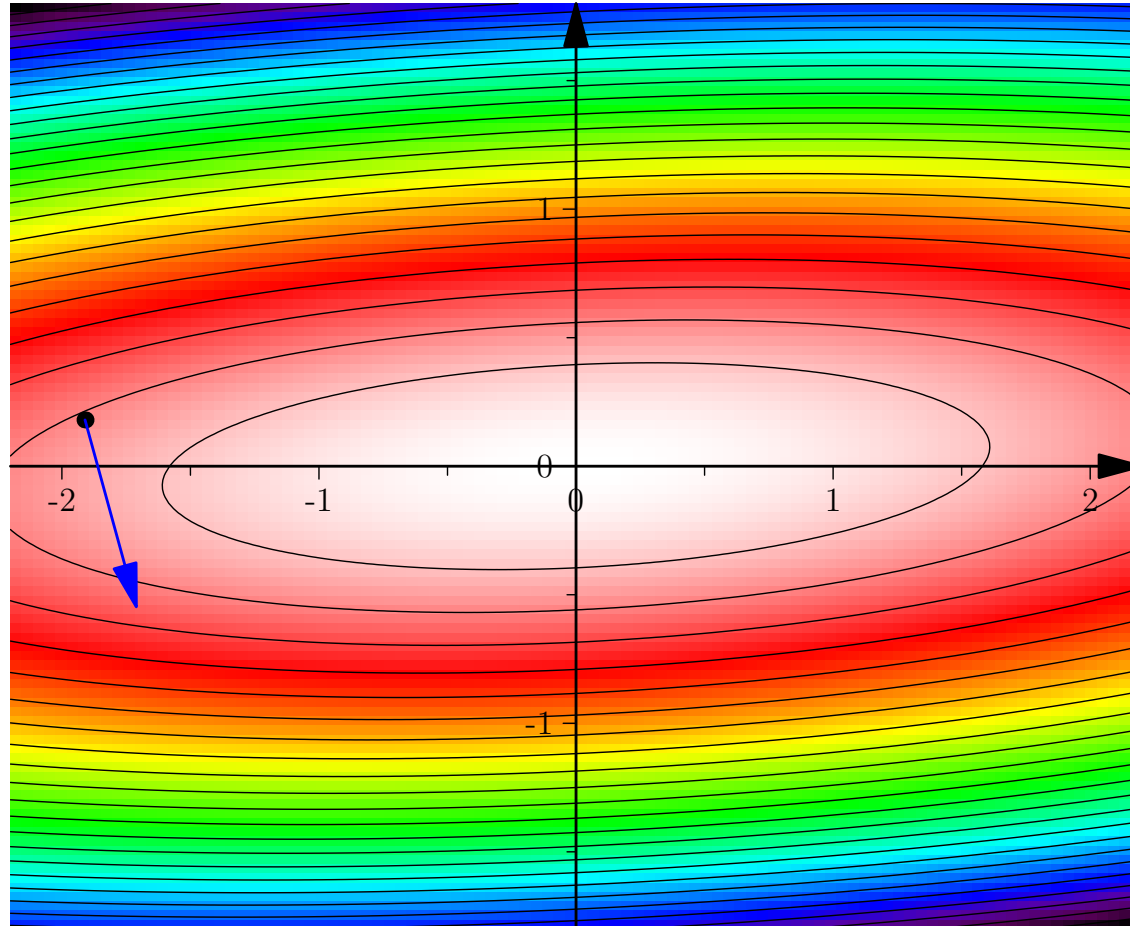
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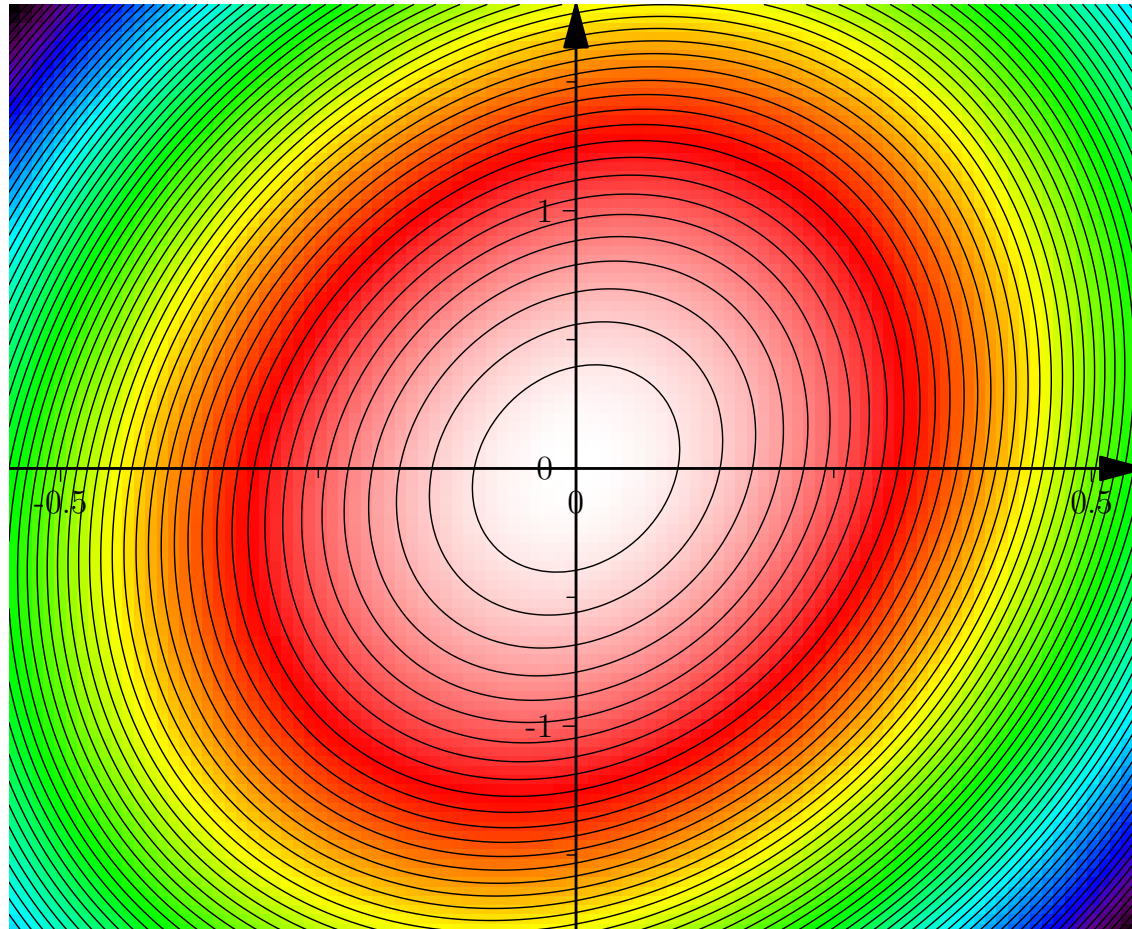
Rescaling Co-ordinates



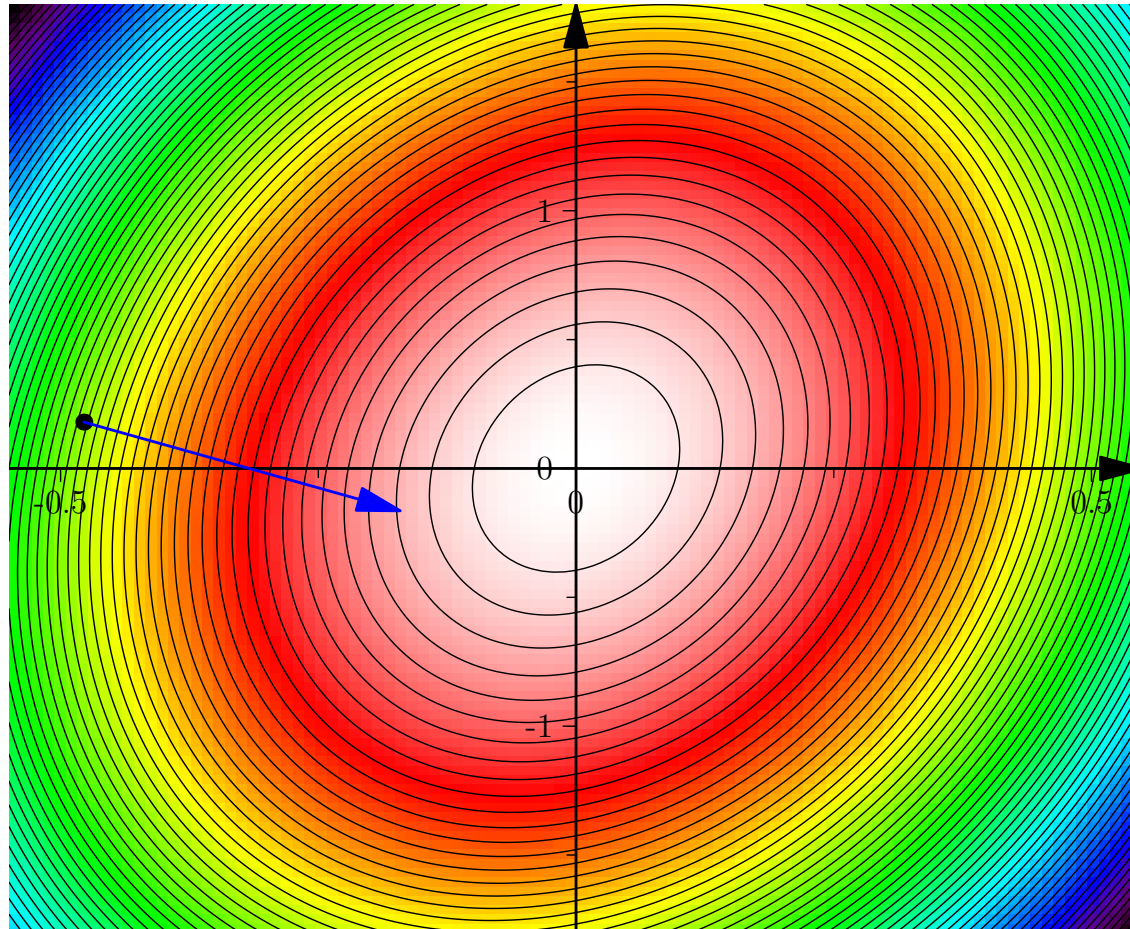
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AdaDelta

- We want to rescale the coordinates in proportion to the curvature in that direction
- To estimate the curvature we compute a running average of the squared gradients

$$S_i^g(t+1) = (1 - \gamma)S_i^g(t) + \gamma \left(\frac{\partial L_{\mathcal{B}}(\mathbf{w}(t))}{\partial w_i(t)} \right)^2$$

- To estimate the relative scale of the weights we compute a running average of the squared weights

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- The AdaDelta algorithm uses the update

$$w_i(t+1) = w_i(t) - \eta \sqrt{\frac{S_i^w(t+1) + \epsilon}{S_i^g(t+1) + \epsilon}} \frac{\partial L_{\mathcal{B}}(\mathbf{w}(t))}{\partial w_i(t)}$$

- Note that we can rescale the weights or rescale the gradients yet the update would be the same
- Because we are adaptively changing our step size we can use a single step size throughout the optimisation

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Adding in Momentum

- AdaDelta doesn't use momentum (there is an argument this isn't so important as it has "regularised the landscape")
- Nevertheless we can learn a momentum term

$$M_i(t+1) = (1 - \beta)M_i(t) + \beta \frac{\partial L_{\mathcal{B}}(\mathbf{w}(t))}{\partial w_i(t)}$$
$$S_i(t+1) = (1 - \gamma)S_i(t) + \gamma \left(\frac{\partial L_{\mathcal{B}}(\mathbf{w}(t))}{\partial w_i(t)} \right)^2$$

- These running averages have a lag time which we can remove

$$\hat{M}_i(t+1) = \frac{M_i(t+1)}{1 - (1 - \beta)^t} \quad \hat{S}_i(t+1) = \frac{S_i(t+1)}{1 - (1 - \gamma)^t}$$

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- Nevertheless we can learn a momentum term

$$M_i(t+1) = (1 - \beta)M_i(t) + \beta \frac{\partial L_{\mathcal{B}}(\mathbf{w}(t))}{\partial w_i(t)}$$

$$S_i(t+1) = (1 - \gamma)S_i(t) + \gamma \left(\frac{\partial L_{\mathcal{B}}(\mathbf{w}(t))}{\partial w_i(t)} \right)^2$$

- These running averages have a lag time which we can remove

$$\hat{M}_i(t+1) = \frac{M_i(t+1)}{1 - (1 - \beta)^t} \quad \hat{S}_i(t+1) = \frac{S_i(t+1)}{1 - (1 - \gamma)^t}$$

Adding in Momentum

- AdaDelta doesn't use momentum (there is an argument this isn't so important as it has "regularised the landscape")
- Nevertheless we can learn a momentum term

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ADAM

- Adaptive Moment Estimation (Adam) adapts the scale of the parameters, but also uses momentum
- Update weights

$$w_i(t+1) = w_i(t) - \frac{\eta}{\sqrt{\hat{S}_i(t+1) + \epsilon}} \hat{M}_i(t+1)$$

- ADAM and its variants are very successful in deep learning
- It is very robust so often results in a network learning which might not learn under standard SGD (with momentum)

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Covariant Equations

- Vector arithmetic, (matrix multiplication, addition, multiplying by a scale, computing gradients) has a covariant property that it is invariant to the coordinate system
- That is we can translate and rotate our coordinates and get the same result
- That is not true of element-wise multiplication of vectors
- Many ML algorithms do this (including ADAM and adaDelta), but they aren't invariant if we rotate our coordinates

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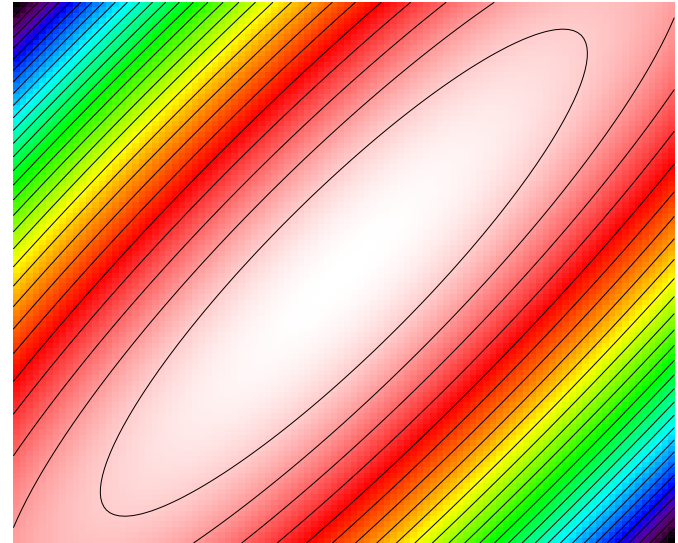
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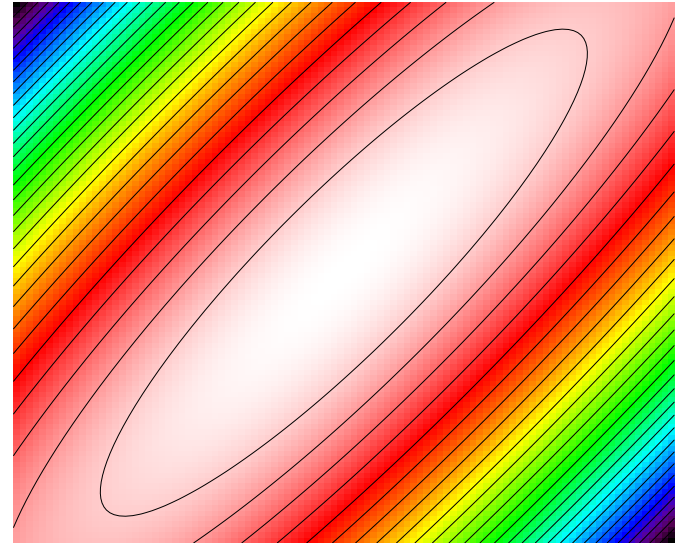
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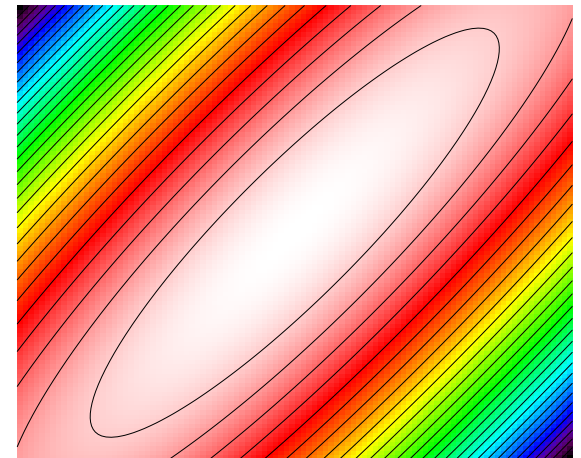
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Can't rescale
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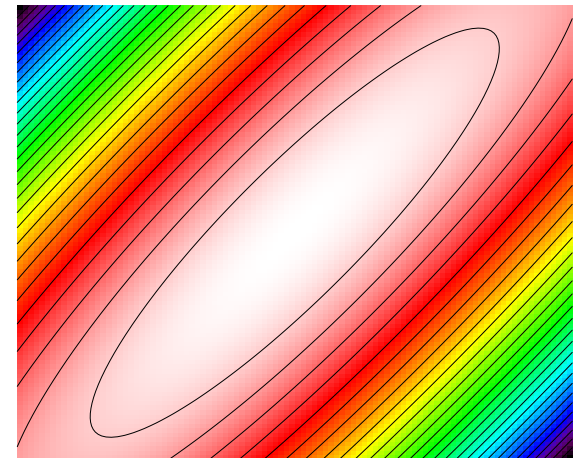
Correlated Weights

- Correlated weights will lead to skewed valleys
- Adaptive methods would be more efficient if they were rotationally invariant
- However, this would slow down the algorithm
- ADAM is a compromise between speed of implementation and effectiveness



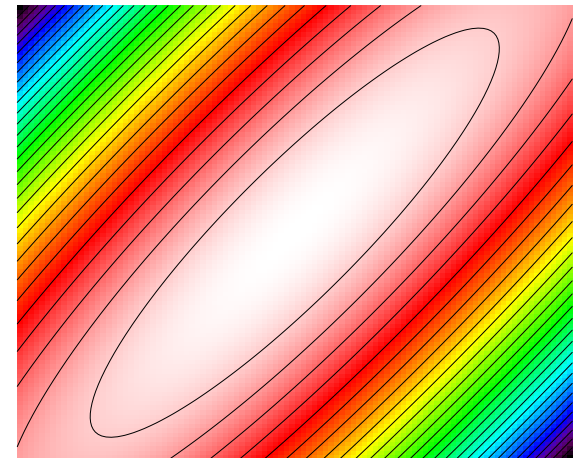
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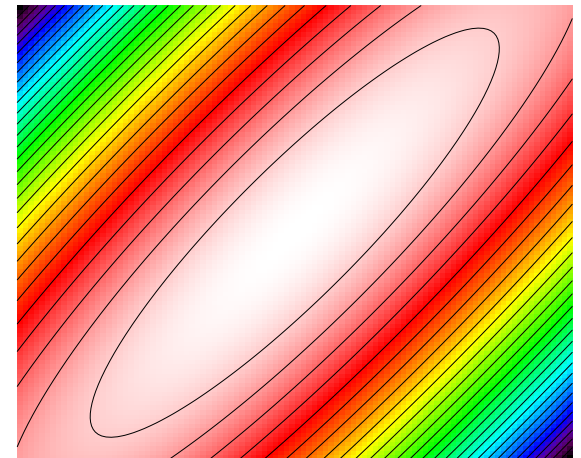
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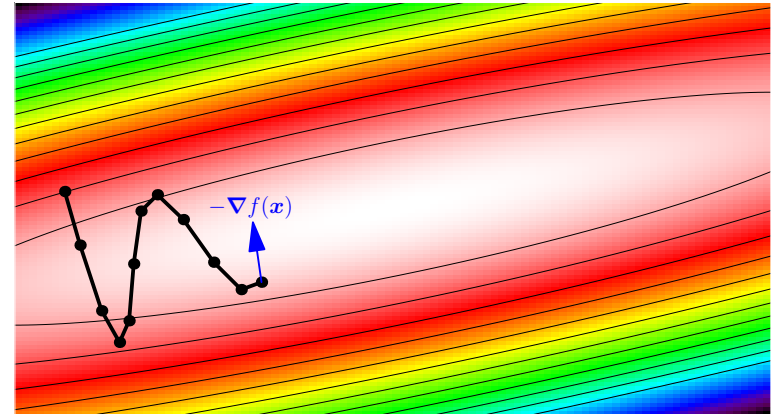
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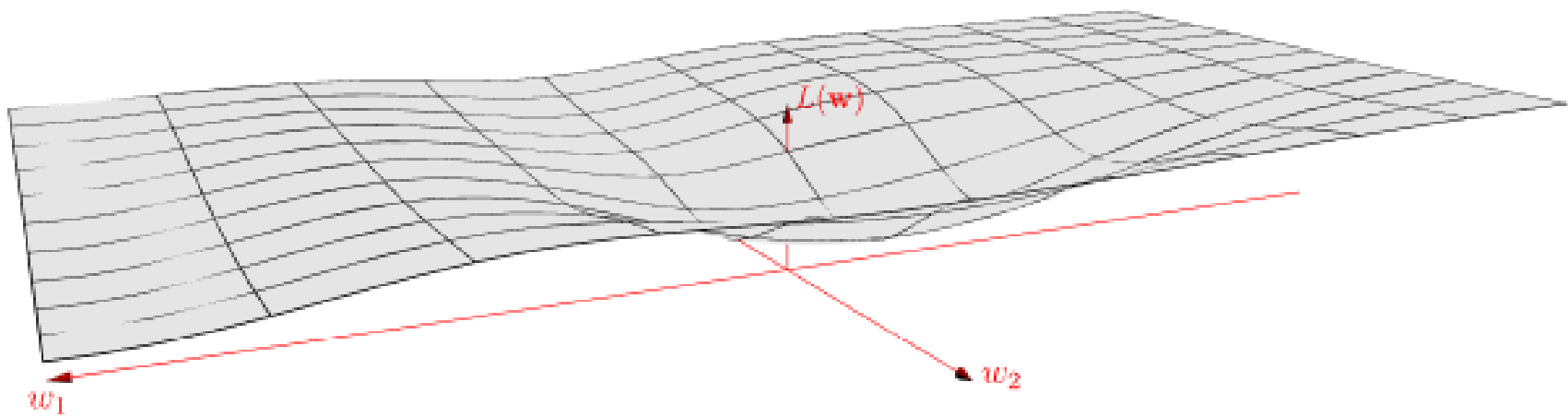
Outline

1. SGD
2. Momentum
3. **Loss Landscapes**



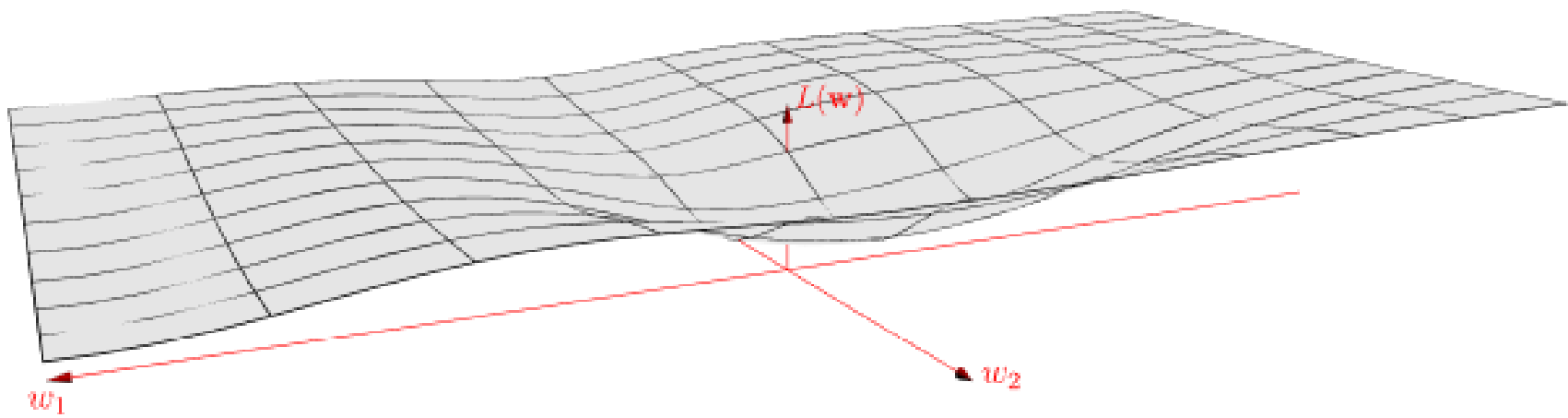
Loss Landscape

- A useful concept for understanding optimisation is the loss landscape
- For every value of the weights w there is some loss associated with it
- Note that this landscape is very high dimensional, so our intuition can be a bit misleading
- The landscape is also huge



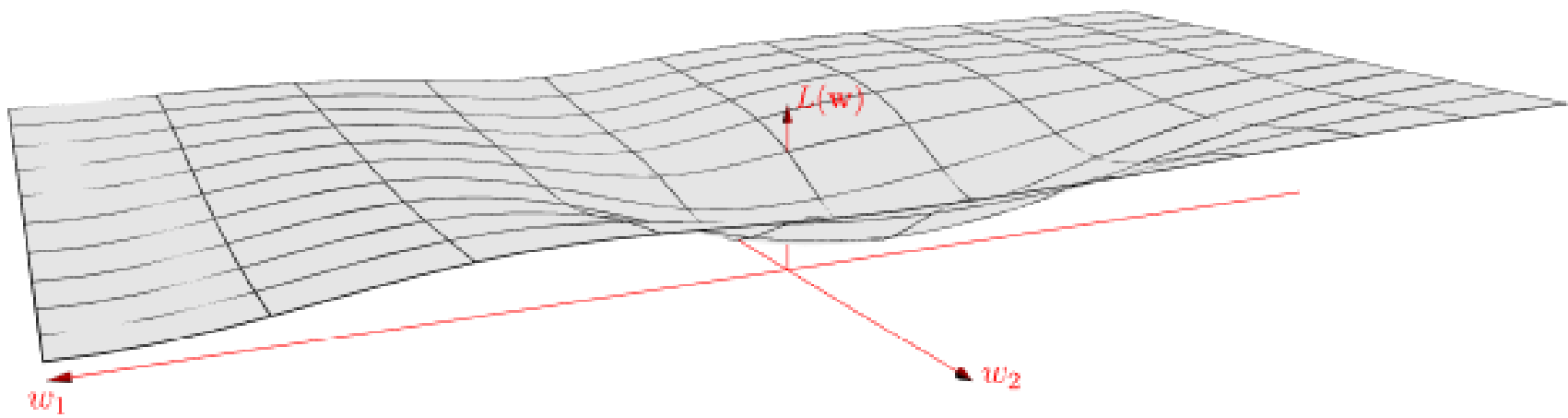
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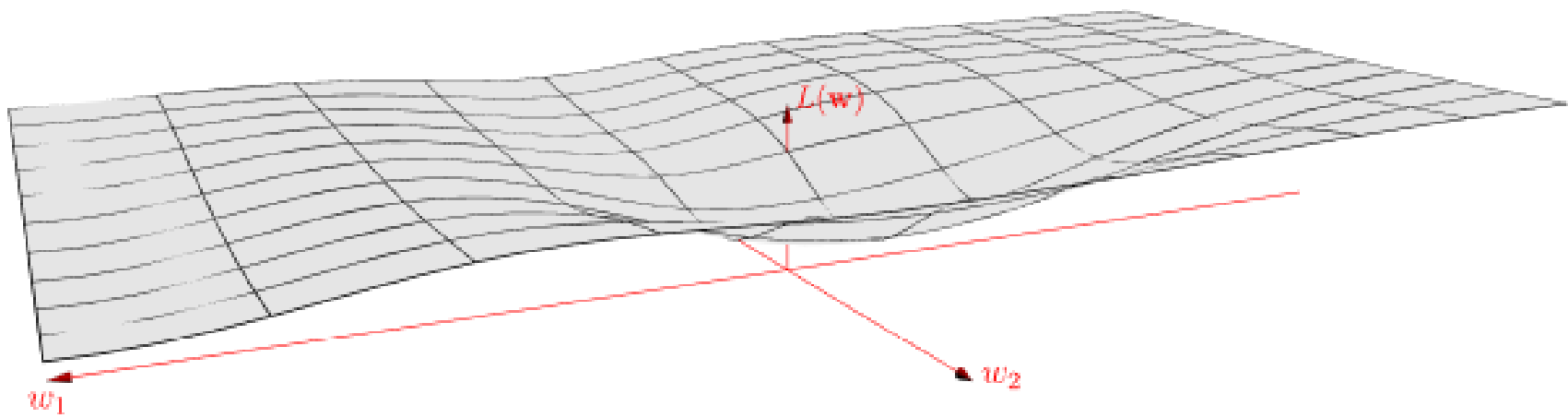
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Global and Local Minima

- Our objective is to minimise our loss
- This would mean finding the global minimum
- There are no algorithms that are guaranteed to find the global minimum
- The best we can hope is to find a local minimum by doing gradient descent
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- Because we are using mini-batches we aren't actually following the correct gradients
- If we reduce our step size then our average gradient is closer to the true gradient
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Valleys in Valleys in Valleys

- My view of the loss landscape is that we have valleys inside bigger valleys inside bigger valleys, and so on
- If our noise is high (we use large step size), we can navigate away from local valleys and move towards the big valley
- But, we can't explore the small valleys
- Often when the rate of improvement slows down, practitioners will reduce their learning rate by a factor of 10 and the rate of improvement jumps up
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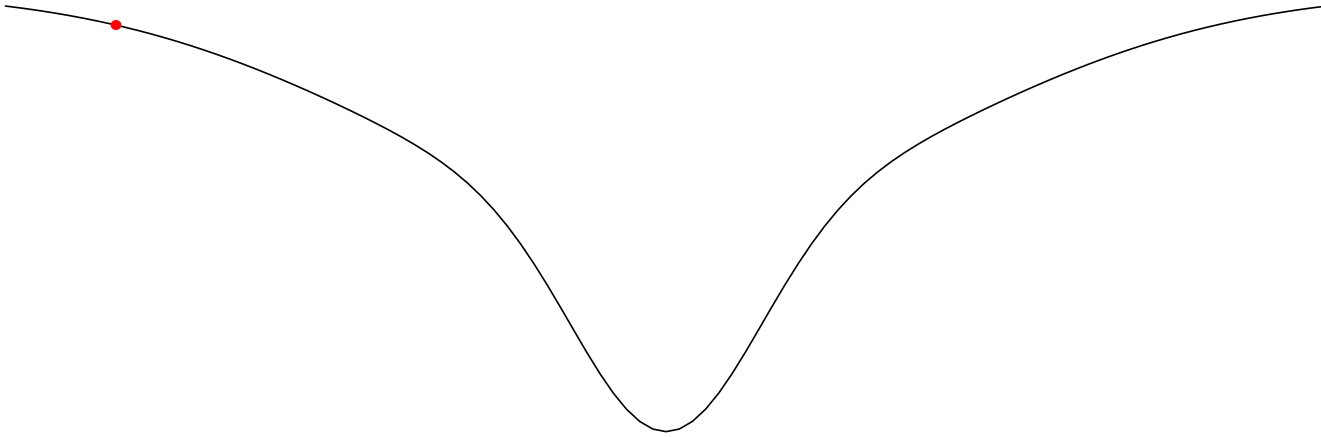
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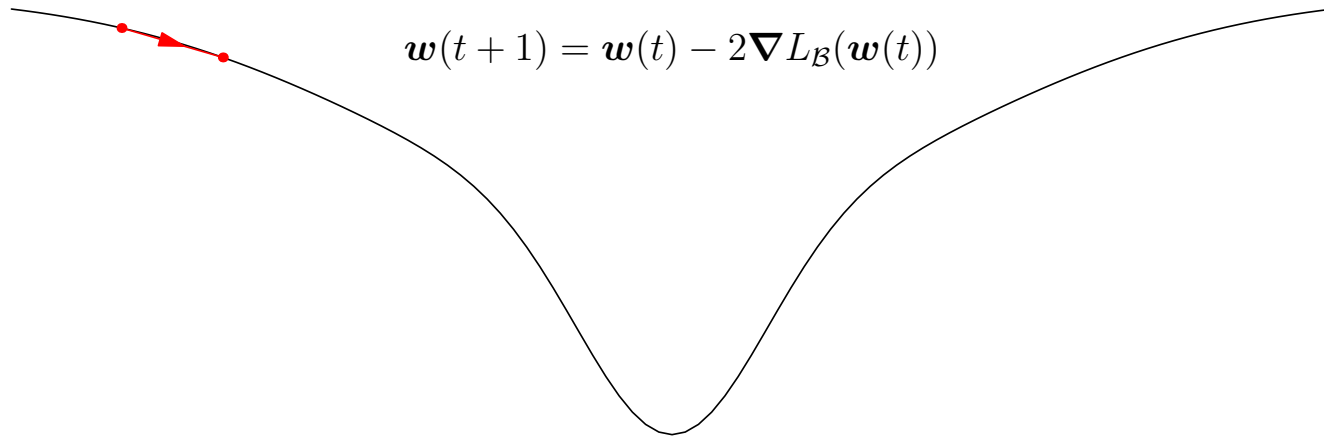
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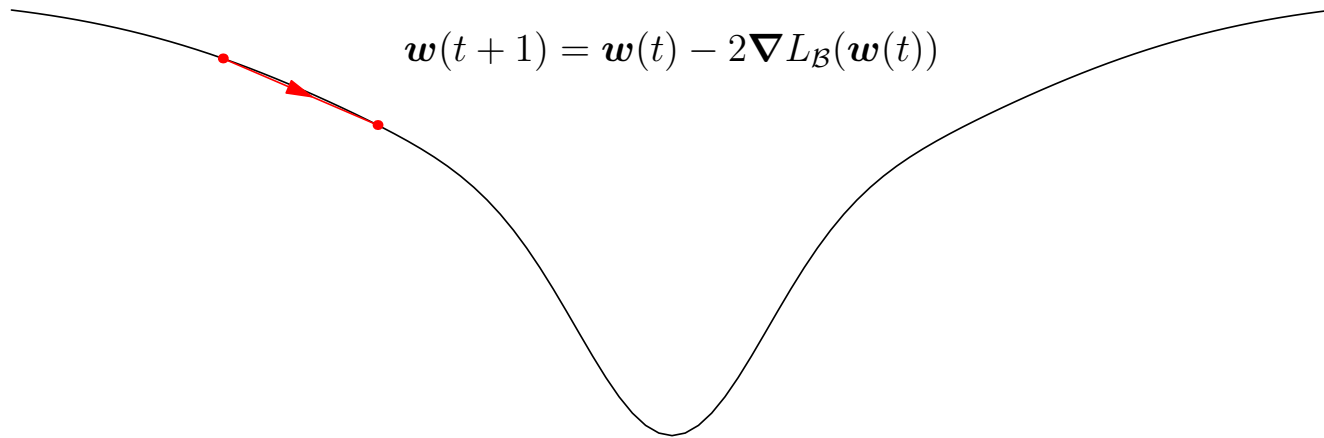
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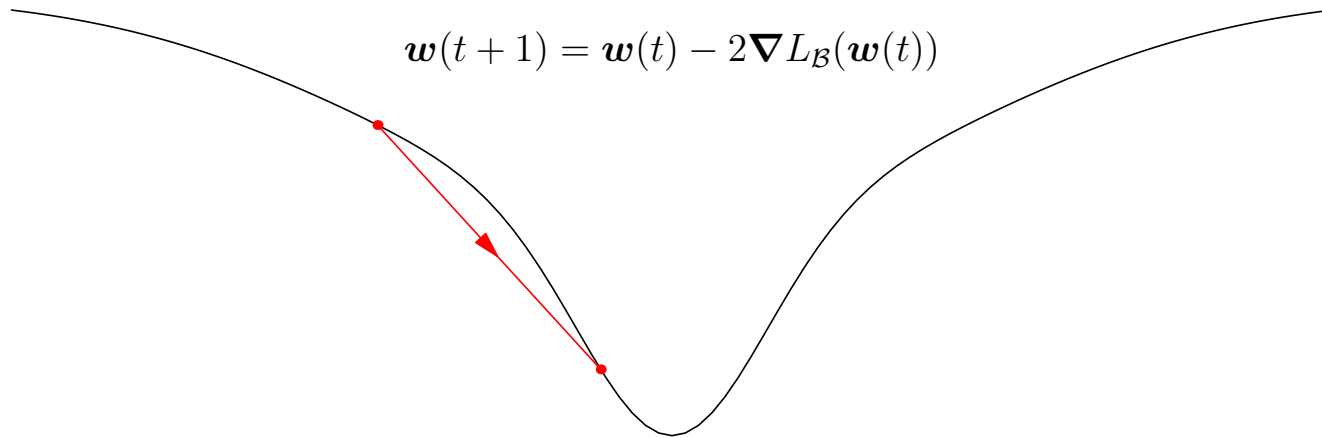
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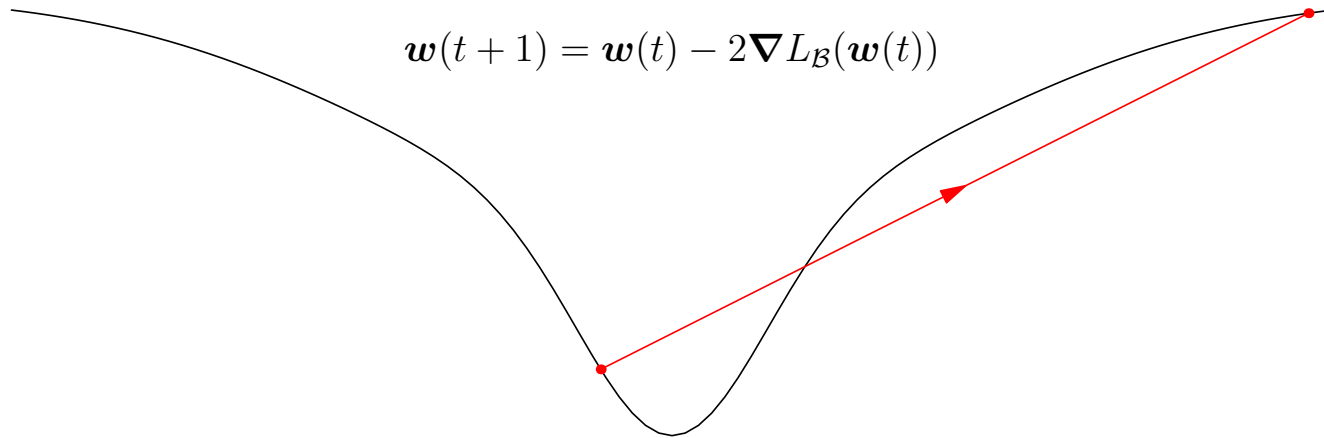
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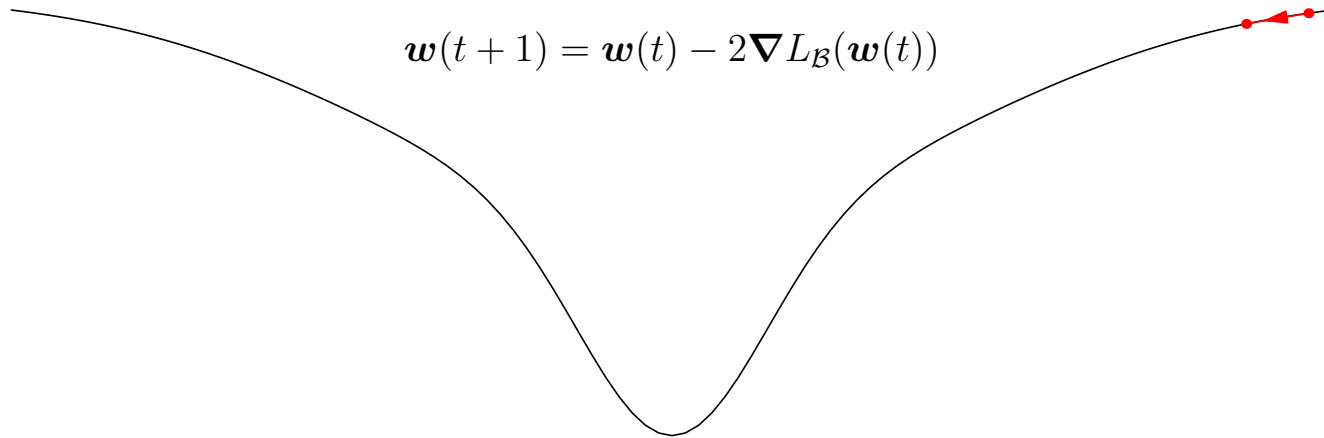
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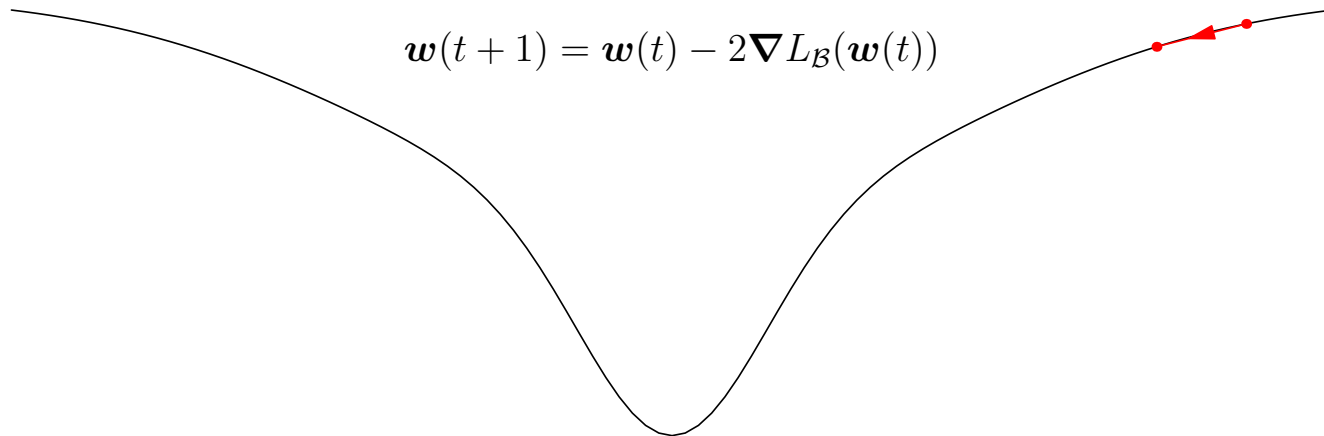
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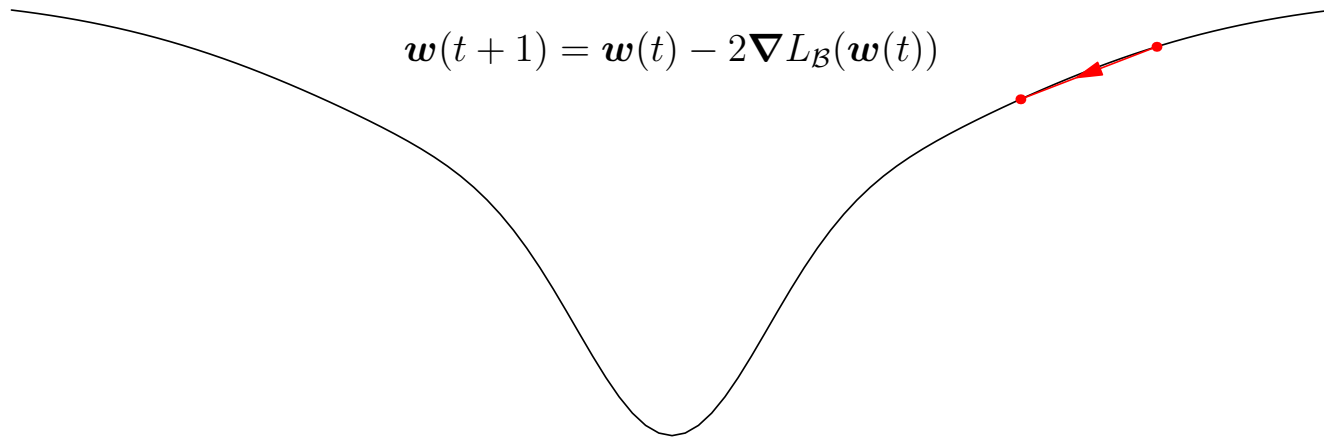
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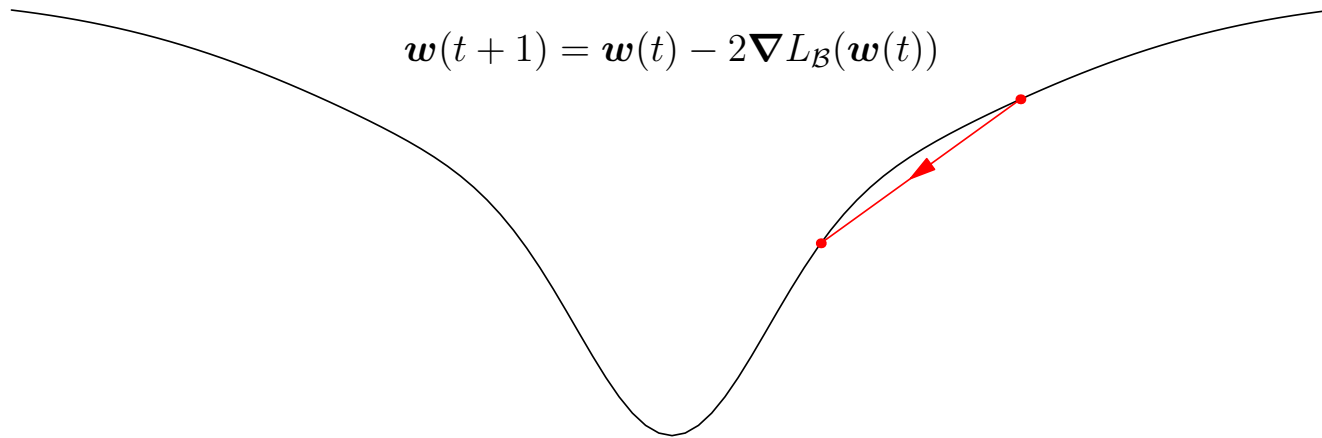
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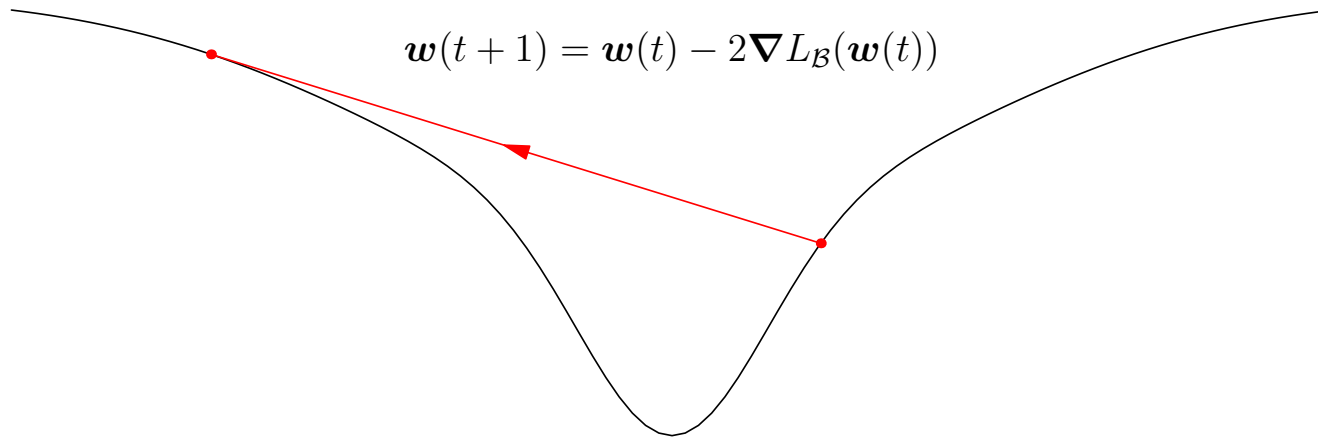
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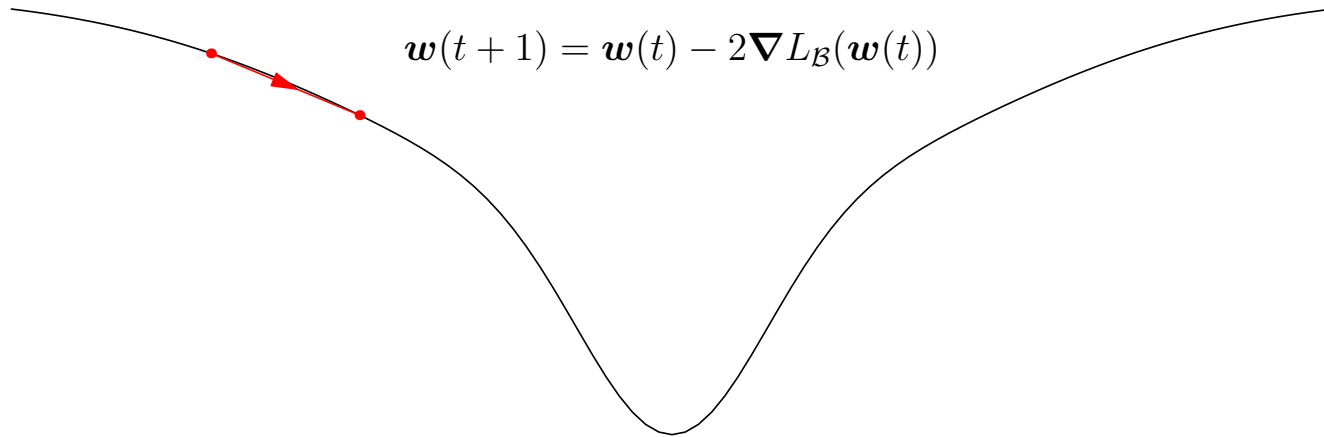
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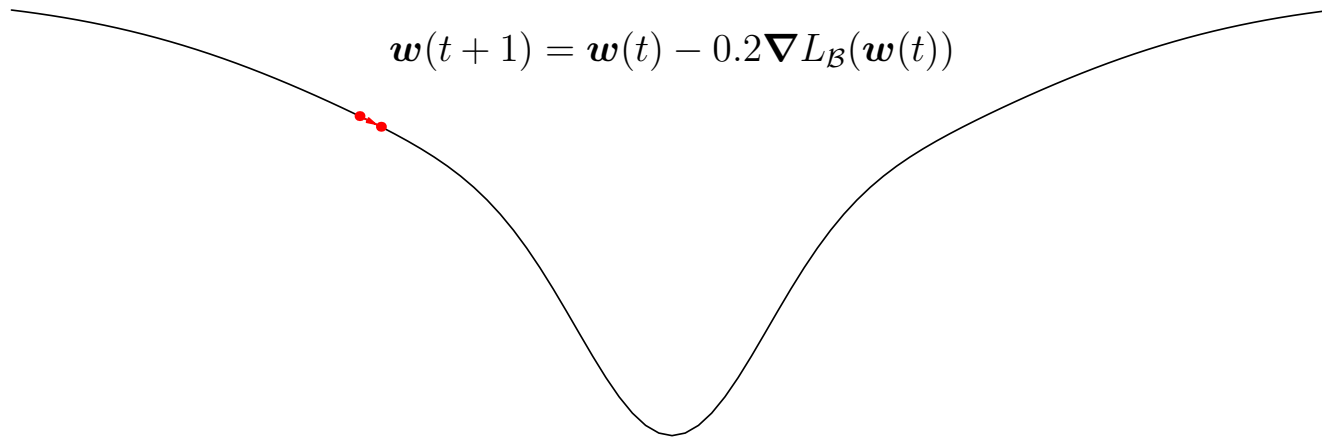
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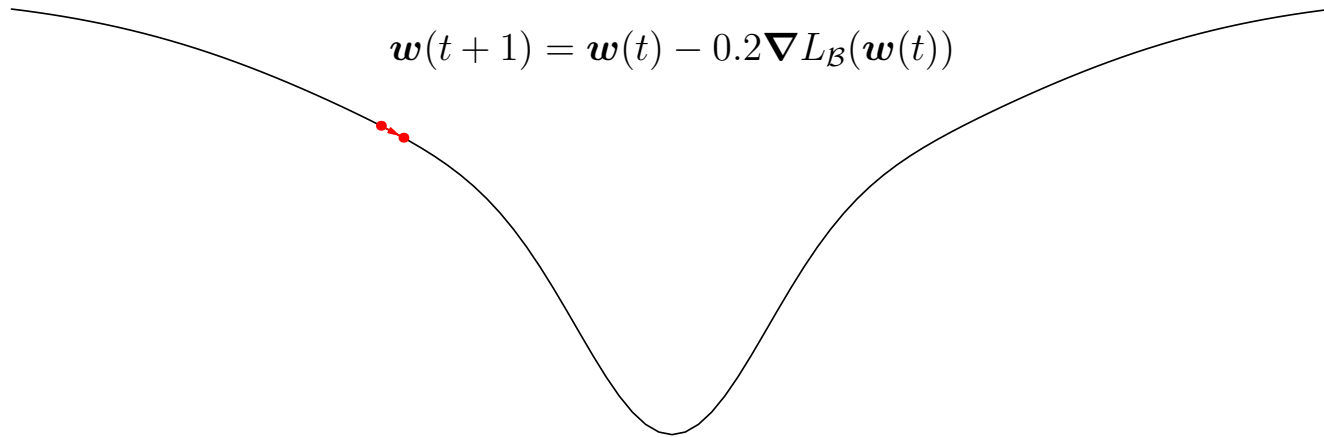
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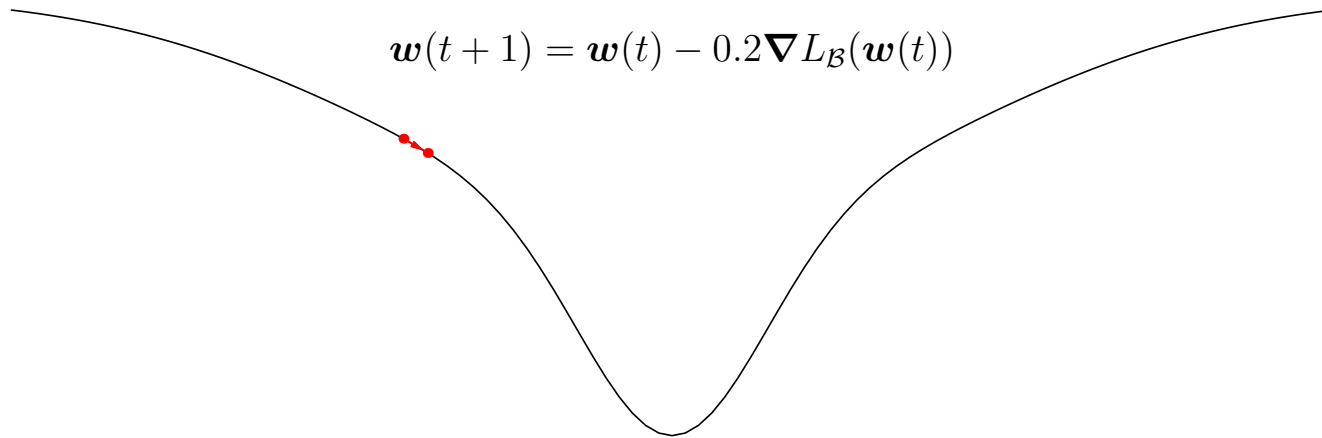
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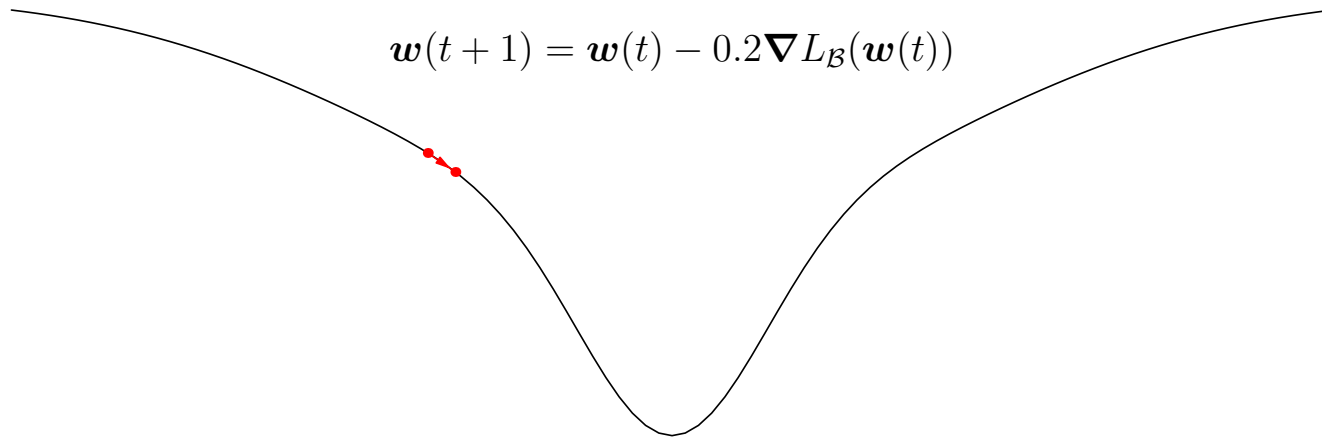
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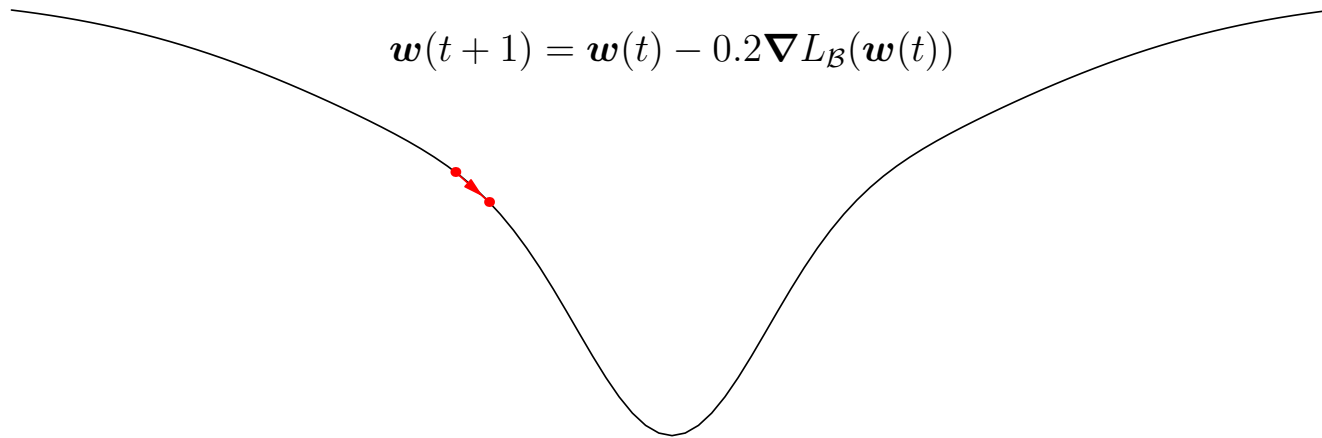
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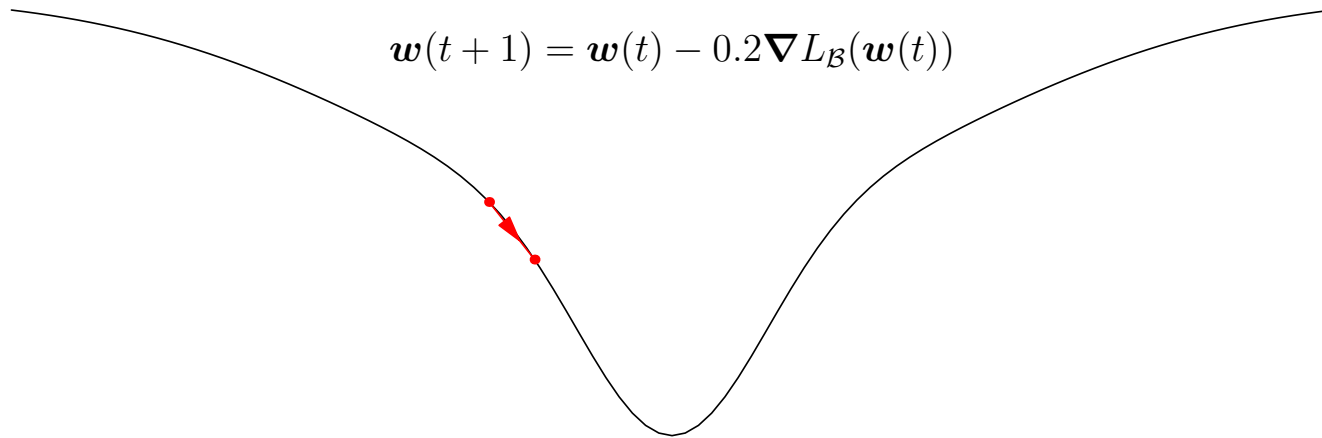
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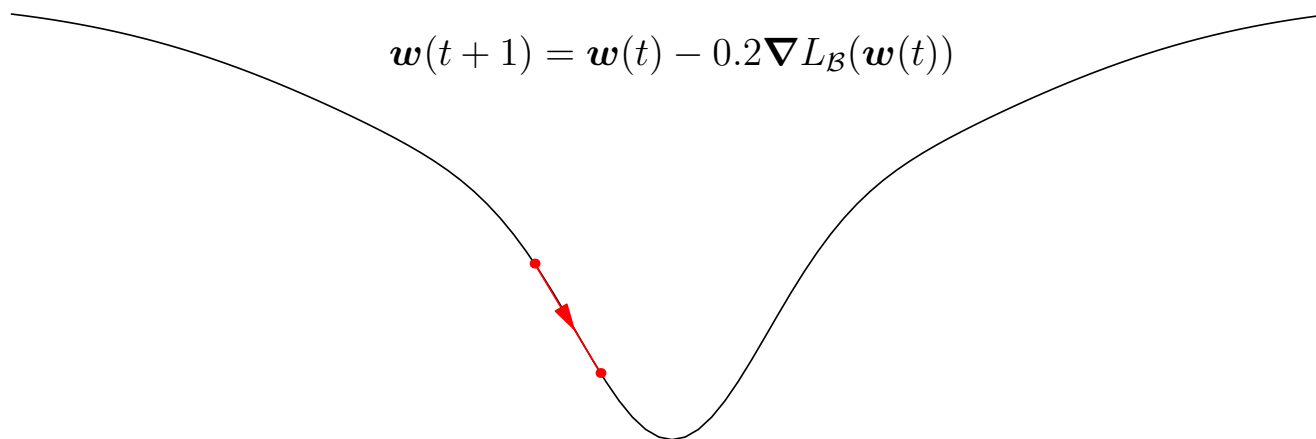
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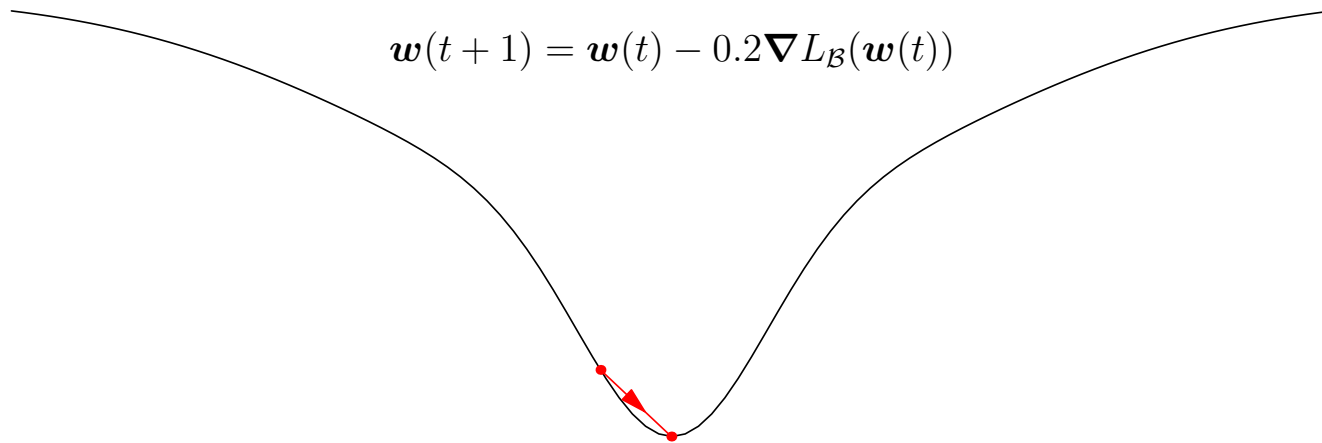
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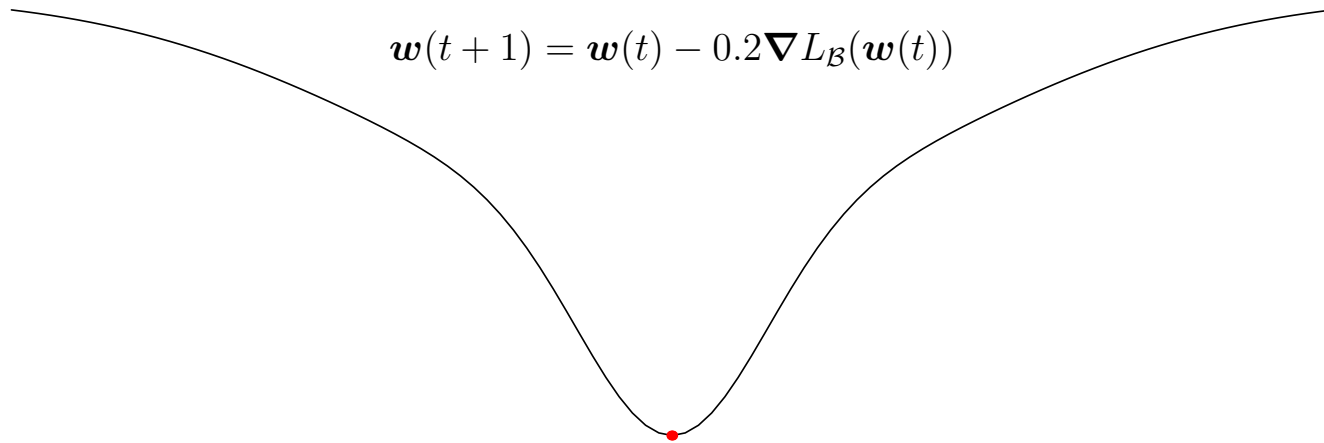
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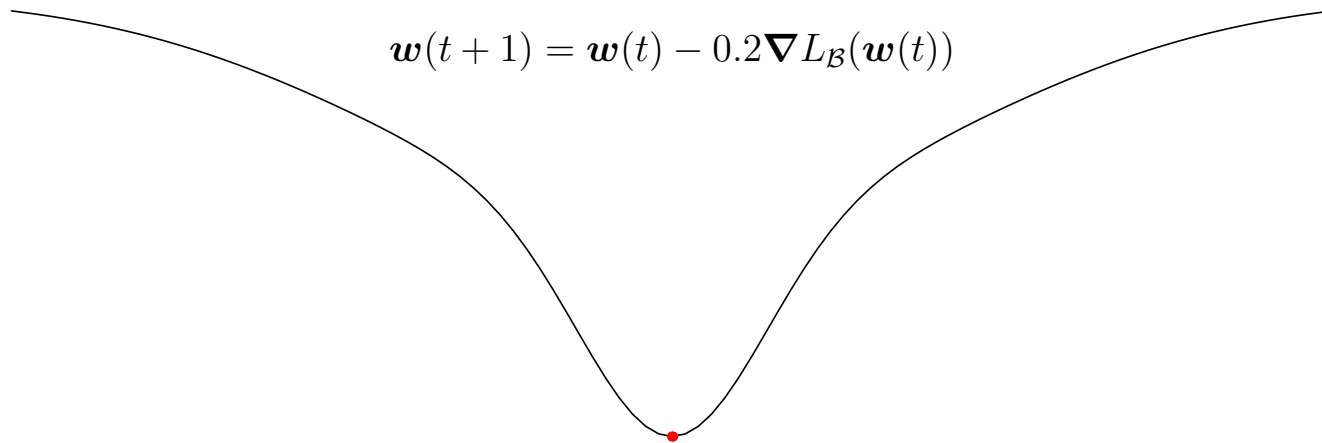
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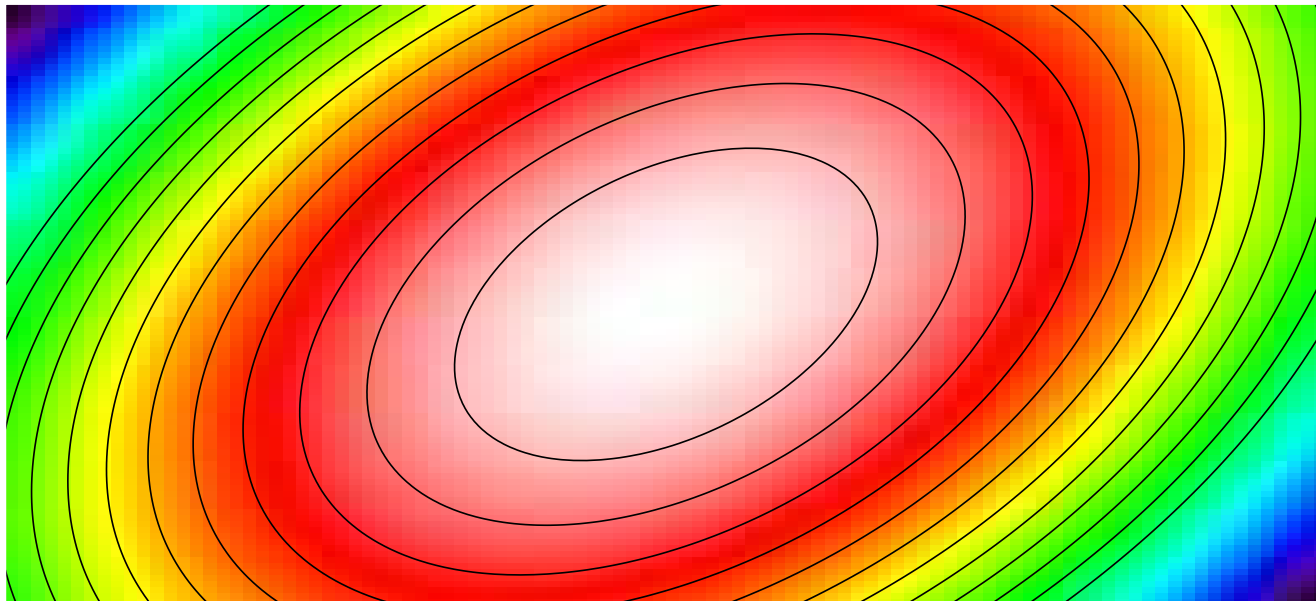
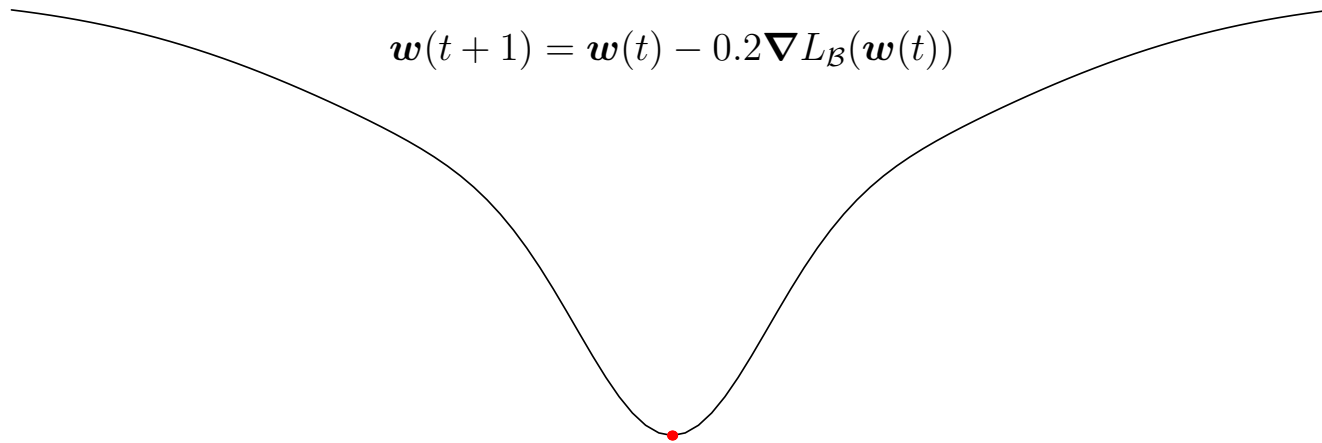
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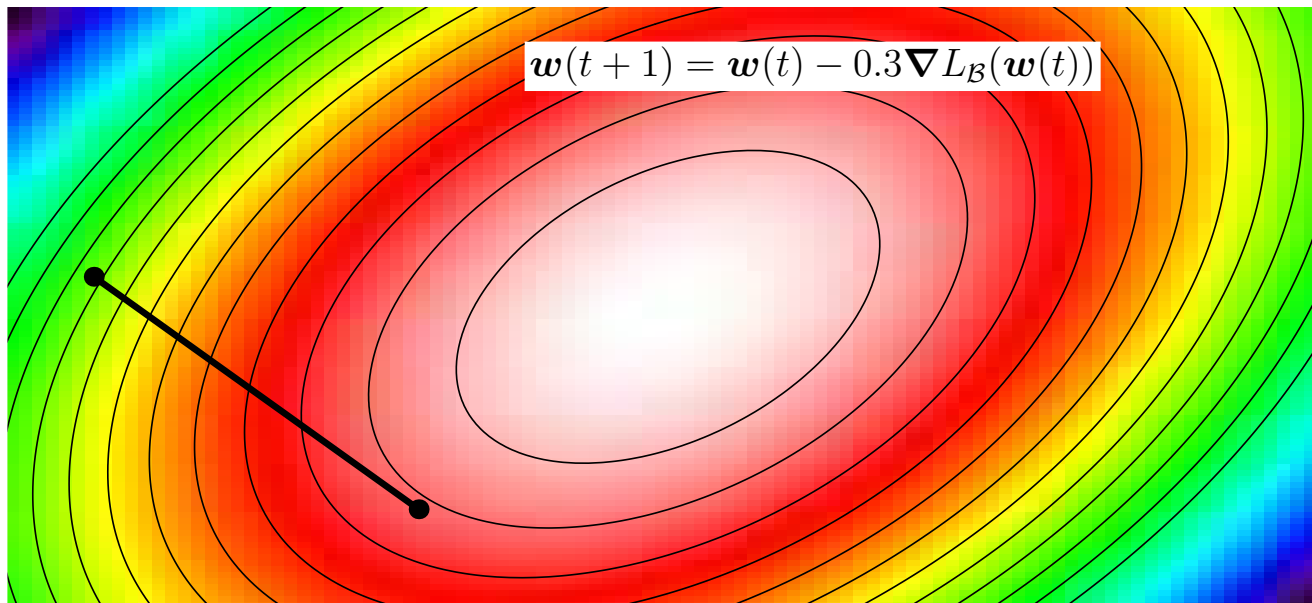
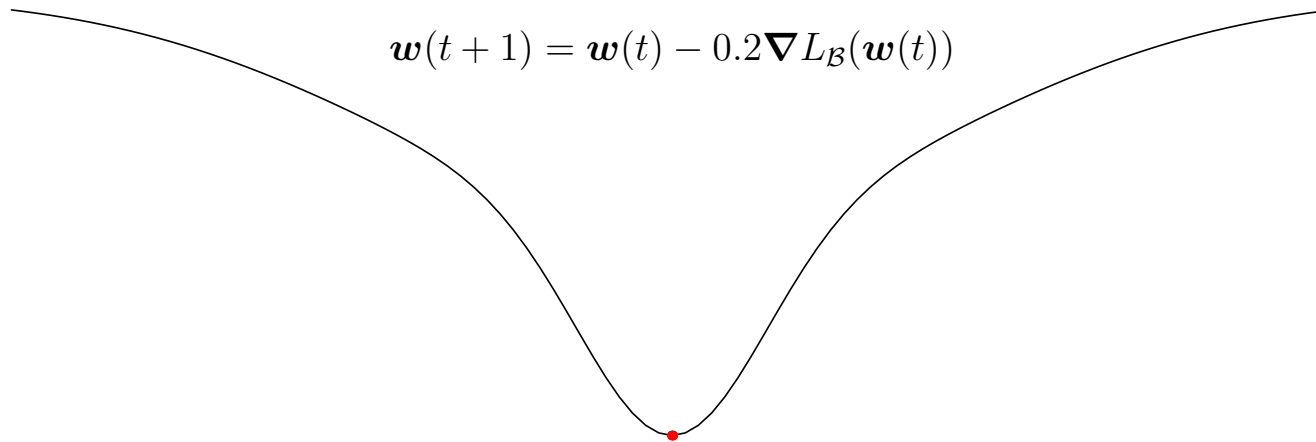
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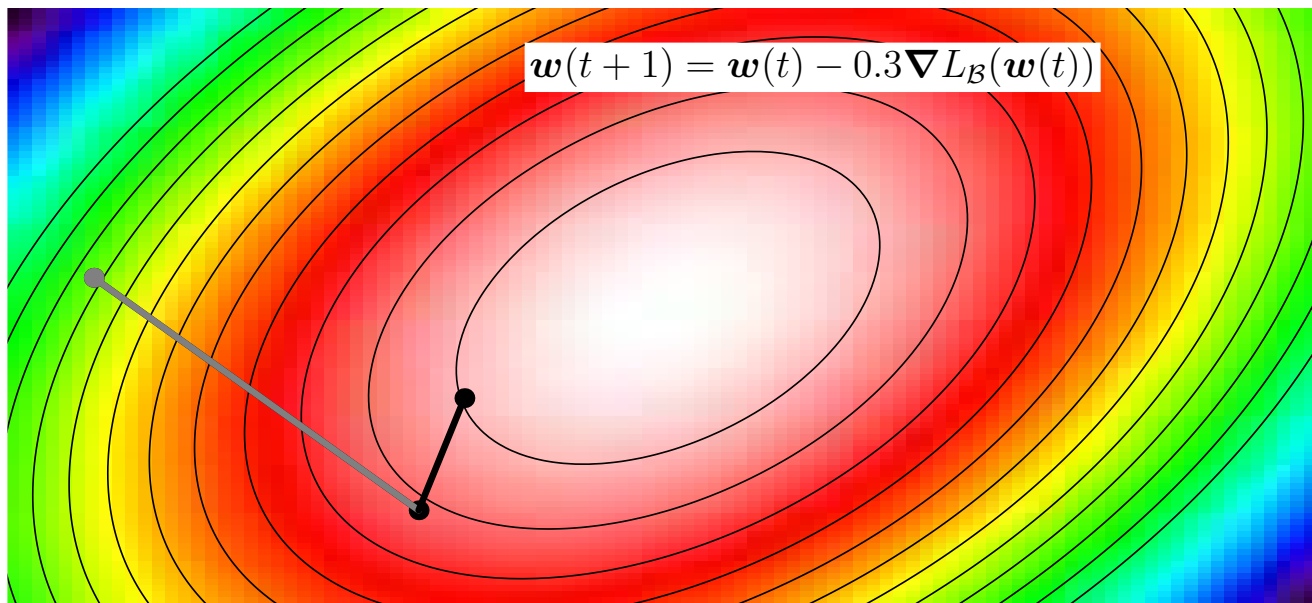
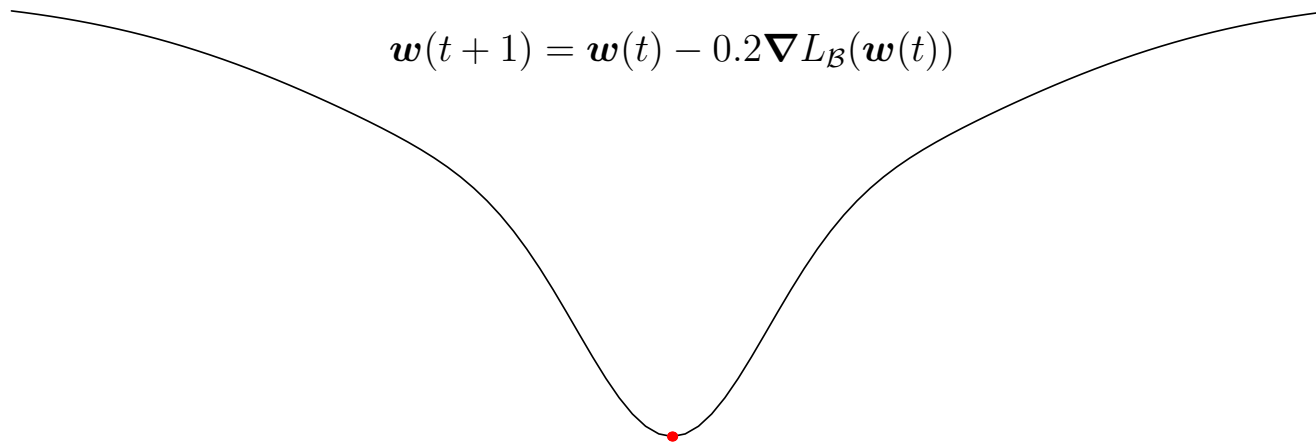
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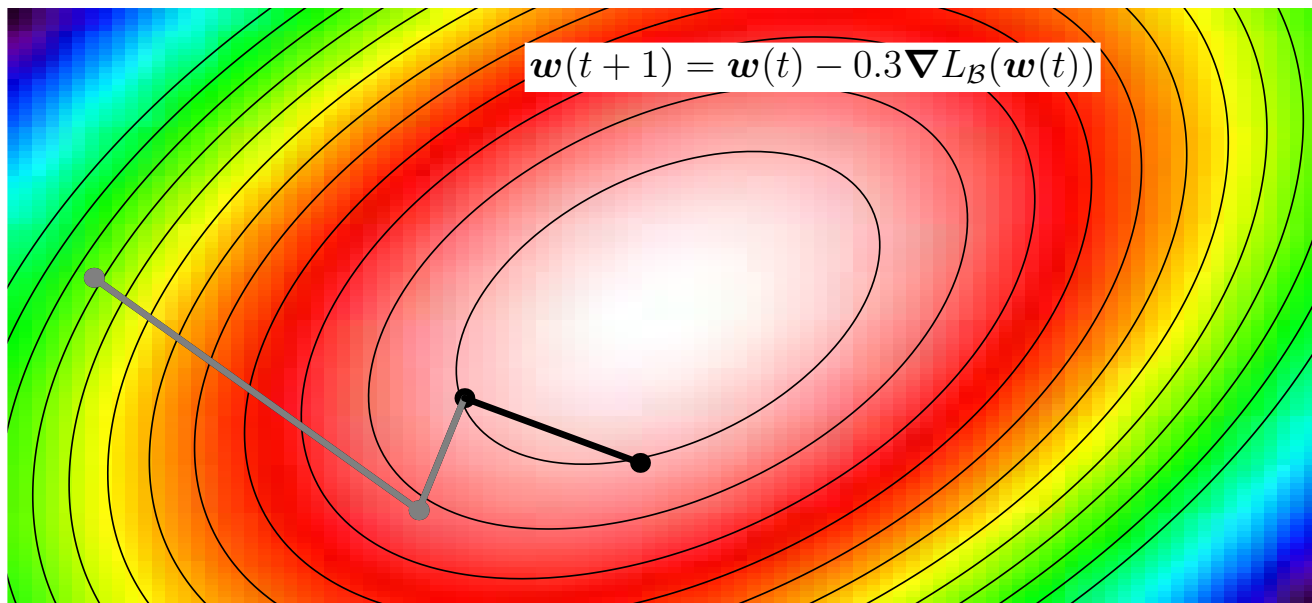
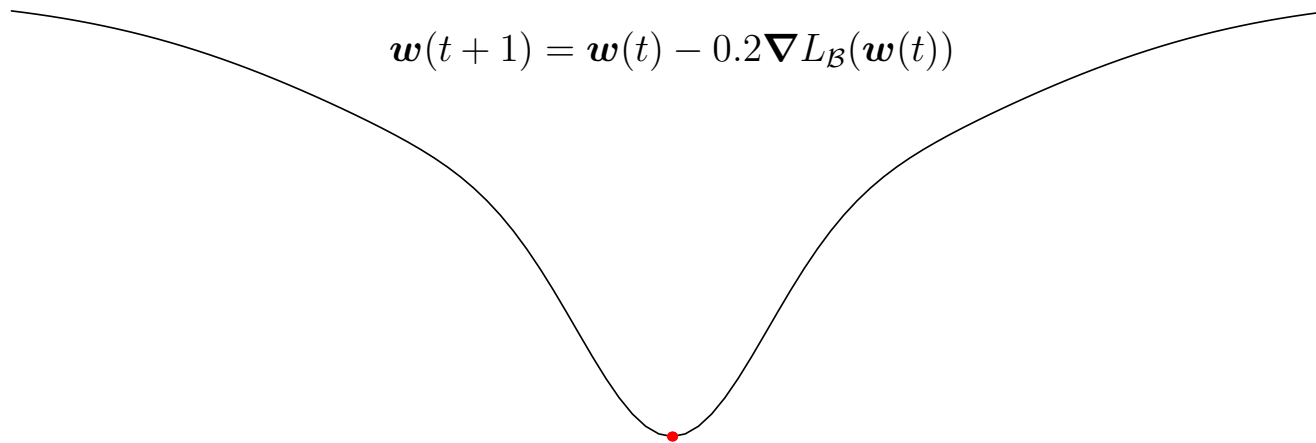
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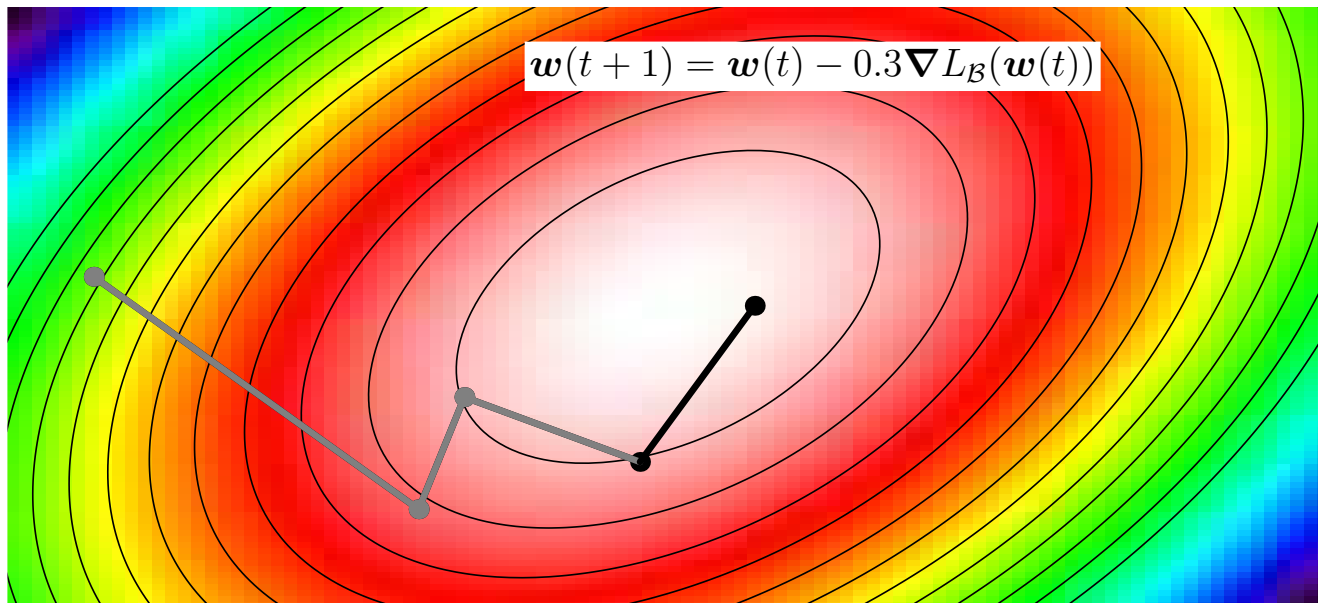
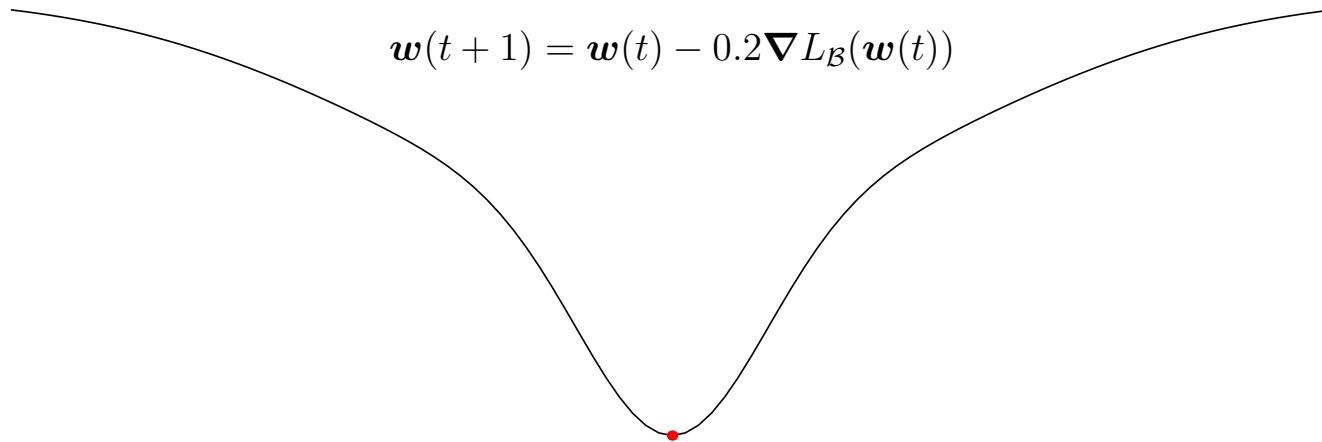
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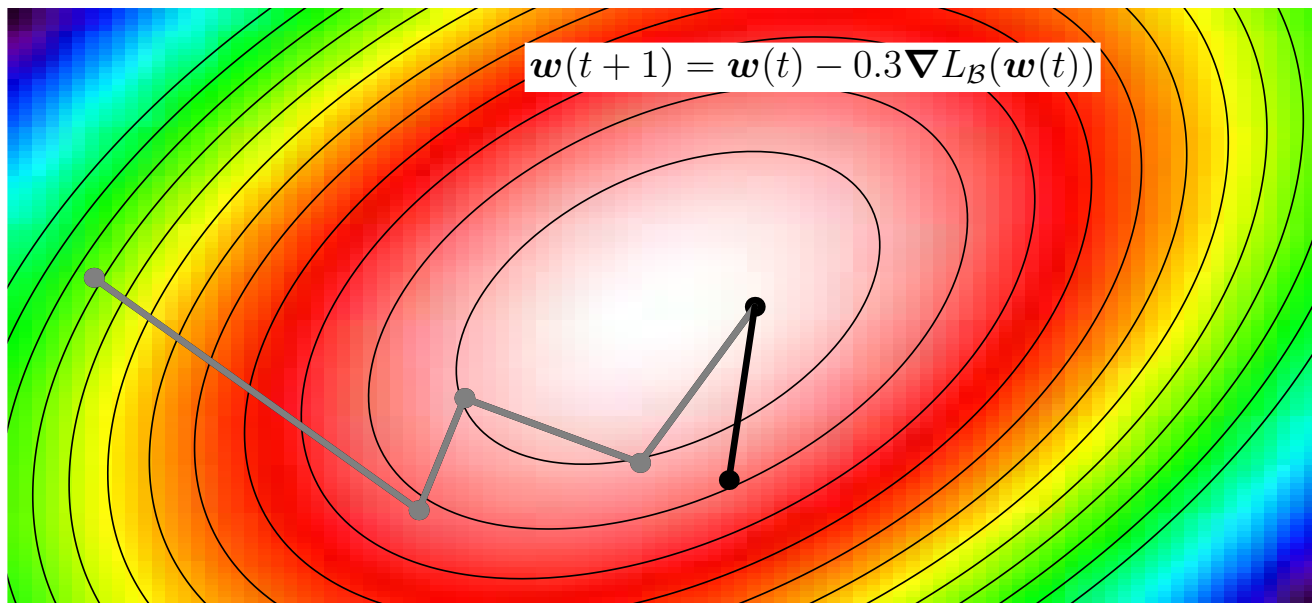
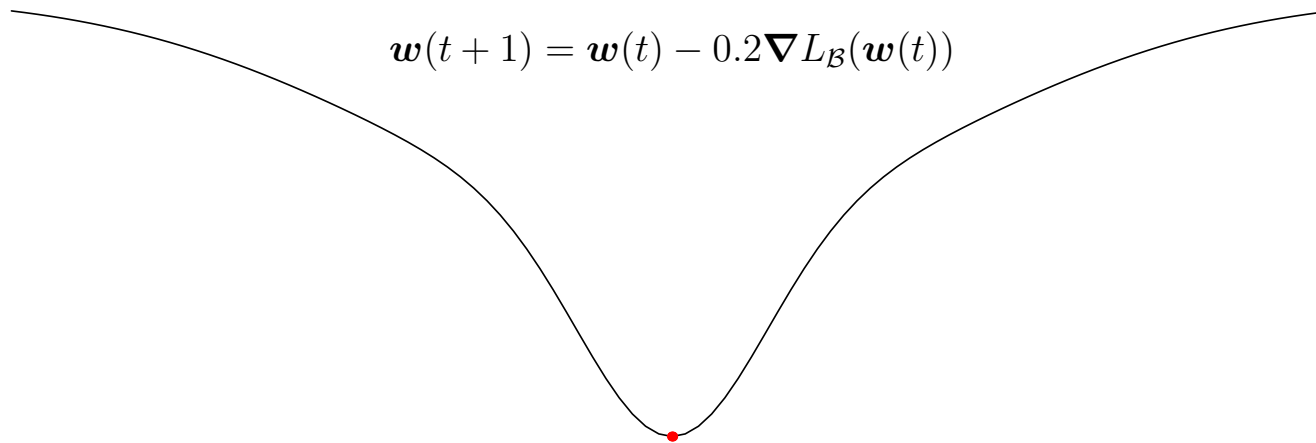
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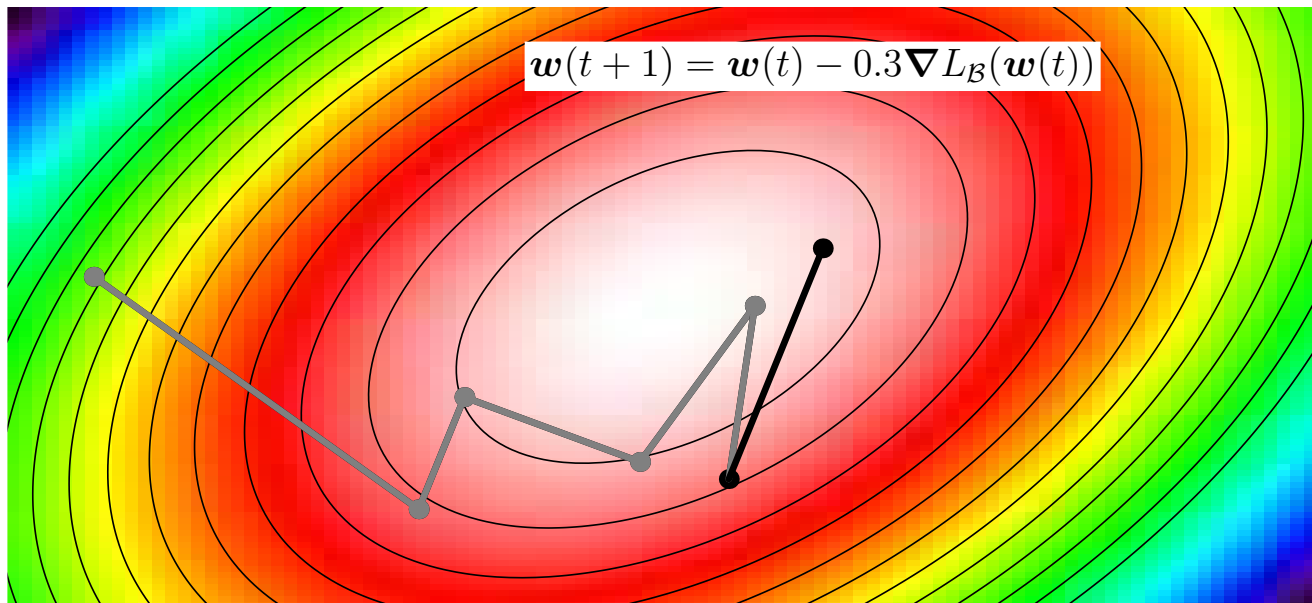
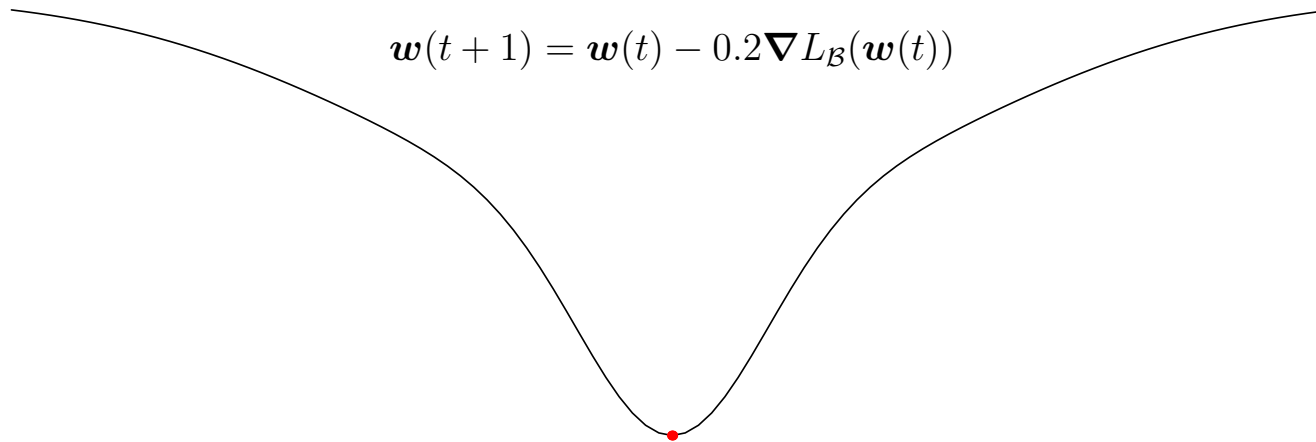
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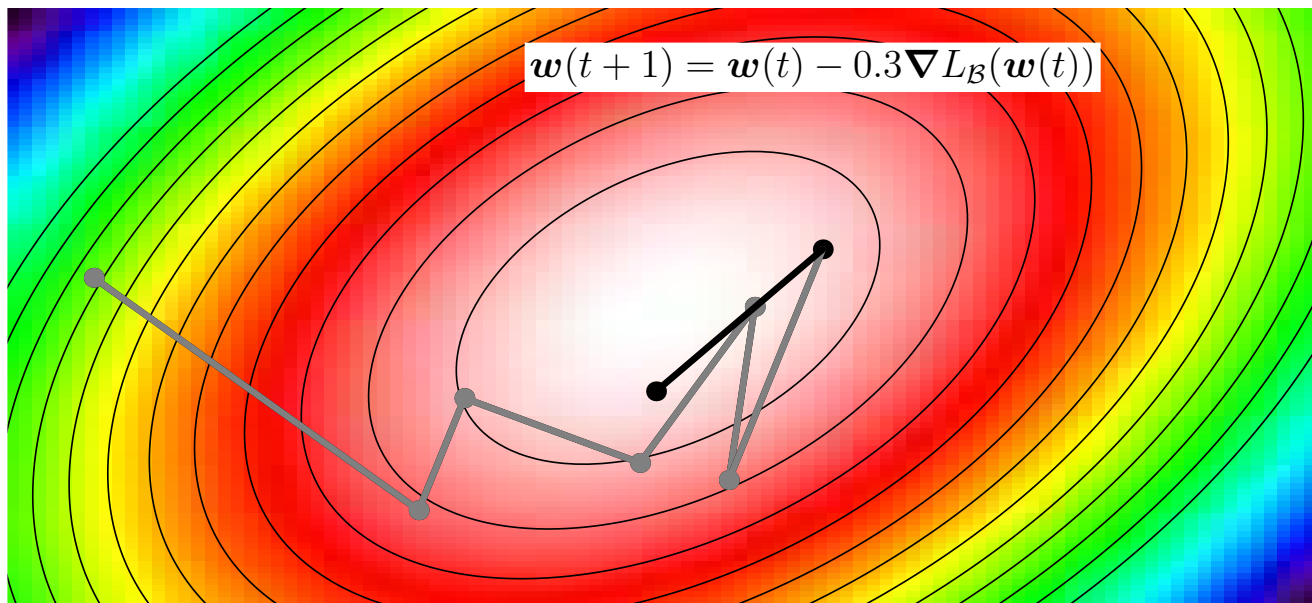
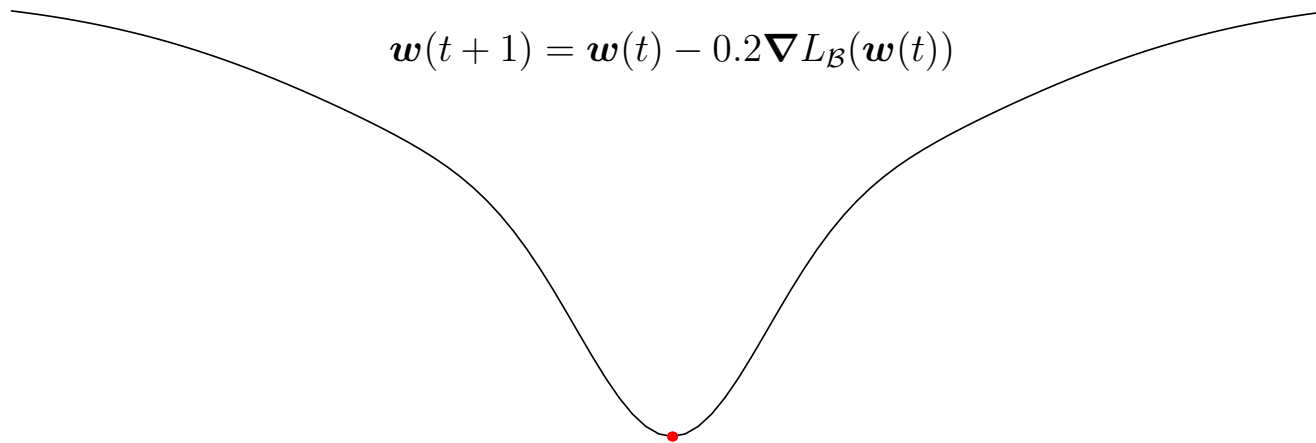
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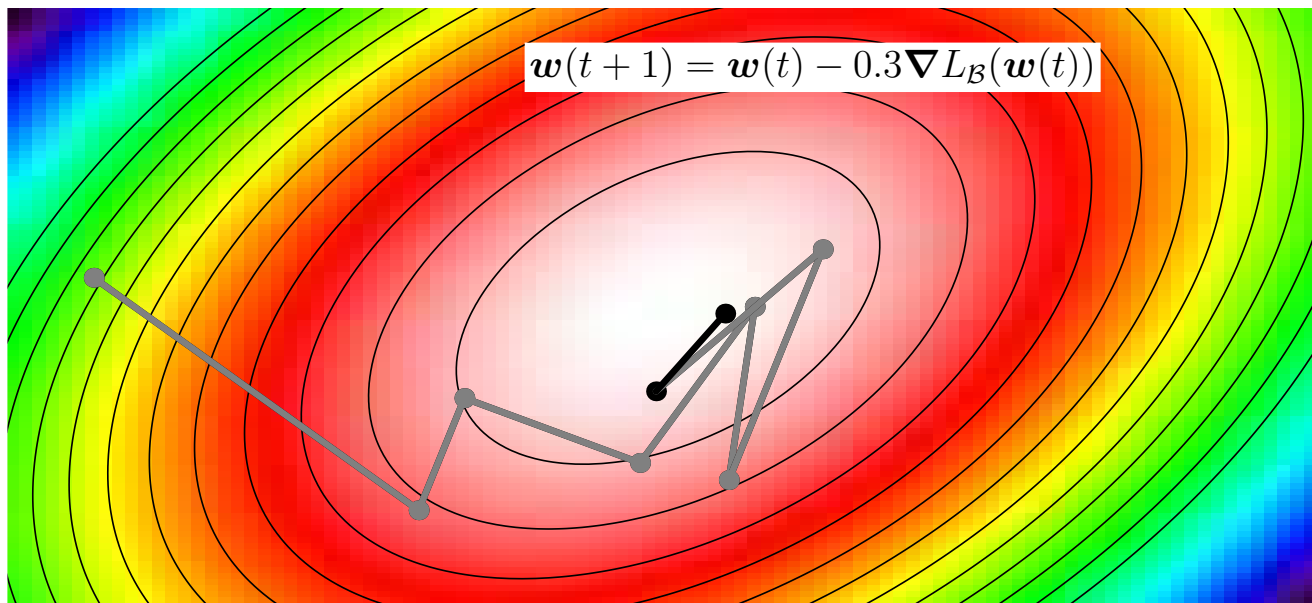
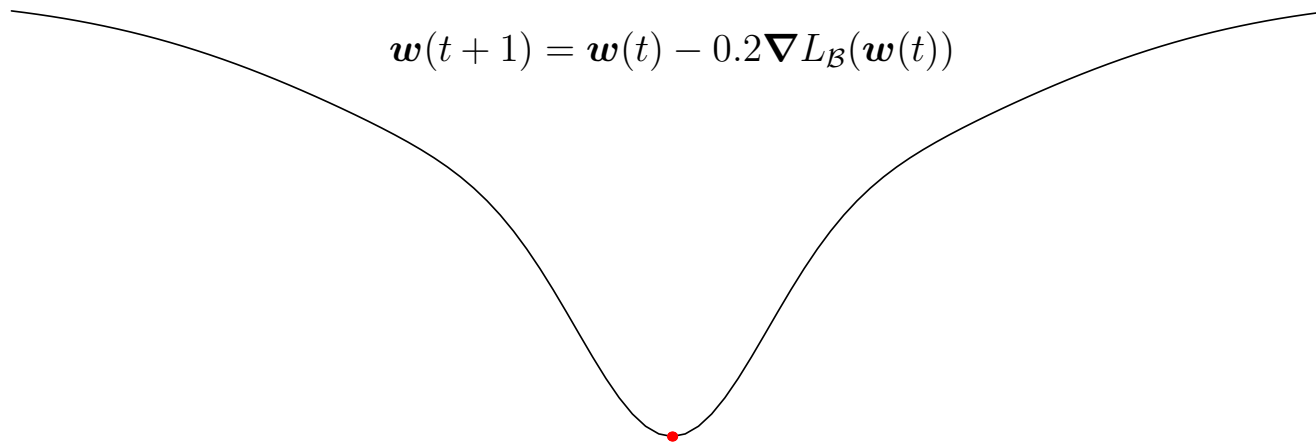
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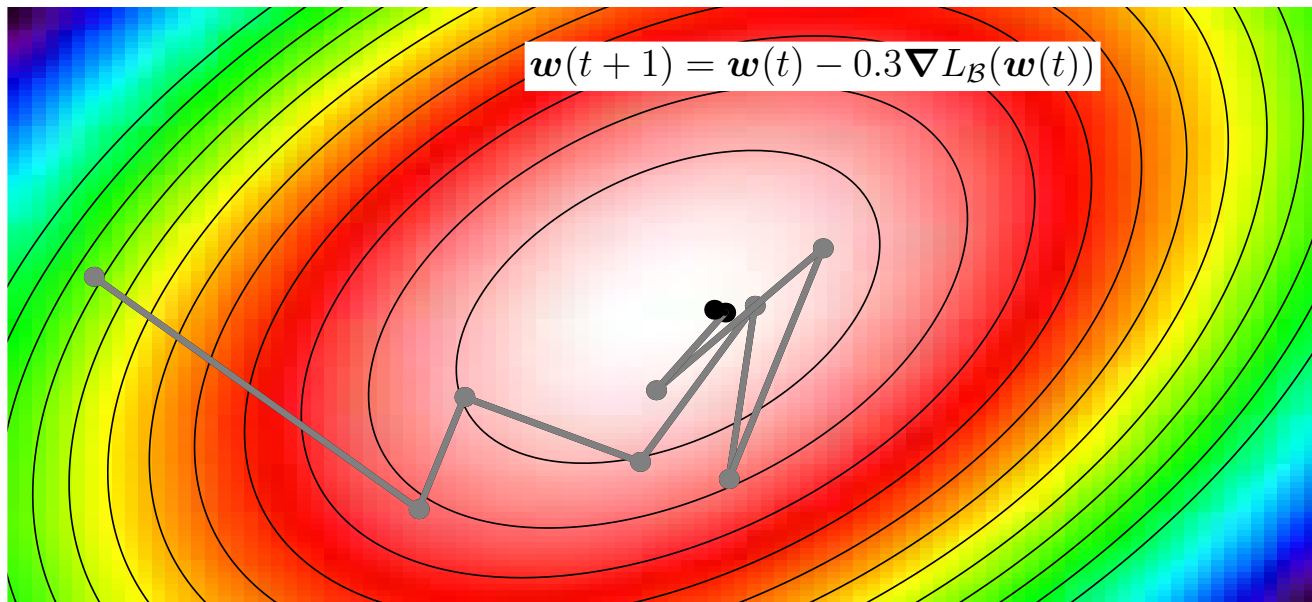
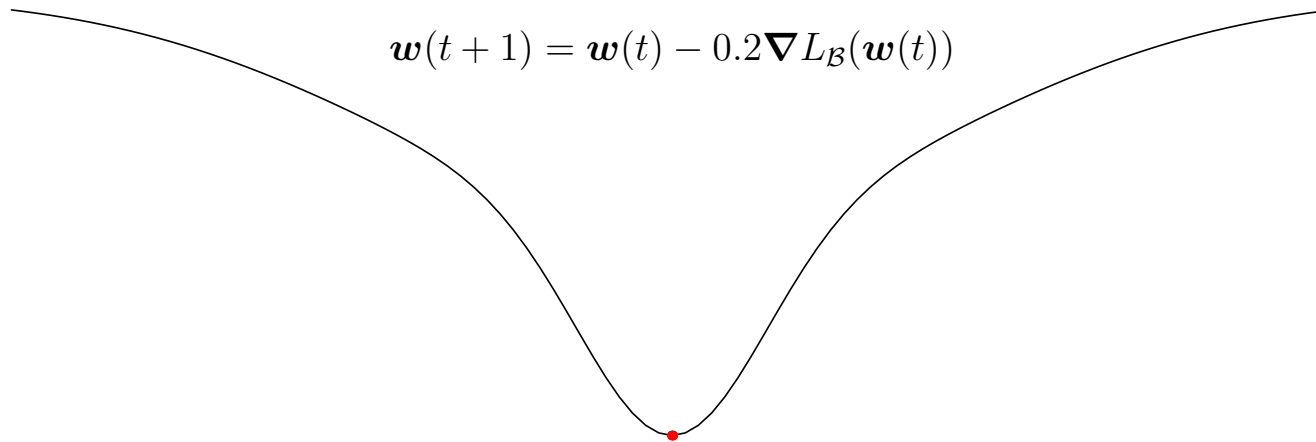
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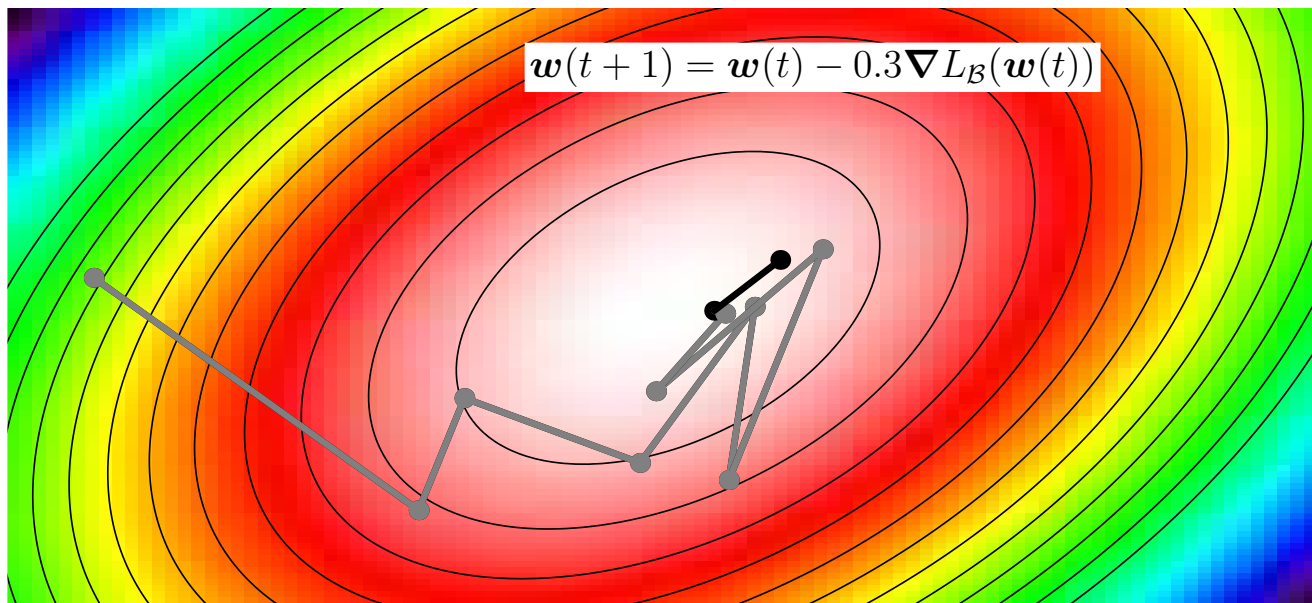
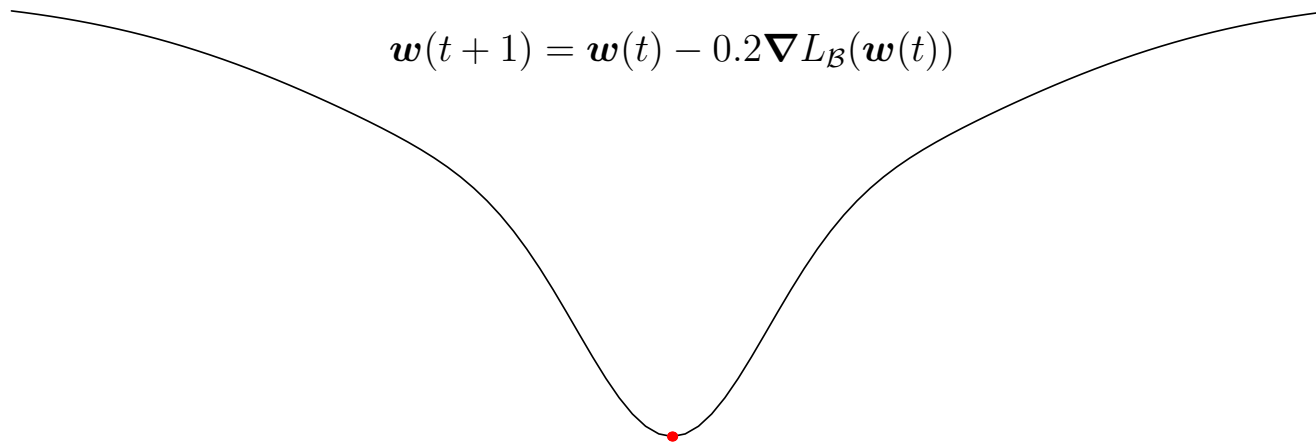
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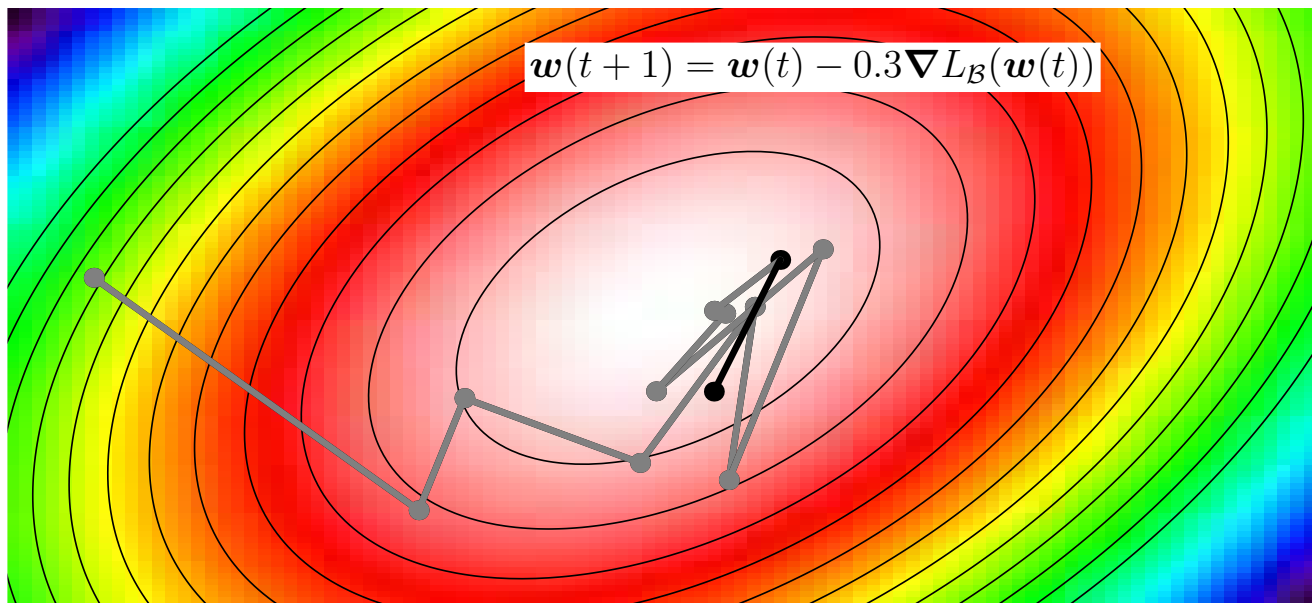
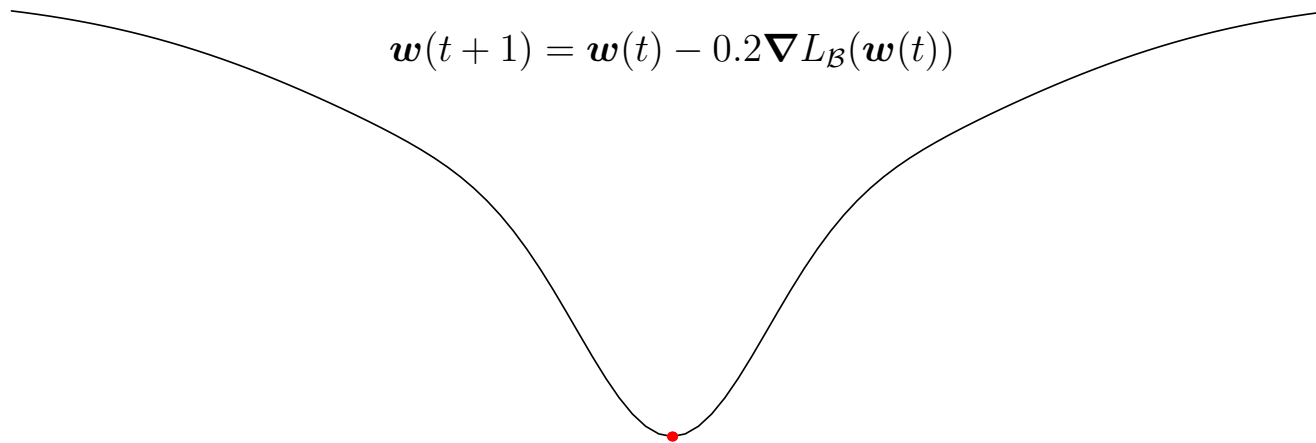
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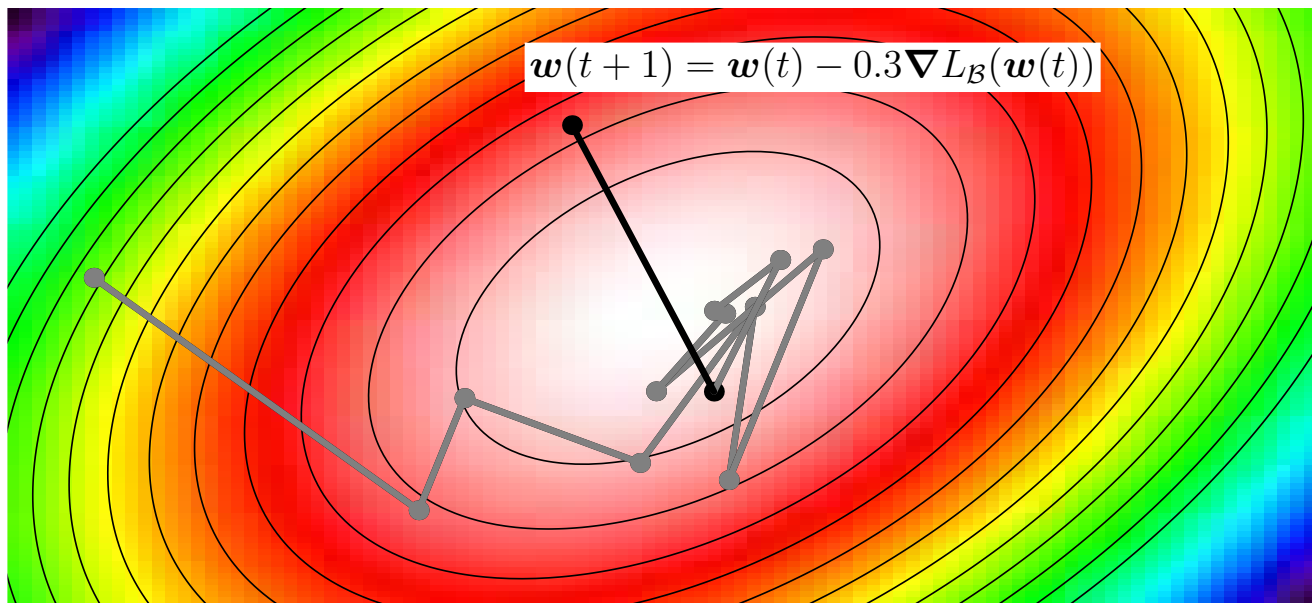
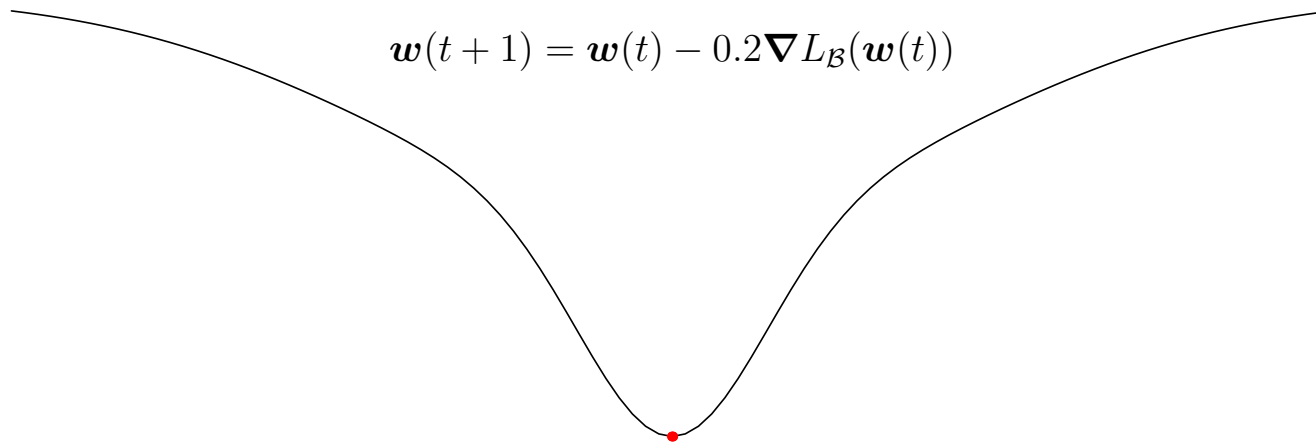
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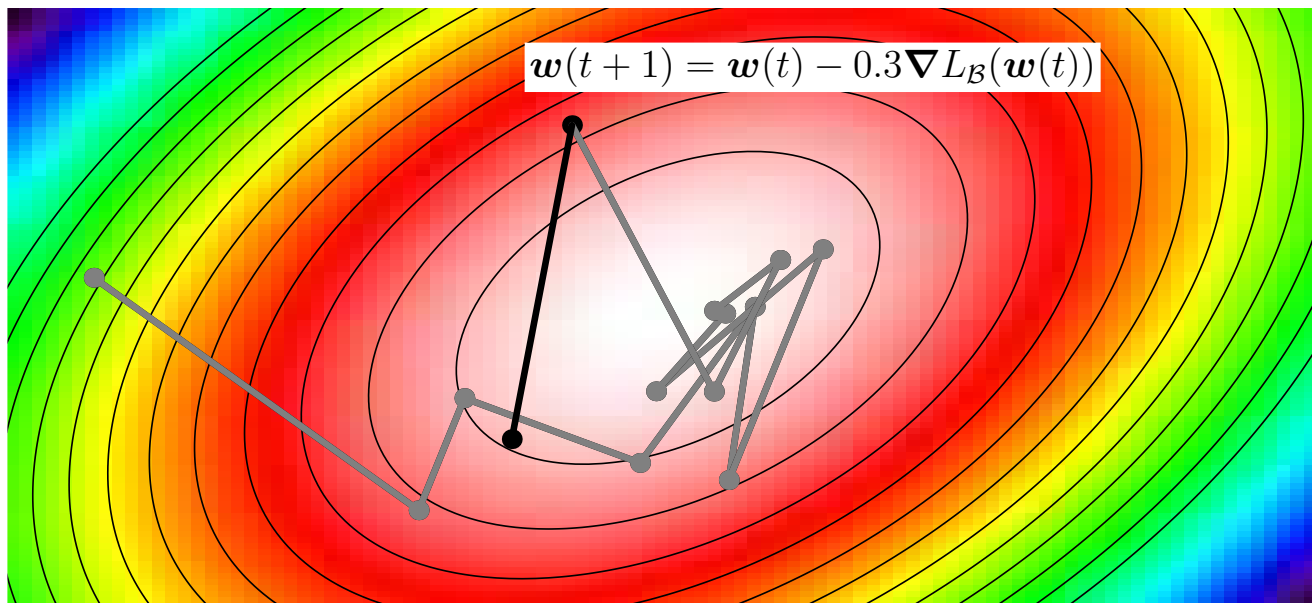
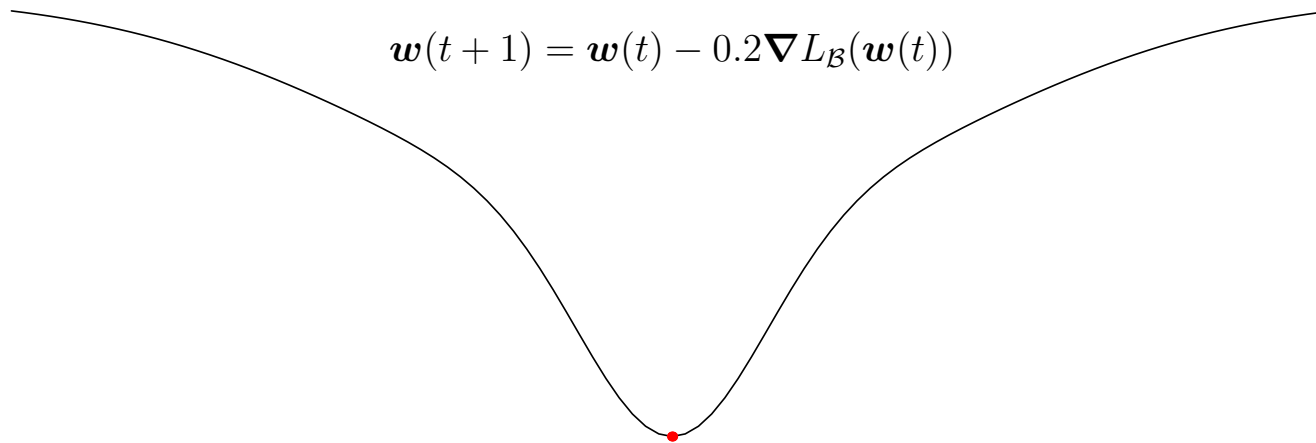
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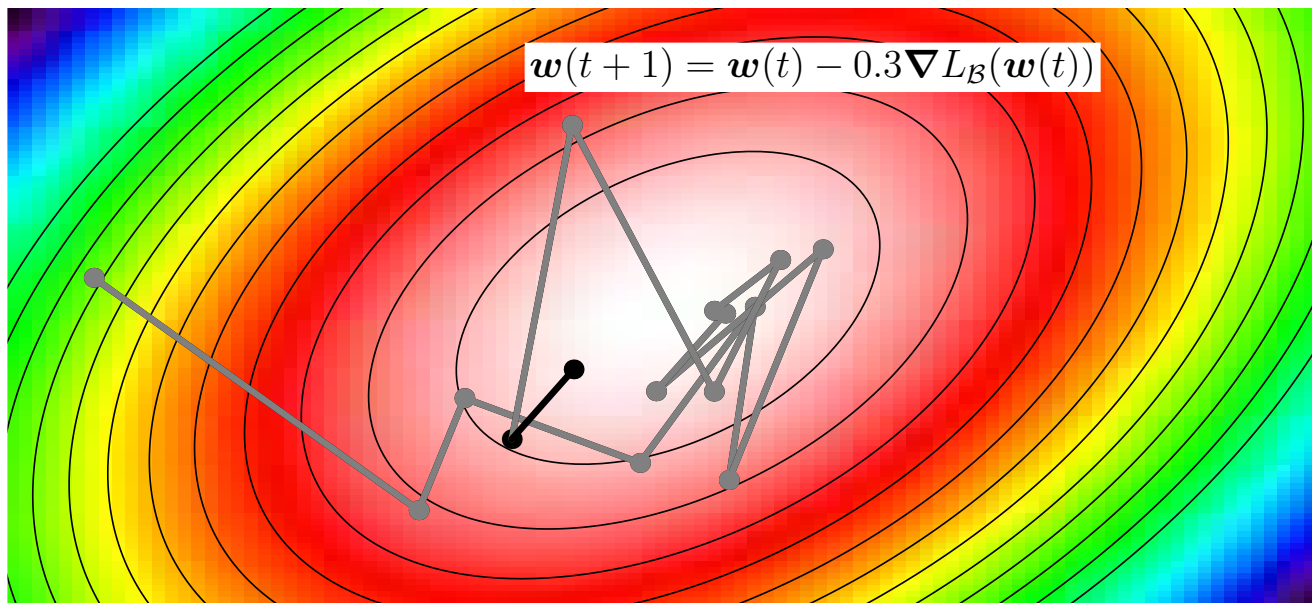
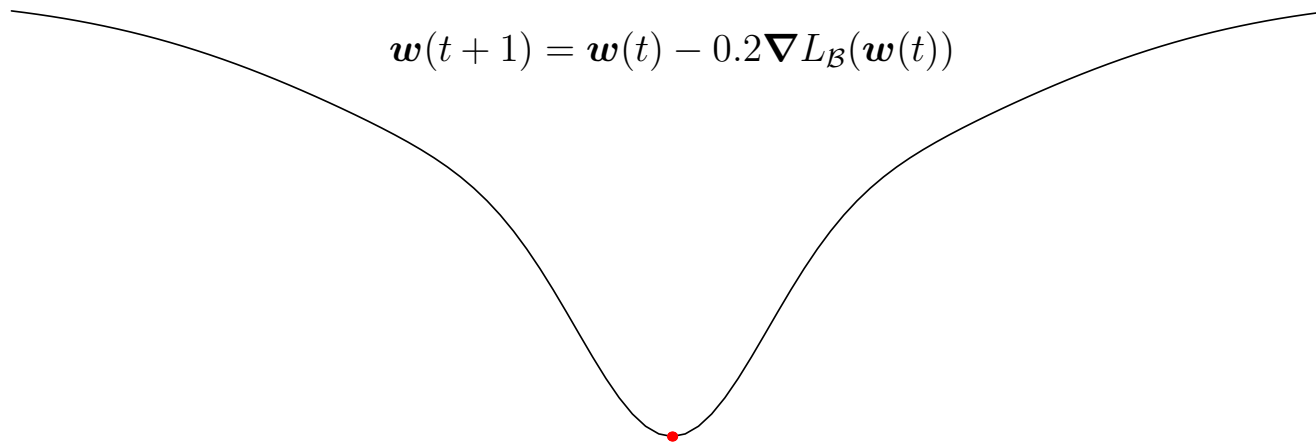
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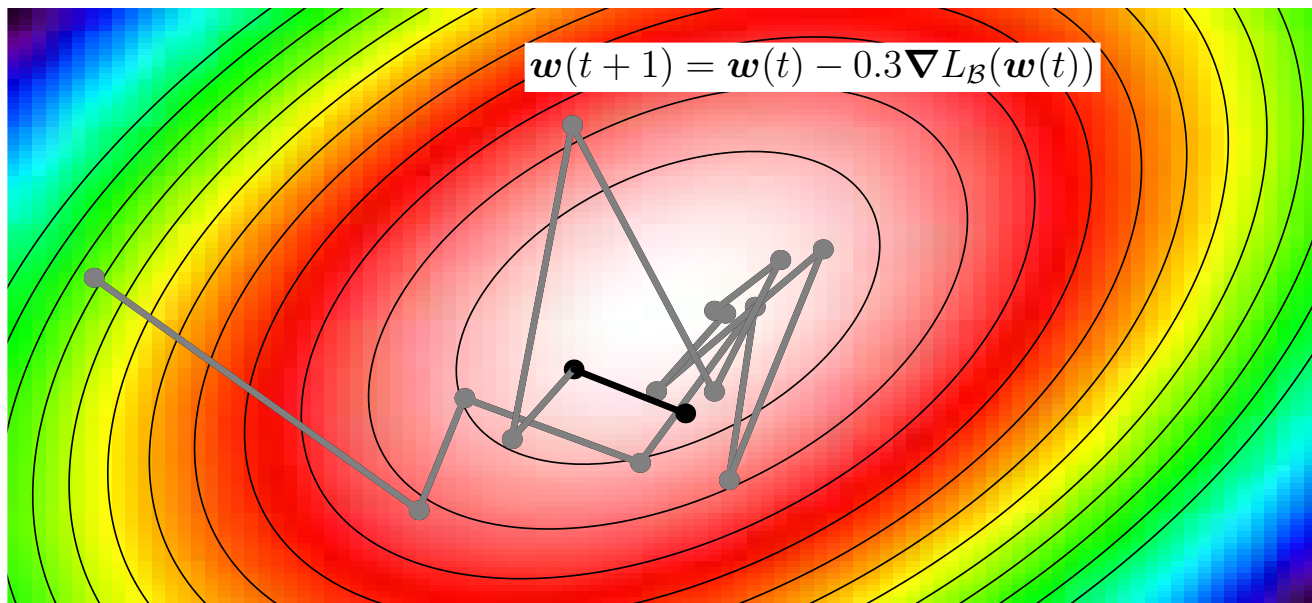
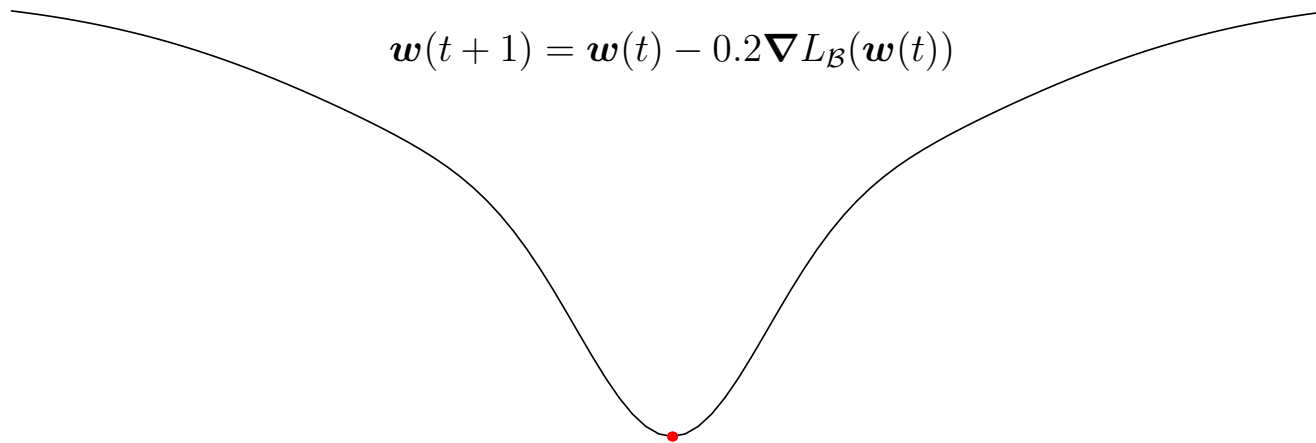
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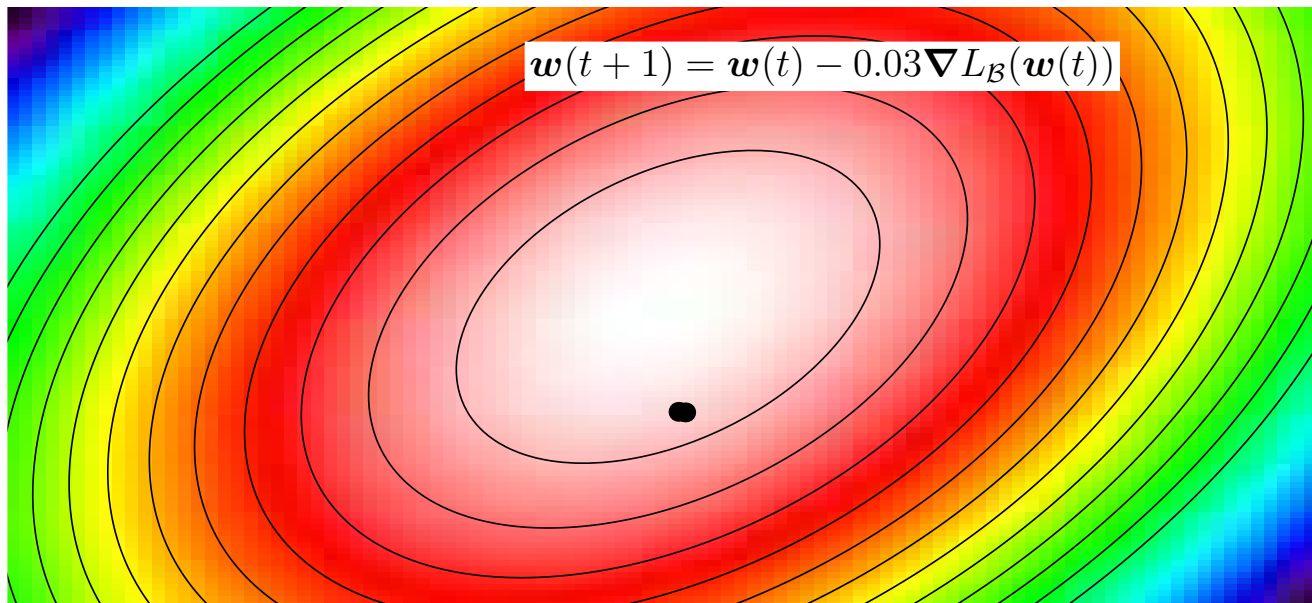
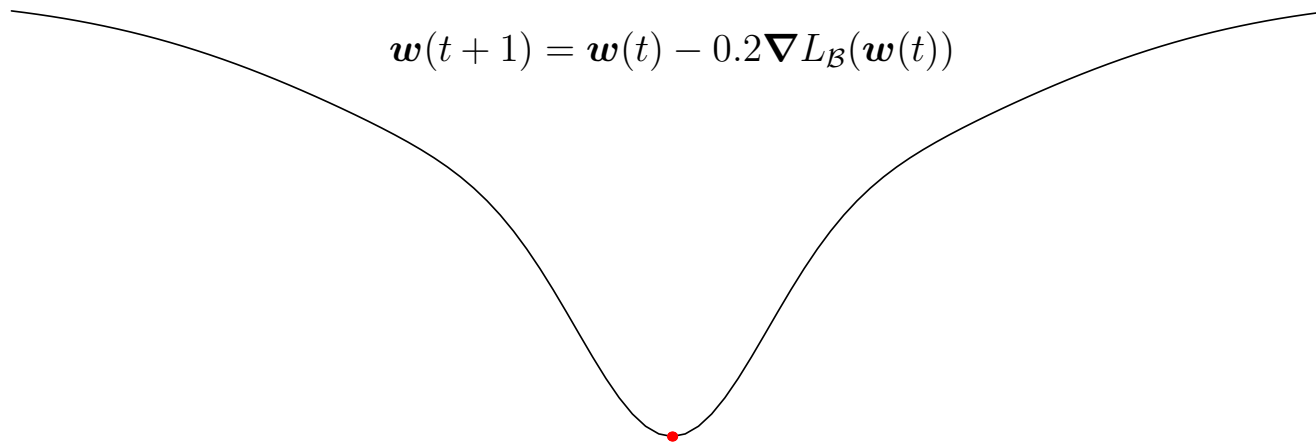
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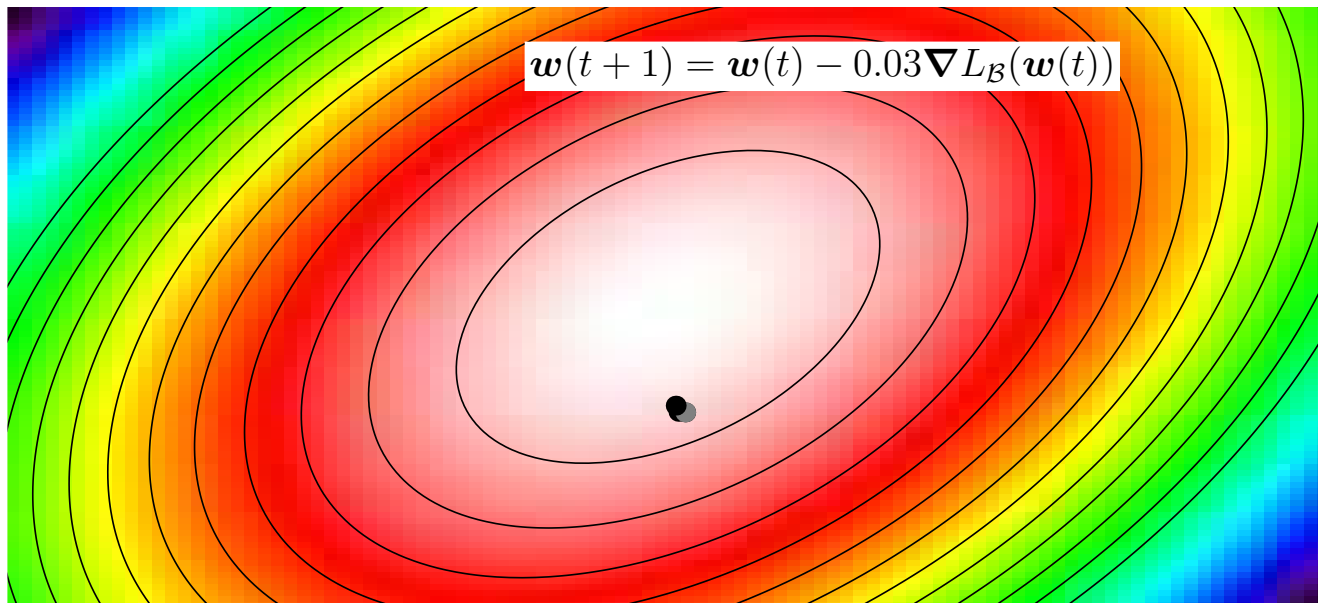
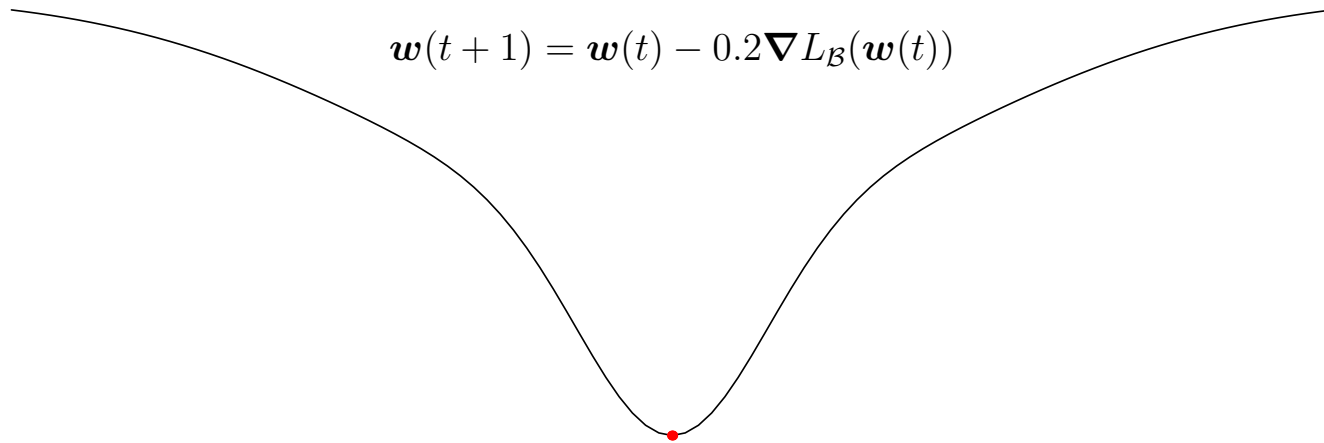
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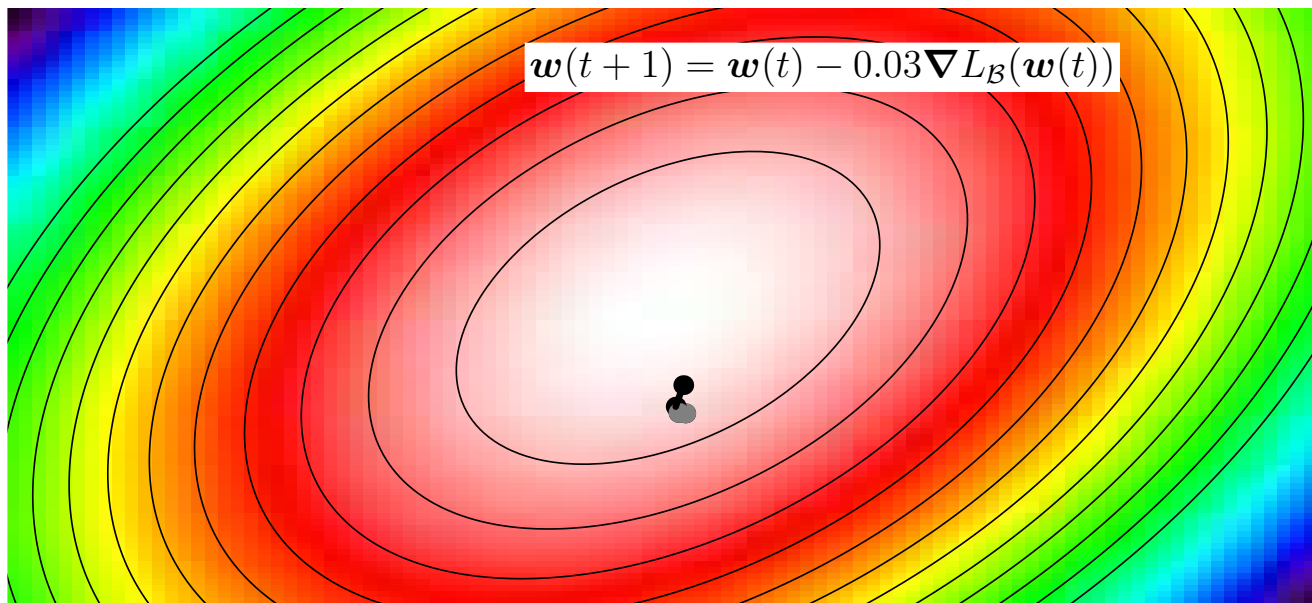
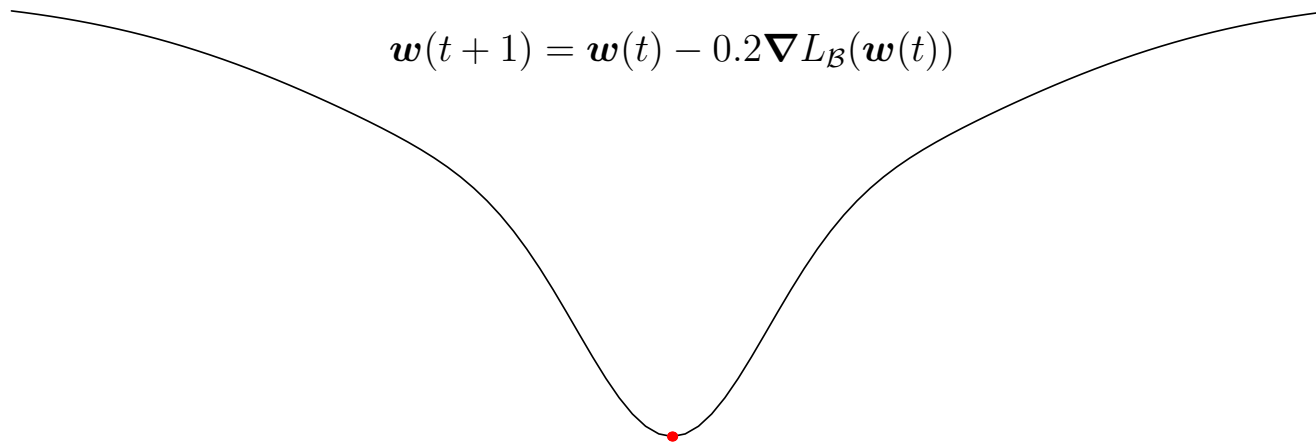
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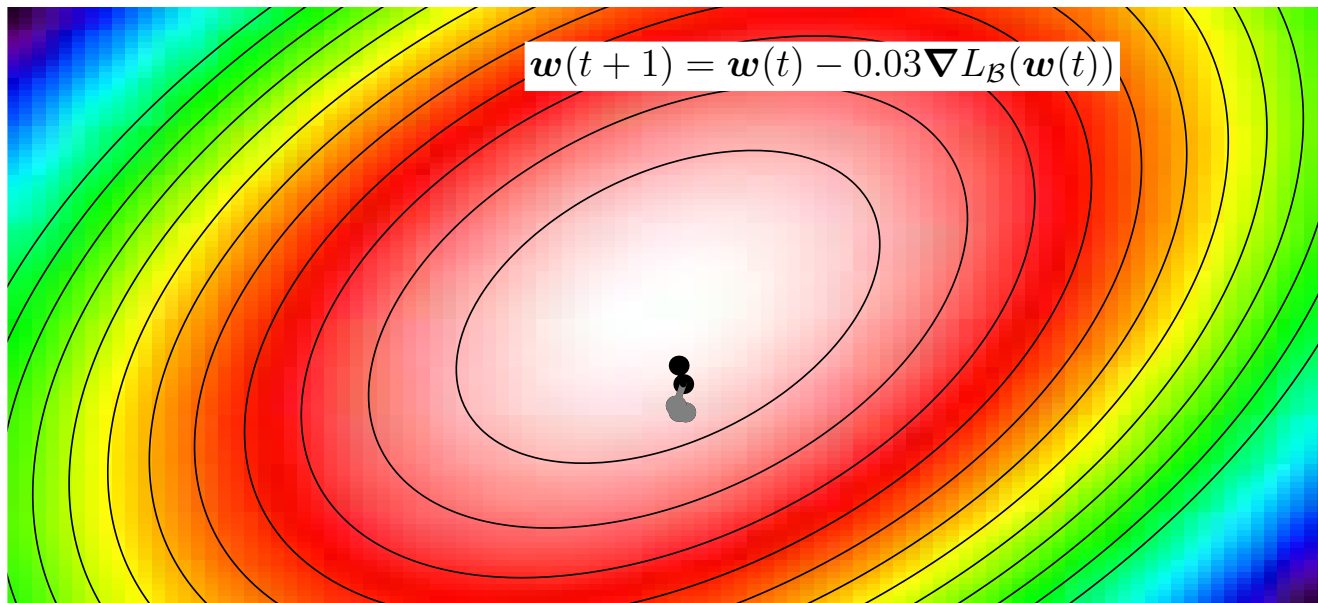
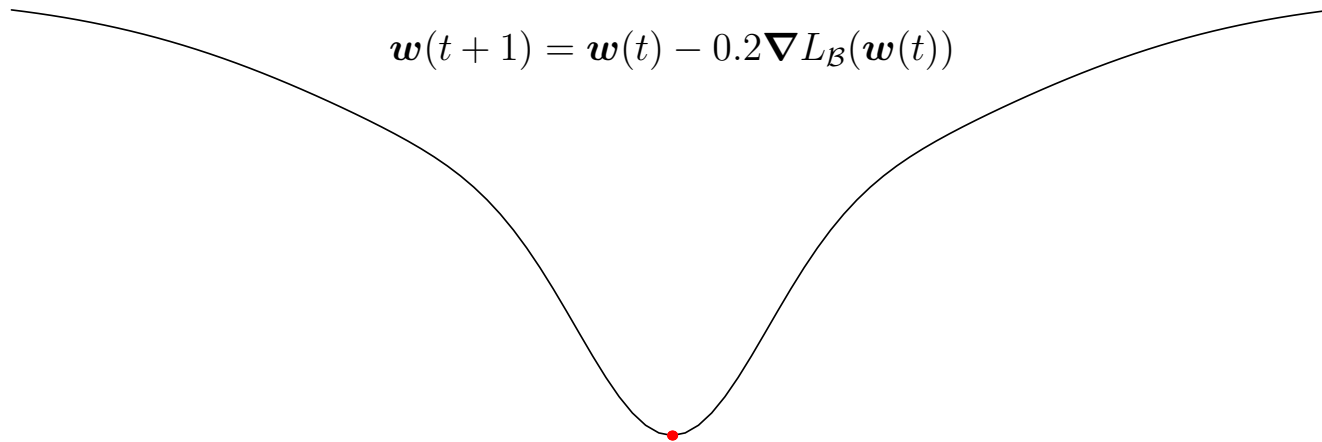
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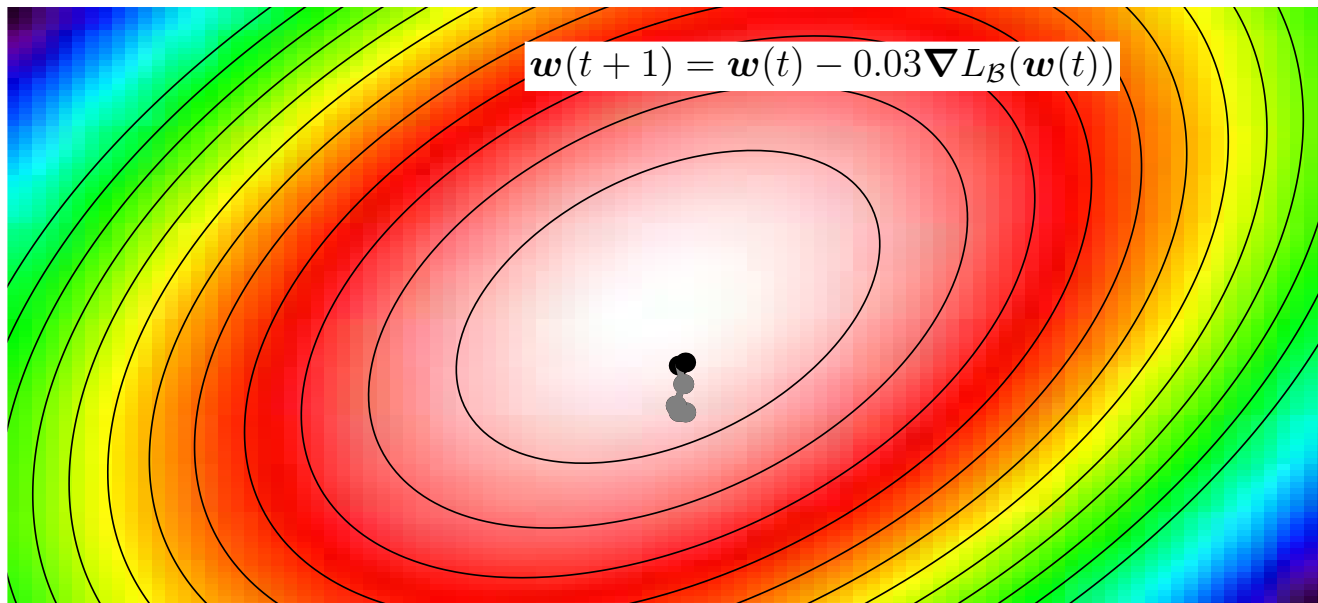
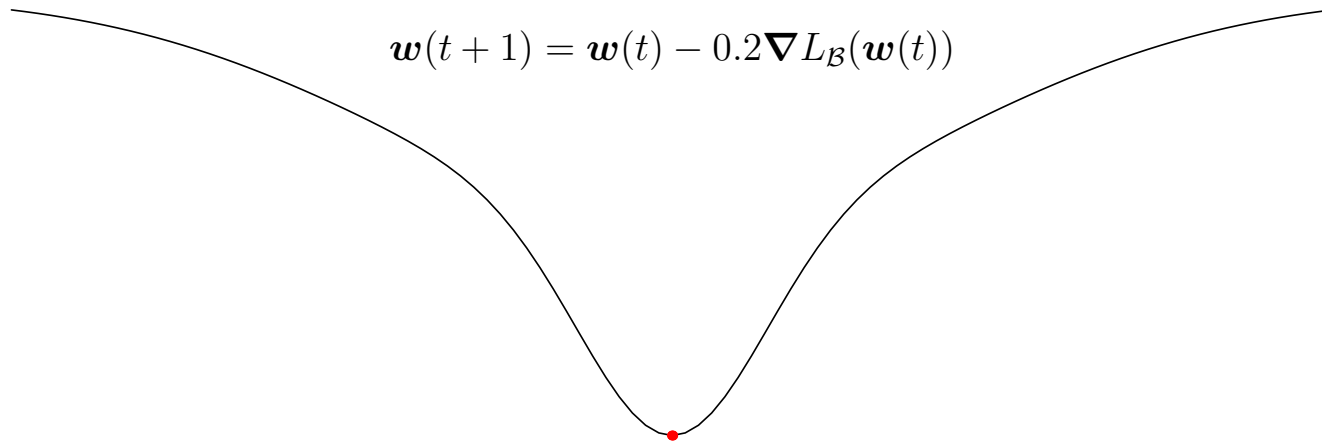
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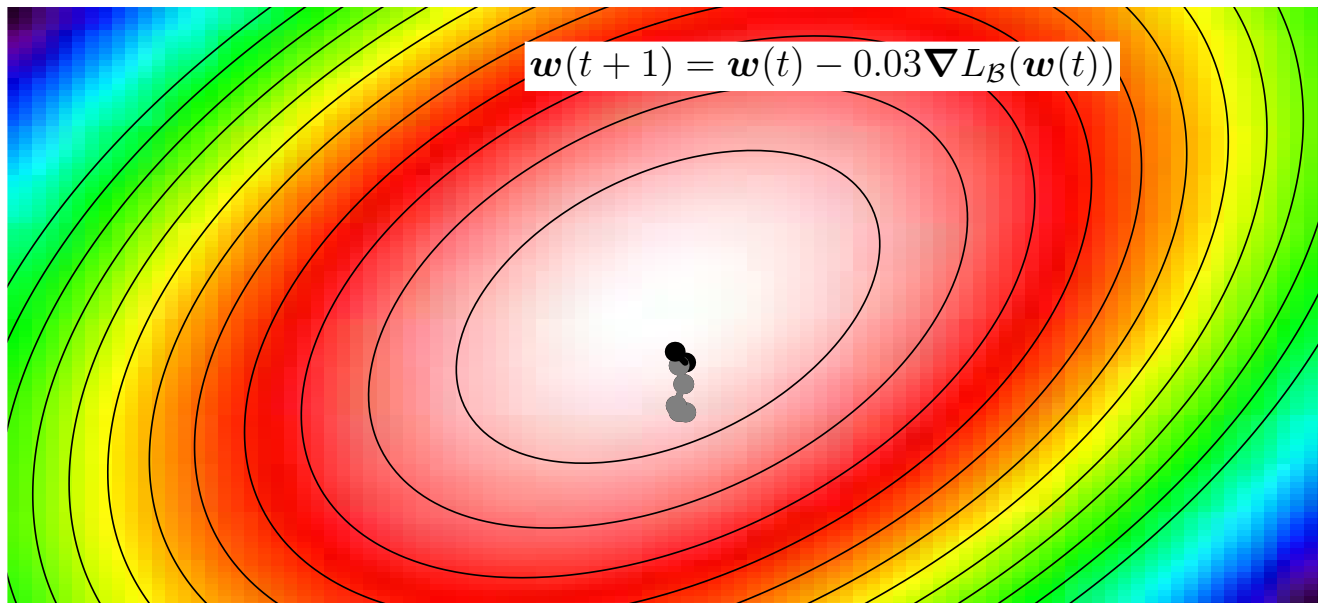
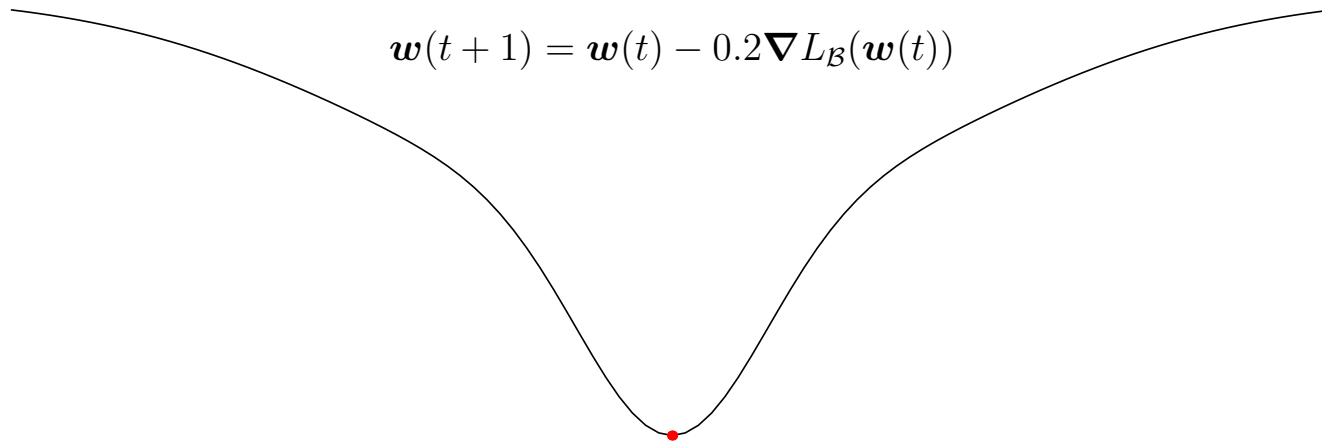
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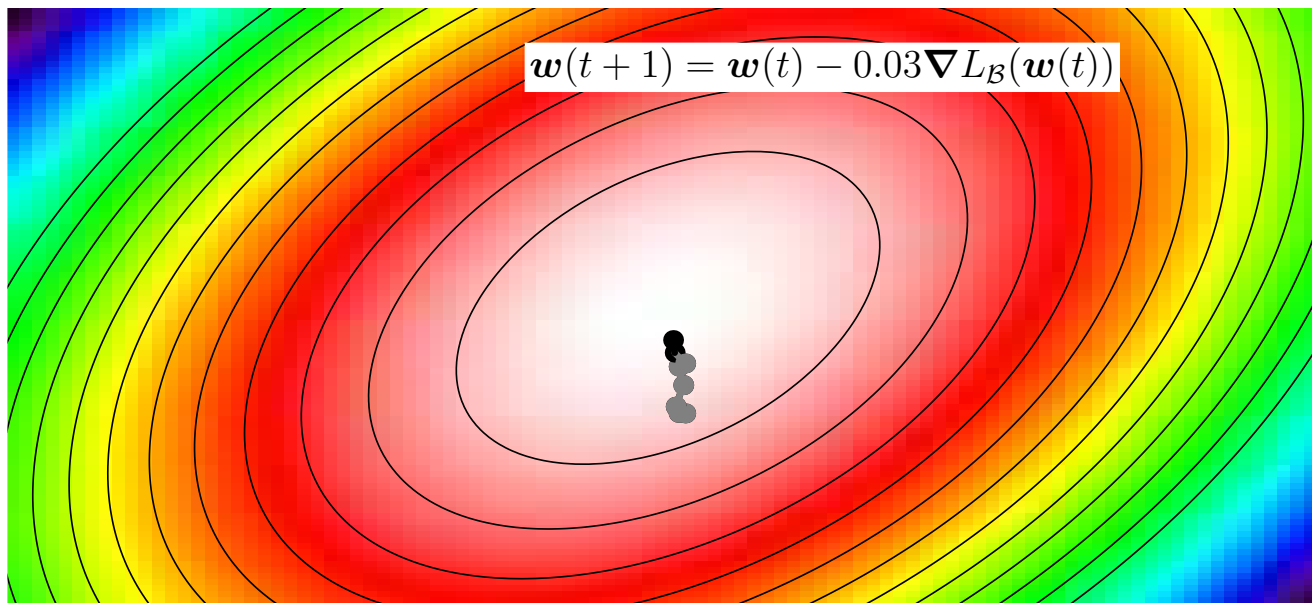
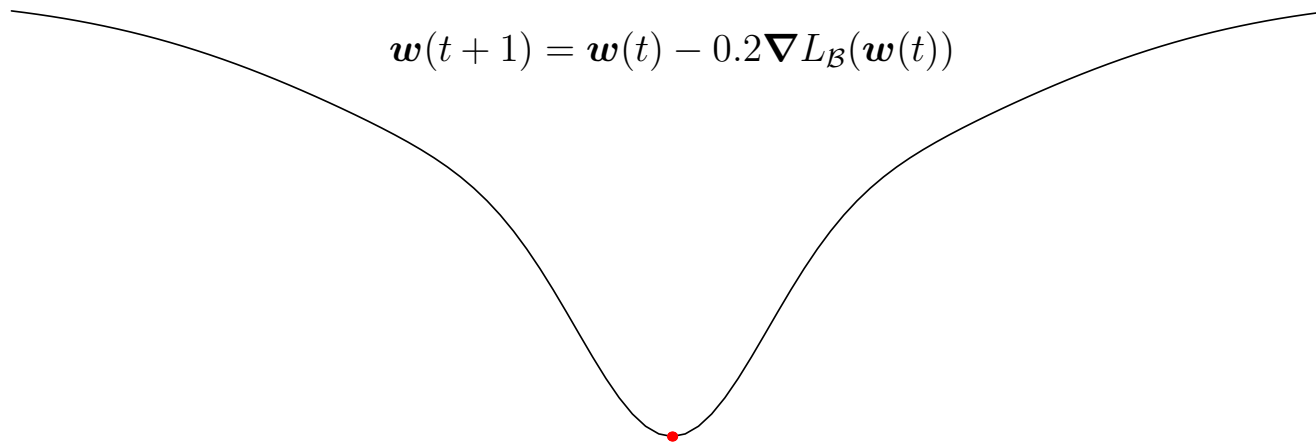
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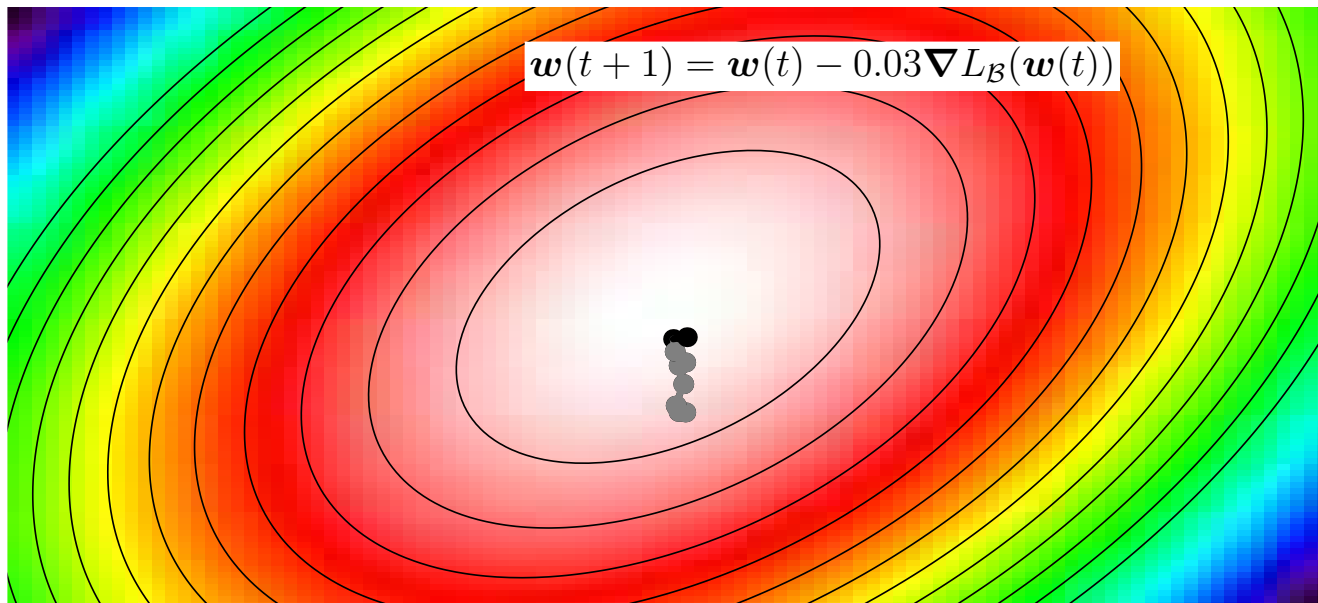
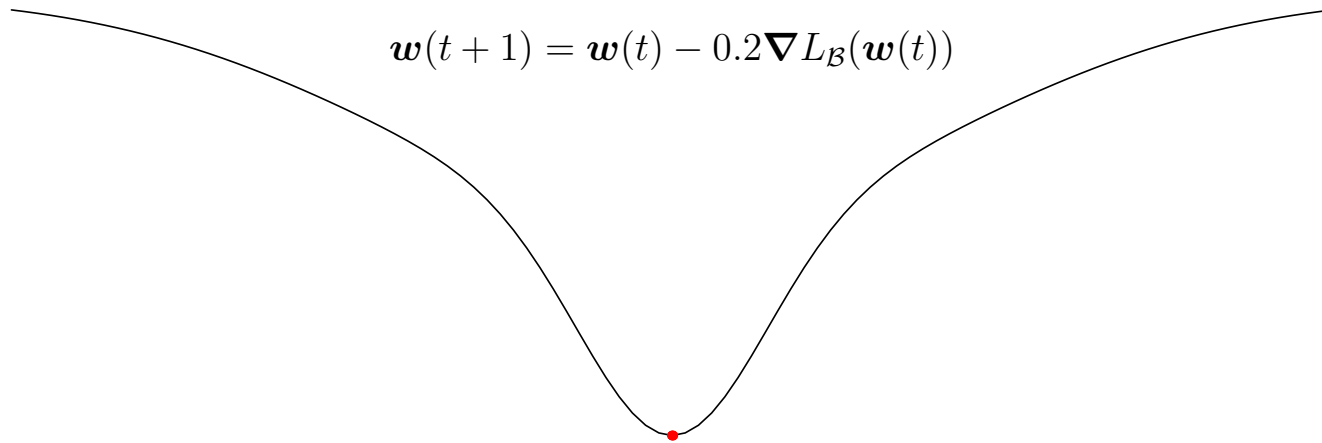
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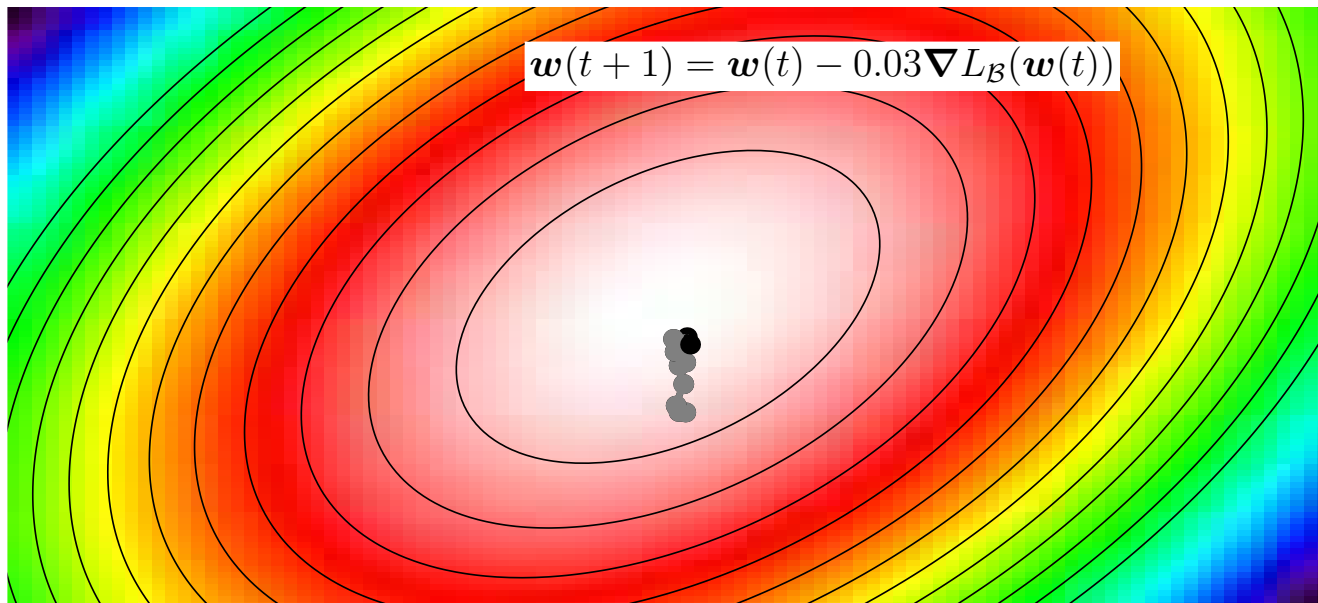
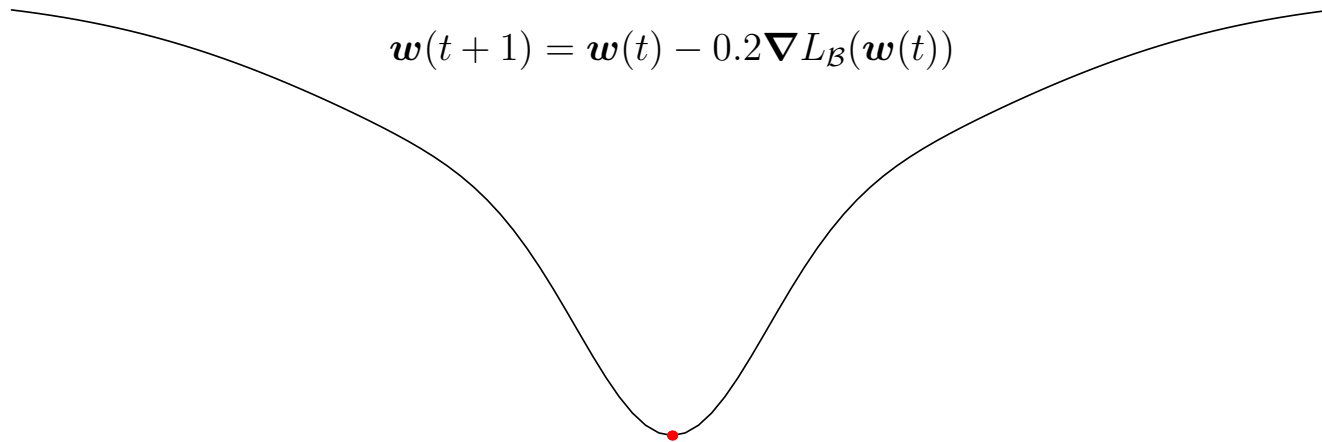
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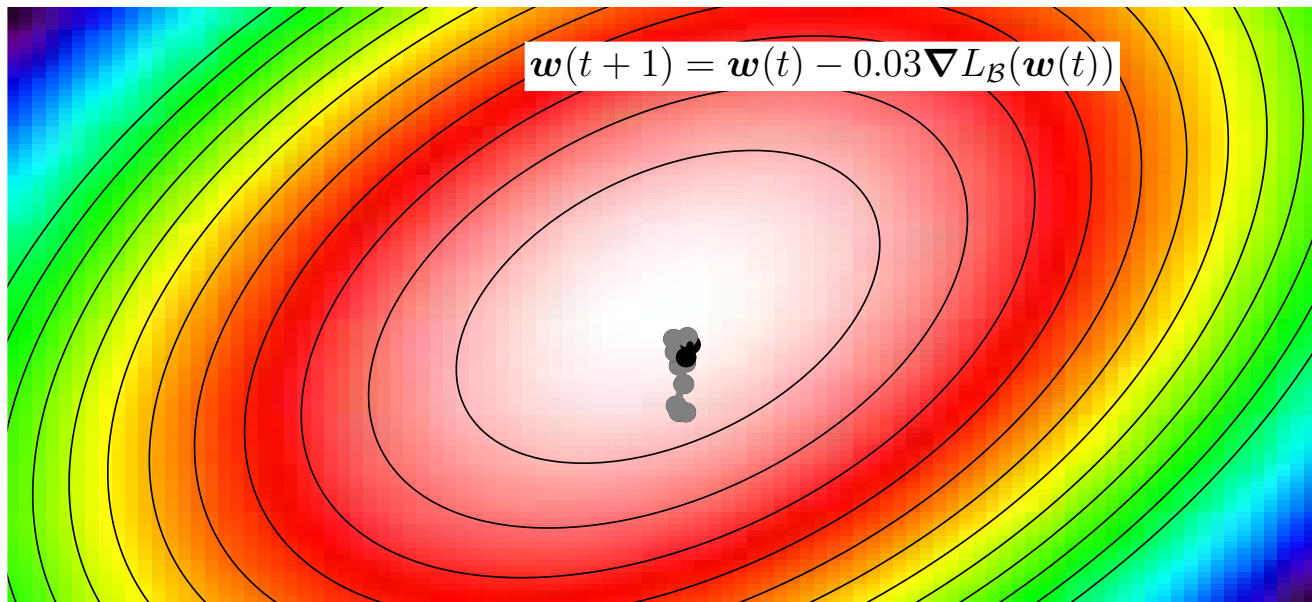
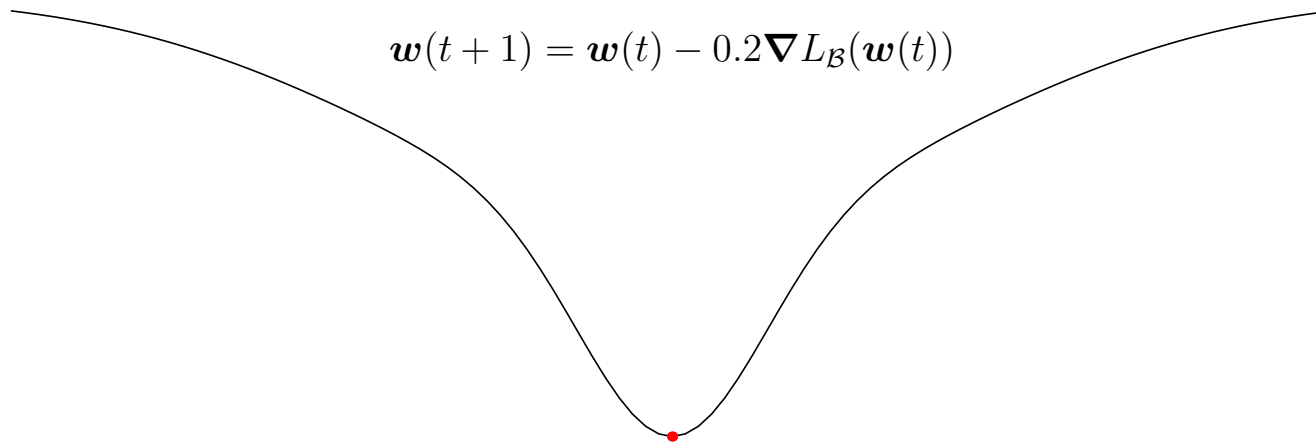
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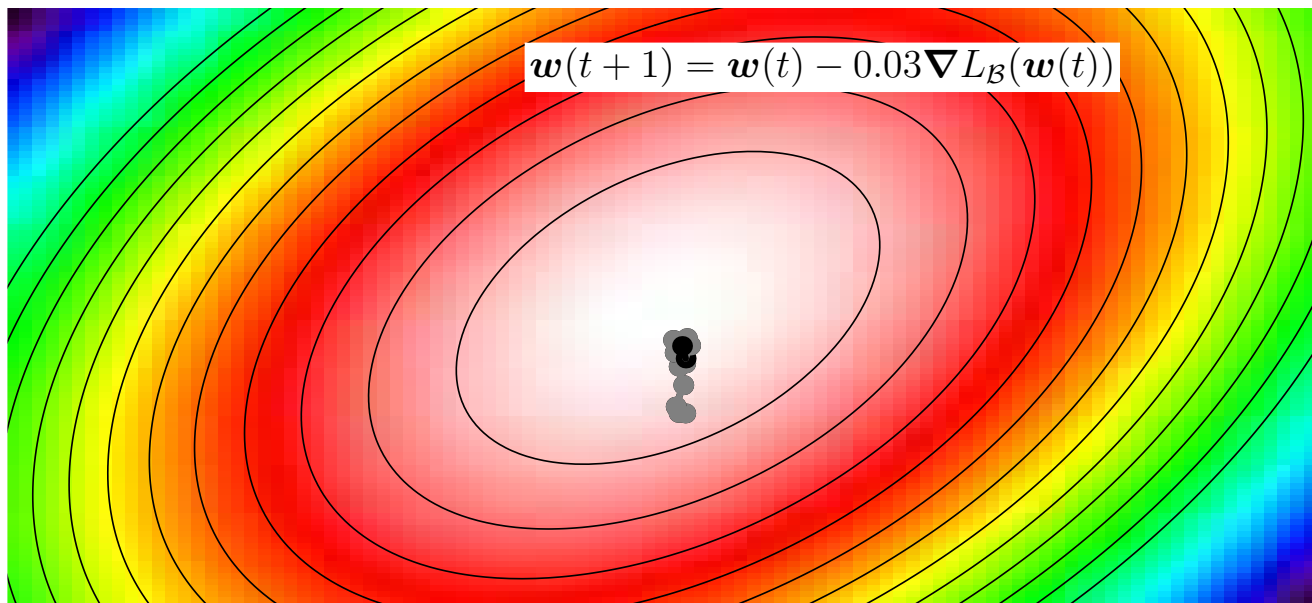
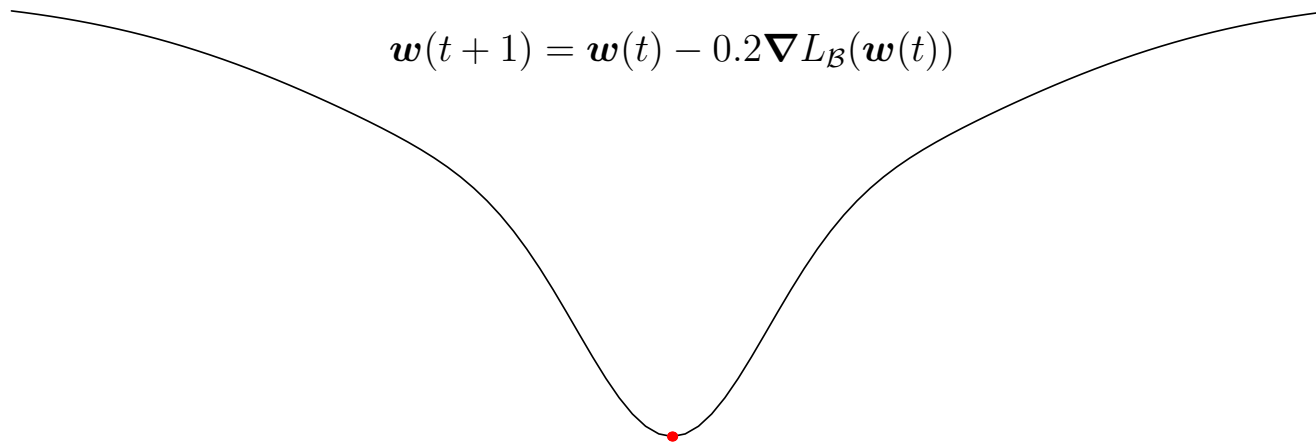
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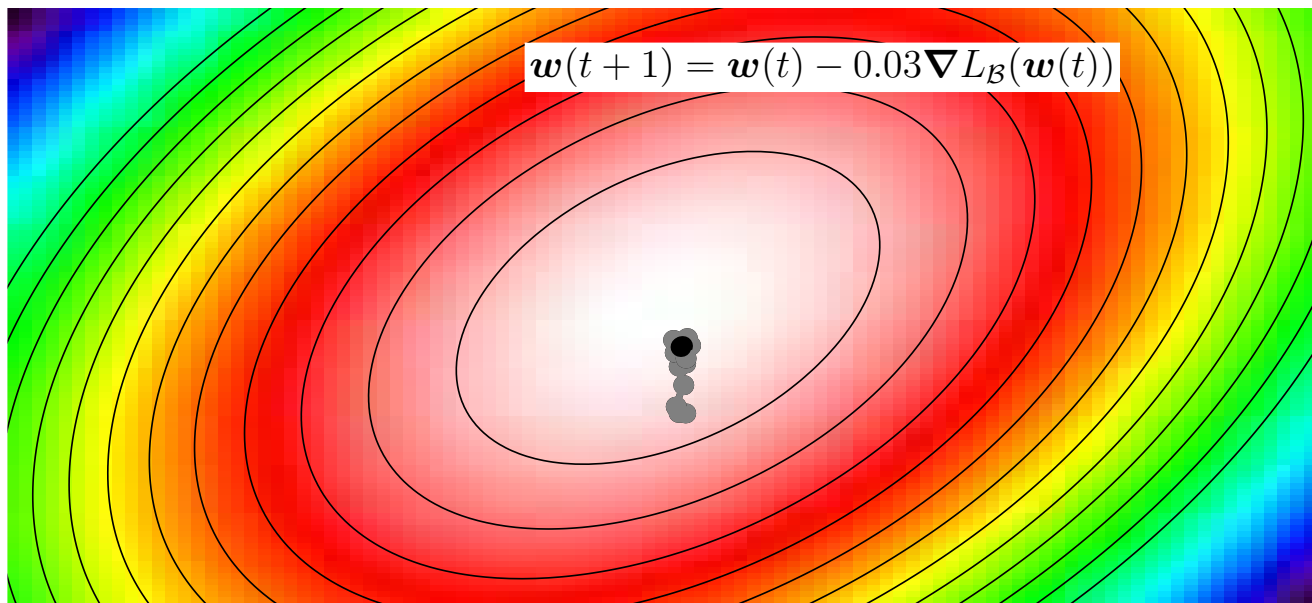
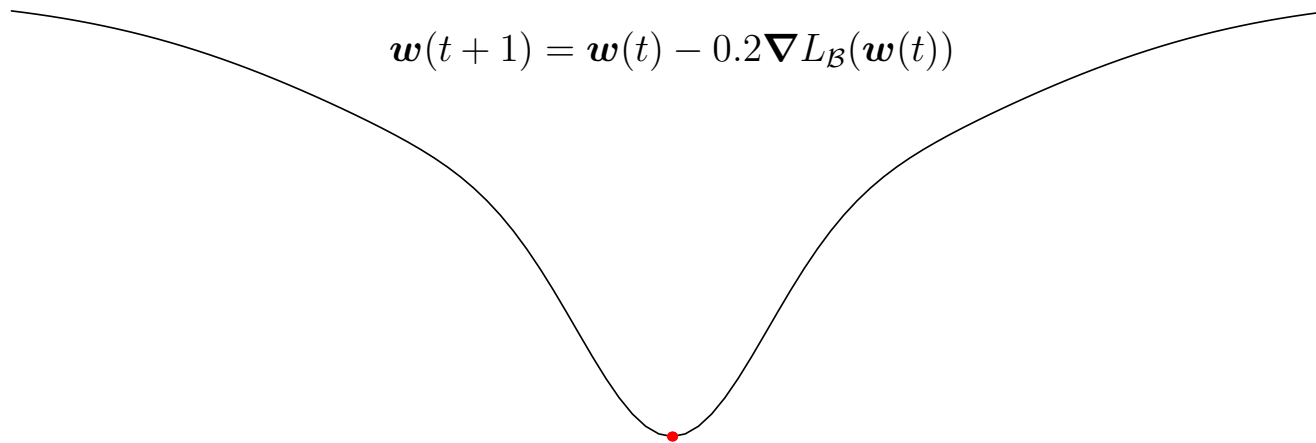
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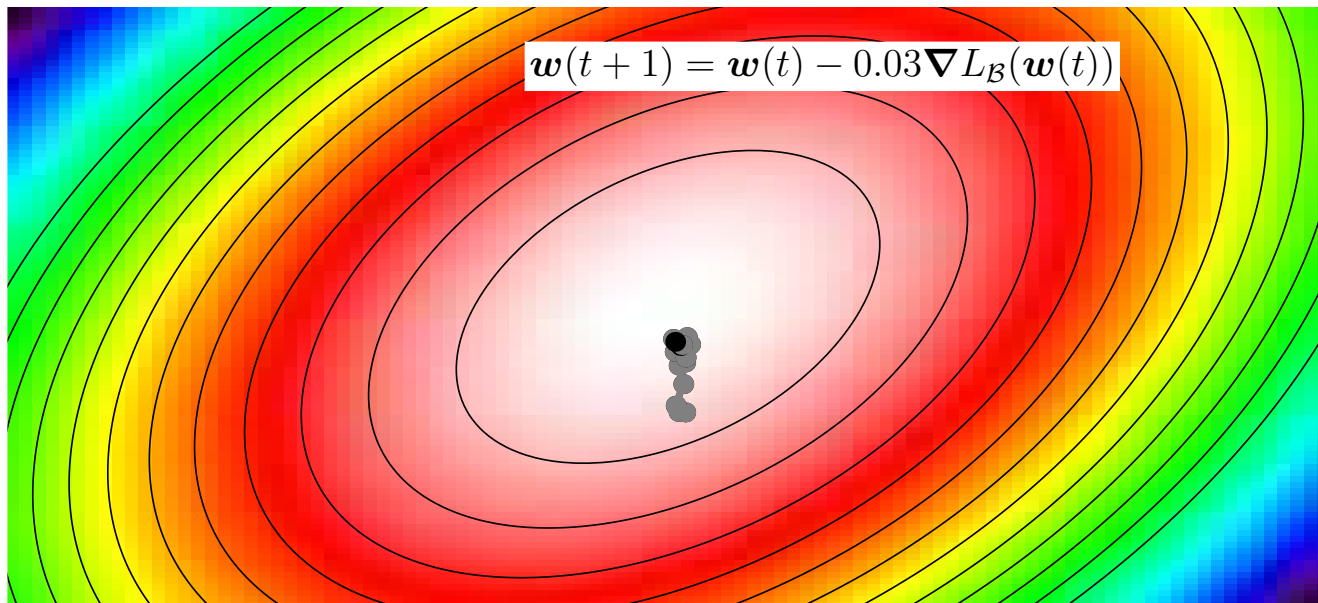
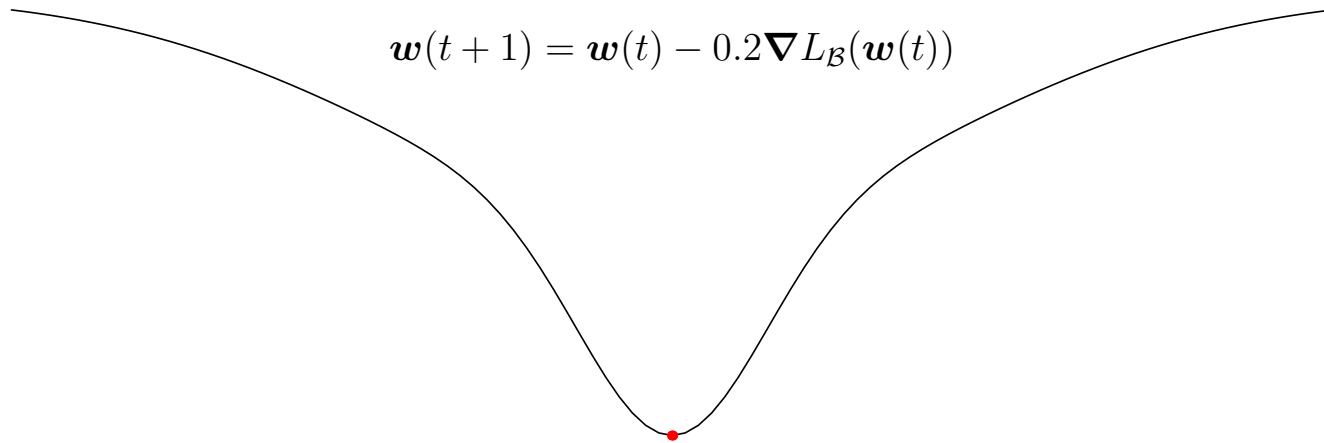
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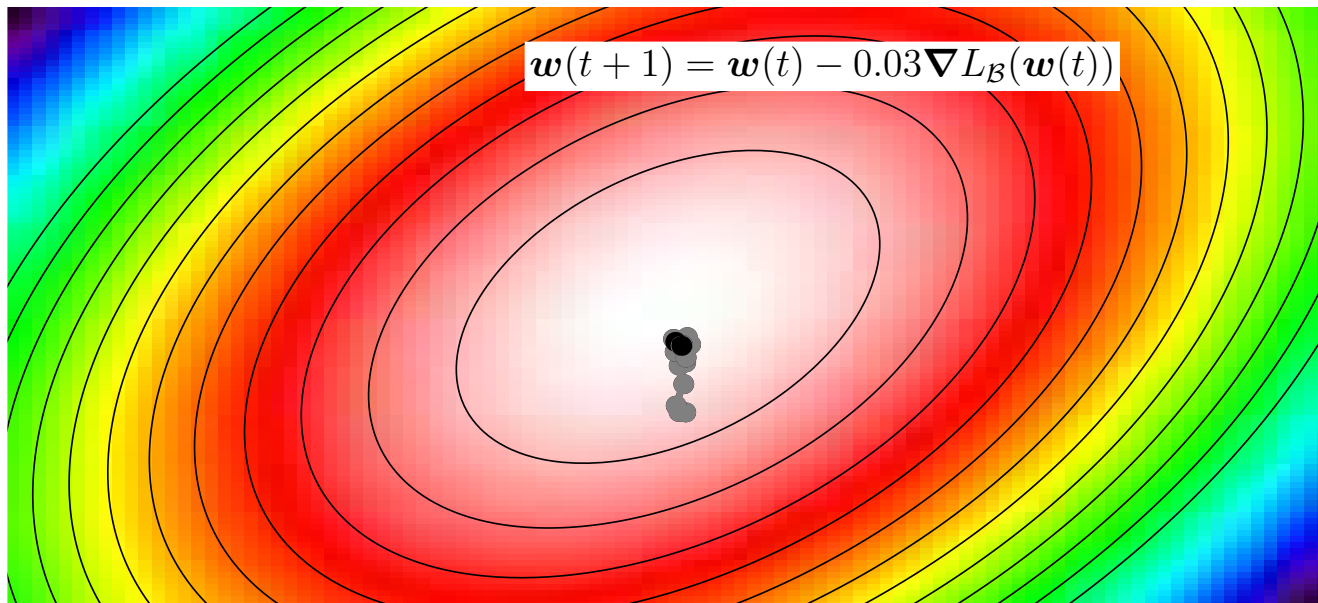
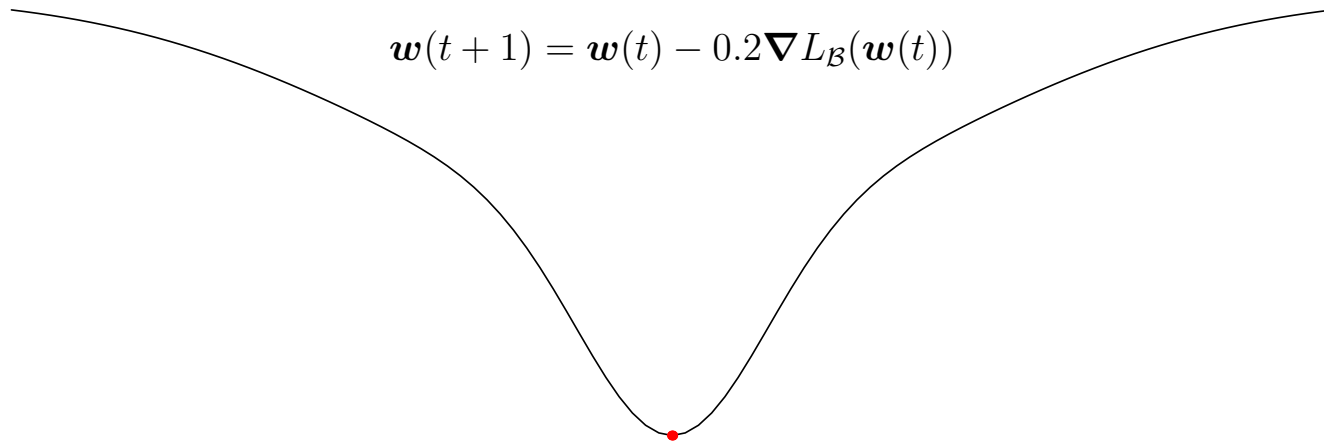
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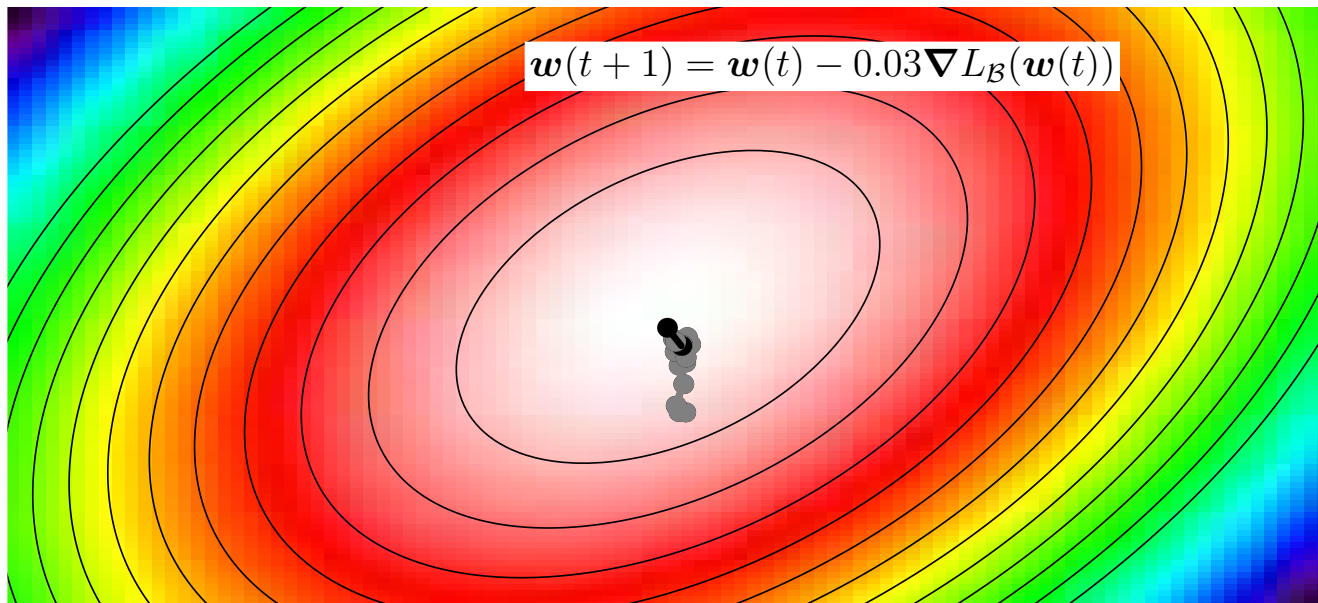
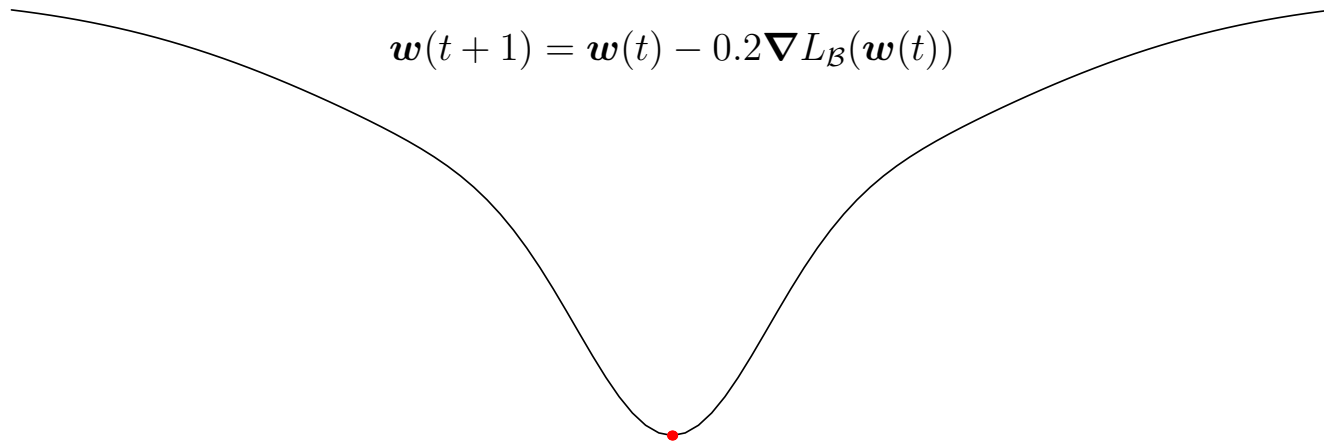
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Symmetries

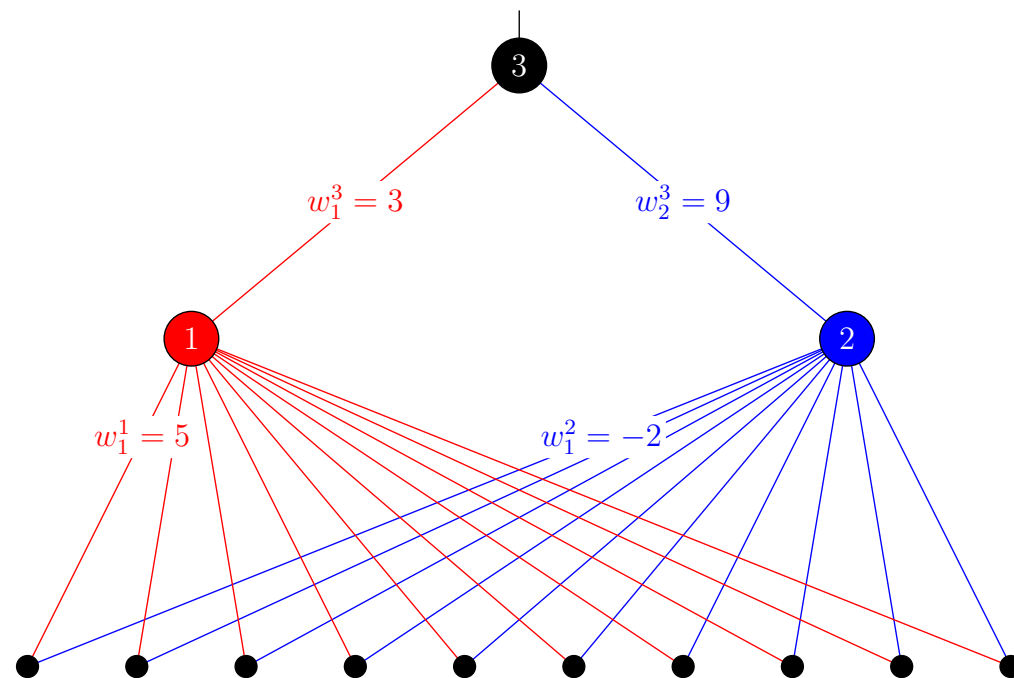
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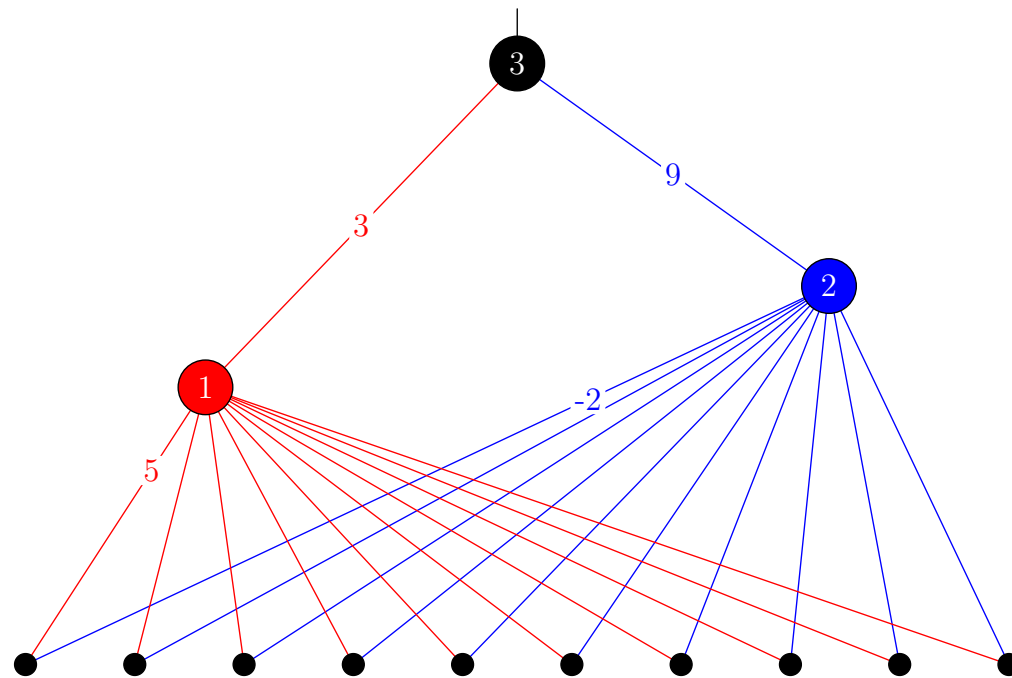
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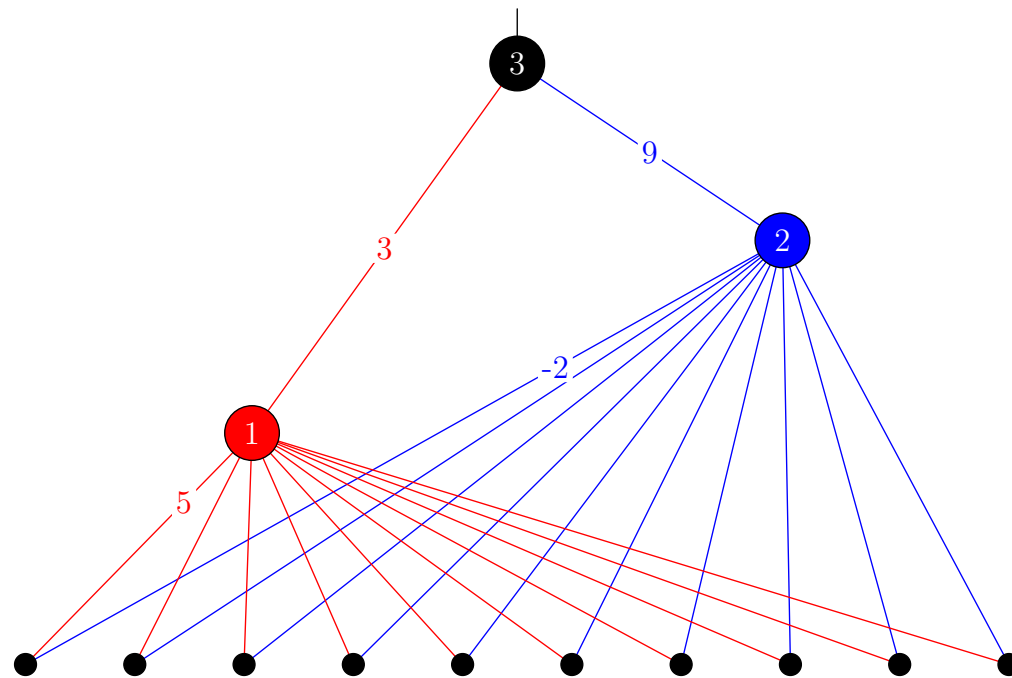
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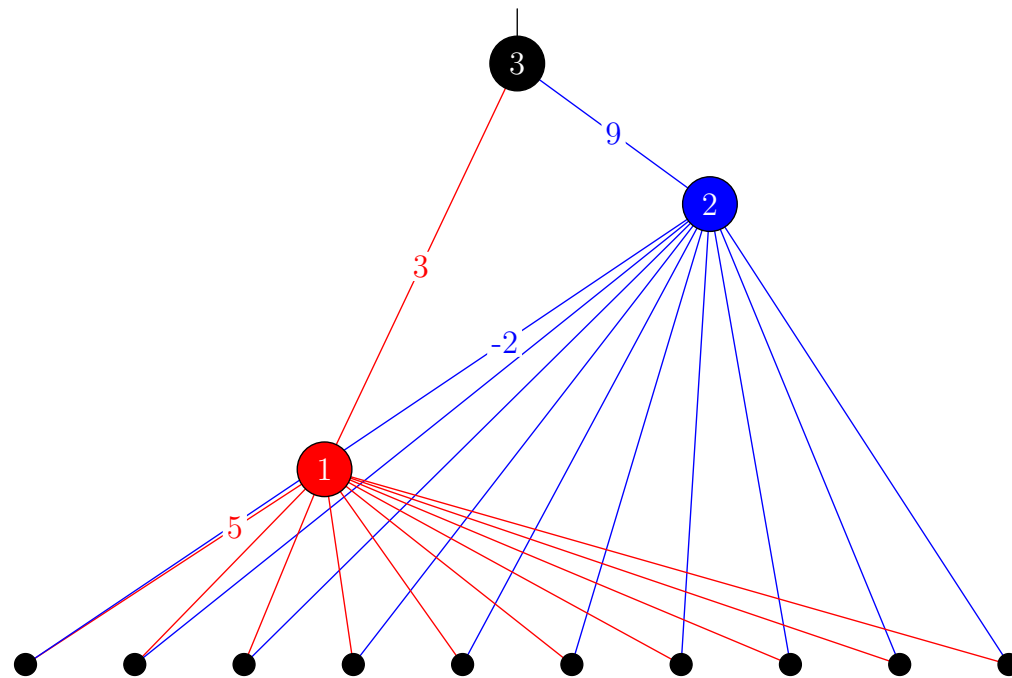
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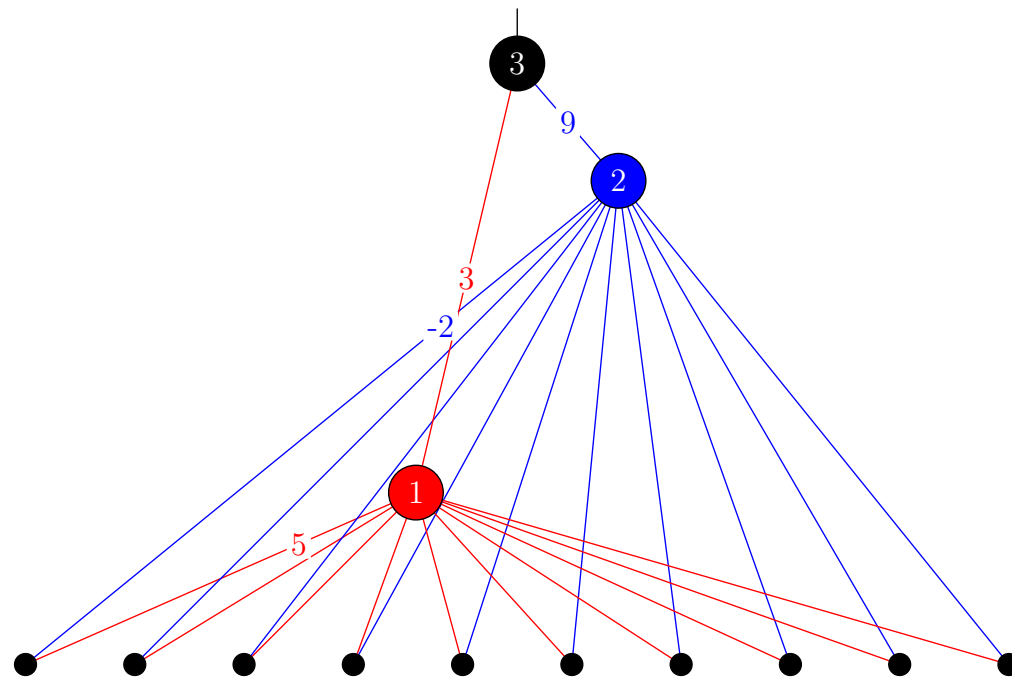
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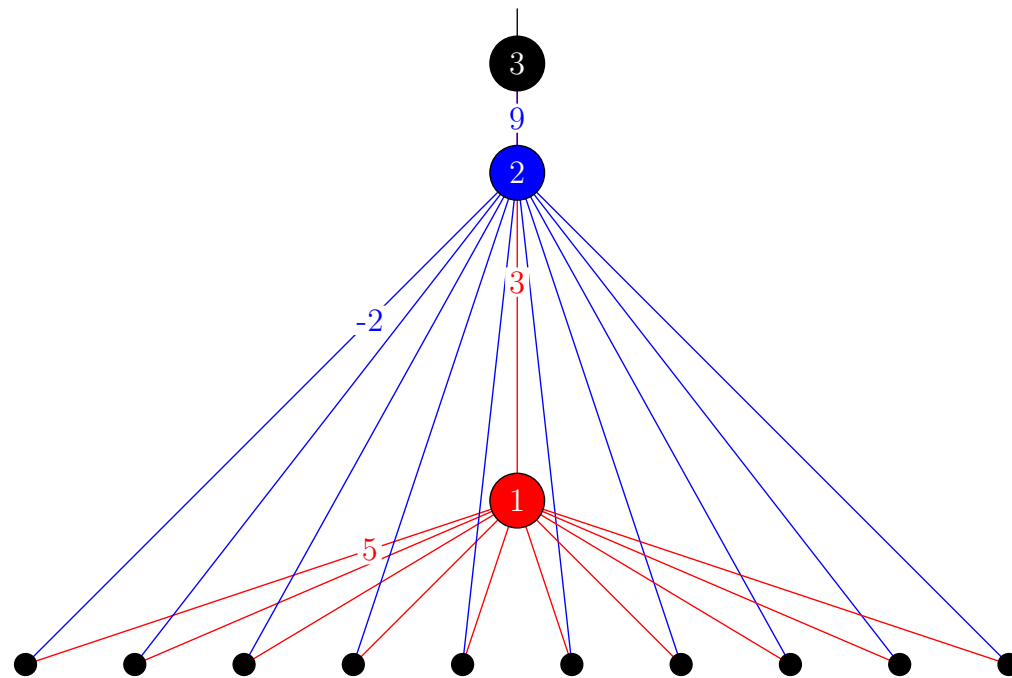
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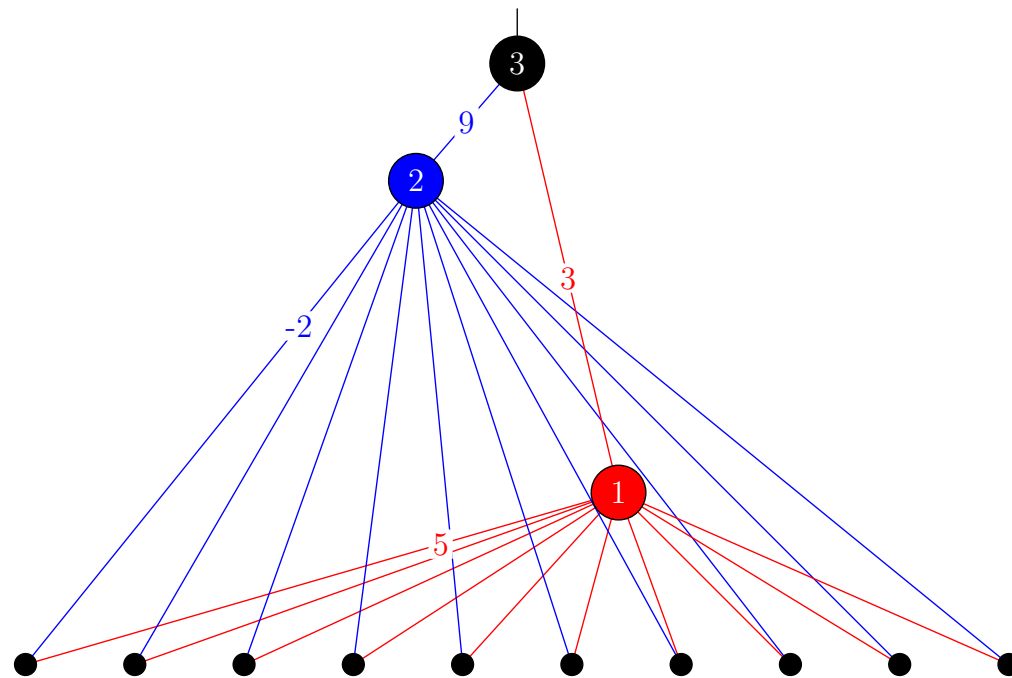
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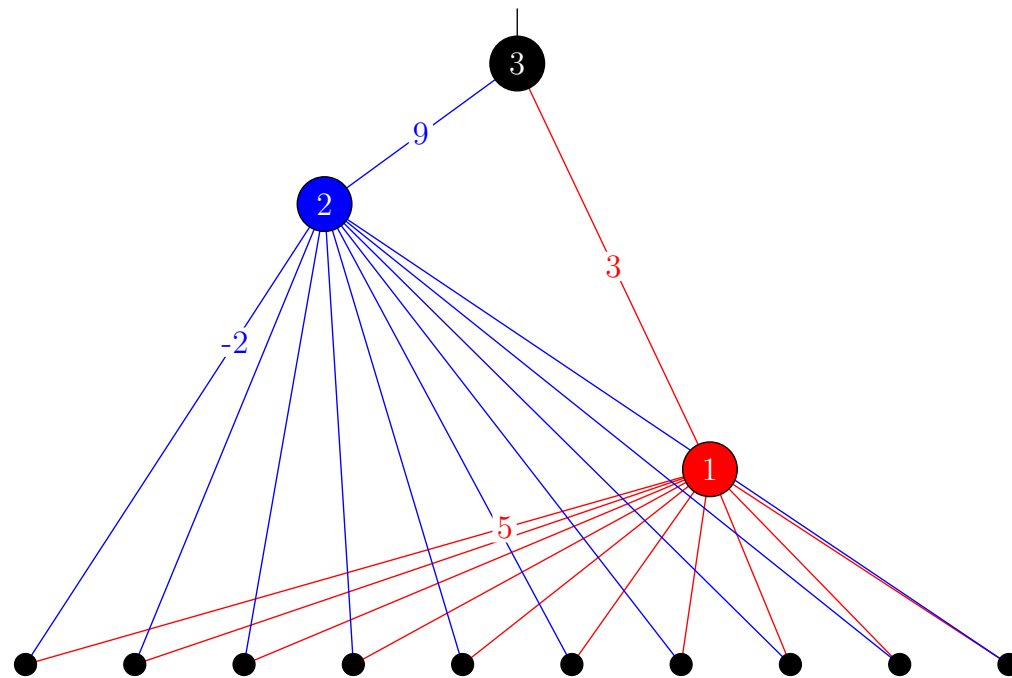
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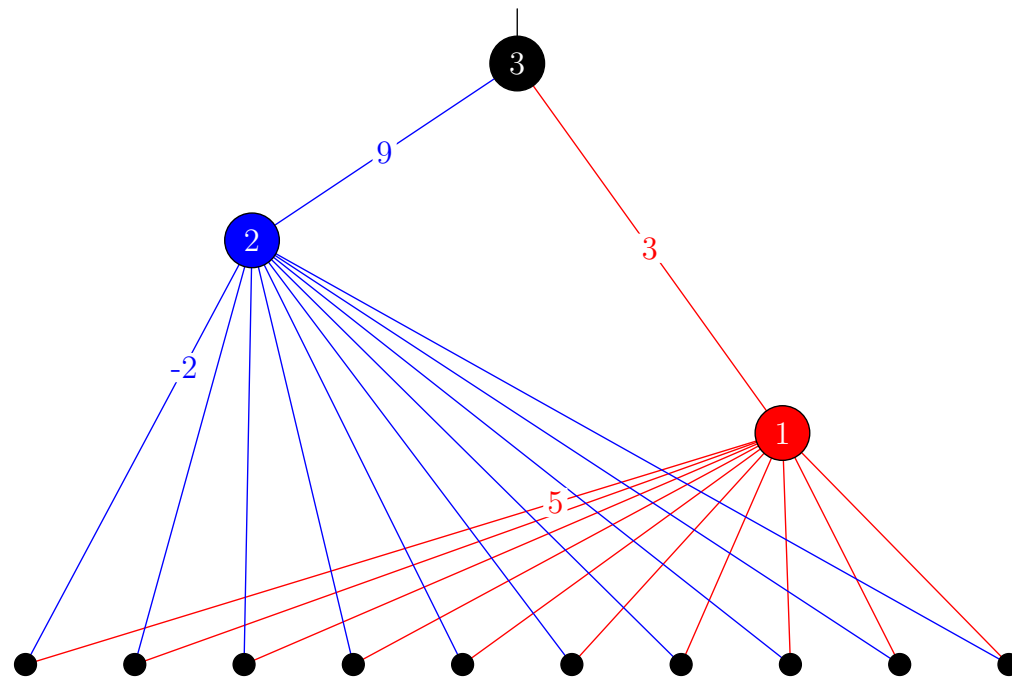
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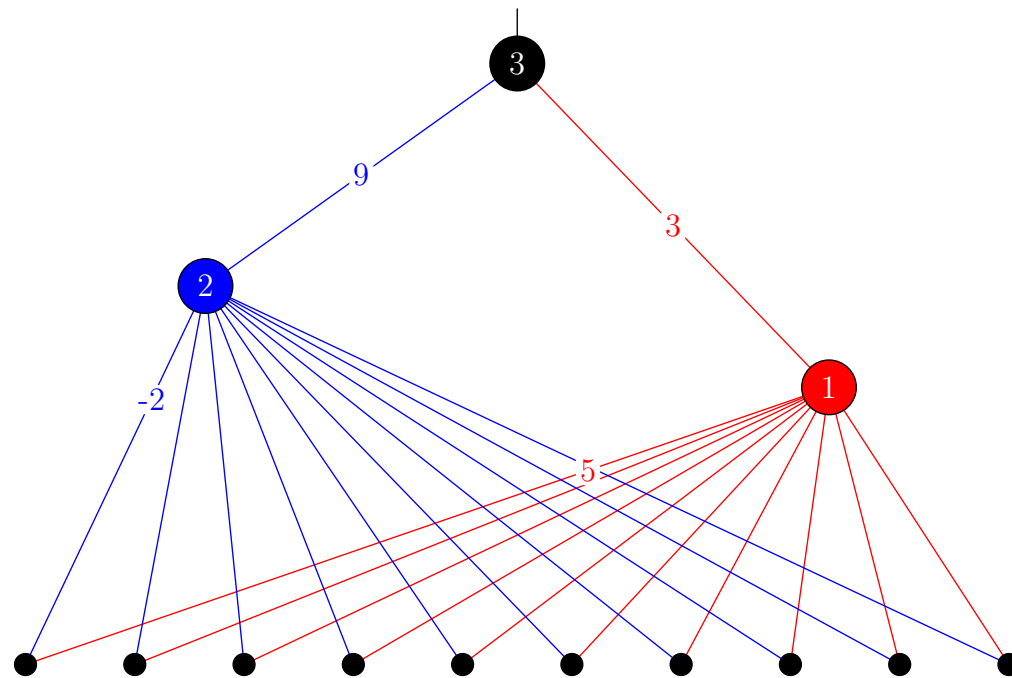
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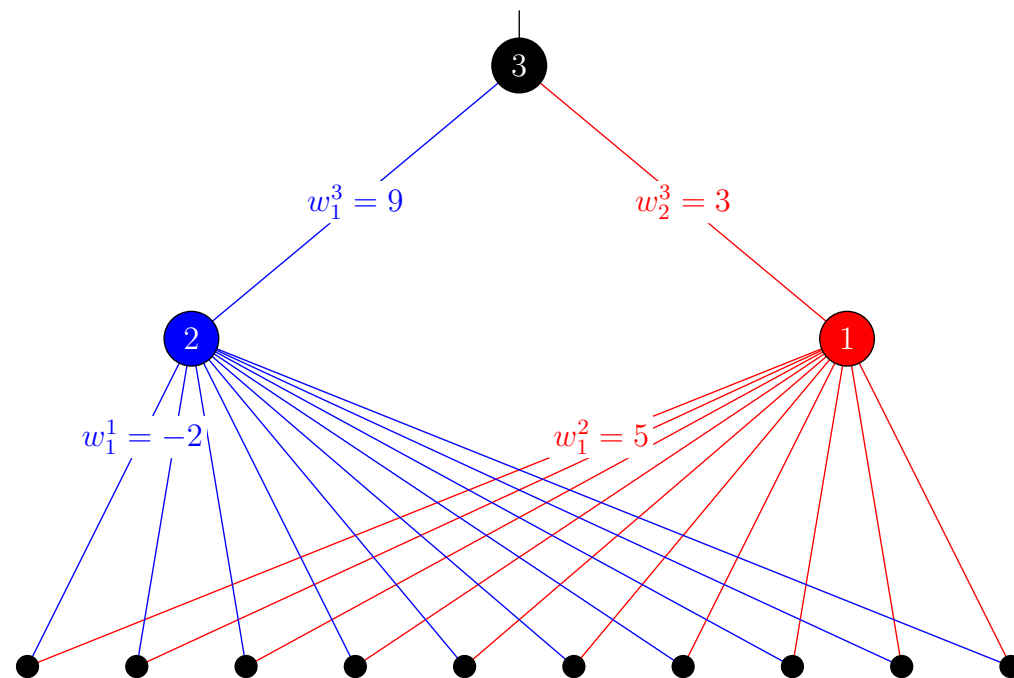
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More Symmetries

- With n hidden nodes there are $n!$ symmetries
- Often if I multiply all the input and output weights to a node by -1 the network will be the same
- There are also continuous symmetries. For linear networks with two layers performing a mapping $\mathbf{W}_2\mathbf{W}_1$ then for any invertible matrix $\mathbf{U}$ (where $\mathbf{U}\mathbf{U}^{-1} = \mathbf{I}$)

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- There will be a huge space of equivalent networks

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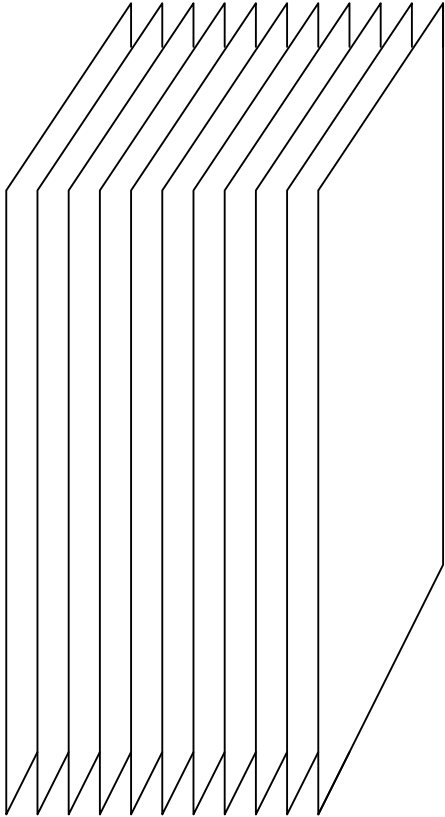
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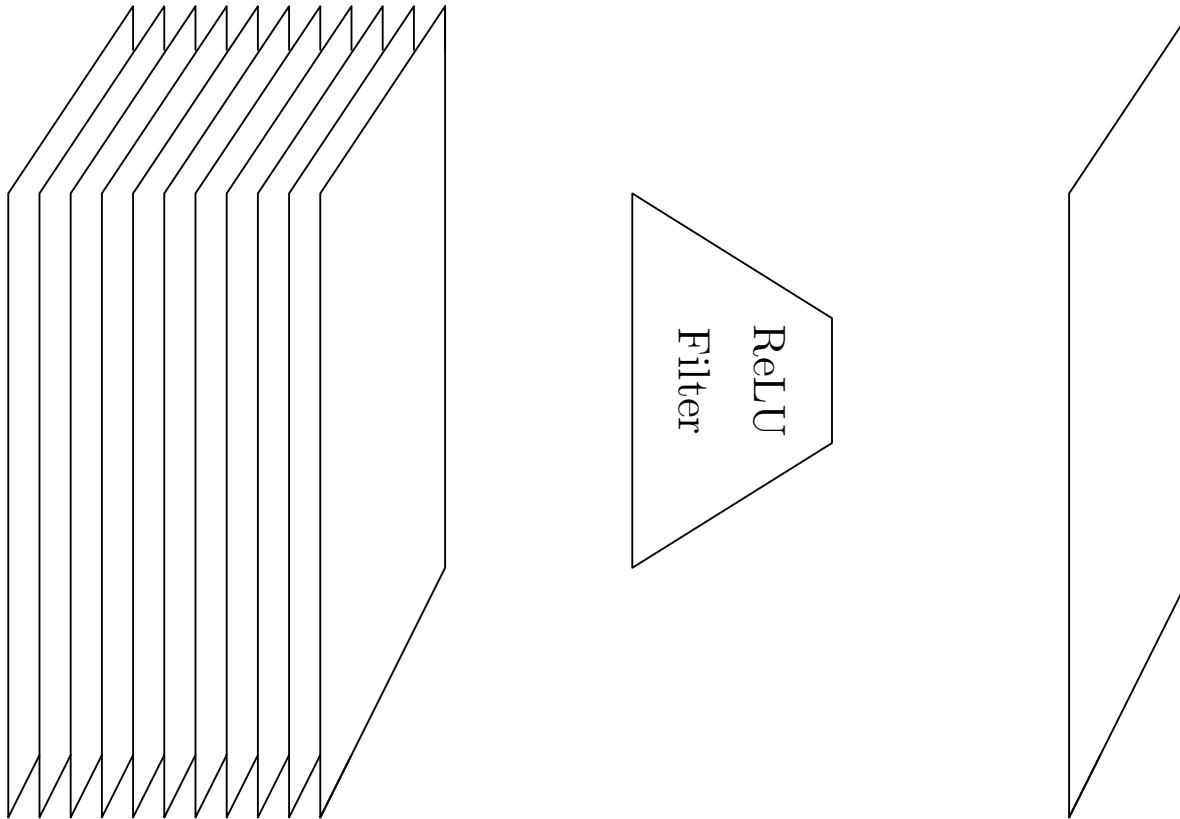
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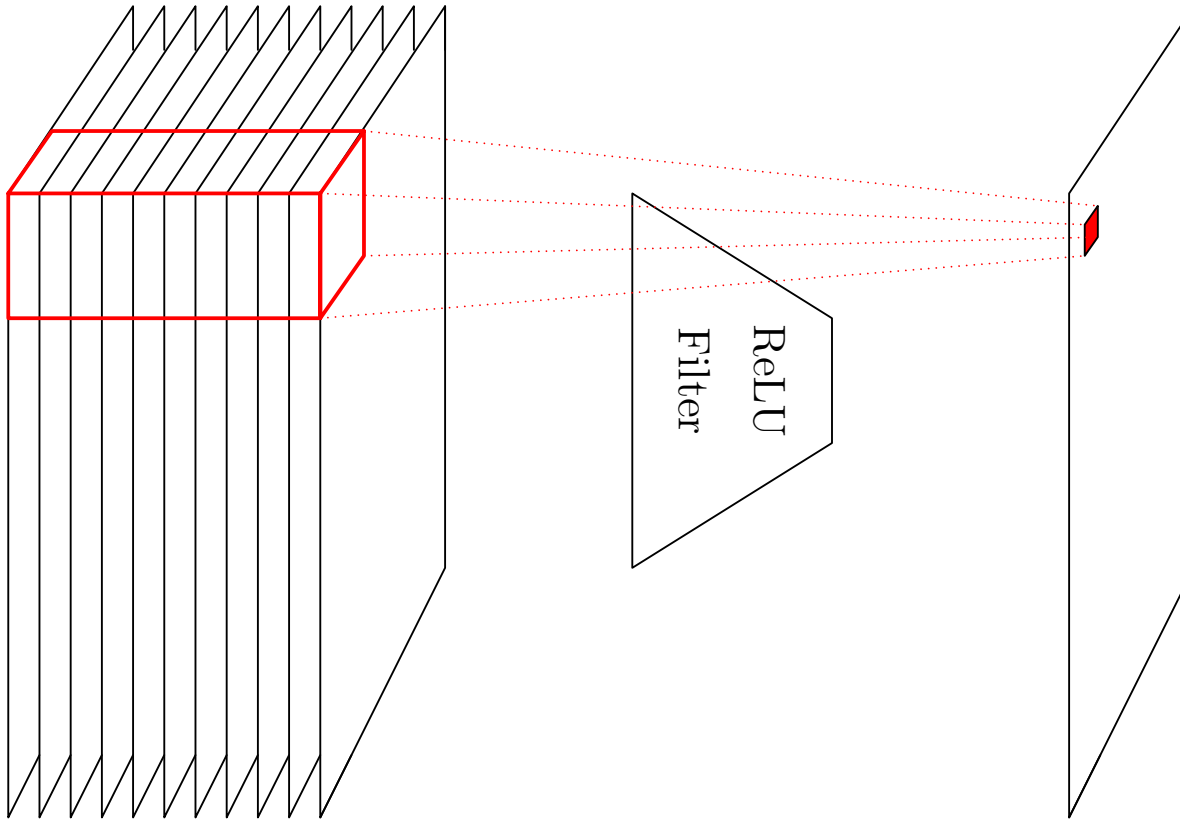
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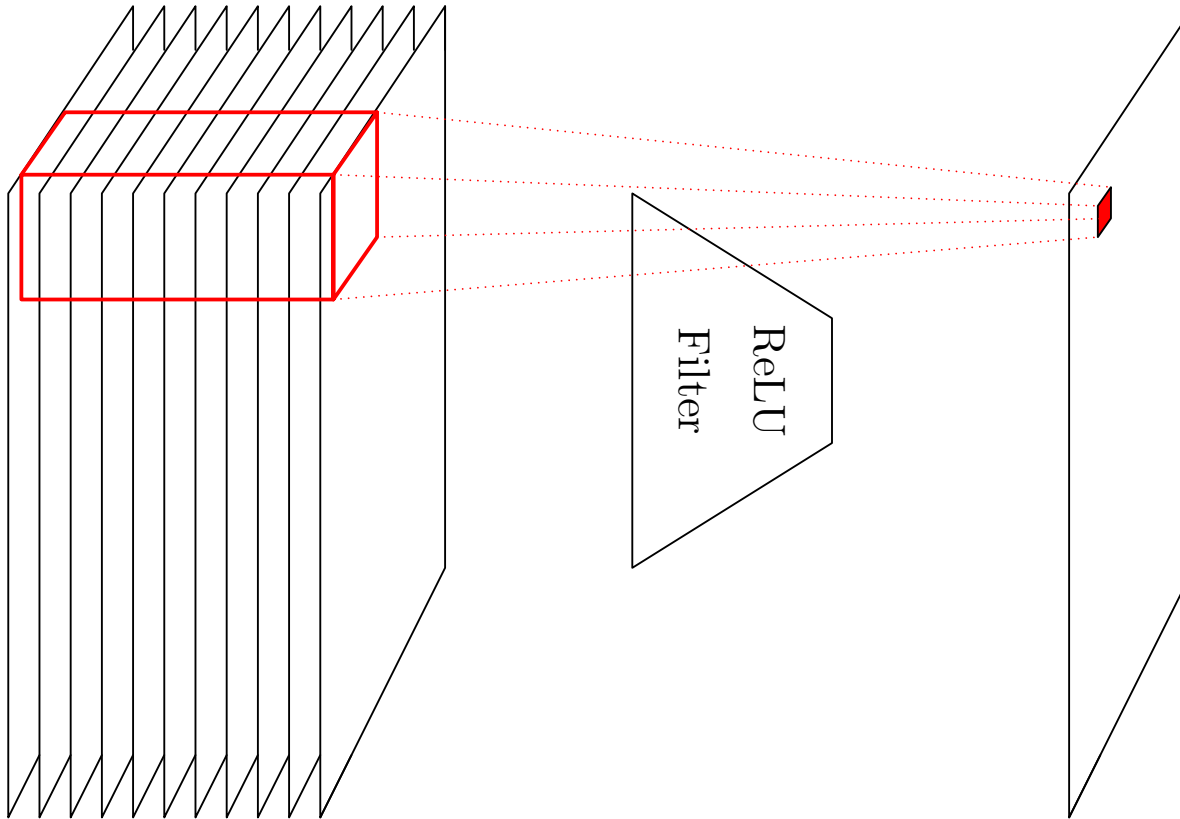
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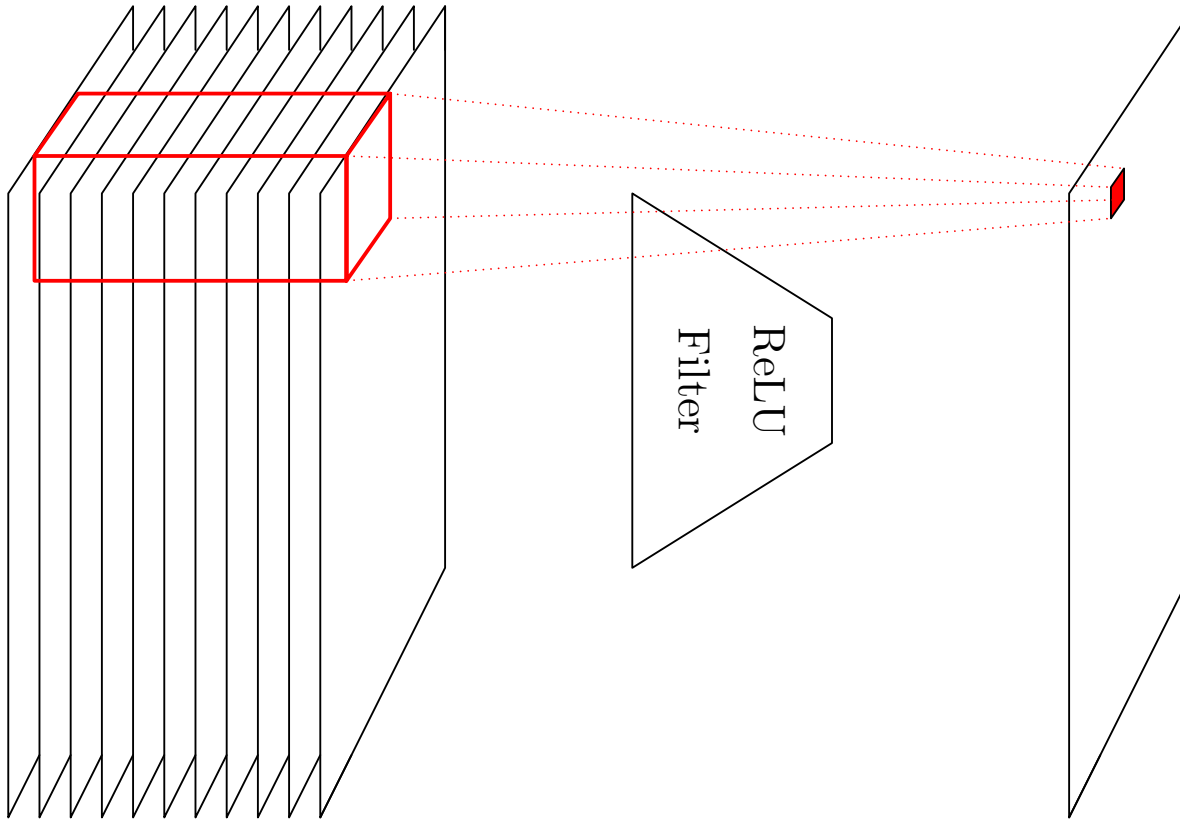
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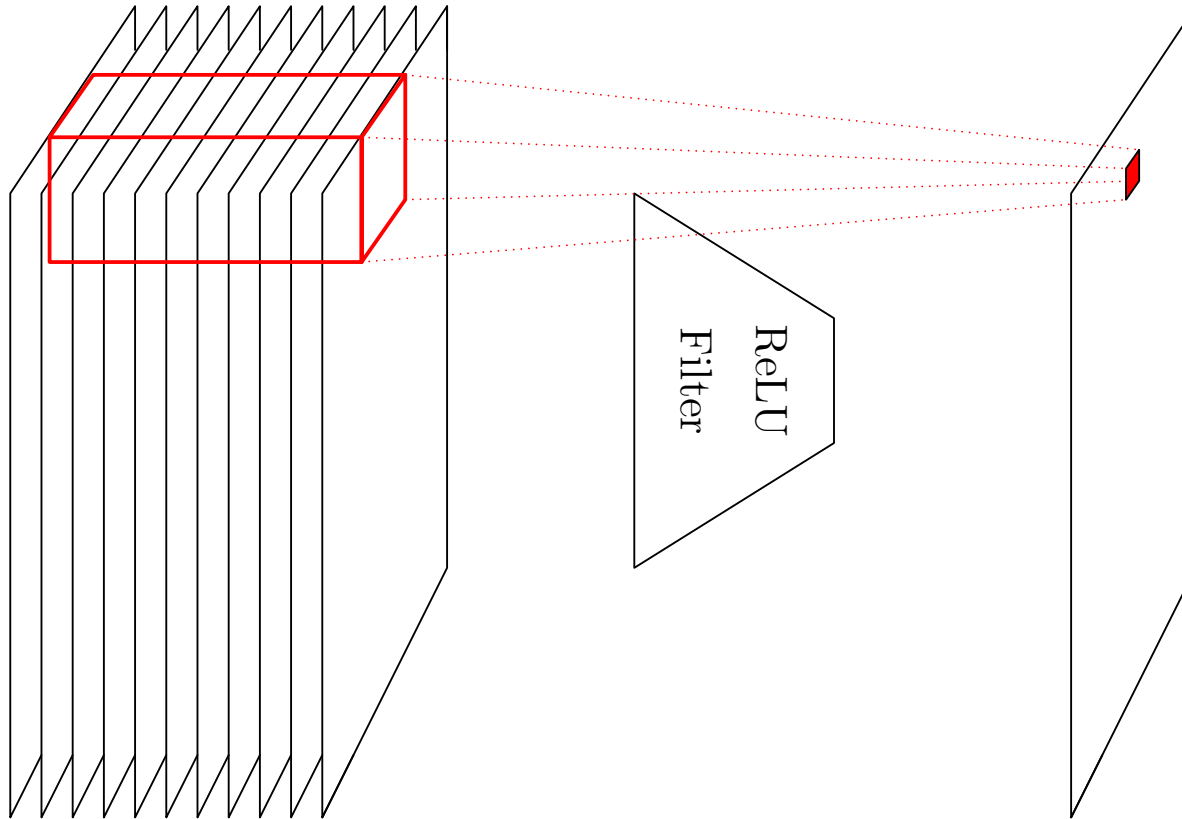
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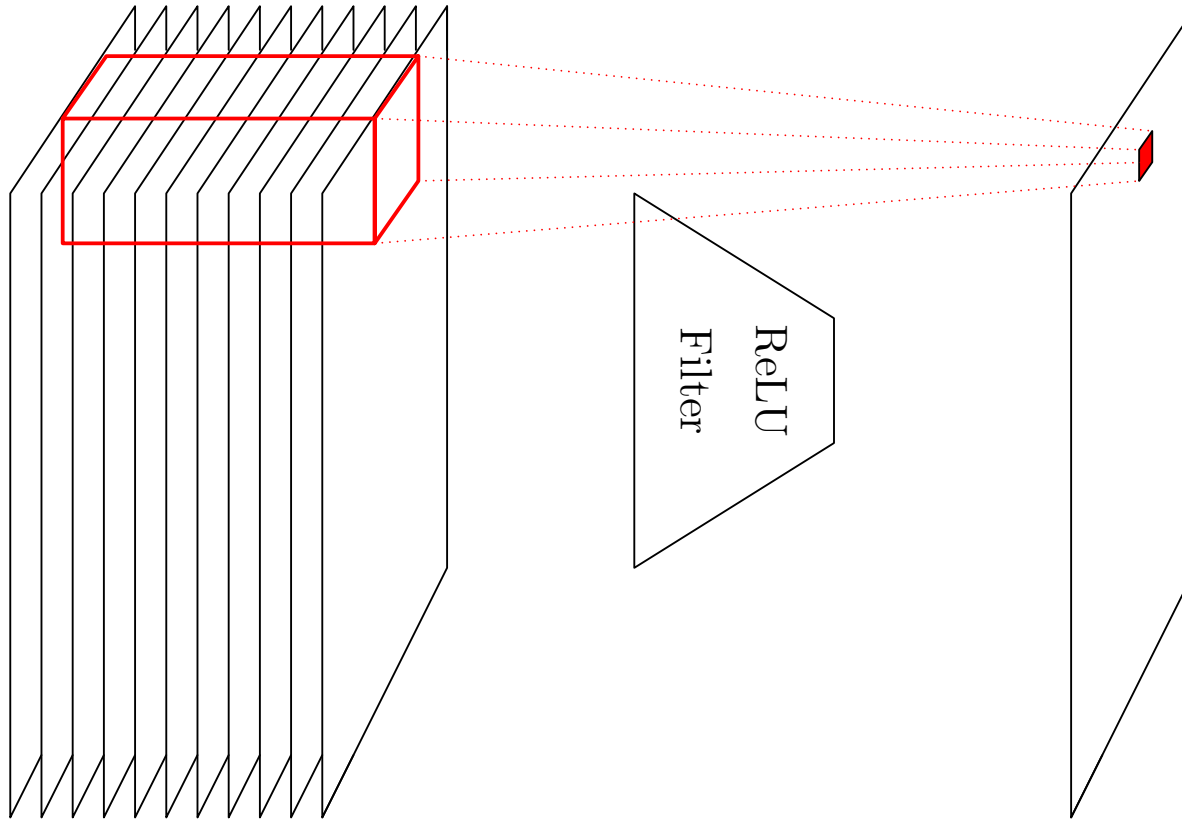
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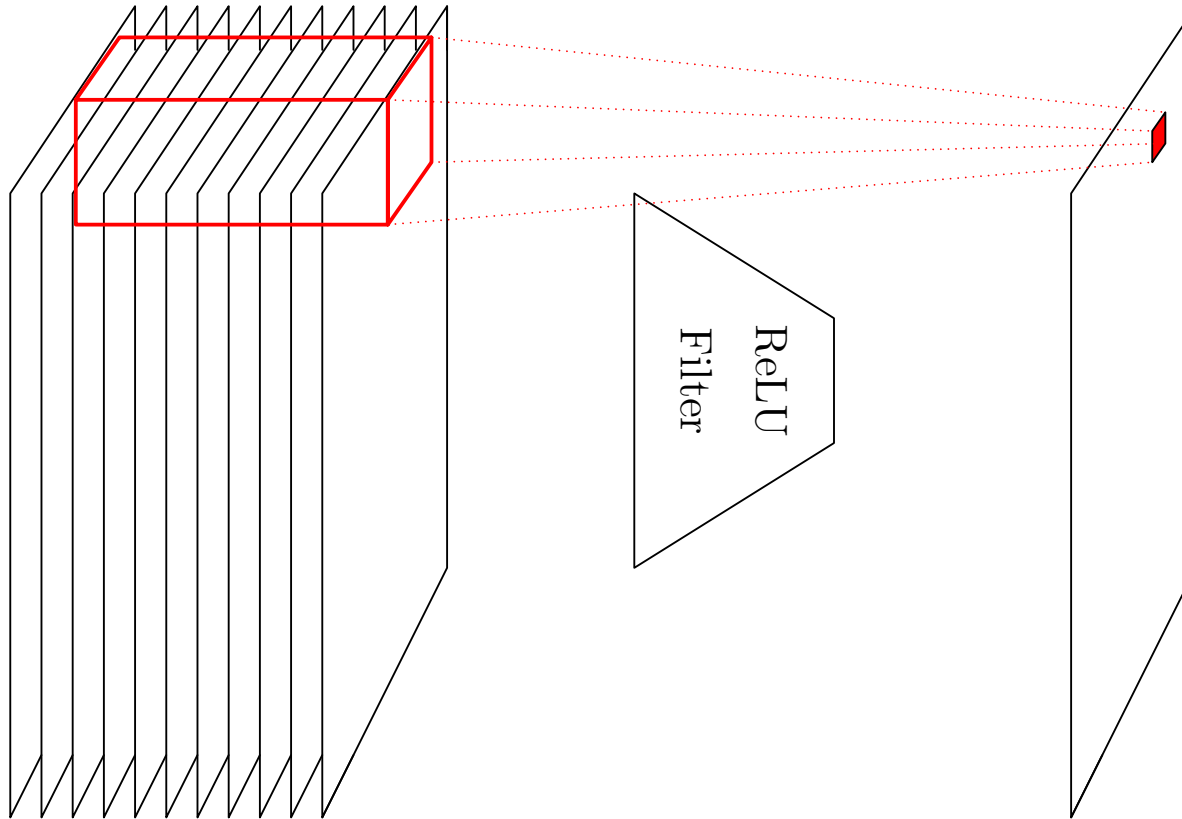
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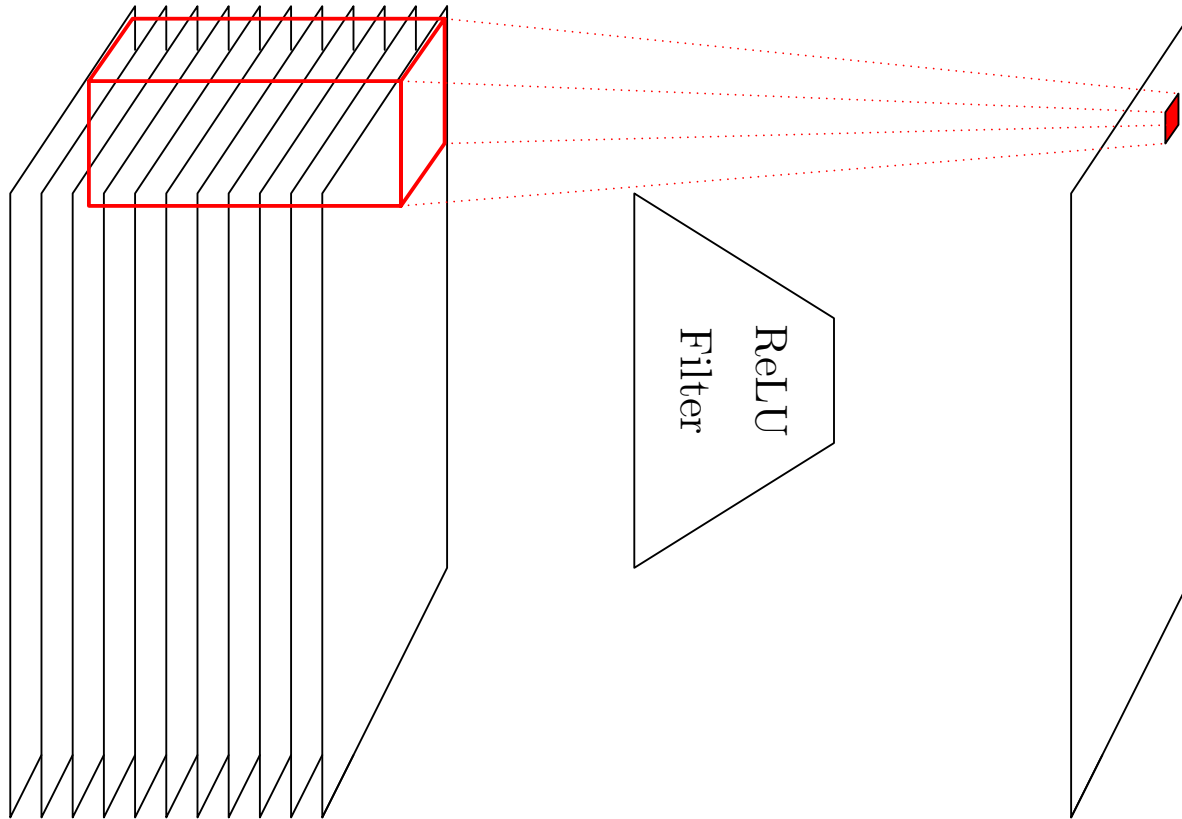
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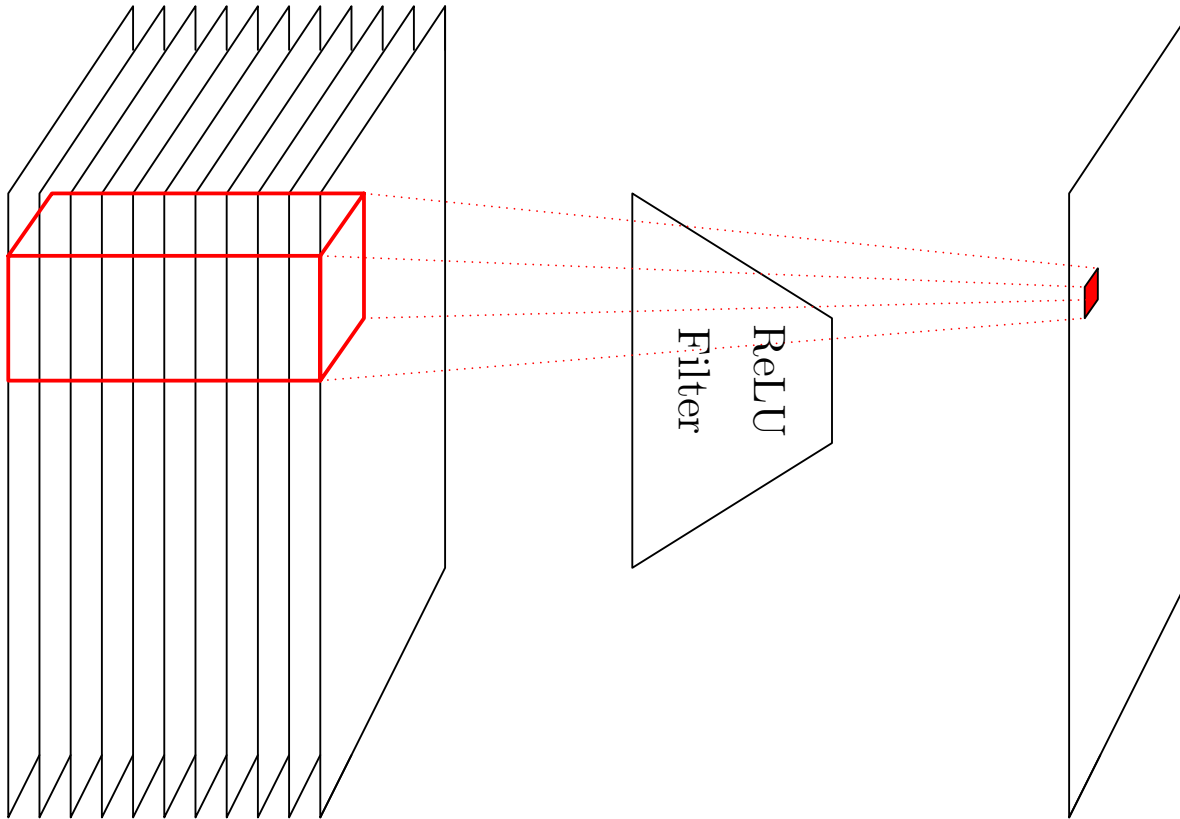
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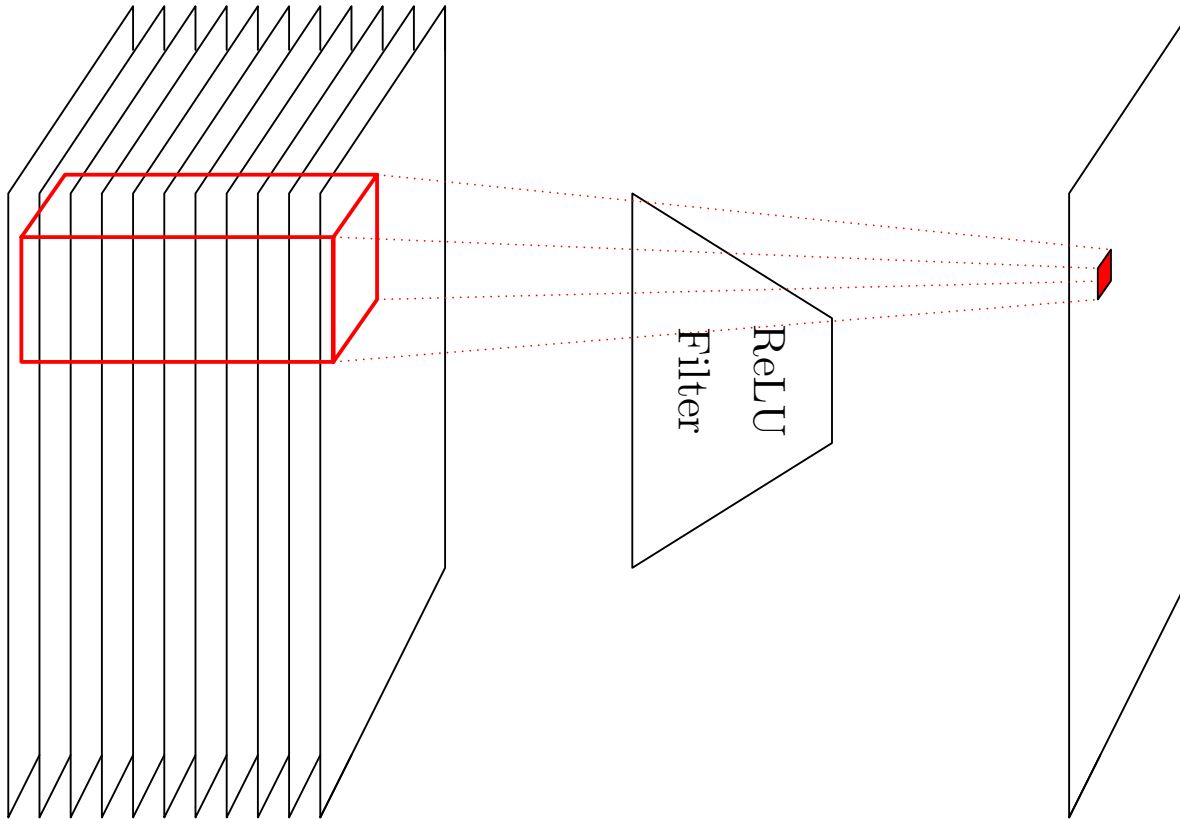
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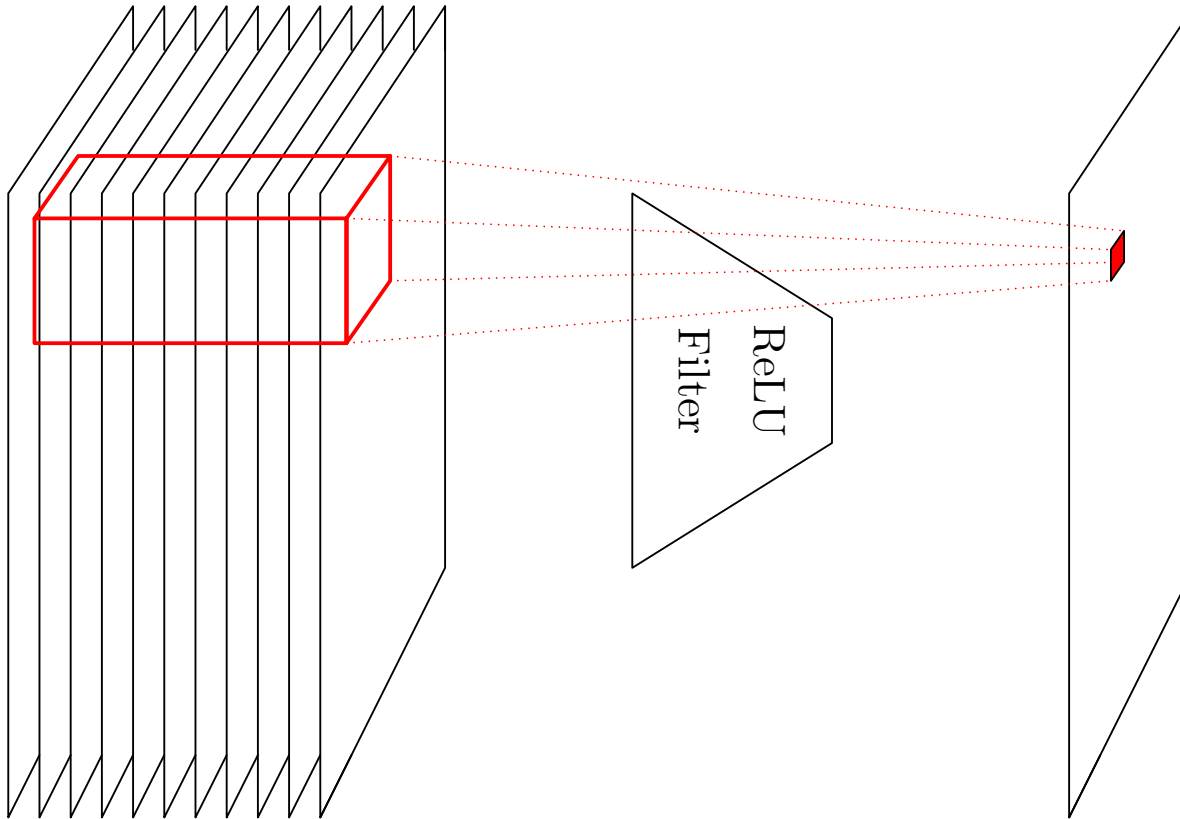
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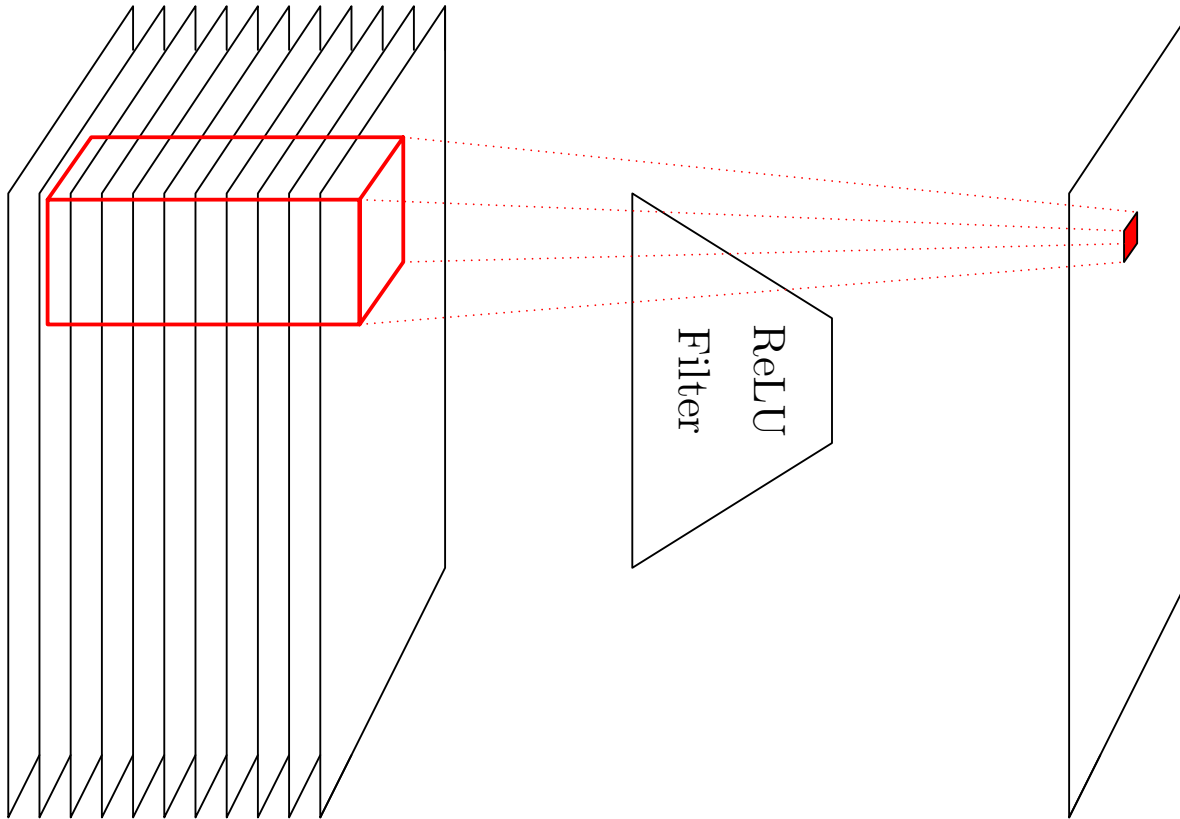
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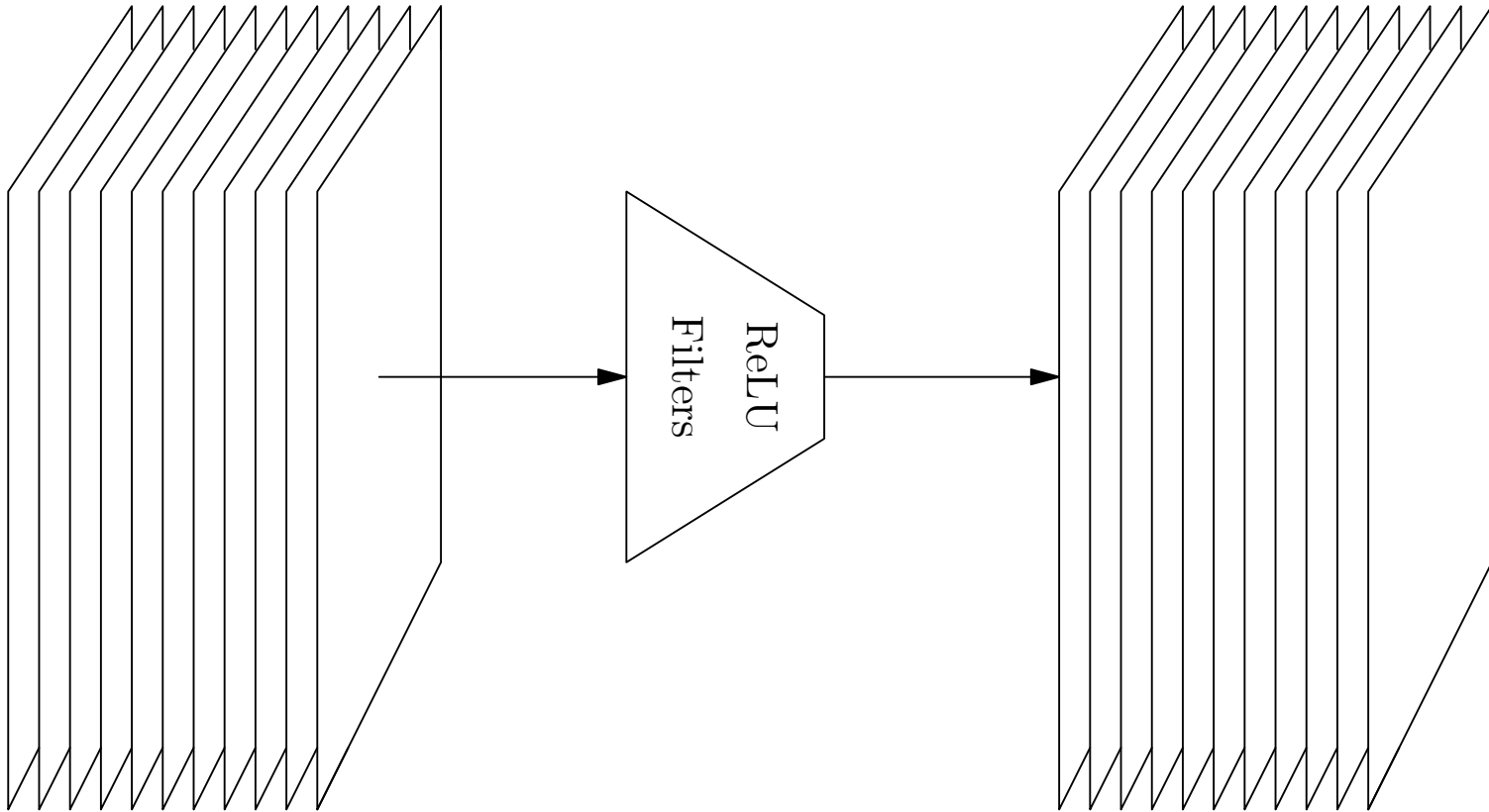
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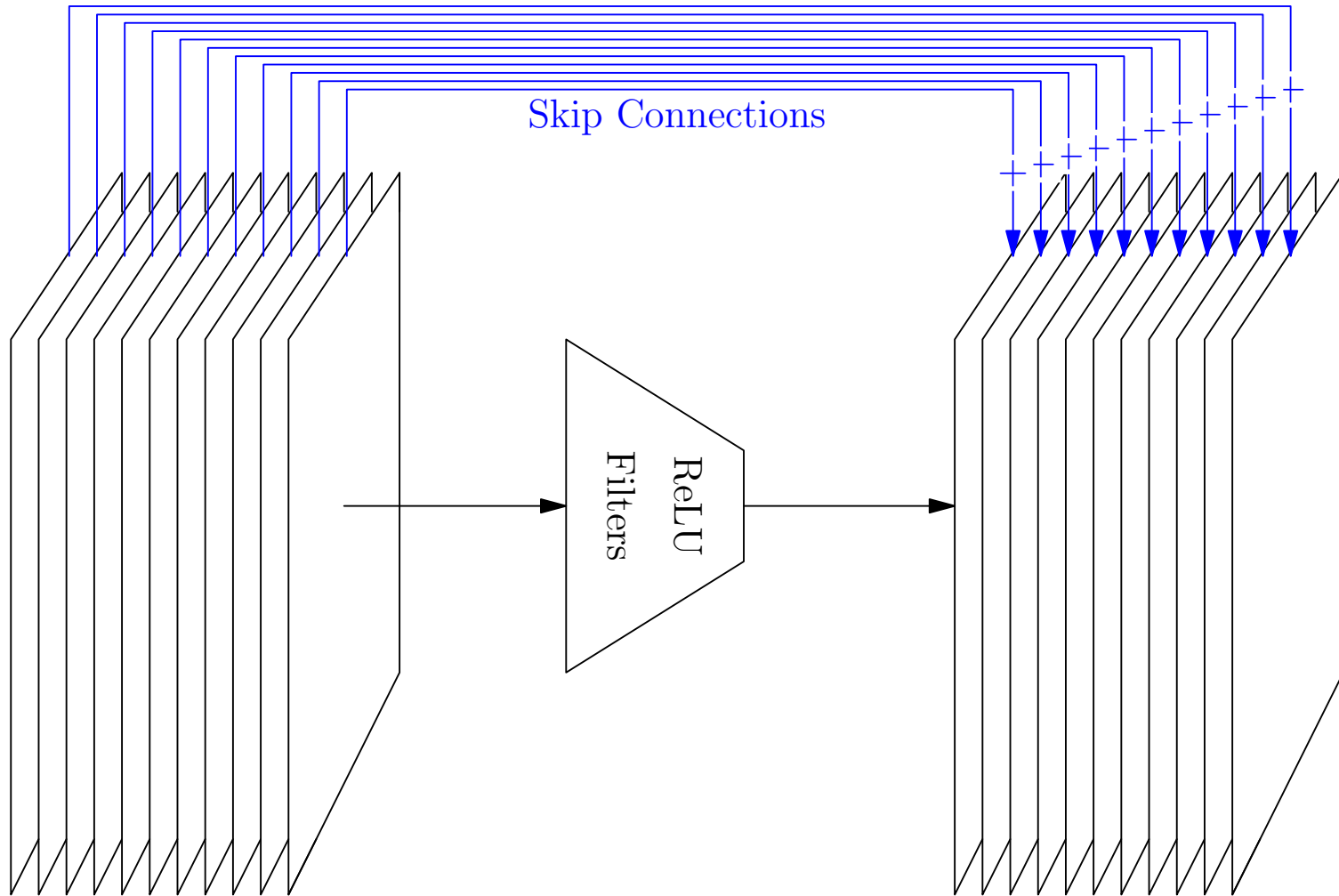
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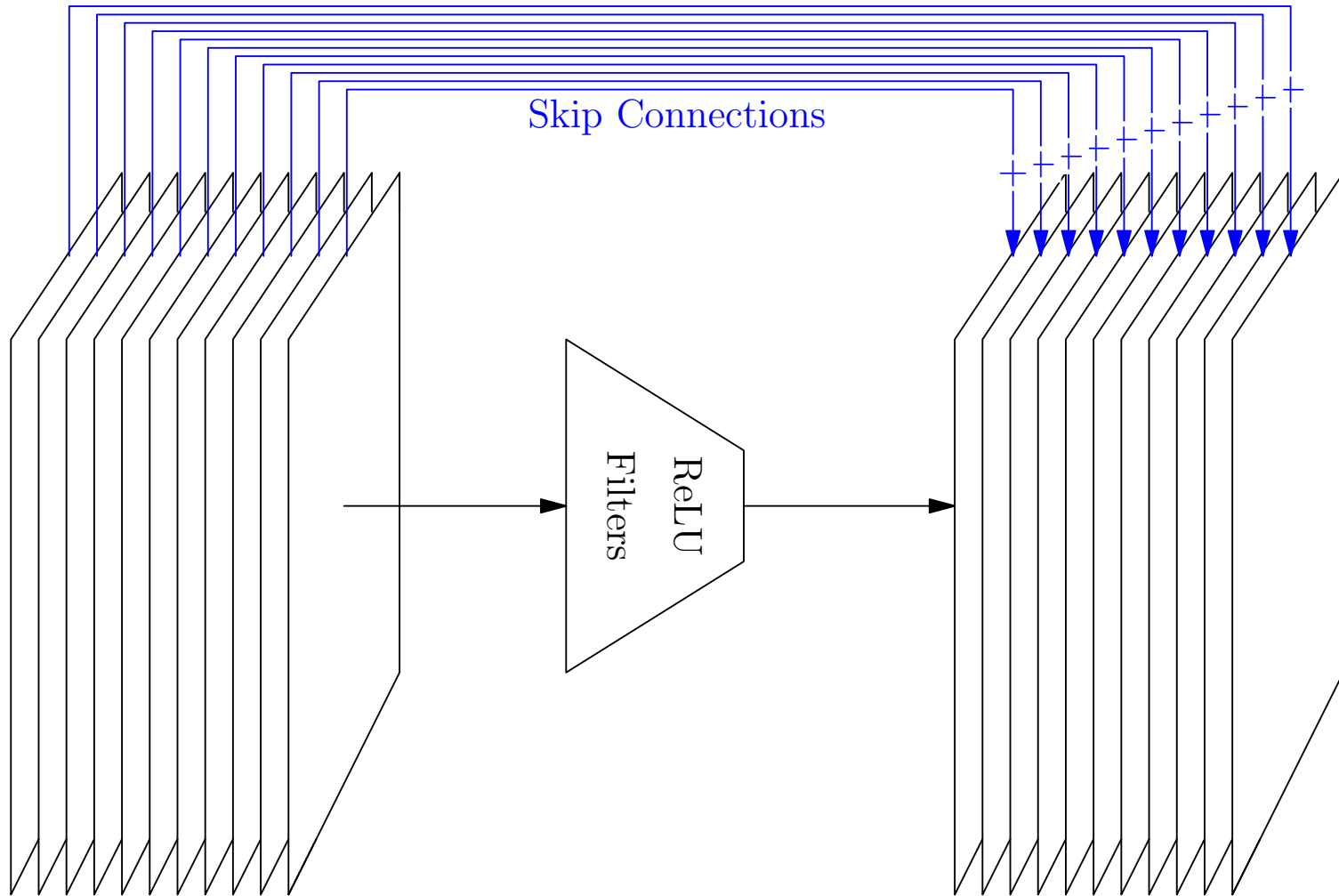
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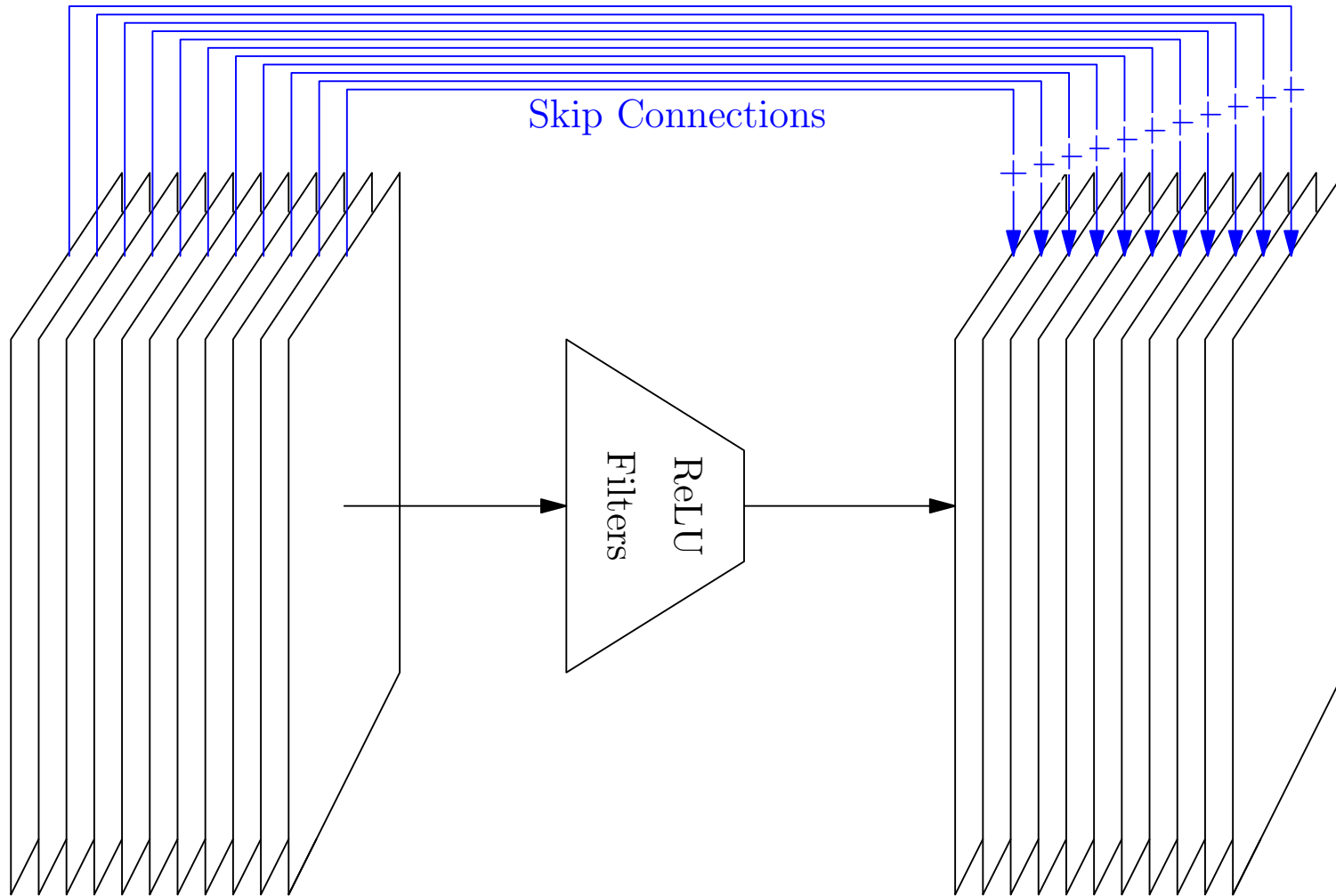


Skip Connections



- Breaks permutation symmetry

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- Breaks permutation symmetry. Used in Resnets, transformer, etc.

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- The landscape potentially has a huge manifold with the same losses (neutral network)
- Find solution close to start
- We should think of optimisation more as a process of travelling than a process of arriving
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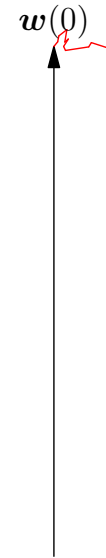
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$w^{(0)}$

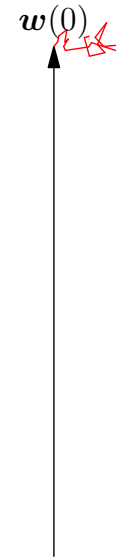
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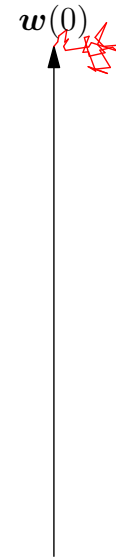
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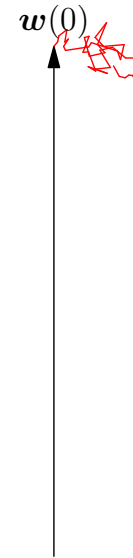
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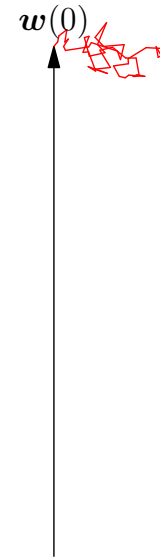
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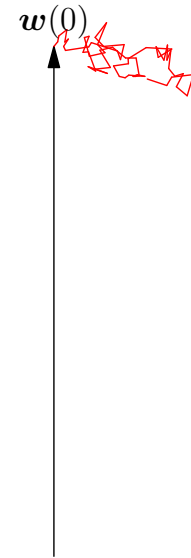
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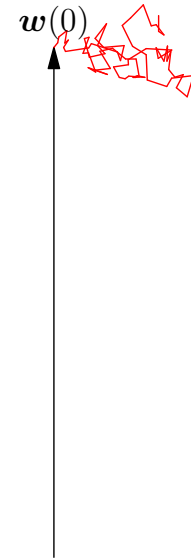
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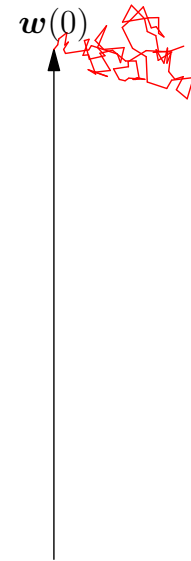
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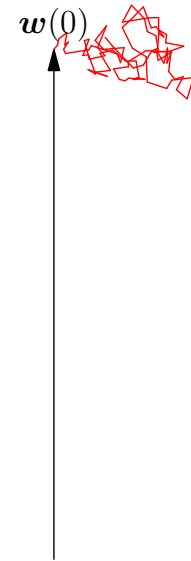
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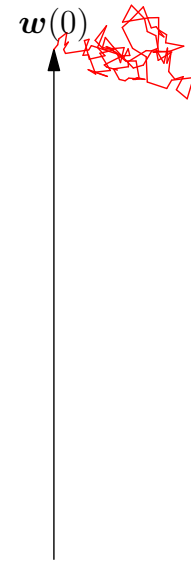
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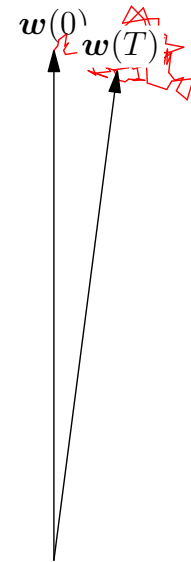
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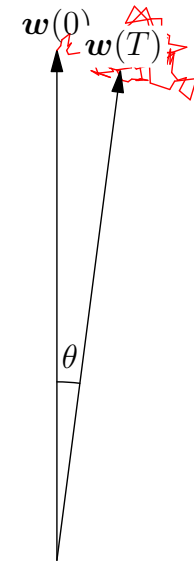
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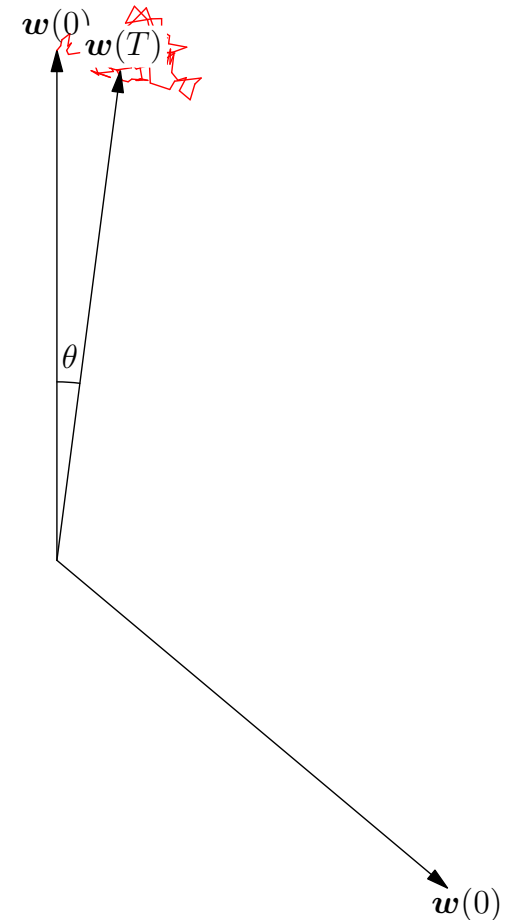
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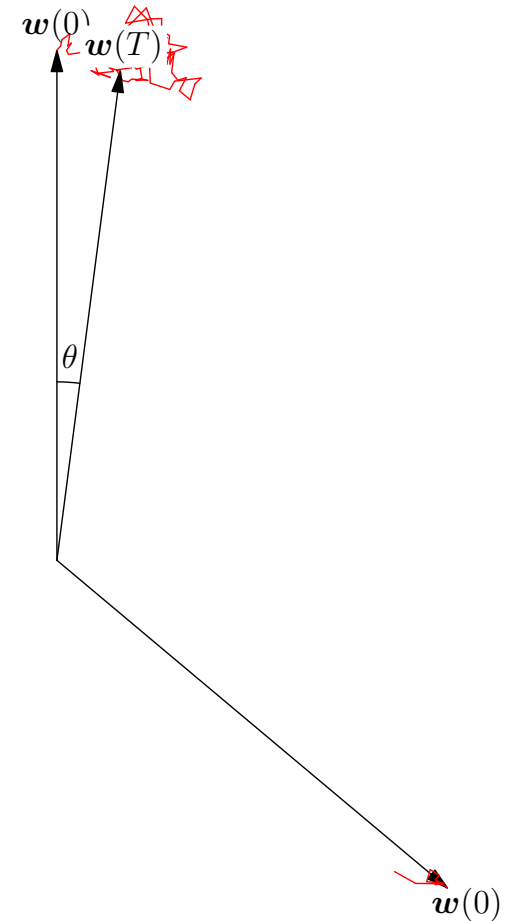
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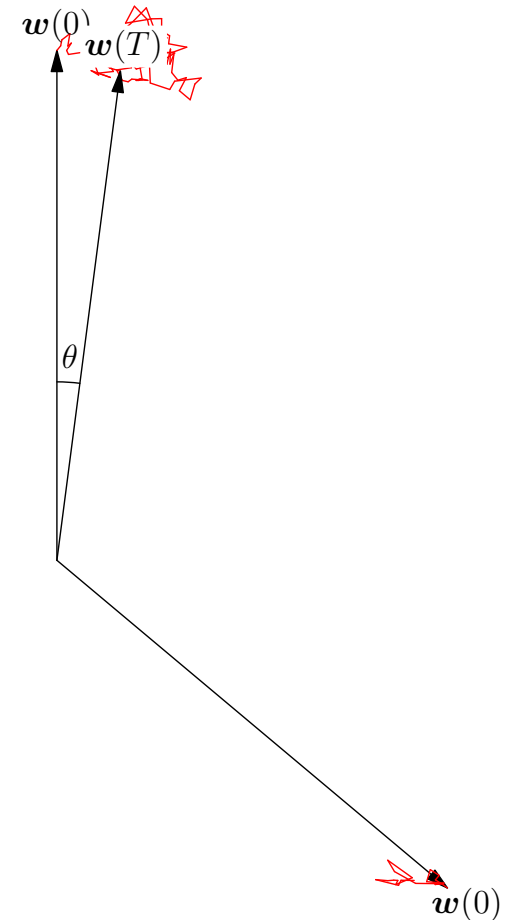
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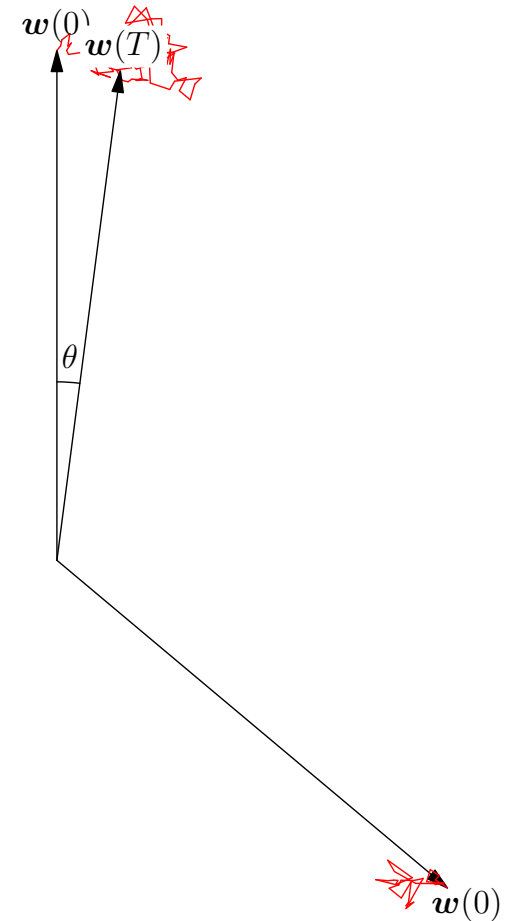
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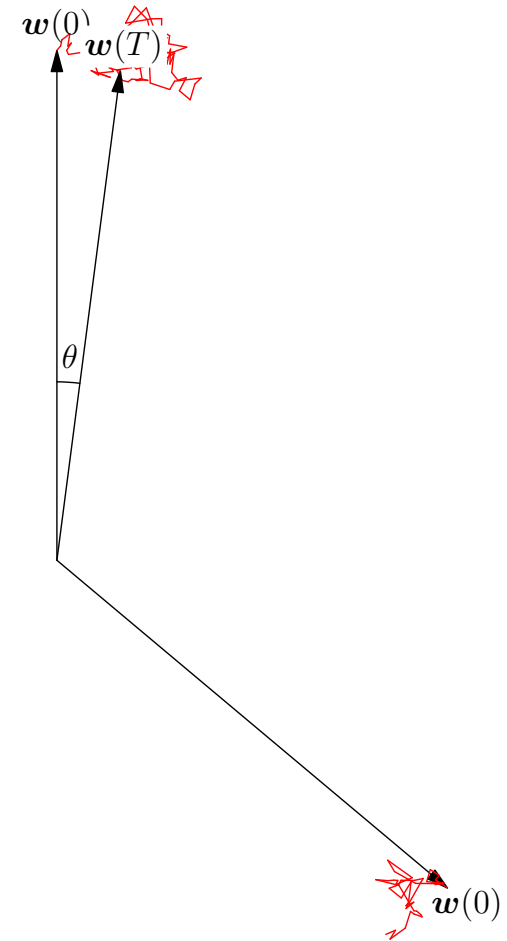
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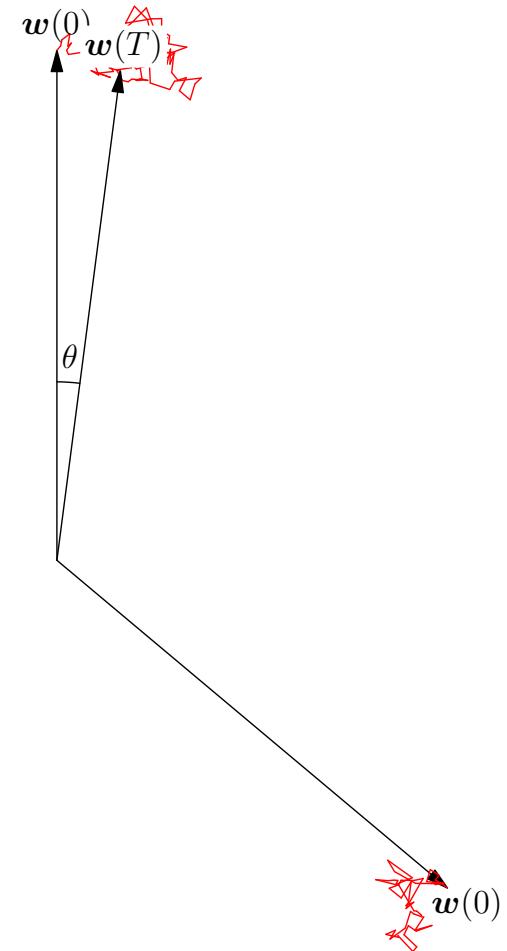
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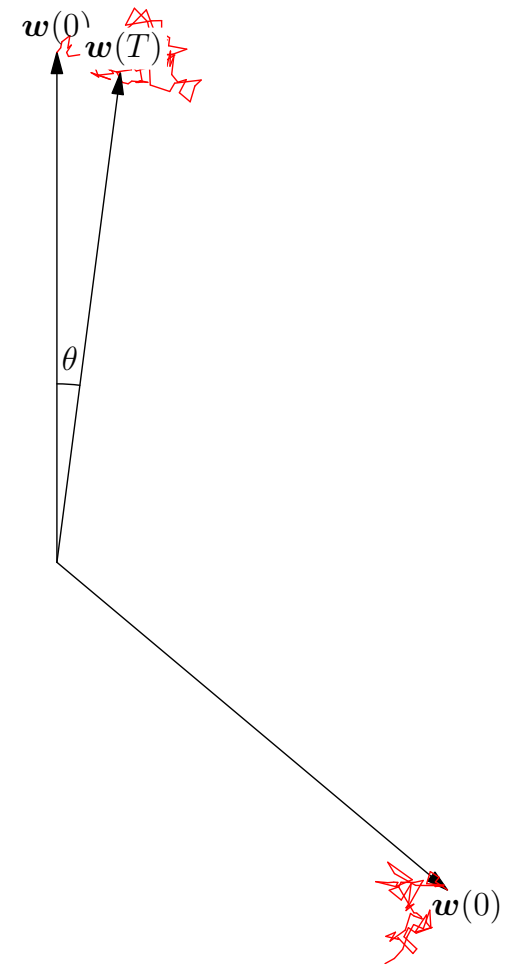
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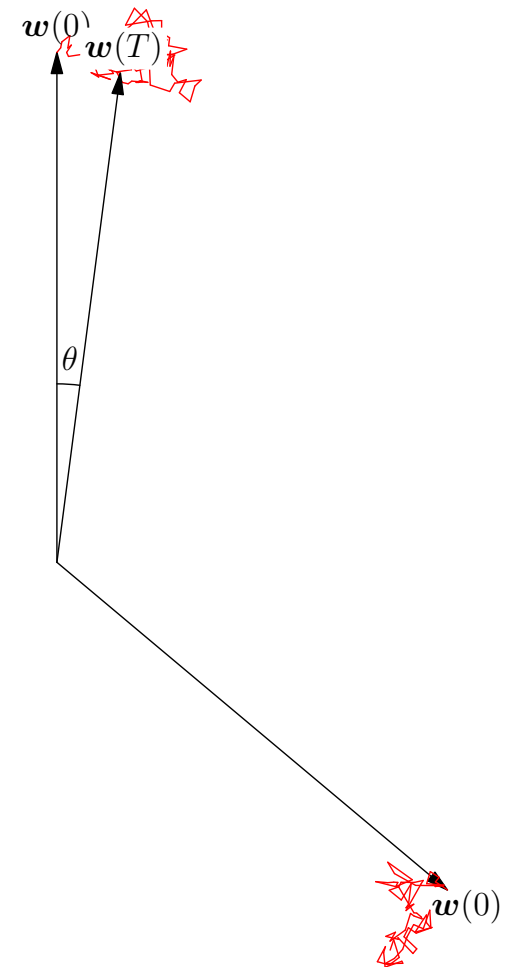
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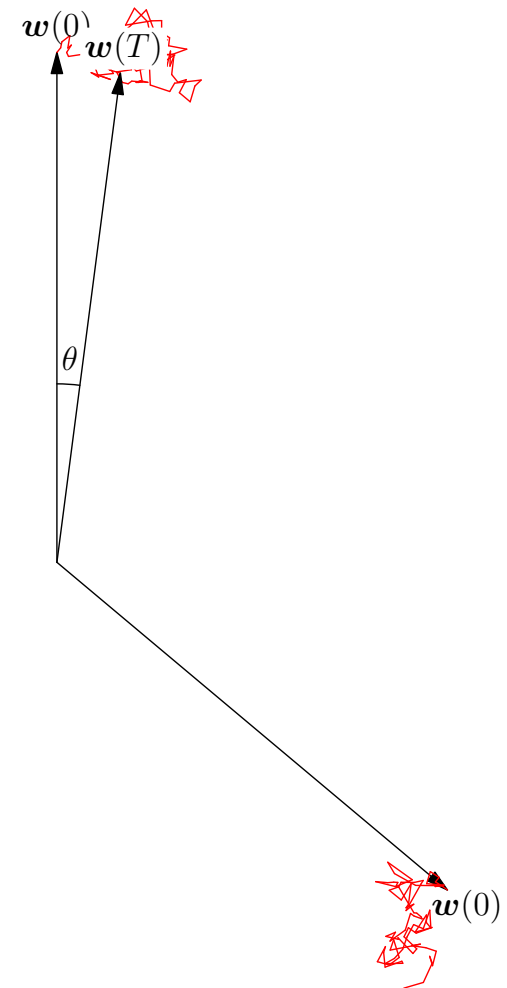
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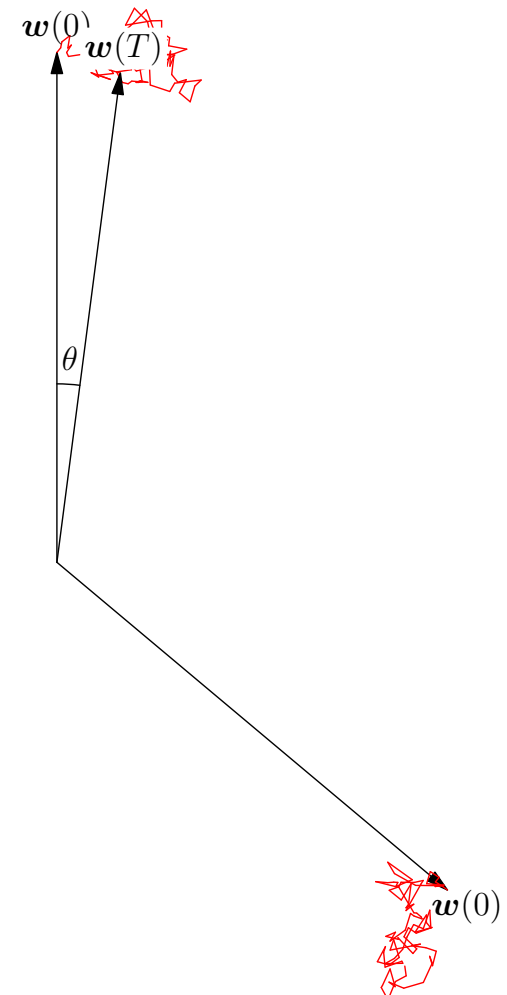
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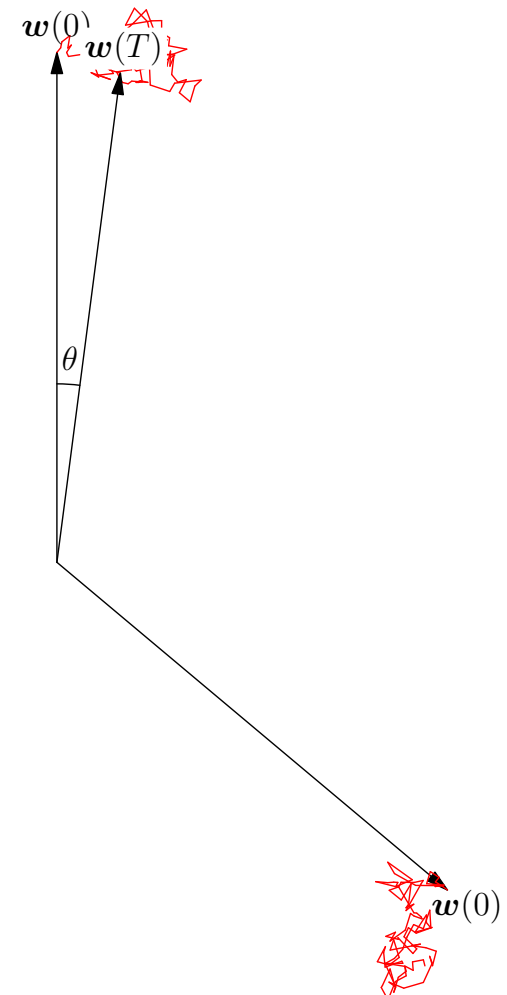
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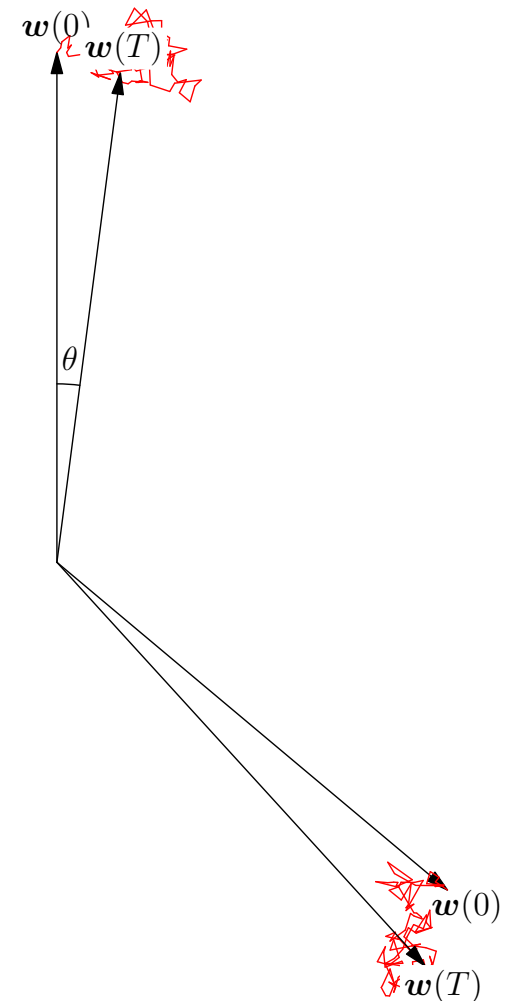
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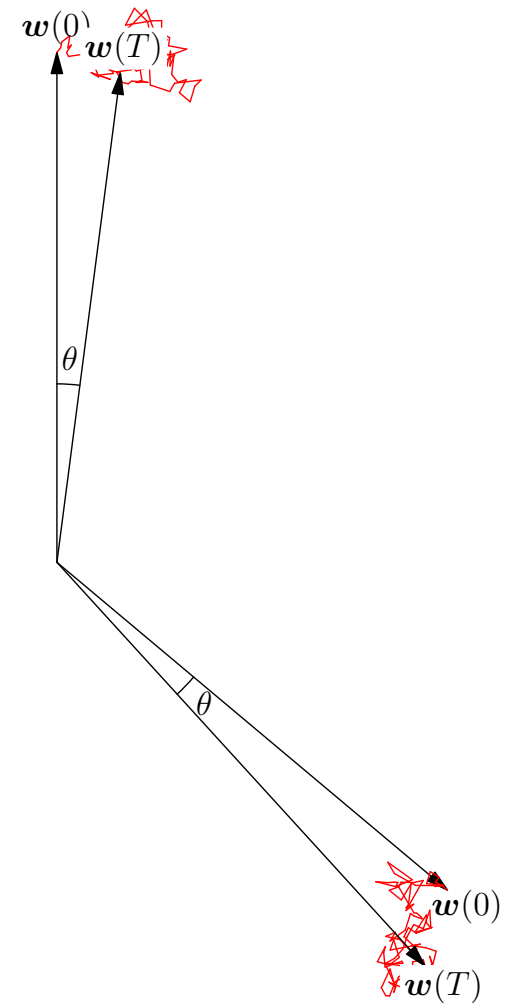
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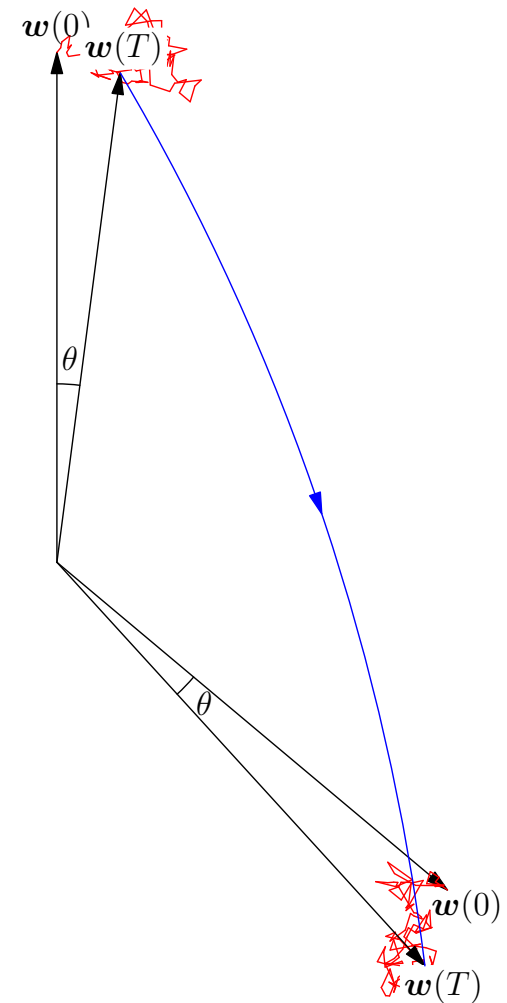
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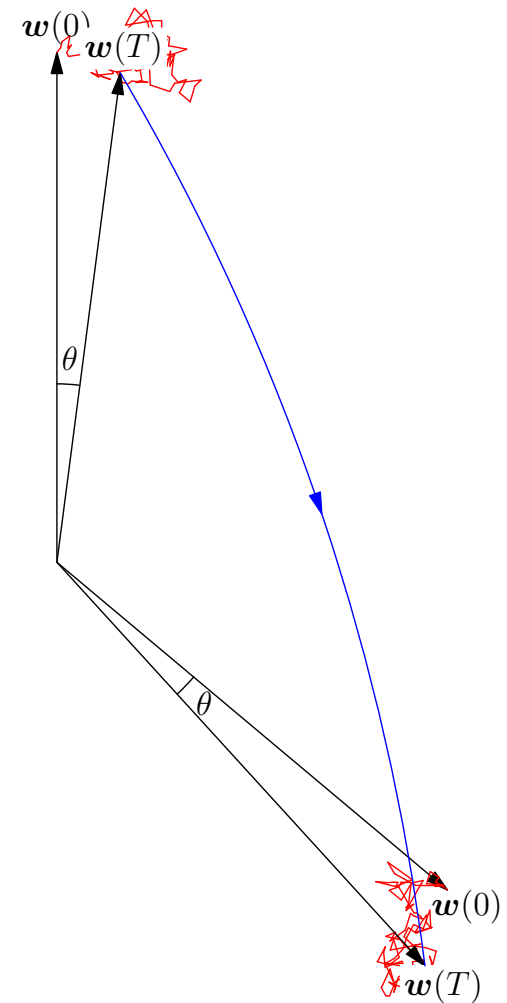
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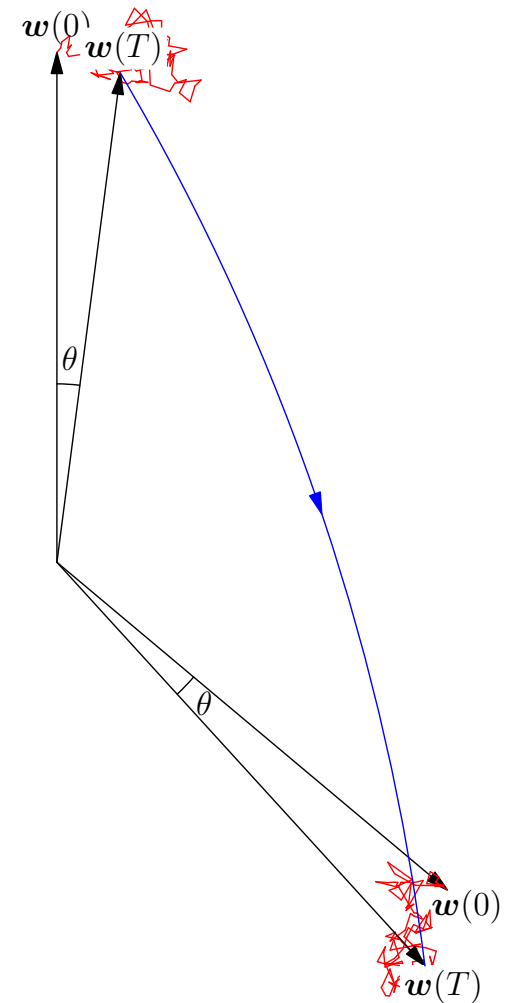
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- Still need to understand that we are exploring a huge and complex loss landscape
- What works is problem specific

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